

Boundary back-propagation verification: complete affected proof bodies

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This verification reader includes complete revised earlier chapters and the complete authoritative boundary proofs. Historical originals and all exact edits are retained beside this source. References to other chapters retain their identifiers for reintegration into the full book.

1 Deligne machinery inside Split-Zero cohomology

1.1 Deligne's image comparison inside the Split-Zero source complexes

This sidebar integrates the delivered specialization-curvature, exponential-comparison, and logarithmic-constituent-control notes. Their logarithmic cohomology and constituent-coupling calculations are reproduced with complete proofs and explicitly joined to the existing Split-Zero complexes; they are attributed to those web contributions. The sidebar supplies the maps connecting the compact-support comparison, its dual, the Split-Zero internal quotient, and the compactified curve of arithmetic representatives. It also calculates the logarithmic de Rham complex of that curve. In particular, it identifies which cohomology groups the exceptional directions occupy and the divisor twist in their support-compatible dual.

The source reference is P. Deligne, *La conjecture de Weil II*, Publications Mathématiques de l'IHÉS **52** (1980), 137–252, https://www.numdam.org/item/PMIHES_1980__52__137_0/. The specific weight comparison is in (3.3.4)–(3.3.6), printed page 206; the preceding geometric argument is in (3.2.8)–(3.2.13). The source-language S20 records for printed pages 201–207 were read in full, including the strict inequality in (3.2.10) and the squared eigenvalue in (3.2.13). The complete S20 typed transcription is retained separately from this integration. The symbol Q_0 in (DS7)–(DS8) denotes a finite-field cardinality; later Q_0 denotes the explicitly defined quotient Gram, as in the supplied logarithmic note; q denotes the dimension of the actual arithmetic packet.

1.1.1 The comparison image, its entire cone, and its dual

Let $c : C^\bullet \rightarrow D^\bullet$ be a cochain map of complexes of vector spaces over a field K . All differentials raise degree by one. Write $Z_C^m = \ker d_C^m$, $B_C^m = \operatorname{im} d_C^{m-1}$, and similarly for D . Define

$$K_c^m = \{x \in Z_C^m : c(x) \in B_D^m\}, \quad I_c^m = \operatorname{im}(H^m(c) : H^m(C) \rightarrow H^m(D)). \quad (\text{DS1})$$

There are exact sequences with the displayed maps:

$$\begin{aligned} 0 \longrightarrow K_c^m / B_C^m &\xrightarrow{[x] \mapsto [x]} H^m(C) \xrightarrow{[x] \mapsto [cx]} I_c^m \longrightarrow 0, \\ 0 \longrightarrow I_c^m &\longrightarrow H^m(D) \longrightarrow H^m(D) / I_c^m \longrightarrow 0. \end{aligned} \quad (\text{DS2})$$

Indeed, $B_C^m \subset K_c^m$ because c commutes with the differentials. The kernel of the second arrow in the first sequence is exactly the classes represented by K_c^m , and every element of I_c^m has the indicated preimage. The second sequence is the defining inclusion and quotient. Thus both the lost source classes and the extra target classes are present.

The single complex retaining both sides is

$$\text{Cone}(c)^m = D^m \oplus C^{m+1}, \quad d(y, x) = (d_D y + cx, -d_C x). \quad (\text{DS3})$$

Expansion gives $d^2(y, x) = (d_D^2 y + d_D cx - cd_C x, d_C^2 x) = 0$. The maps $y \mapsto (y, 0)$ and $(y, x) \mapsto x$ give the exact sequence of complexes $0 \rightarrow D \rightarrow \text{Cone}(c) \rightarrow C[1] \rightarrow 0$, where $C[1]^m = C^{m+1}$ has differential $-d_C$. To make the relevant cohomology segment explicit, a cone cocycle obeys $d_C x = 0$ and $cx = -d_D y$, so its projection represents a class in K_c^{m+1}/B_C^{m+1} . Conversely, if $cx = d_D v$, then $(-v, x)$ is a cone cocycle. A projected class is zero precisely when subtracting a cone boundary leaves a representative $(y, 0)$ with $d_D y = 0$. Finally $(y, 0)$ is a cone boundary precisely when $y = d_D u + cv$ with $d_C v = 0$. This proves the exact segment

$$H^m(C) \xrightarrow{H^m(c)} H^m(D) \longrightarrow H^m(\text{Cone}(c)) \longrightarrow H^{m+1}(C) \xrightarrow{H^{m+1}(c)} H^{m+1}(D). \quad (\text{DS4})$$

Every chain action commuting with c acts on (DS1)–(DS4), since it preserves both cycles and boundaries; on the cone its action is the direct sum of the two specified actions.

For a finite-dimensional linear map $f : V \rightarrow W$, its ordinary linear transpose $f^\vee : W^\vee \rightarrow V^\vee$ gives a perfect pairing

$$\text{im } f \times \text{im } f^\vee \longrightarrow K, \quad (f(v), f^\vee \lambda) \longmapsto \lambda(f(v)). \quad (\text{DS5})$$

If v is changed by $\ker f$, the value is unchanged. If λ is changed by $\ker f^\vee$, its restriction to $\text{im } f$ is unchanged. A nonzero $f(v)$ is detected by some functional on W , and a nonzero $f^\vee \lambda$ is detected by some v , proving nondegeneracy on both sides. In particular

$$(\text{im } f)^\vee \simeq \text{im } f^\vee, \quad \ker f^\vee = (\text{im } f)^\perp, \quad \text{im } f^\vee = (\ker f)^\perp. \quad (\text{DS6})$$

For invertible actions T_V, T_W with $fT_V = T_W f$, the dual actions are $(T_V^{-1})^\vee, (T_W^{-1})^\vee$; their induced actions on the two images make the evaluation in (DS5) invariant. For an additive action A and a specified scalar b , the action $bI - A^\vee$ on the dual instead satisfies $\langle Av, \lambda \rangle + \langle v, (bI - A^\vee)\lambda \rangle = b\langle v, \lambda \rangle$. These are two explicitly defined action conventions, with different types of weight parameter.

Deligne's comparison takes

$$C = R\Gamma_c(X, \mathcal{F}), \quad D = R\Gamma(X, \mathcal{F}), \quad c = c_{X, \mathcal{F}}, \quad I_c^m = \text{im } c_m. \quad (\text{DS7})$$

Here $X = X_0 \otimes_{\mathbb{F}_{Q_0}} \overline{\mathbb{F}_{Q_0}}$, X_0 is smooth of pure dimension N and finite type, and $\mathcal{F}_0$ is a smooth $\overline{\mathbb{Q}_\ell}$ -sheaf, $\ell \neq \text{char } \mathbb{F}_{Q_0}$, pure of weight n . In the source theorem the upper weight bound on H_c^m is $n + m$. The dual of $\mathcal{F}$ has weight $-n$, and Poincaré duality gives the exact degree and twist

$$H^m(X, \mathcal{F})^\vee \simeq H_c^{2N-m}(X, \mathcal{F}^\vee)(N). \quad (\text{DS8})$$

The right side has upper weight $-n + (2N - m) - 2N = -n - m$ by the already established compact-support theorem. Dualizing gives the lower weight bound $n + m$ on H^m . Since c_m is Frobenius equivariant, choose a basis first adapted to its kernel in the source and then to its image in the target. The corresponding action matrices are block triangular. Every eigenvalue on I_c^m therefore occurs on both H_c^m and H^m , with its algebraic conjugates. The two bounds force its weight to equal $n + m$. This proves the image step (3.3.6) from the two stated source bounds and proves exactly why (DS7) retains the cone (DS3).

The tensor comparison has an equally explicit algebraic map. For finite-dimensional $f_a : V_a \rightarrow W_a$ ($1 \leq a \leq r$),

$$\text{im} \left(\bigotimes_{a=1}^r f_a \right) = \bigotimes_{a=1}^r \text{im } f_a \subseteq \bigotimes_{a=1}^r W_a. \quad (\text{DS9})$$

The tensor map sends a pure tensor to the pure tensor of its images, so its image is contained in the right side. Conversely these image pure tensors span the right side and each has a preimage;

inclusivity into the target follows by extending a basis of each image to a basis of W_a . Thus tensor powers preserve the same comparison image. For complexes, the tensor differential is $d(x \otimes y) = dx \otimes y + (-1)^{\deg x} x \otimes dy$; swapping the factors has the sign $(-1)^{\deg x \deg y}$. In characteristic zero a specified symmetry projection is $\frac{1}{r!} \sum_{\pi \in \mathfrak{S}_r} \pi$ with those graded signs. Its idempotence follows because each group element occurs exactly $r!$ times in the double sum. No unspecific tensor eigenvalue is substituted for the eigenvalue on this actual image.

1.1.2 The internal Split-Zero lift of the same maps

For a complex vector space V , set $\mathbf{S}(V) = \{\tau\} \sqcup V^\bullet$. Supported addition and scalar multiplication are the original operations: $v^\bullet + w^\bullet = (v + w)^\bullet$ and $a^\bullet v^\bullet = (av)^\bullet$. External τ is the additive identity and is absorbing for scalar multiplication. The supported zero is $e_V = 0_V^\bullet$. Every linear map $f : V \rightarrow W$ has the canonical lift

$$\mathbf{S}(f)(\tau) = \tau, \quad \mathbf{S}(f)(v^\bullet) = f(v)^\bullet. \quad (\text{DS10})$$

Checking the two supported inputs and inputs containing τ proves additivity, the scalar law and composition. For $B \subseteq V$, $\mathbf{S}(V) \rightarrow \mathbf{S}(V/B)$ sends $b^\bullet$ to $e_{V/B}$ for every $b \in B$, and sends τ to τ . Two supported vectors have the same image exactly when their difference lies in B . This is the coequalizer of inclusion $\mathbf{S}(B) \rightarrow \mathbf{S}(V)$ and the map $b^\bullet \mapsto e_V, \tau \mapsto \tau$: any map equalizing these has equal values on $v^\bullet$ and $(v + b)^\bullet$, by additivity, and hence has a unique factorization through the displayed quotient.

Applying (DS10) to all maps in (DS1)–(DS4) lifts their source kernels, images, quotients and actions. A differential squared is the supported-zero map on represented vectors and still sends τ to τ . The ordinary kernel $\ker H^m(c)$ is carried by the represented fibre of $\mathbf{S}(H^m(c))^{-1}(e)$; the preimage of external absence is only $\{\tau\}$. These two fibres are disjoint. Their exact common carrier is $\mathbf{S}(\ker H^m(c)) = \{\tau\} \sqcup (\ker H^m(c))^\bullet$, with the inclusion of each of the two fibres. If $p_V : \mathbf{S}(V) \rightarrow V$ sends $v^\bullet$ to v and τ to 0 , this carrier is precisely $p_V^{-1}(\ker H^m(c))$. Thus the amplitude projection proves the two inclusions while retaining the two distinct receiving zeros. With a join semilattice of supports, carry out the construction in each fibre and use the original transition maps; (DS1)–(DS4) commute with those transitions because their definitions use the commuting cochain maps. No support label is discarded.

The coefficient comparison remains

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad p_R(a^\bullet) = a, \quad p_R(\tau) = 0. \quad (\text{DS11})$$

It commutes on every supported integer and on τ . In particular the linear/coherent calculation below is the amplitude image of a specified Split-Zero diagram; its supported exceptional kernel remains available before applying p .

1.1.3 The original source relation produces the curve

Fix the actual cyclic object $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi \geq 1$, and the original action $A = M_S$. The monic source spaces $\mathcal{P}_N$, their positive Gram matrices O_N , and their full remainder maps $B_N : \mathcal{P}_N \rightarrow E$ give

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N. \quad (\text{DS12})$$

Take $i \leq j$ with both remainder maps onto, and retain the source inclusion L_{ij} with $B_j L_{ij} = B_i$ and $L_{ij}^* O_j L_{ij} = O_i$. Put

$$\Delta = L_{ij} R_i - R_j, \quad C = \Delta^* O_j \Delta = G_i - G_j, \quad R(z) = R_j + z \Delta. \quad (\text{DS13})$$

The equalities follow directly: $B_N R_N = I$, $R_j^* O_j \Delta = G_j B_j \Delta = 0$, and expansion of $(L_{ij} R_i)^* O_j (L_{ij} R_i)$ gives $G_j + \Delta^* O_j \Delta$. For the kernel order, the actual source projection $P = L_{ij} O_i^{-1} L_{ij}^* O_j$ obeys $P^2 = P$ and $P^* O_j = O_j P$ by the retained source isometry. Therefore $K_j - K_i = B_j (I - P) O_j^{-1} (I - P)^* B_j^* \succeq 0$; expanding the right side gives $B_j O_j^{-1} B_j^* - B_i O_i^{-1} B_i^*$. Therefore

$$B_j R(z) = I, \quad R(z)^* O_j R(z) = G_j + |z|^2 C. \quad (\text{DS14})$$

The source is the original theta source with its arithmetic coefficient unit and every full jet; the new variable z parametrizes only the comparison and does not replace S .

Let $E_0 = \ker \Delta$, $E_1 = E_0^{\perp G_j}$, $r = \dim E_1$, and let P_0, P_1 be the specified projections. On $X = \mathbb{P}_{\mathbb{C}}^1$ with coordinates $[u : v]$, finite coordinate $z = v/u$ and infinity coordinate $w = u/v$, set $D = \{\infty\}$ and

$$\mathcal{E} = (E_0 \otimes \mathcal{O}_X) \oplus (E_1 \otimes \mathcal{O}_X(-D)). \quad (\text{DS15})$$

The source morphism is

$$\Psi(x_0 + x_1 \otimes (u, v)) = R_j x_0 + u R_j x_1 + v \Delta x_1. \quad (\text{DS16})$$

It is fibrewise injective. At $u \neq 0$, apply B_j and use $E_0 \cap E_1 = 0$; at $u = 0$, apply B_j to force $x_0 = 0$ and then use injectivity of $\Delta|_{E_1}$. Thus the original source Gram defines a smooth positive metric on this bundle, including infinity. The observation $\alpha = B_j \Psi$ is identity in the finite frame and $P_0 + w P_1$ in the infinity frame. Consequently

$$0 \rightarrow \mathcal{E} \xrightarrow{\alpha} E \otimes \mathcal{O}_X \rightarrow i_* E_1 \rightarrow 0, \quad (\text{DS17})$$

where $i : D \hookrightarrow X$. In fact its image lattice is $\mathcal{M} = E_0 \mathcal{O}_X + E \mathcal{O}_X(-D)$; at infinity it is $E_0 \mathcal{O} + w E \mathcal{O}$.

Tensoring the two locally free terms of (DS17) with the residue field at infinity gives $[E \xrightarrow{P_0} E]$ in degrees $-1, 0$. Its cohomology is E_1 in both degrees. Intrinsically the negative term is $E_1 \otimes \mathcal{O}(-D)|_D$. This calculation is the derived fibre of the skyscraper cokernel, and explicitly retains the Tor term that ordinary right-exact specialization would omit. Dualizing (DS17) gives

$$0 \rightarrow E^* \mathcal{O}_X \rightarrow \mathcal{E}^\vee \rightarrow i_* (E_1^* \otimes \mathcal{O}(D)|_D) \rightarrow 0. \quad (\text{DS18})$$

To check its terminal term, resolve $\mathcal{O}_D$ by $[\mathcal{O}(-D) \rightarrow \mathcal{O}]$ and apply $\mathcal{H}om(-, \mathcal{O})$. The resulting cokernel is $\mathcal{O}(D)|_D$. The two boundary factors pair by evaluation and $\mathcal{O}(-D)|_D \otimes \mathcal{O}(D)|_D \rightarrow \mathbb{C}_D$.

1.1.4 A calculated compact-support/ordinary de Rham comparison

Transport the trivial connection through α on $X \setminus D$. All connection complexes here are complexes of sheaves of $\mathbb{C}$ -vector spaces: the differentials satisfy the Leibniz rule and are not asserted to be $\mathcal{O}_X$ -linear. In the infinity frame it is

$$\nabla = d + P_1 \frac{dw}{w}, \quad \mathcal{K} = [\mathcal{E} \xrightarrow{\nabla} \mathcal{E} \otimes \Omega_X^1(\log D)] \quad \text{in degrees } 0, 1. \quad (\text{DS19})$$

Indeed $\alpha^{-1} d\alpha = P_1 dw/w$. Its residue is P_1 , its horizontal monodromy is $\exp(-2\pi i P_1) = I$, and the map α is a morphism to $\mathcal{K}_{\text{ord}} = [E \mathcal{O}_X \xrightarrow{d} E \Omega_X^1(\log D)]$.

Here is the complete local calculation of the extension at infinity. On analytic power series the E_0 differential, with dw/w as the one-form frame, is $\sum a_n w^n \mapsto \sum n a_n w^n$. Its kernel and cokernel are the constant terms, both E_0 . The E_1 differential is

$$\sum_{n \geq 0} b_n w^n \mapsto \sum_{n \geq 0} (n+1) b_n w^n. \quad (\text{DS20})$$

It is invertible: its inverse divides the n th coefficient by $n+1$, which preserves every disk of convergence. Thus the E_1 part has zero derived stalk at infinity. On a punctured disk, integrating

a Laurent series term by term divides its w^n coefficient by n for $n \neq 0$; the only one-form class remaining is dw/w with integral $2\pi i$ around a positively oriented w -circle. These statements, together with the usual local primitive on a disk, identify the E_0 part with $Rj_*\underline{E}_0$ and the E_1 part with $j_!\underline{E}_1$, where $j : X \setminus D \hookrightarrow X$. The identification for $j_!$ is the extension-by-zero map of flat sections; its stalk cohomology is an isomorphism on the open disk and is zero on both sides at infinity by (DS20). The identification for Rj_* is the restriction map to the punctured disk and the two Laurent calculations. This constructs the comparison on the present curve rather than assigning compact support by name.

Its entire global cohomology is

$$\mathbb{H}^0(X, \mathcal{K}) = E_0, \quad \mathbb{H}^1(X, \mathcal{K}) = 0, \quad \mathbb{H}^2(X, \mathcal{K}) = E_1 \otimes \mathbb{T}, \quad \mathbb{T} = H^1(X, \Omega_X^1). \quad (\text{DS21})$$

For a direct verification, $\Omega_X^1(\log D) = \mathcal{O}_X(-1)$. The two terms of $\mathcal{K}$ are therefore $E_0\mathcal{O} \oplus E_1\mathcal{O}(-1)$ and $E_0\mathcal{O}(-1) \oplus E_1\mathcal{O}(-2)$. Using the two affine charts, polynomial division of Laurent polynomials proves

$$\begin{aligned} H^0(\mathcal{O}) = \mathbb{C}, \quad H^1(\mathcal{O}) = 0, \quad H^0(\mathcal{O}(-1)) = H^1(\mathcal{O}(-1)) = 0, \\ H^0(\mathcal{O}(-2)) = 0, \quad H^1(\mathcal{O}(-2)) = \mathbb{C}. \end{aligned} \quad (\text{DS22})$$

More explicitly, after identifying $\mathcal{O}(-2)$ with Ω_X^1 , a Laurent one-form on the overlap has every term except dw/w removable by a one-form on one of the charts. The coefficient of dw/w is not removable and is its stated one-dimensional quotient. Retain $\mathbb{T}$ as this quotient line, with its original generator $[dz/z] = -[dw/w]$. The double complex for these two terms has only the entries E_0 in total degree zero and $E_1 \otimes \mathbb{T}$ in total degree two. There is no differential to or from them, proving (DS21) without an uncomputed higher differential.

Taking the quotient of the two complexes in (DS19) gives

$$\mathcal{Q}_D = [i_*E_1 \xrightarrow{0} i_*E_1 dw/w] \quad \text{in degrees } 0, 1. \quad (\text{DS23})$$

The zero differential follows by reducing $w\partial_w$ modulo w . The termwise short exact sequence $0 \rightarrow \mathcal{K} \rightarrow \mathcal{K}_{\text{ord}} \rightarrow \mathcal{Q}_D \rightarrow 0$ and the fact $\mathbb{H}^0(\mathcal{K}_{\text{ord}}) = E$, $\mathbb{H}^{>0}(\mathcal{K}_{\text{ord}}) = 0$ give

$$0 \rightarrow E_0 \rightarrow E \rightarrow E_1 \rightarrow 0, \quad E_1 dw/w \xrightarrow{\delta} E_1 \otimes \mathbb{T} \quad \text{an isomorphism.} \quad (\text{DS24})$$

Fix the Čech convention $s_V - s_U$ for the overlap, with U the finite chart and V the infinity chart, and the total-complex differential $\delta + (-1)^p \nabla$ on Čech degree p . Lifting a boundary one-form as $(0, v dw/w)$ makes its connecting class $v[dw/w]$; thus $\delta(v dw/w) = v \otimes [dw/w] = -v \otimes [dz/z]$. In the original finite coordinate $dw/w = -dz/z$. Both that sign and the chosen positive orientation of the w -circle are retained. Under (DS21), the ordinary comparison image is exactly E_0 in degree zero and zero in degrees one and two. The full E_1 occurs in its boundary comparison and in degree two.

1.1.5 The divisor twist in the support-compatible dual

The bundle dual in (DS18) and the dual de Rham comparison are joined by the exact additional map of line bundles $\mathcal{O}(-D) \hookrightarrow \mathcal{O}$. Define

$$\mathcal{E}^{\text{cd}} = \mathcal{E}^\vee \otimes \mathcal{O}(-D), \quad \mathcal{K}^{\text{cd}} = [\mathcal{E}^{\text{cd}} \xrightarrow{\nabla^{\text{cd}}} \mathcal{E}^{\text{cd}} \otimes \Omega_X^1(\log D)]. \quad (\text{DS25})$$

The dual connection has residue $-P_1^\vee$. Multiplication of its local frame by w , the local generator of $\mathcal{O}(-D)$, adds I to the residue. Hence the residue of (DS25) is $P_0^\vee$, and

$$\mathcal{E}^{\text{cd}} = E_0^*\mathcal{O}(-D) \oplus E_1^*\mathcal{O}, \quad \begin{array}{c|ccc} m & 0 & 1 & 2 \\ \hline \mathbb{H}^m(\mathcal{K}^{\text{cd}}) & E_1^* & 0 & E_0^* \otimes \mathbb{T}. \end{array} \quad (\text{DS26})$$

The proof is exactly the power-series calculation (DS20) and the four line-bundle groups (DS22), now with the two summands exchanged.

Evaluation is a morphism $\mathcal{E} \otimes \mathcal{E}^{\text{cd}} \rightarrow \mathcal{O}(-D)$. The connections obey the Leibniz identity for evaluation, since the dual connection cancels the original connection term. Wedge product therefore gives a cochain map

$$\mathcal{K} \otimes_{\mathbb{C}} \mathcal{K}^{\text{cd}} \longrightarrow [\mathcal{O}(-D) \xrightarrow{d} \Omega_X^1] =: \mathcal{K}_c(\mathbb{C}). \quad (\text{DS27})$$

On a degree-zero tensor it is evaluation; on degree-one tensors it is evaluation times the retained one-form; on two one-forms it is zero because $\Omega_X^2 = 0$. These prescriptions and the tensor sign in (DS9) verify the cochain identity in every degree. The degree-wise evaluation and wedge maps factor through their $\mathcal{O}_X$ -balanced tensor products; the displayed complex tensor product is over $\mathbb{C}$, so its differential is defined. Define the trace on $\mathbb{H}^2(\mathcal{K}_c(\mathbb{C}))$ by

$$\text{Tr}_D([dz/z]) = 1, \quad \text{Tr}_D([dw/w]) = -1 = -\frac{1}{2\pi i} \int_{|w|=\epsilon}^{\text{counterclockwise}} \frac{dw}{w}. \quad (\text{DS28})$$

Holomorphic one-forms on either chart have zero overlap residue, so this is well-defined on (DS22). The induced pairings are

$$E_0 \times E_0^* \rightarrow \mathbb{C}, \quad E_1 \times E_1^* \rightarrow \mathbb{C}, \quad (\text{DS29})$$

ordinary evaluation after applying the retained trace on $\mathbb{T}$, in complementary degrees 0, 2 and 2, 0. They are perfect, since a nonzero vector is detected by a linear functional. Thus (DS25)–(DS29) prove the support-compatible duality, including the dimension-one degree change and its actual boundary line factor. The finite-field analogue of the trace target is $\overline{\mathbb{Q}}_\ell(-1)$; the characteristic-zero calculation here has the explicit residue normalization (DS28).

1.1.6 Retaining the arithmetic action through the exceptional fibre

Transporting the original A through α gives

$$\hat{A}_\infty = (P_0 + w^{-1}P_1)A(P_0 + wP_1) = \begin{pmatrix} A_{00} & wA_{01} \\ w^{-1}A_{10} & A_{11} \end{pmatrix}. \quad (\text{DS30})$$

It extends regularly on $\mathcal{E}$ exactly when $AE_0 \subseteq E_0$, since its sole possible negative power is $w^{-1}P_1AP_0$. The explicit repair on the same ambient arithmetic target is

$$F_A = \sum_{a=0}^{q-1} A^a E_0, \quad \mathcal{M}_A = F_A \mathcal{O}_X + E \mathcal{O}_X(-D). \quad (\text{DS31})$$

Cayley–Hamilton puts $A^q E_0$ in the displayed sum, proving invariance; every invariant subspace containing E_0 contains this sum. Similarly every coherent A -stable submodule of $E\mathcal{O}$ containing $\mathcal{M}$ contains $\sum_{a=0}^{q-1} A^a \mathcal{M}$, which is exactly $\mathcal{M}_A$ in (DS31). There are exact sequences

$$\begin{aligned} 0 \rightarrow \mathcal{M} \rightarrow \mathcal{M}_A \rightarrow i_*(F_A/E_0) \rightarrow 0, \\ 0 \rightarrow \mathcal{M}_A \rightarrow E\mathcal{O}_X \rightarrow i_*(E/F_A) \rightarrow 0. \end{aligned} \quad (\text{DS32})$$

The second is A equivariant on every term. The connection d on $E\mathcal{O}$ preserves $\mathcal{M}_A$ logarithmically, because $d(wv) = w dv + v dw$ lies in $\mathcal{M}_A \otimes \Omega^1(\log D)$. It commutes with the constant original A . Define its two-term complex $\mathcal{K}_A$ with the same degree convention as (DS19). Calculation with any vector-space complement of F_A proves, intrinsically,

$$\mathbb{H}^0(\mathcal{K}_A) = F_A, \quad \mathbb{H}^1(\mathcal{K}_A) = 0, \quad \mathbb{H}^2(\mathcal{K}_A) = (E/F_A) \otimes \mathbb{T}, \quad (\text{DS33})$$

with actions $A|_{F_A}$ and $\overline{A}$, respectively. The identifications are independent of that auxiliary complement: the degree-zero map is inclusion into E and the degree-two map is the connecting quotient map in (DS32), as in (DS24).

Its dual lattice has a coordinate-free formula. Put $F_A^\perp = \{\lambda \in E^* : \lambda(F_A) = 0\}$. Then

$$\mathcal{M}_A^\vee(-D) = F_A^\perp \mathcal{O}_X + E^* \mathcal{O}_X(-D). \quad (\text{DS34})$$

Indeed a meromorphic functional belongs to the left side precisely when its value on every element of $F_A \mathcal{O} + wE\mathcal{O}$ lies in $w\mathcal{O}$. This first forces it to be holomorphic, by testing wE , and then forces its constant term to annihilate F_A . This is exactly the right side. Fix the additive dual action as $-A^\vee$, so evaluation has the zero action on its one-dimensional target. The displayed lattice is invariant under this action because F_A is invariant. Equations (DS27)–(DS29) therefore become the perfect equivariant pairings $F_A \times F_A^*$ and $(E/F_A) \times F_A^\perp$. For the original real integer k , replacing the dual action by $kI - A^\vee$ puts the scalar k action on the evaluation target, as checked in (DS6). The passage to the metric antidual is the specified conjugate-linear map $\mathfrak{c} : E^\vee \rightarrow E^\vee$, $\mathfrak{c}(\lambda)(v) = \overline{\lambda(v)}$. In the fixed polynomial basis it conjugates the functional's coefficient column, and hence $\mathfrak{c}(kI - A^T) = (kI - A^*)\mathfrak{c}$. This proves the exact coefficient map between the linear dual and the original metric antidual convention. For every polynomial p , an adapted basis to F_A proves

$$\text{Tr } p(A) = \text{Tr } p(A|_{F_A}) + \text{Tr } p(\overline{A}). \quad (\text{DS35})$$

This retains the complete arithmetic trace across the calculated cohomological degrees. There is no need to presume that the metric kernel E_0 was an arithmetic constituent.

The source cost of (DS31) remains visible. If $s_A = \dim F_A - \dim E_0$, choose a basis first for E_0 , then for F_A/E_0 , and then for E/F_A . The local inclusion $\mathcal{M} \rightarrow \mathcal{M}_A$ has diagonal blocks I, wI_{s_A}, I and determinant w^{s_A} . If the metric on $\mathcal{M}_A$ is transported on the punctured chart from the original source metric on $\mathcal{M}$, the determinant identity is

$$\det G_A(w) = |w|^{-2s_A} \det G_{\mathcal{M}}(w). \quad (\text{DS36})$$

It follows by taking determinants of $G_{\mathcal{M}} = J^* G_A J$. Thus the action-stable complex (DS33) retains the exceptional contribution through an explicitly measured metric pole.

1.1.7 The same curvature, its source map, and the weight interface

For $\mathcal{L} = (\det \mathcal{E})^\vee$ with its original source metric, put

$$Z = G_j^{-1}(G_i - G_j) = (K_j - K_i)G_i. \quad (\text{DS37})$$

The second equality is $K_j G_i - I = (K_j - K_i)G_i$. The first expression proves self-adjointness and positivity in G_j ; the second proves the same in G_i , because $G_i Z = G_i(K_j - K_i)G_i \succeq 0$ and $K_j - K_i \succeq 0$. No zero eigenvalue is removed. Differentiating $\log \det(G_j + |z|^2 C)$ yields

$$c_1(\mathcal{L}) = \frac{i}{2\pi} \kappa(|z|^2) dz \wedge d\bar{z}, \quad \kappa(t) = \text{Tr}(Z(I + tZ)^{-2}). \quad (\text{DS38})$$

To verify every coefficient, set $f(t) = \log \det(I + tZ)$. Then $f'(t) = \text{Tr}(Z(I + tZ)^{-1})$ and $f'(t) + t f''(t) = \kappa(t)$. Since $\partial \bar{\partial} f(|z|^2) = (f'(|z|^2) + |z|^2 f''(|z|^2)) dz \wedge d\bar{z}$, this proves (DS38) with the stated sign. For a positive eigenvalue λ , $\int_0^\infty \lambda(1 + t\lambda)^{-2} dt = 1$; for a zero eigenvalue the integrand is zero. Since $i dz \wedge d\bar{z} = 2dx \wedge dy$, polar integration gives the degree and the logarithmic moment

$$\int_X c_1(\mathcal{L}) = r, \quad \log \frac{\det G_i}{\det G_j} = \int_0^1 \log(1/t) \kappa(t) dt. \quad (\text{DS39})$$

For the second equality, write $M(t) = t f'(t)$, so $M' = \kappa$ and $M(0) = 0$. Nonnegative Fubini gives $\int_0^1 \log(1/t) \kappa(t) dt = \int_0^1 t^{-1} M(t) dt = f(1) - f(0)$.

The full source observation is also retained. With

$$P_z = R(z)G(z)^{-1}R(z)^*O_j, \quad \mathcal{N}_z = (I - P_z)\Delta, \quad G(z) = G_j + |z|^2 C, \quad (\text{DS40})$$

orthogonal projection gives

$$\mathcal{N}_z^* O_j \mathcal{N}_z = C - |z|^2 C G(z)^{-1} C, \quad \text{Tr}(G(z)^{-1} \mathcal{N}_z^* O_j \mathcal{N}_z) = \kappa(|z|^2). \quad (\text{DS41})$$

Indeed $R(z)^* O_j \Delta = \bar{z} C$; substitute into $\Delta^* O_j (I - P_z) \Delta$. Factoring $G(z) = G_j(I + |z|^2 Z)$ in the trace proves the second equality. Applying the actual remainder map gives

$$B_j P_z \Delta = \bar{z} G(z)^{-1} C, \quad B_j \mathcal{N}_z = -\bar{z} G(z)^{-1} C. \quad (\text{DS42})$$

Their sum is the original theta relation. In the split lift its quotient value is the supported zero e , with both prequotient pieces still given by (DS42).

The curvature jets are the retained relation moments:

$$\kappa^{(n)}(0) = (-1)^n (n+1)! \text{Tr}(Z^{n+1}), \quad t_1 = \kappa(0), \quad t_2 = -\frac{1}{2} \kappa'(0), \quad \mathcal{A}_2 = \frac{1}{2} (t_1^2 - t_2). \quad (\text{DS43})$$

These follow by differentiating each rational eigenvalue term in (DS38); finiteness of the matrix makes the differentiation exact. The dimension-one divisor degree in (DS39), the arithmetic trace in (DS35), and the source metric pole in (DS36) are now related by specified maps on the same compactified curve.

For completeness, exponentiation supplies the precise multiplicative observer of the original additive action, for every real t :

$$F_t = \exp(tA) : E \rightarrow E, \quad F_t|_{F_A} = \exp(tA|_{F_A}), \quad \bar{F}_t = \exp(t\bar{A}). \quad (\text{DS44})$$

On the full generalized eigenspace of A at ρ , write $A = \rho I + N$, $N^m = 0$ with its actual order. Then $F_t = e^{t\rho} \sum_{a=0}^{m-1} t^a N^a / a!$; in particular it retains the nilpotent terms and has eigenvalue modulus $e^{t \text{Re} \rho}$. Choosing $t = \log Q_0$ gives the exact number $Q_0^{\text{Re} \rho}$. This is the proved operator map connecting the additive spectral parameter and a multiplicative weight observer. It does not provide a finite-field descent or an étale Frobenius structure. Those structures in (DS7) have the specific types $X_0/\mathbb{F}_{Q_0}$, a smooth $\overline{\mathbb{Q}}_\ell$ -sheaf and its geometric Frobenius action; the constructed characteristic-zero object has instead the fully calculated complexes (DS19), (DS25), (DS33) and the commuting action (DS44). Its complete comparison to those source operations is (DS1)–(DS9), (DS24), (DS27), and (DS34), with their actual kernels.

Finally the original positive metric supplies its precise additive weight measurement on every invariant inclusion $I_F : F \hookrightarrow E$:

$$G_F = I_F^* G(z) I_F, \quad W_F = A_F^* G_F + G_F A_F - k G_F, \quad \text{Tr}(G_F^{-1} W_F) = 2 \text{Re} \text{Tr} A_F - k \dim F. \quad (\text{DS45})$$

The identity follows by cyclicity of the finite trace, with $A I_F = I_F A_F$. Thus k remains the original tensor parameter, and all constituents of E are retained through (DS32)–(DS35). Equations (DS38)–(DS43) make the current endpoint estimate an estimate on a specified source-induced curvature measure. Its total degree alone does not bound its logarithmic moment: for the explicitly declared matrix family $G_j = I_r$, $C = L I_r$, $L > 0$, these quantities are respectively r and $r \log(1 + L)$, and the mass on $0 \leq t \leq 1/L$ is exactly $r/2$ by direct integration. The original arithmetic source, its relation map (DS13), its full dual comparison (DS25)–(DS34), and the actual moment observers (DS43) therefore remain the concrete data for the continuing curvature calculation.

1.1.8 Every divisor depth and the unchanged conormal directions

For a specified A -invariant subspace $F \subseteq E$ and an integer $m \geq 1$, retain

$$\mathcal{M}_{F,m} = F \mathcal{O}_X + E \mathcal{O}_X(-mD), \quad \mathcal{K}_{F,m} = [\mathcal{M}_{F,m} \xrightarrow{d} \mathcal{M}_{F,m} \otimes \Omega_X^1(\log D)]. \quad (\text{DS46})$$

The local coefficient differential on $w^m(E/F)$ is $w\partial_w + m$. The inclusion $\mathcal{K}_{F,m} \hookrightarrow \mathcal{K}_{F,1}$ has quotient supported at D , with coefficient basis $w, w^2, \dots, w^{m-1}$ in each of degrees zero and one and differential

$$w^a v \mapsto aw^a v dw/w \quad (1 \leq a < m). \quad (\text{DS47})$$

Define $h(w^a v dw/w) = a^{-1}w^a v$ on degree one and $h = 0$ on degree zero. Then $dh + hd = I$ on each displayed basis element, so this entire quotient is contractible. Thus the inclusion is a quasi-isomorphism, equivariant for the original A , and

$$\mathbb{H}^0(\mathcal{K}_{F,m}) = F, \quad \mathbb{H}^1(\mathcal{K}_{F,m}) = 0, \quad \mathbb{H}^2(\mathcal{K}_{F,m}) = (E/F) \otimes \mathbb{T}. \quad (\text{DS48})$$

This preserves every higher coefficient until its explicit differential (DS47) contracts it. In original arithmetic degree k the shifted complex is $\mathcal{K}_{F,m}[-k]$, so the two surviving degrees are $k, k+2$ and

$$\text{Str}(p(A) \mid R\Gamma \mathcal{K}_{F,m}[-k]) = (-1)^k \text{Tr}_E p(A). \quad (\text{DS49})$$

The equality follows from (DS35); multiplication by $p(A)$ acts trivially on the factor $\mathbb{T}$.

The exact higher-depth dual is

$$\mathcal{M}_{F,m}^\vee(-D) = F^{\text{ann}} \mathcal{O}_X((m-1)D) + E^* \mathcal{O}_X(-D) \subset E^* \otimes \mathcal{O}_X(*D). \quad (\text{DS50})$$

At infinity, testing a meromorphic functional on $w^m E$ permits poles of order at most $m-1$, and testing on F forces every coefficient of powers $w^{1-m}, \dots, w^0$ to annihilate F . Its remaining coefficients are $wE^* \mathcal{O}$. This proves (DS50), including its ambient meromorphic sheaf. In a coefficient complement its residues are 1 on F^* and $1-m$ on $(E/F)^*$. The natural inclusion of the depth-one dual into (DS50) has quotient basis $w^{1-m}, \dots, w^{-1}$ in both degrees, with differential $w^a v \mapsto aw^a v dw/w$ and inverse a^{-1} , so it is contractible. Therefore its dual cohomology is F^{ann} in degree zero and $F^* \otimes \mathbb{T}$ in degree two. Evaluation into $\mathcal{O}(-D)$ and (DS27) give the same perfect complementary-degree pairings. All negative powers, residues and the divisor twist have been retained.

The independent conormal depth has the explicit map

$$\frac{\mathbb{C}[w] \otimes \mathcal{P}}{(w^m) + I^a(\mathbb{C}[w] \otimes \mathcal{P})} \xrightarrow{\sim} (\mathbb{C}[w]/(w^m)) \otimes (\mathcal{P}/I^a), \quad [f(w)P] \mapsto [f] \otimes [P], \quad (\text{DS51})$$

where $\mathcal{P} = \mathbb{C}[s_1, \dots, s_k]$ and $I = (h(s_1), \dots, h(s_k))$. Reduction of the w -degree writes the left side uniquely as $\sum_{b=0}^{m-1} w^b [P_b]_{I^a}$. This supplies the inverse and proves injectivity as well as surjectivity. Restriction in either depth commutes with the other. Differentiation in w has the well-defined type from depth $m+1$ to depth m ; differentiation in s_j has type from I^{a+1} quotients to I^a quotients, because $\partial_{s_j} I^{a+1} \subset I^a$. On pure tensors the two derivatives commute, hence on their full spans. Lifting these maps by (DS10) retains the same support label and the receiving supported zero in both depth directions.

Pullback of (DS38) under the actual map $z \mapsto z^m$ has radial density

$$\kappa_m(t) = m^2 t^{m-1} \kappa(t^m), \quad t = |z|^2. \quad (\text{DS52})$$

Indeed differentiate $f(t^m)$ twice in the combination $f'_m + t f''_m$ used in (DS38). Substitution $s = t^m$ proves $\int_0^\infty \kappa_m(t) dt = mr$ and $\int_0^1 \log(1/t) \kappa_m(t) dt = \int_0^1 \log(1/s) \kappa(s) ds$. Thus increasing divisor depth multiplies degree by m while leaving the original endpoint logarithmic moment fixed.

1.1.9 The invariant quotient gives an exact coupling subtraction

The following quantitative calculation is the delivered logarithmic constituent-control result, now proved in the same source notation. Take $I : F \hookrightarrow E$ and $\pi : E \twoheadrightarrow E/F$ with $AI = IA_F$ and $\pi A = A_Q \pi$, and put $G(t) = G_0 + tC$, $G_0 = G_j$, $0 \leq t \leq 1$. The cases $F = 0, E$ use the empty

determinant 1, empty trace 0, and the unique maps on the zero-dimensional space; all formulas below still hold. For any initial coefficient lift $\overline{H}$, define

$$H_0 = \overline{H} - I(I^*G_0I)^{-1}I^*G_0\overline{H}, \quad \pi H_0 = I_{E/F}, \quad I^*G_0H_0 = 0. \quad (\text{DS53})$$

These last two identities follow from $\pi I = 0$ and matrix multiplication. Changing $\overline{H}$ by IU cancels the change in (DS53); thus the orthogonal lift itself is independent of the initial lift. In the original bases define

$$\begin{aligned} G_{F,0} &= I^*G_0I, \quad C_F = I^*CI, \quad D_F = I^*CH_0, \\ Q_0 &= H_0^*G_0H_0, \quad C_Q = H_0^*CH_0, \quad G_F(t) = G_{F,0} + tC_F, \\ H_t &= H_0 - tIG_F(t)^{-1}D_F. \end{aligned} \quad (\text{DS54})$$

Here Q_0 is the original quotient Gram, not the finite-field cardinality used in (DS7). Every inverse is defined: I and H_0 are injective on their respective coefficient spaces, and $G(t) > 0$. In coordinates $[I, H_0]$ the original metric is exactly

$$\begin{pmatrix} G_F(t) & tD_F \\ tD_F^* & Q_0 + tC_Q \end{pmatrix}. \quad (\text{DS55})$$

The replacement $H_0 \mapsto H_t$ is the upper triangular coefficient change $\begin{pmatrix} I & -tG_F(t)^{-1}D_F \\ 0 & I \end{pmatrix}$ of determinant 1. It gives

$$I^*G(t)H_t = 0, \quad \pi H_t = I_{E/F}, \quad Q(t) := H_t^*G(t)H_t = Q_0 + tC_Q - t^2D_F^*G_F(t)^{-1}D_F > 0. \quad (\text{DS56})$$

Expansion proves the formula, and injectivity of H_t proves positivity. It is also the quotient norm: any other lift of v is $H_tv + Ix$, whose squared norm is $v^*Q(t)v + x^*G_F(t)x$ by orthogonality. The unique minimum is therefore attained at H_tv .

Define the actual coupling map

$$\beta_t = G_F(t)^{-1}I^*CH_t : E/F \rightarrow F. \quad (\text{DS57})$$

Differentiate $\pi H_t = I$ to write $H'_t = IU_t$; differentiating $I^*G(t)H_t = 0$ then gives $I^*CH_t + G_F(t)U_t = 0$. Therefore $H'_t = -I\beta_t$. In the derivative of $Q(t) = H_t^*G(t)H_t$ the two terms containing H'_t vanish by orthogonality. Differentiating the remaining term gives

$$Q'(t) = H_t^*CH_t, \quad Q''(t) = -2\beta_t^*G_F(t)\beta_t. \quad (\text{DS58})$$

The factor two follows because $I^*CH_t = G_F(t)\beta_t$ and its adjoint supply the two equal terms. The exact integral remainder for $Q(1)$ is $Q(1) = Q(0) + Q'(0) + \int_0^1 (1-t)Q''(t)dt$; insert (DS56) at the two endpoints to get

$$\Pi_F := D_F^*G_F(1)^{-1}D_F = 2 \int_0^1 (1-t)\beta_t^*G_F(t)\beta_t dt. \quad (\text{DS59})$$

This identifies the finite subtraction with an integrated squared map on the original quotient source.

The action defect is the unique map $a_t : E/F \rightarrow F$ with

$$AH_t - H_tA_Q = Ia_t, \quad a'_t = \beta_t A_Q - A_F \beta_t. \quad (\text{DS60})$$

The first left side has zero image under π , so it lies in $\text{im } I$, which proves existence and uniqueness. Differentiating and inserting $H'_t = -I\beta_t$ proves the second formula, using injectivity of I . Thus the nilpotent extension data are retained in a_t even when the two constituent actions have equal eigenvalues.

For clarity the entire Sylvester operator has a finite exact description. On a pair of primary spaces let $A_F = \lambda I + N_F$, $A_Q = \mu I + N_Q$, with $N_F^a = 0$ and $N_Q^b = 0$. Put $\mathcal{N}(X) = XN_Q - N_F X$. The two multiplications commute, so the binomial expansion makes $\mathcal{N}^{a+b-1} = 0$. If $\mu \neq \lambda$, the inverse of $X \mapsto XA_Q - A_F X$ is

$$Y \mapsto \sum_{r=0}^{a+b-2} \frac{(-1)^r}{(\mu - \lambda)^{r+1}} \mathcal{N}^r(Y). \quad (\text{DS61})$$

Multiplying the finite geometric series by $(\mu - \lambda)I + \mathcal{N}$ leaves identity and a zero terminal power. At $\mu = \lambda$, the operator is exactly $\mathcal{N}$; its kernel is the space $\{X : XN_Q = N_F X\}$ and its image is the span of the explicit matrices $E_{uv}N_Q - N_F E_{uv}$. Matrix-unit expansion proves that this span is its whole image. This keeps the kernel and every coefficient of the image without a nonexistent inverse.

The same original control $W(t) = A^*G(t) + G(t)A - kG(t)$ obeys

$$\begin{aligned} I^*W(t)I &= A_F^*G_F(t) + G_F(t)A_F - kG_F(t), \\ H_t^*W(t)H_t &= A_Q^*Q(t) + Q(t)A_Q - kQ(t). \end{aligned} \quad (\text{DS62})$$

The first identity uses invariance of F . For the second insert (DS60); both terms containing Ia_t vanish by (DS56). On the original theta source write $r(z) = \mathcal{V}R(z)$ and retain its actual primitive $D^{(k)}r(z) - r(z)A = dK(z)$, with $K(z) = (1 - z)K_j + zK_i$ in the inherited primitive notation. Then exact multiplication gives

$$D^{(k)}r(z)H_t - r(z)H_tA_Q = dK(z)H_t + r(z)Ia_t, \quad t = |z|^2. \quad (\text{DS63})$$

Here $D^{(k)}$ differentiates the original source variables, not the comparison parameter z or t . The first term is killed by the original theta quotient. The second has its retained image in the arithmetic constituent F and is killed by the further quotient $E \rightarrow E/F$. Formula (DS63) proves the precise map between these two successive supported-zero operations.

Put $Q_+ = Q_0 + C_Q$, $\Xi_F = Q_+^{-1}\Pi_F$. Then Ξ_F is self-adjoint and nonnegative for Q_+ , and $Q(1) = Q_+ - \Pi_F > 0$ proves that all its eigenvalues belong to $[0, 1)$. Taking determinants in (DS55)–(DS56), with the fixed coefficient determinant $|\det[I, H_0]|^2$ cancelled only between the same two original endpoints, gives

$$\begin{aligned} \log \frac{V_i}{V_j} &= \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} + \log \det(I - \Xi_F) \\ &= \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r}. \end{aligned} \quad (\text{DS64})$$

For each eigenvalue $0 \leq x < 1$, integrating the geometric series $\sum_{a \geq 0} x^a = (1 - x)^{-1}$ from zero to x proves the logarithm series. There are finitely many eigenvalues, so their sum yields the second equality and every term is nonnegative. Its first p terms therefore give the stated upper estimate, and these estimates decrease to the exact original endpoint volume.

There is also an explicit remainder from the same matrices. If $\rho = \lambda_{\max}(\Xi_F) < 1$, then for every $p \geq 0$,

$$0 \leq \sum_{r=p+1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r} \leq \frac{\text{Tr}(\Xi_F^{p+1})}{(p+1)(1-\rho)}. \quad (\text{DS65})$$

Indeed $\text{Tr}(\Xi_F^{p+1+a}) \leq \rho^a \text{Tr}(\Xi_F^{p+1})$ on each nonnegative eigenvalue, and $p+1+a \geq p+1$; summing the resulting geometric series proves the inequality. The value ρ is the largest generalized eigenvalue of (Π_F, Q_+) of this same source pair, with no independently assumed uniform gap. On the zero-dimensional quotient $E/F = 0$, define $\rho = 0$; every trace and both sides of (DS65) are then zero, so the same formula includes that endpoint.

The two positive terms in (DS64) are the logarithmic determinants of the compressed positive operators $Z_F = G_{F,0}^{-1}C_F$ and $Z_Q = Q_0^{-1}C_Q$. Consequently the already proved two-trace majorant, applied to each of these specified matrices in its own full dimension, can be combined with every finite subtraction in (DS64). This integration retains the extension map (DS60), the quotient source (DS63), and the original total curvature (DS39) together; the cohomological constituent decomposition does not remove its measurable coupling.

1.1.10 The degree-one exponential family joins the same coefficient object

The delivered exponential-comparison note constructs the additional degree-one family from the unchanged polynomial $\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a$. Its immediate join to the present cohomology calculation is as follows. Use new independent parameters u, t and retain every coefficient of

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1}, \quad \Phi_t(S) = \Phi(S) - tS, \quad R = \mathbb{C}[u, t]. \quad (\text{DS66})$$

The two-term coefficient complex is

$$\mathcal{C} = [R[S] \xrightarrow{\mathcal{D}} R[S]dS], \quad \mathcal{D}P = (uP' + (\chi - t)P)dS. \quad (\text{DS67})$$

For a nonzero polynomial P of degree a and leading coefficient $b \in R$, $\mathcal{D}P/dS$ has degree $a + q$ and leading coefficient b . Successively subtracting $\mathcal{D}(bS^a)/dS$ cancels each highest term in a polynomial of degree at least q and terminates. Uniqueness follows from the same degree calculation. Hence $R[S] = \text{im}(\mathcal{D}/dS) \oplus R[S]_{<q}$ as coefficient modules, $H^0(\mathcal{C}) = 0$, and $\mathcal{H} = H^1(\mathcal{C})$ is free with the original basis $[1], \dots, [S^{q-1}]$. At $u = t = 0$ the differential is multiplication by the literal χ , so $\mathcal{H}/(u, t)\mathcal{H} = E$ with every nilpotent jet.

The commutators are $[\mathcal{D}/dS, \partial_t] = I$ and $[\mathcal{D}/dS, -S/u] = -I$ after inverting u . They prove that $\nabla_t = \partial_t - S/u$ descends through the differential. Define $\mathcal{R} : E \rightarrow E$ by sending a remainder to its S^{q-1} coefficient times $[1]$. Reducing S^q with the relation $[\chi - t] = 0$ shows in the displayed basis that

$$\nabla_t = \partial_t - \frac{A + t\mathcal{R}}{u}, \quad (-u\nabla_t) \bmod (u, t) = A. \quad (\text{DS68})$$

The regular operator $-u\partial_t + A + t\mathcal{R}$ preserves $(u, t)\mathcal{H}$: on uv its derivative contribution is $-u^2\partial_tv$, and on tv it is $-uv - ut\partial_tv$. This verifies the asserted specialization as an actual induced operator, rather than specializing a singular connection symbol.

The same family now also has the full u -connection. Use the literal division (XD.u1) for this χ , with matrices $C_\chi(t)$, B_χ and $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$. The connections on the two terms of (DS67) are

$$\partial_u - \Phi_t/u^2 + 1/u \quad \text{and} \quad \partial_u - \Phi_t/u^2, \quad (\text{DS70})$$

in degrees zero and one. Their chain identity is the commutator $[u\partial_S + \chi - t, \partial_u - \Phi_t/u^2] = -(u\partial_S + \chi - t)/u$, proved before quotienting in (XD.u4). On an ordered product of k such polynomial factors, the same calculation in cochain degree j gives

$$\nabla_u^{(j)} = \partial_u - \frac{\sum_i \Phi_{t_i}(S_i)}{u^2} + \frac{k-j}{u}. \quad (\text{DS71})$$

Each summand of the exterior differential has commutator $-L_i/u$; the adjacent-degree scalar difference is $+L_i/u$, so all terms cancel with their original exterior signs. This includes the non-top degrees rather than imposing the top formula on them.

There is an exact source lift for the fixed cyclic observer already used in (DS69). On parameter-dependent vectors of its original admitted theta-source domain define

$$\nabla_{u,N}^{\text{src}} = \partial_u + r_N \Omega_u j_E, \quad \nabla_{t,N}^{\text{src}} = \partial_t + r_N \Omega_t j_E, \quad \Omega_t = -(A + t\mathcal{R})/u. \quad (\text{DS72})$$

The domain is the original one on which j_E , $D^{(k)}$, r_N and the boundary primitive are defined, with holomorphic parameter coefficients on a branch of $u \neq 0$. The maps j_E, r_N are constant in these parameters. Multiplying by $j_E r_N = I$ gives

$$j_E \nabla_{a,N}^{\text{src}} = (\partial_a + \Omega_a) j_E, \quad \nabla_{a,N}^{\text{src}} r_N = r_N (\partial_a + \Omega_a), \quad a = u, t. \quad (\text{DS73})$$

The source splitting is the exact identity $x = r_N j_E x + (x - r_N j_E x)$, with its second term in $\ker j_E$. On that kernel (DS72) is ordinary parameter differentiation. Since $r_N j_E r_N j_E = r_N j_E$, the full source curvature equals $r_N (\partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t]) j_E = 0$ by (XD.u6). The original source differential still gives, for each $a = u, t$,

$$[D^{(k)}, \nabla_{a,N}^{\text{src}}] = r_N [A, \Omega_a] j_E + dK_N \Omega_a j_E. \quad (\text{DS74})$$

To prove this, insert $D^{(k)} r_N = r_N A + dK_N$ and $j_E D^{(k)} = A j_E$ into the commutator; $D^{(k)}$ commutes with parameter differentiation. Every term has the original source target, and $K_N \Omega_a j_E$ is the displayed full primitive. Thus the parameter-flat source observation and the original Euler action are related by their exact commutator, retaining its coefficient part and theta boundary. No source norm or full Taylor unit in r_N has been changed.

The logarithmic comparison retains the original exact coefficient sequence $0 \rightarrow F \rightarrow E \rightarrow E/F \rightarrow 0$, with the outer terms occurring in (DS48). The common full E is obtained through its actual outer comparison complexes: $\mathbb{H}^0(\mathcal{K}_{E,1}) = E$ and $\mathbb{H}^2(\mathcal{K}_{0,1}) = E \otimes \mathbb{T}$. The maps to and from the mixed complex are $0 \rightarrow F \rightarrow E$ and $E \otimes \mathbb{T} \rightarrow (E/F) \otimes \mathbb{T} \rightarrow 0$. At the exponential special fibre the cochain map $[\mathbb{C}[S] \xrightarrow{\chi} \mathbb{C}[S]dS] \rightarrow E[-1]$ is zero in degree zero and $P dS \mapsto [P]_\chi$ in degree one. Its kernel complex is the isomorphism $\mathbb{C}[S] \xrightarrow{\chi} \chi \mathbb{C}[S]dS$ and is contractible by division by χ on its image. This proves the special-fibre quasi-isomorphism. The resulting map to the full coefficient object of the logarithmic outer complexes is the identity on $[1], \dots, [S^{q-1}]$, intertwining A . Composing with $E \rightarrow E\mathcal{O}_X$, $v \mapsto v \otimes 1$, gives the map into the original constant target of (DS32). It retains the coefficient extension and its map a_t in (DS60). The original theta comparison $j_E r_N = I$, $j_E D^{(k)} = A j_E$ makes this identity a source comparison as well:

$$D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E, \quad j_E D_t^{(k)} = (A + t \mathcal{R}) j_E, \quad D_t^{(k)} r_N - r_N (A + t \mathcal{R}) = dK_N. \quad (\text{DS69})$$

Both equalities follow by multiplying and using $j_E r_N = I$; the added terms cancel exactly in the second equality. Thus the source primitive is unchanged and the full kernel $\ker j_E$ remains. The exact period equation (XD.u5), chain terms (DS70)–(DS71), and source connection and commutator (DS72)–(DS74) now develop this same family in both parameters. The finite-field boundary extension and the full single-primary calculation are included with their exact specialization hypotheses. Equations (DS66)–(DS74) give the coefficient, cohomological and source-level join used here.

2 Exact determinant of the polynomial-exponential period comparison

2.1 Exact determinant of the polynomial–exponential period comparison

Fix the original monic cyclic polynomial

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad d = q + 1, \quad \Phi_t(S) = \frac{S^d}{d} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS.$$

Fix $u \in \mathbb{C}^*$ and one argument of u . All the following comparisons retain this u , the original coordinate S , and the ordered remainder basis $1, S, \dots, S^{q-1}$. Set

$$D_{u,t} P = u P' + (\chi - t) P, \quad H_{u,t} = \mathbb{C}[S] / D_{u,t} \mathbb{C}[S].$$

This is a quotient of coefficient vector spaces. Let $\text{red}_{u,t}$ be its unique representative of degree less than q . Let $A(t)$ be the multiplication matrix of S on $\mathbb{C}[S]/(\chi - t)$ in the same ordered basis, so

$$A(t)e_b = e_{b+1} \quad (0 \leq b < q-1), \quad A(t)e_{q-1} = (-c_0 + t)e_0 - \sum_{a=1}^{q-1} c_a e_a. \quad (\text{XD1})$$

The empty sum convention in this display retains the case $q = 1$.

Use precisely the oriented rays and differences

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \ell_j : S = re^{i\theta_j}, \quad 0 \leq r < \infty, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq q). \quad (\text{XD2})$$

Write

$$\Pi_{jb}(u, c, t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{XD3})$$

2.1.1 Uniform estimates on the fixed unbounded contours

For a compact set $K \subset \mathbb{C}^{q+1}$ of the parameters $(c_0, \dots, c_{q-1}, t)$, put

$$M_K = \sum_{a=0}^{q-1} \frac{\sup_K |c_a|}{a+1} + \sup_K |t|, \quad R_K = \max\{1, 2dM_K\}, \quad \alpha = \frac{1}{2d|u|}. \quad (\text{XD4})$$

For every ray in (XD2) its leading term satisfies $S^d/(du) = -r^d/(d|u|)$. For $r \geq 1$ every other exponent has degree at most $d-1$, so

$$\text{Re} \frac{\Phi_t(re^{i\theta_j})}{u} \leq -\frac{r^d}{d|u|} + \frac{M_K r^{d-1}}{|u|} \leq -\alpha r^d \quad (r \geq R_K). \quad (\text{XD5})$$

Every parameter derivative of the integrand is its original exponential times a polynomial in S : a derivative in c_a contributes $S^{a+1}/((a+1)u)$ and a derivative in t contributes $-S/u$. For any fixed derivative order its absolute value on the tail is bounded by $C r^L e^{-\alpha r^d}$, with C, L uniform on K . The required majorant is integrable, since substitution $x = \alpha r^d$ gives

$$\int_0^\infty r^L e^{-\alpha r^d} dr = \frac{1}{d} \alpha^{-(L+1)/d} \Gamma\left(\frac{L+1}{d}\right) < \infty. \quad (\text{XD6})$$

The remaining finite interval is uniformly bounded by continuity. Thus (XD3) converges absolutely and is entire in all the lower coefficients and t ; every parameter derivative may be taken under the integral. The same bound gives $e^{\Phi_t(S)/u} P(S) \rightarrow 0$ at the infinite endpoint of each ray, for every polynomial P . Consequently

$$\int_{\Gamma_j} e^{\Phi_t/u} D_{u,t} P dS = u \int_{\Gamma_j} d(e^{\Phi_t/u} P) = 0. \quad (\text{XD7})$$

Indeed the integral of that derivative on each outward ray is $-e^{\Phi_t(0)/u} P(0) = -P(0)$; the coefficients $-1, +1$ of the two rays cancel these values. These estimates handle the noncompact contours directly, including parameter values at which critical points coincide.

2.1.2 An exact trace comparison for twisted reduction

For $1 \leq m \leq q$ and $0 \leq b < q$, ordinary monic division gives unique polynomials $Q_{m,b}, R_{m,b}$ satisfying

$$S^{m+b} = (\chi - t)Q_{m,b} + R_{m,b}, \quad \deg R_{m,b} < q. \quad (\text{XD8})$$

If $m + b < q$, take $Q_{m,b} = 0$. Otherwise $\deg Q_{m,b} = m + b - q \leq q - 1$, and the differential itself gives

$$\text{red}_{u,t}(S^{m+b}) = R_{m,b} - uQ'_{m,b}. \quad (\text{XD9})$$

Both terms on the right have degree below q . In every nonzero case,

$$\deg Q'_{m,b} \leq m + b - q - 1 \leq b - 1. \quad (\text{XD10})$$

Thus the coefficient of S^b in the derivative correction vanishes. Let $T_m(u, c, t)$ have column b equal to the coordinates of the left side of (XD9). The columns $R_{m,b}$ are exactly those of $A(t)^m$. Although (XD9) retains a possibly nonzero correction matrix, (XD10) proves the exact trace identity

$$\text{Tr } T_m(u, c, t) = \text{Tr } A(t)^m \quad (1 \leq m \leq q). \quad (\text{XD11})$$

Set $B_a = T_{a+1}/((a+1)u)$ for $0 \leq a < q$. Differentiation of (XD3), followed by (XD7)–(XD9), now proves

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u, \quad \text{Tr } B_a = \frac{\text{Tr } A(t)^{a+1}}{(a+1)u}. \quad (\text{XD12})$$

All the coefficient matrices here have polynomial entries in c, t for this fixed nonzero u .

2.1.3 The full singular parameter connection on this same cyclic family

The original determinant (XD19) is retained. Its complete parameter connection is now computed on the same coefficient quotient, before taking traces. For every original column $0 \leq b < q$, perform the unique monic division

$$\Phi_t(S)S^b = (\chi(S) - t)Q_b^\Phi(S, t) + R_b^\Phi(S, t), \quad \deg R_b^\Phi < q, \quad \deg Q_b^\Phi \leq b + 1. \quad (\text{XD.u1})$$

Write $C_\chi(t)$ for the matrix of the remainders and B_χ for the matrix of the derivatives $(Q_b^\Phi)'$, in the unchanged basis. The differential gives the exact, already reduced identity

$$\text{red}_{u,t}(\Phi_t S^b) = R_b^\Phi - u(Q_b^\Phi)', \quad C_\chi(t) = \Phi_t(A(t)). \quad (\text{XD.u2})$$

The second equality uses ordinary multiplication in $\mathbb{C}[S]/(\chi - t)$; the first retains its differential correction. Comparing division coefficients above degree q shows that only the constant coefficient of Q_b^Φ can depend on t : both $-tS^{b+1}$ and $-tQ_b^\Phi$ have degree at most q . Differentiation removes that constant. Its leading term is $S^{b+1}/(q+1)$. Therefore

$$(B_\chi)_{bb} = \frac{b+1}{q+1}, \quad (B_\chi)_{rb} = 0 \ (r > b), \quad \partial_t B_\chi = 0, \quad \text{Tr } B_\chi = q/2. \quad (\text{XD.u3})$$

On degrees zero and one of the original two-term complex the connections are respectively

$$\nabla_u^0 = \partial_u - \Phi_t/u^2 + 1/u, \quad \nabla_u^1 = \partial_u - \Phi_t/u^2. \quad (\text{XD.u4})$$

Indeed, writing $L = u\partial_S + \chi - t$, direct commutators give $[L, \partial_u] = -\partial_S$ and $[L, -\Phi_t/u^2] = -(\chi - t)/u$. Hence $\nabla_u^1 L = L \nabla_u^0$. Reduction by (XD.u2) proves the full top connection and period differential:

$$\Omega_u = -C_\chi(t)/u^2 + B_\chi/u, \quad \nabla_u = \partial_u + \Omega_u, \quad \partial_u \Pi = \Pi \Omega_u, \quad \Omega_t = -A(t)/u, \quad \partial_t \Pi = \Pi \Omega_t. \quad (\text{XD.u5})$$

All these statements hold locally on every retained branch of $u \neq 0$. One may deform the labelled contours within the same decay sectors; the tail estimate (XD5), locally uniform in $u \neq 0$, removes the connecting arcs. Differentiation in u inserts $-\Phi_t/u^2$, so (XD.u2) proves the asserted period equation without changing a ray orientation. The scalar commutators $[\partial_u - \Phi_t/u^2, \partial_t - S/u] = 0$, with the same degree-zero correction, prove flatness. Equivalently the retained matrices obey

$$\partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t] = 0. \quad (\text{XD.u6})$$

Taking the exterior power of (XD.u5) gives $\partial_u \det \Pi = (-\mathcal{F}/u^2 + q/(2u)) \det \Pi$, where $\mathcal{F}$ is precisely (XD13). If $\zeta = e^{2\pi i/(q+1)}$, $\beta_b = (b+1)/(q+1)$ and $W_{jb} = \zeta^{j(b+1)} - 1$, its retained constant is

$$K_q = (q+1)^{-q/2} e^{\pi i q/2} \det W \prod_{b=0}^{q-1} \Gamma(\beta_b), \quad \det \Pi = K_q u^{q/2} e^{\mathcal{F}/u}. \quad (\text{XD.u7})$$

This is exactly (XD16)–(XD19): each original column factor is $[(q+1)u]^{\beta_b} e^{\pi i \beta_b} \Gamma(\beta_b)/(q+1)$ on the specified branch, and $\sum_b \beta_b = q/2$. It adds the full singular connection to the earlier determinant proof; the determinant itself is the same proved result.

The packet polynomial h in the marked-product calculation and the cyclic polynomial $\chi_{h,k}$ have distinct specified coefficient maps. Formula (XD.u1) applies literally to the present χ . For $k=1$, $\chi_{h,1} = h$ via $S \mapsto s$, and every displayed matrix is the matrix C_h, B_h under that exact variable isomorphism. For general k the product lattice is instead connected to E at the marked fibre by the original injection $\eta_k[P] = v_h^{\otimes k} P(\sum_i s_i)1$. Its full source identity is $q_{\mathcal{B}}^{\otimes k} \mathcal{V}_{h,k} P = \sigma_h^{\otimes k} \eta_k[P]$. In the unweighted polynomial product family its cyclic relation has the explicit parameter image $[\chi(S)P(S)] = \sum_i [t_i Q_i - u \partial_i Q_i]$ whenever $\chi(S)P(S) = \sum_i h(s_i)Q_i$ in the original polynomial ring. Thus the two connections have a common specified marked-fibre map; no equality of the two families or their ranks is inserted.

2.1.4 The determinant, including every monomial factor

Define the scalar polynomial in the original coefficients

$$\mathcal{F}(c, t) = \text{Tr } \Phi_t(A(t)) = \frac{\text{Tr } A(t)^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a \text{Tr } A(t)^{a+1}}{a+1} - t \text{Tr } A(t). \quad (\text{XD13})$$

There is no choice or enumeration of roots in this definition. For any differentiable matrix A and integer $n \geq 1$, the product rule and cyclic invariance of trace give

$$d \text{Tr } A^n = \sum_{r=0}^{n-1} \text{Tr}(A^r(dA)A^{n-1-r}) = n \text{Tr}(A^{n-1}dA). \quad (\text{XD14})$$

Because $A(t)$ acts on the actual quotient by $\chi - t$, it satisfies $\chi(A(t)) - tI_q = 0$. Applying (XD14) to (XD13) therefore gives

$$d\mathcal{F} = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt + \text{Tr}((\chi(A(t)) - tI_q)dA(t)) = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt. \quad (\text{XD15})$$

At $c = t = 0$ the period matrix is

$$\Pi_{j,l-1}^{\text{mon}}(u) = a_l(u) e^{il\theta_0} (\zeta^{jl} - 1), \quad a_l(u) = \frac{(d|u|)^{l/d}}{d} \Gamma(l/d), \quad \zeta = e^{2\pi i/d}, \quad 1 \leq j, l \leq q. \quad (\text{XD16})$$

This follows by substituting $x = r^d/(d|u|)$ on each oriented ray in (XD3). In particular all $a_l(u)$ are positive real numbers. For $V_{jl} = \zeta^{jl} - 1$ one may include $j=0$ in the Gram sum because its summand is zero. The complete root-of-unity sums give

$$(V^*V)_{lm} = \sum_{j=0}^{d-1} (\zeta^{-jl} - 1)(\zeta^{jm} - 1) = d\delta_{lm} + d. \quad (\text{XD17})$$

Here $1 \leq l, m < d$, so the two individual nonconstant sums vanish, and the first sum is $d\delta_{lm}$. The all-ones rank-one matrix has eigenvalue q on the all-ones vector and zero on its $(q-1)$ -dimensional perpendicular complement. It follows that

$$\det(V^*V) = d^q(1+q) = d^d > 0, \quad C_q(u) := |\det \Pi^{\text{mon}}(u)|^2 = d^d \prod_{l=1}^q \left[\frac{(d|u|)^{l/d}}{d} \Gamma(l/d) \right]^2. \quad (\text{XD18})$$

The complex determinant itself retains the oriented phase: $\det \Pi^{\text{mon}} = \det V \prod_{l=1}^q a_l(u) e^{il\theta_0}$.

We claim the exact global formula

$$\boxed{\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp(\mathcal{F}(c, t)/u)}. \quad (\text{XD19})$$

To prove it without assuming invertibility along the path, join $(0, 0)$ to (c, t) by (sc, st) , $0 \leq s \leq 1$. Equations (XD12) give $\Pi' = \Pi B(s)$, with continuous $B(s)$, and (XD15) gives $\text{Tr } B(s) = \mathcal{F}'(sc, st)/u$. For every square matrix, including a singular one,

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi) \Pi B) = \det \Pi \text{Tr } B. \quad (\text{XD20})$$

Consequently $\det \Pi \exp(-\mathcal{F}/u)$ has derivative zero. At the starting point A is nilpotent and $\mathcal{F}(0, 0) = 0$, so its value is $\det \Pi^{\text{mon}}(u)$. This proves (XD19). Equation (XD18) proves its nonvanishing for every c, t , including every repeated-root fibre. In particular the period map

$$H_{u,t} \longrightarrow \mathbb{C}^q, \quad [P] \longmapsto \Pi \text{red}_{u,t}(P), \quad (\text{XD21})$$

is an isomorphism with inverse given by Π^{-1} followed by the specified degree-below- q section.

2.1.5 Exact determinant transport to the arithmetic representative

Retain the original positive Gram G_N on the cyclic coefficient space E , its determinant $V_N = \det G_N$, and the original maps $r_N : E \longrightarrow \mathcal{B}^{\otimes k}$ and j_E on the admitted test-function domain with $j_E r_N = I_E$. Identify E with $H_{u,t}$ by their specified ordered remainder bases. Then the actual maps

$$r_N(E) \longrightarrow \mathbb{C}^q, \quad x \longmapsto \Pi j_E x, \quad \mathbb{C}^q \longrightarrow r_N(E), \quad y \longmapsto r_N \Pi^{-1} y \quad (\text{XD22})$$

are inverse: their composites are respectively $r_N j_E$ on $r_N(E)$ and $\Pi(j_E r_N) \Pi^{-1}$ on $\mathbb{C}^q$. On the full admitted domain the first observation has precisely $\ker j_E$ as its kernel; all original theta boundaries killed by j_E remain in that kernel.

With the explicitly fixed coordinate metric I_q on the target, put

$$H_{\Pi} = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_{\Pi}. \quad (\text{XD23})$$

These operators satisfy $G_N B_N^{\text{per}} = (B_N^{\text{per}})^* G_N = H_{\Pi} > 0$. Taking determinants in (XD19) gives the exact identities

$$\det H_{\Pi} = C_q(u) \exp\left(2 \text{Re} \frac{\mathcal{F}(c, t)}{u}\right), \quad (\text{XD24})$$

$$\boxed{\log \det B_N^{\text{per}} = \log C_q(u) + 2 \text{Re} \frac{\mathcal{F}(c, t)}{u} - \log V_N}. \quad (\text{XD25})$$

Every determinant in (XD24)–(XD25) is positive real; these logarithms therefore require no branch choice. At the same u, c, t for two admitted degrees $i \leq j$, subtraction proves

$$\boxed{\log \det B_j^{\text{per}} - \log \det B_i^{\text{per}} = \log(V_i/V_j)}. \quad (\text{XD26})$$

Thus the common polynomial–exponential factor is now calculated exactly while the full original endpoint cost is retained by the displayed pair of positive comparison operators. No uniform upper estimate on V_i/V_j is inferred from this determinant identity.

3 Translation of the exponential comparison: full coefficient, period, metric, and finite-field maps

Translation of the exponential comparison:

full coefficient, period, metric, and finite-field maps 13 September 2026

3.1 The original fibre and the auxiliary translation

The September 13 exponential comparison starts with the actual arithmetic polynomial and sum operator

$$\chi(S) = S^q + \sum_{b=0}^{q-1} c_b S^b, \quad q > 0, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S,$$

and the specified primitive

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{b=0}^{q-1} \frac{c_b S^{b+1}}{b+1}.$$

Its cohomology is the cokernel of $D_{u,t} = u\partial_S + \chi(S) - t$, with the omitted differential symbol understood as the coefficient identification with one-forms. All formulas below retain this original object at $a = 0$. The parameter $a \in \mathbb{C}$ defines an auxiliary comparison family; a translated polynomial is not asserted to be a new zeta zero packet.

Set

$$\chi_a(S) = \chi(S - a), \quad \Phi_a(S) = \Phi(S - a) - \Phi(-a), \quad T_a P(X) = P(X + a). \quad (\text{DT.1})$$

Thus $\Phi_a(0) = 0$, and no additive constant in the potential is lost. Use the same substitution on coefficients of one-forms, with $dS = dX$.

Proposition 3.1 (The coefficient and connection maps). *The map T_a is an isomorphism of the two-term polynomial complexes for χ_a and χ , over $\mathbb{C}[u, t]$. It induces an isomorphism of their free rank- q cohomology modules. On the special fibre $(u, t) = (0, 0)$,*

$$T_a : E_a = \mathbb{C}[S]/(\chi_a) \xrightarrow{\sim} E, \quad T_a A_a T_a^{-1} = A + aI. \quad (\text{DT.2})$$

Every primary multiplicity and nilpotent Jordan block is retained. For $u \neq 0$, the descended parameter connections satisfy

$$T_a \nabla_t^a T_a^{-1} = \nabla_t - \frac{a}{u} I, \quad \nabla_t = \partial_t - \frac{A(t)}{u}, \quad (\text{DT.3})$$

where $A(t)$ is the original companion matrix for $\chi - t$.

Proof. Substitution gives, for every polynomial P ,

$$T_a(uP' + (\chi_a - t)P) = u\partial_X(T_a P) + (\chi(X) - t)T_a P.$$

The inverse is substitution by $-a$. A polynomial of degree n with leading coefficient $b \neq 0$ is sent by either differential to a polynomial of degree $n + q$ with the same leading coefficient. Successive subtraction of such images therefore gives a unique representative of degree less than q , over the coefficient ring. This proves the stated freeness and the base-change identification. On the special fibre, $T_a(SP) = (X + a)T_a P$, which proves DT.2. In particular, $(A_a - (\lambda + a)I)^m$ is conjugate to $(A - \lambda I)^m$ for every m , proving the multiplicity statement without discarding the nilpotent part. On arbitrary polynomial representatives, the connection is $\partial_t - S/u$. Since T_a is independent of t , its conjugate is $\partial_t - (X + a)/u$. The commutator with the differential vanishes, so this equality descends and yields DT.3. $\square$

In the fixed bases $1, S, \dots, S^{q-1}$ and $1, X, \dots, X^{q-1}$, the matrix of T_a is

$$(C_a)_{rb} = \begin{cases} \binom{b}{r} a^{b-r}, & 0 \leq r \leq b < q, \\ 0, & r > b, \end{cases} \quad C_a^{-1} = C_{-a}, \quad \det C_a = 1. \quad (\text{DT.4})$$

The inverse and composition rules follow by composing the literal substitutions. In particular, $C_a A_a(t) C_a^{-1} = A(t) + aI$.

3.2 Periods and the retained exponential factor

Fix $u \neq 0$, put $d = q + 1$, and use the original rays

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \Gamma_j = -\ell_0 + \ell_j, \quad 1 \leq j \leq q,$$

oriented as in the exponential comparison. Let $\Pi(u, t)$ denote its period matrix, whose (j, b) entry is $\int_{\Gamma_j} e^{(\Phi(X) - tX)/u} X^b dX$. Let Π_a use the identical labelled ray prescription for Φ_a in the S -plane.

Proposition 3.2 (Exact period translation). *Define*

$$h_a(t) = -\Phi(-a) - ta, \quad f_a(u, t) = \exp(h_a(t)/u). \quad (\text{DT.5})$$

Then

$$\Pi_a(u, t) = f_a(u, t) \Pi(u, t) C_a. \quad (\text{DT.6})$$

The map $f_a T_a$ intertwines the two parameter connections in DT.3.

Proof. The substitution $S = X + a$ gives the exact identity

$$\Phi_a(S) - tS = \Phi(X) - tX + h_a(t).$$

It changes the contour to $\Gamma_j - a$. This translated contour may be moved to Γ_j without changing the integral. Indeed, truncate the two rays at radius R , join their translated endpoints by segments of length $|a|$, and apply Cauchy's theorem to the resulting closed path. The entire integrand has no finite singularity. On each joining segment its leading exponent is $-R^d/(d|u|) + O(R^{d-1})$, uniformly along that segment; a polynomial factor grows at most polynomially. Its integral tends to zero. The two portions at the finite starting point have opposite orientation and cancel. Absolute convergence on the original and translated rays follows from the same estimate. Finally, $(X + a)^b$ is the column expansion DT.4. This proves DT.6 including the scalar and all phases.

Since $\partial_t f_a = -(a/u) f_a$, for every section v one has

$$\nabla_t(f_a T_a v) = f_a \left(\nabla_t T_a v - \frac{a}{u} T_a v \right) = f_a T_a \nabla_t^a v.$$

The scalar f_a is nonzero on $u \neq 0$; no extension of this exponential gauge to $u = 0$ is used. DT.2 is the separately proved coefficient specialization at that fibre. $\square$

3.3 Translation of the full singular connection and boundary character

The same retained scalar $h_a(t) = -\Phi(-a) - ta$ also controls the u direction. Write ∇_u^a for the degree-one connection of the translated phase and ∇_u for that of the original phase, both including their exact matrices (XD.u1)–(XD.u5). Literal substitution in the full polynomial complex gives

$$T_a \nabla_u^a T_a^{-1} = \nabla_u - \frac{h_a(t)}{u^2} I, \quad \nabla_u(f_a T_a v) = f_a T_a \nabla_u^a v. \quad (\text{DT.14})$$

Indeed $T_a \Phi_t^a T_a^{-1} = \Phi_t + h_a(t)$ and T_a is independent of u . Differentiating the unchanged exponential gives $\partial_u f_a = -h_a(t) f_a / u^2$, proving the second identity term by term. In degree zero each

side additionally contains the same $1/u$ term. Together with DT.3 and DT.6 this proves a flat two-parameter intertwiner on $u \neq 0$, with its singular scalar retained. In matrix form, if Ω_u^a, Ω_u are the two original remainder matrices,

$$C_a \Omega_u^a C_a^{-1} = \Omega_u - h_a(t)I/u^2. \quad (\text{DT.15})$$

It follows equally by differentiating DT.6 in u , using invertibility of the original period matrix. Its determinant has the scalar f_a^q , in agreement with the complete constant and phase in (XD.u7).

Before fixing $u = u_0$ in the finite-field calculation, the same polynomial substitution is a morphism over the actual parameter scheme. Use the specified coefficient map $R_0[a] \rightarrow \mathbb{F}_Q$ of characteristic $p > d = q + 1$, the same $\ell \neq p$, and the nontrivial additive character ψ fixed in the finite-field comparison below. Write $\mathcal{U} = \mathbb{G}_{m, \mathbb{F}_Q, u} \times \mathbb{A}_{\mathbb{F}_Q, t}^1$, and let $\pi_X : \mathbb{A}_X^1 \times \mathcal{U} \rightarrow \mathcal{U}$ and $\pi_S : \mathbb{A}_S^1 \times \mathcal{U} \rightarrow \mathcal{U}$ be the literal projections. Define the full derived complexes

$$\mathcal{F} = R\pi_{X!} \mathcal{L}_\psi((\Phi(X) - tX)/u), \quad \mathcal{F}_a = R\pi_{S!} \mathcal{L}_\psi((\Phi_a(S) - tS)/u).$$

The base-defined isomorphism $\sigma_a(X, u, t) = (X + a, u, t)$ satisfies $\pi_S \sigma_a = \pi_X$. Its exact derived compact-support direct-image identity, in every cohomology degree, is

$$\mathcal{F}_a \simeq \mathcal{F} \otimes \mathcal{L}_\psi(h_a(t)/u). \quad (\text{DT.16})$$

The Artin–Schreier torsor addition proving DT.12 works over this coefficient ring without any change, and the character line is pulled back from the parameter scheme, so it factors out of the compact-support complex. Restricting to a specified (u_0, t_0) is exactly DT.12–DT.13. At fixed t_0 , its possible singular boundary character is therefore the full $\mathcal{L}_\psi(h_a(t_0)/u)$; it cannot be replaced by its value at a nonzero u_0 . This is the precise family map used when passing from the translated single-primary phase to its local boundary representation.

3.4 The original positive forms and their control operators

The period matrices used below are invertible on the stated domain $u \neq 0$. Here is a complete verification needed for the determinant ratios. At the monomial potential, the substitution $x = r^d/(d|u|)$ in the absolutely convergent ray integral gives

$$\Pi_{j,b}^{\text{mon}} = \frac{(d|u|)^{(b+1)/d}}{d} \Gamma\left(\frac{b+1}{d}\right) e^{i(b+1)\theta_0} (\zeta^{j(b+1)} - 1), \quad \zeta = e^{2\pi i/d},$$

where $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx > 0$ for $s > 0$. The column factors are all nonzero. A dependence among the remaining columns would give a polynomial $\sum_{l=1}^q b_l(Z^l - 1)$ of degree at most q , vanishing at all $q + 1$ roots of unity, including 1. This polynomial is zero, so every $b_l = 0$.

Join the monomial coefficients to the given (c, t) by a straight path with real parameter $s \in [0, 1]$, keeping u fixed. On this compact coefficient path the leading negative ray term dominates all lower terms uniformly: for sufficiently large r the absolute value of any fixed coefficient derivative of the integrand is at most $Cr^L \exp[-r^d/(2d|u|)]$. This is integrable by the same gamma substitution; hence differentiation under every contour integral is valid. A derivative in c_b multiplies the integrand by $S^{b+1}/((b+1)u)$, and one in t multiplies it by $-S/u$. The unique reduction proved above, together with the vanishing of $\int_{\Gamma_j} e^{(\Phi-tS)/u} D_{u,t} P dS$, thus gives $\Pi'(s) = \Pi(s)B(s)$ with a continuous matrix $B(s)$. That integral vanishes because it is u times an exact derivative; both infinite endpoint values are zero and the two finite endpoint values cancel with the specified contour signs. The determinant identity

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi) \Pi B) = (\det \Pi) \text{Tr} B$$

holds also at singular matrices. Therefore $\det \Pi(s) = \det \Pi(0) \exp(\int_0^s \text{Tr } B(v) dv) \neq 0$. This proves invertibility without removing repeated-root parameters. Equation DT.6 proves it for the translated matrix as well.

Let G_N be any one of the original canonical arithmetic Gram matrices, and retain its index N , cochain degree k , and control form $W_N = A^* G_N + G_N A - k G_N$. The exact pullback along DT.2 is

$$G_N^{[a]} = C_a^* G_N C_a. \quad (\text{DT.7})$$

This is a transported specified form. No claim identifies it with a newly computed canonical theta form for a translated polynomial.

Proposition 3.3 (Control and determinant transport). *For the transported form and the original value of k ,*

$$W_N^{[a]} = A_a^* G_N^{[a]} + G_N^{[a]} A_a - k G_N^{[a]} = C_a^* (W_N + (a + \bar{a}) G_N) C_a, \quad (\text{DT.8})$$

$$\text{Tr}((G_N^{[a]})^{-1} W_N^{[a]}) = \text{Tr}(G_N^{-1} W_N) + 2q \text{Re } a. \quad (\text{DT.9})$$

Write $H_\Pi = \Pi^* \Pi$ and $B_N^{\text{per}} = G_N^{-1} H_\Pi$. Then

$$B_N^{\text{per}, [a]} := (G_N^{[a]})^{-1} \Pi_a^* \Pi_a = |f_a|^2 C_a^{-1} B_N^{\text{per}} C_a. \quad (\text{DT.10})$$

Consequently, for the same (u, t) and the same original indices,

$$\frac{\det B_j^{\text{per}, [a]}}{\det B_i^{\text{per}, [a]}} = \frac{\det B_j^{\text{per}}}{\det B_i^{\text{per}}} = \frac{\det G_i}{\det G_j}. \quad (\text{DT.11})$$

Proof. Insert $C_a A_a C_a^{-1} = A + aI$ into the first definition of DT.8. Taking the inverse Gram and using invariance of trace under similarity gives DT.9. Equation DT.6 gives $\Pi_a^* \Pi_a = |f_a|^2 C_a^* H_\Pi C_a$, proving DT.10. Its determinant contains the factor $|f_a|^{2q}$; this same factor occurs at both indices and cancels in their ratio. The remaining period Gram also cancels, giving DT.11. Thus the original endpoint determinant cost is retained by this exact comparison. $\square$

3.5 The finite-field comparison with its scalar character

Let $R_0 \subset \mathbb{C}$ be the finitely generated coefficient ring of the original exponential construction, and adjoin a . For a specified map $R_0[a] \rightarrow \mathbb{F}_Q$ of characteristic $p > d$, and specified $u_0 \in \mathbb{F}_Q^\times$, $t_0 \in \mathbb{F}_Q$, reduce every displayed coefficient through that map. Put

$$f(X) = \frac{\Phi(X) - t_0 X}{u_0}, \quad \gamma_a = \frac{-\Phi(-a) - t_0 a}{u_0} \in \mathbb{F}_Q.$$

Use the Artin–Schreier rank-one sheaf convention in which the geometric-Frobenius trace at $x \in \mathbb{F}_{Q^n}$ is $\psi(\text{Tr}_{\mathbb{F}_{Q^n}/\mathbb{F}_p} f(x))$, for a fixed nontrivial additive character ψ valued in $\overline{\mathbb{Q}_\ell}$, $\ell \neq p$.

Proposition 3.4 (Translation and the constant character line). *For the isomorphism $\sigma_a : X \mapsto S = X + a$,*

$$\sigma_a^* \mathcal{L}_\psi((\Phi_a - t_0 S)/u_0) \simeq \mathcal{L}_\psi(f) \otimes \mathcal{L}_\psi(\gamma_a). \quad (\text{DT.12})$$

Hence compactly supported cohomology in every degree is identified with that of $\mathcal{L}_\psi(f)$, tensored with the displayed constant character line. Each geometric-Frobenius eigenvalue is multiplied by

$$\zeta_a = \psi(\text{Tr}_{\mathbb{F}_Q/\mathbb{F}_p} \gamma_a). \quad (\text{DT.13})$$

This is a root of unity, so it preserves each complex absolute value and hence the purity weights of this finite-field realization.

Proof. The potential identity in the period proof holds after the specified coefficient reduction. The tensor identity of Artin–Schreier character sheaves can be obtained directly from their covers: if $y_1^p - y_1 = f$ and $y_2^p - y_2 = \gamma_a$, then $(y_1 + y_2)^p - (y_1 + y_2) = f + \gamma_a$. The addition map identifies the contracted product of the two $\mathbb{F}_p$ -torsors with the third torsor. On the character lines, the addition action becomes the product of their characters, which gives DT.12. Pullback by the base-defined isomorphism σ_a preserves compact supports and commutes with Frobenius. Tensoring with a line pulled back from the base tensors the compact-support complex with that line: this can be checked on a cochain resolution before taking cohomology. Our stated trace convention makes its Frobenius scalar DT.13. The additive group of $\mathbb{F}_p$ has exponent p , so $\zeta_a^p = 1$, proving the absolute-value assertion. $\square$

These formulas compose with the exponential-family application of Deligne, *La conjecture de Weil II*, §§3.7.2–3.7.4, to the degree- d polynomial with nonzero leading coefficient. The source for that application, including the boundary chart and its hypotheses, is retained in the exponential comparison note; the present proof establishes the additional translation maps themselves.

3.6 Specialization and split support

The polynomial chain isomorphism T_a is defined before either complex or finite-field realization. Its coefficient map, the specialization $u = t = 0$, and the inclusion of the original fibre $a = 0$ induce maps under the same split-zero functor G . For a ring map $\varphi : B \rightarrow B'$, this functor acts by

$$G(\varphi)(\tau) = \tau, \quad G(\varphi)(b^\bullet) = \varphi(b)^\bullet.$$

Therefore a supported coefficient specialized to zero maps to $e = 0^\bullet$, and an absent coefficient maps to τ . The square with the amplitude maps $p_B, p_{B'}$ commutes, since both composites take τ to zero and $b^\bullet$ to $\varphi(b)$. On each already specified receiving module fibre, the coefficient chain relation specializes to the relation $\chi^P = 0$ without changing its label. The functorial map thus retains both the supported kernel and external absence.

The translation formulas specify exactly how the two realizations respond to the same coefficient operation: DT.2 shifts the arithmetic sum operator, DT.9 changes its metric trace by $2q \operatorname{Re} a$, DT.10 retains the analytic exponential scalar, and DT.13 supplies the finite-field character scalar. These are complete comparison maps. They do not identify a finite-field Frobenius with A , and they leave the original theta metric estimate as the same determinant quantity in DT.11. The original arithmetic construction remains the specified fibre $a = 0$ throughout.

4 SGA trace, exact constituent curvature, and the derived collision fibre

4.1 SGA trace, exact constituent curvature, and the derived collision fibre

This calculation extends the constituent period map in the delivered *SGA Trace Period Control*, equations (54a)–(54c). Its period determinant is the same determinant already proved in (XD1)–(XD26), with every original coefficient, ray and phase retained. We calculate the full normal derivative, its kernel and its norm. The result applies to the actual invariant constituents, including repeated primary factors.

4.1.1 The retained period map and arithmetic source

Keep the original monic polynomial and its increasing-power frame:

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S, \quad q \geq 1, \quad e_0 = [1], \quad \ell([P]) = [S^{q-1}] \operatorname{rem}_\chi P. \quad (\text{SC1})$$

Thus $\ell : E \rightarrow \mathbb{C}$ is a specified linear functional and $\mathcal{R} = e_0 \ell : E \rightarrow E$ has rank one. Fix $u \in \mathbb{C}^*$, one argument of u , and the actual potential and contours

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS, \quad \theta_j = \frac{\arg u + \pi + 2\pi j}{q+1}, \quad \Gamma_j = -\ell_0^{\text{ray}} + \ell_j^{\text{ray}}. \quad (\text{SC2})$$

Here ℓ_j^{ray} is the outward ray $S = re^{i\theta_j}$, $r \geq 0$; it is distinct from the coefficient functional ℓ . The period matrix is

$$\Pi_{jb}(t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{SC3})$$

The full convergence and determinant proof is (XD4)–(XD21). To state its precise retained inputs here, on a compact coefficient set K its tail is bounded by $r^L \exp(-r^{q+1}/(2(q+1)|u|))$ after $r \geq \max(1, 2(q+1)M_K)$, where $M_K = \sum_a \sup_K |c_a|/(a+1) + \sup_K |t|$. Differentiation is therefore valid in every coefficient direction. Integration by parts cancels the two origin terms, and reduction of S^q uses $[\chi - t] = 0$ with no derivative correction. It follows that

$$\Pi'(t) = -\frac{1}{u} \Pi(t)(A + t\mathcal{R}). \quad (\text{SC4})$$

The proven determinant is

$$\det \Pi(t) = \det \Pi_{\text{mon}}(u) \exp\left(\frac{\text{Tr } \Phi_t(A + t\mathcal{R})}{u}\right), \quad |\det \Pi_{\text{mon}}(u)|^2 = (2\pi|u|)^q > 0. \quad (\text{SC5})$$

The complex initial matrix is explicitly

$$(\Pi_{\text{mon}})_{j,b} = \frac{((q+1)|u|)^{(b+1)/(q+1)}}{q+1} \Gamma\left(\frac{b+1}{q+1}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/(q+1)} - 1). \quad (\text{SC6})$$

The product form and the full complex phase in (SC5)–(SC6) are reused from (XD18)–(XD19). The evaluated squared constant is the additional gamma-product evaluation in the delivered SGA note. Here is every factor in that evaluation. Set $d = q + 1$. The matrix $V_{jl} = e^{2\pi i j l/d} - 1$ has $V^*V = d(I_q + J_q)$ by the root-of-unity sum in (XD17), so $|\det V|^2 = d^{q+1}$. Since $2 \sum_{l=1}^{d-1} l/d = q$, its column factors give

$$|\det \Pi_{\text{mon}}|^2 = d^{q+1} \frac{(d|u|)^q}{d^{2q}} \left(\prod_{l=1}^{d-1} \Gamma(l/d) \right)^2. \quad (\text{SC6a})$$

The gamma multiplication formula, DLMF (5.5.6), is

$$\Gamma(dz) = (2\pi)^{(1-d)/2} d^{dz-1/2} \prod_{a=0}^{d-1} \Gamma(z + a/d). \quad (\text{SC6b})$$

At $z = 1/d$, its left side and the final product factor are $\Gamma(1) = 1$, so the remaining product equals $(2\pi)^{q/2} d^{-1/2}$, which is DLMF (5.5.7). In (SC6a) the complete power of d is $(q+1) + q - 2q - 1 = 0$. This proves $|\det \Pi_{\text{mon}}|^2 = (2\pi|u|)^q$ without a frame change or an omitted phase. Thus every $\Pi(t)$ is invertible, including at collisions. The full reference is <https://dlmf.nist.gov/5.5.E7>.

For every original admitted source degree N , retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad r_N = \mathcal{V} R_N, \quad j_E r_N = I_E. \quad (\text{SC7})$$

The source identities and primitive have their original types $j_E D^{(k)} = A j_E$ and $D^{(k)} r_N - r_N A = dK_N^{\text{prim}}$. The exact parameter action on the admitted source domain is

$$D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E, \quad j_E D_t^{(k)} = (A + t\mathcal{R}) j_E, \quad D_t^{(k)} r_N - r_N (A + t\mathcal{R}) = dK_N^{\text{prim}}. \quad (\text{SC8})$$

Indeed insertion of $j_E r_N = I_E$ proves each equality by multiplication; the added terms cancel in the last difference. In particular the normal map calculated below has an explicit lift to this source, with the complete original primitive and kernel of j_E retained.

4.1.2 Every invariant constituent has a nonzero top coefficient

Let $I : F \hookrightarrow E$ be an actual injective linear map whose image is A -invariant, with $AI = IA_F$, and put $p = \dim F$. For the moment suppose $0 < p < q$. The image $I(F)$ is an ideal of E : invariance under A implies invariance under every polynomial in A , and every element of E is such a polynomial applied to e_0 . Its inverse image in $\mathbb{C}[S]$ is an ideal containing (χ) . It is generated by a unique monic divisor g of χ ; writing $\chi = gh$ gives $\deg h = p$ and $\deg g = q - p$. The exact map is

$$\mathbb{C}[S]/(h) \longrightarrow I(F), \quad [a]_h \longmapsto [ga]_\chi. \quad (\text{SC9})$$

It is onto by the definition of (g) , and is injective because $gh \mid ga$ in the polynomial domain implies $h \mid a$. Consequently $[g], [Sg], \dots, [S^{p-1}g]$ is an ordered basis of $I(F)$. Its last member is monic of degree $q - 1$, while every earlier member has smaller degree. Hence the restricted functional $L = \ell I : F \rightarrow \mathbb{C}$ is onto. Also $e_0 \notin I(F)$, since an ideal containing the unit is the whole algebra. We have proved

$$\begin{aligned} \mathcal{R}I = e_0 L, \quad \text{rank}(\mathcal{R}I) = 1, \quad \ker(\mathcal{R}I) = \ker L, \quad \dim \ker L = p - 1, \\ \mathbb{C}e_0 \cap I(F) = \{0\} \quad \text{inside } E. \end{aligned} \quad (\text{SC10})$$

These arguments never replace a repeated factor by its square-free part. In the basis (SC9), L is exactly the last coordinate; in the original chosen basis of F it remains the literal row ℓI .

4.1.3 The exact normal derivative at every parameter

Give the target $\mathbb{C}^q$ the period-coordinate Hermitian metric I_q specified in the delivered note. Define

$$\begin{aligned} Y(t) = \Pi(t)I, \quad H(t) = Y(t)^* Y(t), \quad P(t) = Y(t)H(t)^{-1}Y(t)^*, \\ N(t) = I_q - P(t), \quad z(t) = N(t)\Pi(t)e_0. \end{aligned} \quad (\text{SC11})$$

Every $Y(t)$ is injective, so $H(t) > 0$. Multiplication shows $P(t)^2 = P(t) = P(t)^*$ and $P(t)Y(t) = Y(t)$; thus $N(t)$ is the orthogonal projection onto the complement of the displayed period subspace. The vector $z(t)$ is nonzero for every t : otherwise $\Pi(t)e_0$ lies in $\Pi(t)I(F)$, and invertibility of $\Pi(t)$ would give $e_0 \in I(F)$, contradicting (SC10).

Using invariance under the original A , equation (SC4) yields

$$Y'(t) = -\frac{1}{u}Y(t)A_F - \frac{t}{u}\Pi(t)e_0L, \quad \boxed{N(t)Y'(t) = -\frac{t}{u}z(t)L.} \quad (\text{SC12})$$

This holds for every complex t , with the same fixed constituent F of the original A . We have not assumed that F is invariant under $A + t\mathcal{R}$. Formula (SC12) computes exactly its failure of invariance after passing through the original periods. For $t \neq 0$ the normal derivative has image $\mathbb{C}z(t)$, kernel $\ker L$ and rank one. At $t = 0$ it vanishes. Differentiating (SC4) once more gives

$$Y''(0) = \frac{1}{u^2}\Pi(0)A^2I - \frac{1}{u}\Pi(0)\mathcal{R}I, \quad N(0)Y''(0) = -\frac{1}{u}z(0)L. \quad (\text{SC13})$$

The first term has image inside $Y(0)F$ because $A^2I = IA_F^2$. The normal acceleration therefore has rank exactly one, image $\mathbb{C}z(0)$ and kernel $\ker L$, including all repeated-root cases.

The actual original source lift of its parameter defect is $r_N e_0 L : F \rightarrow r_N(E)$. The map $x \mapsto N(t)\Pi(t)j_E x$ sends it to $z(t)L$. Thus its source, coefficient and period forms have the commuting factorization

$$F \xrightarrow{L} \mathbb{C} \xrightarrow{a \mapsto ar_N e_0} r_N(E) \xrightarrow{j_E} E \xrightarrow{N(t)\Pi(t)} \mathbb{C}^q. \quad (\text{SC14})$$

At the original Split-Zero support, every linear arrow lifts by $x^\bullet \mapsto f(x)^\bullet$ and $\tau \mapsto \tau$. In particular $\ker L$ maps to the supported zero; external absence retains its distinct image τ . No new zero-adjunction is used.

4.1.4 Curvature is an exact rank-one expression

Write $\partial_t = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{t}} = \frac{1}{2}(\partial_x + i\partial_y)$ for $t = x + iy$. Since Y is holomorphic, $\partial_t H = Y^* Y'$, $\partial_{\bar{t}} H = Y'^* Y$ and $\partial_{\bar{t}} \partial_t H = Y'^* Y'$. Differentiating $H^{-1} H = I_p$ gives $\partial_{\bar{t}} H^{-1} = -H^{-1} Y'^* Y H^{-1}$. The determinant derivative and cyclic invariance of trace therefore give

$$\begin{aligned}\partial_{\bar{t}} \partial_t \log \det H &= \text{Tr}(H^{-1} Y'^* Y' - H^{-1} Y'^* Y H^{-1} Y^* Y') \\ &= \text{Tr}(H^{-1} Y'^* N Y').\end{aligned}\tag{SC15}$$

There is no logarithm-branch choice: $\det H > 0$. Insert the full normal map (SC12), use $N^* = N = N^2$, and retain the rank-one factors to obtain

$$\partial_{\bar{t}} \partial_t \log \det H(t) = |t|^2 c_F(t), \quad c_F(t) = \frac{\|z(t)\|_{I_q}^2}{|u|^2} L H(t)^{-1} L^* > 0.\tag{SC16}$$

Strict positivity follows because $z(t) \neq 0$, $L \neq 0$ and $H(t)^{-1} > 0$. Thus this curvature vanishes exactly at $t = 0$ and vanishes there to precisely quadratic order; it is strictly positive at every $t \neq 0$. For the dual determinant line with metric $(\det H)^{-1}$ the convention $c_1 = -(i/(2\pi)) \partial \bar{\partial} \log h$ gives

$$c_1 = \frac{i}{2\pi} |t|^2 c_F(t) dt \wedge d\bar{t}.\tag{SC17}$$

In these coordinates $i dt \wedge d\bar{t} = 2 dx \wedge dy$; this records the orientation and factor explicitly.

For completeness the exact formula supplies a convergent Taylor statement. Entries of Π have convergent complex power series; entries of H , P , N and c_F are real analytic because $\det H$ is everywhere positive. On any closed disk $|t| \leq \varepsilon$, continuity of the first real derivatives gives a finite bound $M_\varepsilon = \sup \|\nabla c_F\|$. Integrating the derivative along the real segment $s \mapsto st$ proves $|c_F(t) - c_F(0)| \leq M_\varepsilon |t|$. Consequently

$$|\partial_{\bar{t}} \partial_t \log \det H(t) - |t|^2 c_F(0)| \leq M_\varepsilon |t|^3.\tag{SC18}$$

This estimate follows from the actual analytic functions in (SC11), and (SC16) supplies the stronger identity at all parameters.

One may check the projector's first derivatives directly: $\partial_t P = N Y' H^{-1} Y^*$ and $\partial_{\bar{t}} P = Y H^{-1} Y'^* N$. They vanish at zero by (SC12), explaining the retained stationary tangent space at that parameter without deleting its acceleration.

4.1.5 Both original theta and period metrics remain explicit

On F retain the positive original theta Gram $G_{N,F} = I^* G_N I$ and the positive comparison operator

$$B_{N,F}(t) = G_{N,F}^{-1} H(t), \quad \mathcal{O}_{N,F}(t) = \log \frac{\det H(t)}{\det G_{N,F}}.\tag{SC19}$$

Here $G_{N,F}$ is fixed as t varies. Thus (SC15)–(SC18) hold unchanged with $\mathcal{O}_{N,F}$ in place of $\log \det H$. For real t at zero and $u > 0$, (SC12) gives $H'(0) = -(A_F^* H(0) + H(0) A_F)/u$, hence

$$-u \mathcal{O}'_{N,F}(0) - k p = 2 \text{Re Tr } A_F - k p.\tag{SC20}$$

The original constant k and its generalized eigenspace multiplicities are retained. On the full positive-real-defect constituent this is the existing aggregate defect $L_{h,k}$.

The squared Hilbert–Schmidt norm of the normal acceleration in the original domain metric, and its one nonzero squared singular value, are both

$$a_{N,F} = \frac{\|z(0)\|_{I_q}^2}{|u|^2} L G_{N,F}^{-1} L^*.\tag{SC21}$$

Indeed the adjoint of $zL : (F, G_{N,F}) \rightarrow (\mathbb{C}^q, I_q)$ is $G_{N,F}^{-1}L^*z^*$, so the product and its trace give (SC21). The unique nonzero eigenvalue equals its trace because the product has rank one. The curvature coefficient is instead the exact observer

$$c_F(t) = \frac{\|z(t)\|_{I_q}^2}{|u|^2} L B_{N,F}(t)^{-1} G_{N,F}^{-1} L^*. \quad (\text{SC22})$$

This follows from $H = G_{N,F} B_{N,F}$; it is the required map between the two metric measurements. It does not replace either metric by an unweighted coordinate norm.

Finally transport the original metric on the whole period target by $\tilde{G}_N = \Pi^{-*} G_N \Pi^{-1}$. Its projection onto YF is

$$P_N^\theta = Y G_{N,F}^{-1} Y^* \tilde{G}_N = \Pi I G_{N,F}^{-1} I^* G_N \Pi^{-1}. \quad (\text{SC23})$$

Direct multiplication proves its idempotence and its $\tilde{G}_N$ -self-adjointness. With $z_N^\theta = (I_q - P_N^\theta) \Pi e_0$, its squared target norm is the literal theta Schur complement

$$\|z_N^\theta\|_{\tilde{G}_N}^2 = e_0^* G_N e_0 - e_0^* G_N I G_{N,F}^{-1} I^* G_N e_0 > 0. \quad (\text{SC24})$$

The right side is positive because it is the norm of the nonzero G_N -orthogonal projection of e_0 away from $I(F)$. For this moving target metric $Y^* \tilde{G}_N Y = G_{N,F}$ is constant. Its differentiated metric terms are explicitly

$$\partial_t \tilde{G}_N = \Pi^{-*} G_N (A + t\mathcal{R}) \Pi^{-1} / u, \quad \partial_{\bar{t}} \tilde{G}_N = \Pi^{-*} (A + t\mathcal{R})^* G_N \Pi^{-1} / \bar{u}. \quad (\text{SC25})$$

They cancel $Y^* \tilde{G}_N Y'$ and its conjugate derivative in the product rule. Thus the constant pulled-back theta Gram and the nonzero curvature (SC16) are connected by the exact metric congruence (SC23)–(SC25), with every term retained.

For $F = 0$, use the empty Gram determinant 1, empty trace 0 and the zero normal map. For $F = E$, $P = I_q$ and $N = 0$ identically, so (SC15) is zero: the full determinant logarithm in (SC5) is the real part of a holomorphic function plus its stated constant. The strictly positive coefficient in (SC16) belongs precisely to $0 < p < q$. This curvature and the original theta comparison are fully calculated observables; no uniform arithmetic upper bound is asserted by these finite-dimensional identities.

4.1.6 The full two-parameter curvature in the same original frames

Allow the original $u \neq 0$ to vary on a retained contour branch, keeping the same χ , F , I , G_N , and target metric. Use $C_\chi(t)$ and B_χ from the literal division (XD.u1). For $a = u, t$ put

$$\Omega_u = -C_\chi(t)/u^2 + B_\chi/u, \quad \Omega_t = -(A + t\mathcal{R})/u, \quad Y_a = \Pi \Omega_a I, \quad Z_a = N \Pi \Omega_a I. \quad (\text{SC.u1})$$

The convergence and full reduction in (XD.u1)–(XD.u5) prove these are precisely $\partial_a Y$ and its original normal projection. For every $a, b \in \{u, t\}$, the complete Hermitian matrix of curvature coefficients is

$$\boxed{\partial_{\bar{a}} \partial_b \log \det H = \text{Tr}(H^{-1} Z_a^* Z_b)}. \quad (\text{SC.u2})$$

For the proof, holomorphy gives $\partial_b H = Y^* Y_b$, $\partial_{\bar{a}} H = Y_a^* Y$, and $\partial_{\bar{a}} \partial_b H = Y_a^* Y_b$. The derivative of H^{-1} is $-H^{-1} Y_a^* Y H^{-1}$. Substitution in the logarithmic determinant derivative gives $\text{Tr}(H^{-1} Y_a^* (I - Y H^{-1} Y^*) Y_b)$. Since $N^2 = N = N^*$, this is (SC.u2). In particular every linear combination has value $\text{Tr}(H^{-1} (\sum_b v_b Z_b)^* (\sum_b v_b Z_b)) \geq 0$, using the original positive H . This proves the mixed coefficients as well as both diagonal ones, with the original one-form ordering $du \wedge d\bar{u}$, $dt \wedge d\bar{u}$, $du \wedge d\bar{t}$, $dt \wedge d\bar{t}$.

At $t = 0$, $C_\chi(0) = \Phi(A)$ is a polynomial in the original action, so $C_\chi(0)I = I\Phi(A_F)$. Consequently the order-two term has zero normal projection there, while the complete first-order term remains:

$$Z_u(u, 0) = u^{-1} N \Pi B_\chi I, \quad Z_t(u, 0) = 0, \quad \partial_{\bar{u}} \partial_u \log \det H(u, 0) = |u|^{-2} \text{Tr}(H^{-1} I^* B_\chi^* \Pi^* N \Pi B_\chi I). \quad (\text{SC.u3})$$

Thus the earlier stationary t tangent retains the calculated u normal map. No invariance of F under B_χ is assumed. For $t \neq 0$, the complete expression in (SC.u1) retains both $-C_\chi(t)/u^2$ and B_χ/u before taking its squared norm.

The positive original source Gram is constant in both parameters. The transported metric therefore has the exact derivatives

$$\partial_a \tilde{G}_N = -\Pi^{-*} G_N \Omega_a \Pi^{-1}, \quad \partial_a \tilde{G}_N = -\Pi^{-*} \Omega_a^* G_N \Pi^{-1}, \quad a = u, t. \quad (\text{SC.u4})$$

They follow from $\partial_a \Pi^{-1} = -\Omega_a \Pi^{-1}$; holomorphy gives $\partial_a \Pi^{-*} = 0$. In the product rule for $Y^* \tilde{G}_N Y$ these terms cancel the derivatives of Y and Y^* , giving exactly the fixed $I^* G_N I$. Formula (SC25) is its $a = t$ case. The full original source comparison is (DS72)–(DS74); in particular its theta boundary $dK_N \Omega_a j_E$ remains in the Euler commutator. All curvature coefficients (SC.u2) also apply to $\log \det(G_{N,F}^{-1} H)$, since the determinant of $G_{N,F}$ is constant in both parameters.

For every neighboring inclusion J in (SC31), the identity $h_J = v_{N,J} b_{N,J}$ in (SC35) now holds on this same two-parameter branch. Its u and mixed logarithmic derivatives are the corresponding formula (SC.u2) with $I = J$, $H = J^* \Pi^* \Pi J$, and its own actual normal projection. For a zero-column inclusion its determinant is 1 and every derivative is zero; for the full inclusion the normal projection is zero. This propagates the singular connection through all three retained minor maps rather than replacing a rectangular minor by the full determinant.

4.1.7 The full primary matrix now evaluates every original ideal volume

The completed single-primary proof specializes this calculation to the original $h(s) = (s - \rho)^m$, $t = 0$, with $n = m + 1$, $c = -(-\rho)^n/n$ and $y = s - \rho$. Its exact Pascal matrix is $P_{ab} = \binom{b}{a} \rho^{b-a}$ for $a \leq b$, zero otherwise; $e_s = e_y P$ and P^{-1} has $-\rho$ in place of ρ . Retain its full ordered period matrix

$$\Pi_s = e^{c/u} W D(u) P, \quad W_{jb} = e^{2\pi i j(b+1)/n} - 1, \quad D(u)_{bb} = d_b u^{\beta_b}, \quad \beta_b = (b+1)/n, \quad d_b = n^{\beta_b-1} e^{\pi i \beta_b} \Gamma(\beta_b). \quad (\text{SC.u5})$$

The proof SP.10–15 of this formula is included in full: it uses the original contour translation, the decaying endpoint connectors and the gamma radial integral, so both c and all phases in (SC.u5) have their original values.

For an original invariant constituent of dimension p , put $r = m - p$ and let J_r have the literal polynomial columns $y^r, y^{r+1}, \dots, y^{m-1}$ in the original s basis. Every ideal of the primary algebra is $(y^r)/(y^m)$ by division in $\mathbb{C}[y]$, so its specified inclusion has the unique form $I = J_r C_F$ for an invertible constant p -by- p matrix C_F . Let W_r select the columns $r, \dots, m-1$ from W , and D_r the same diagonal entries from D . Since $P J_r$ is the literal inclusion of those translated coordinates,

$$Y = e^{c/u} W_r D_r C_F, \quad H = e^{2\text{Re}(c/u)} C_F^* D_r^* (W_r^* W_r) D_r C_F. \quad (\text{SC.u6})$$

The exact root-of-unity sum gives $W_r^* W_r = n(I_p + \mathbf{1}_p \mathbf{1}_p^*)$: including the zero row adds zero, and expansion of the three nonconstant character sums leaves $n\delta_{ab} + n$. This matrix has eigenvalue $n(p+1)$ on $\mathbf{1}_p$ and eigenvalue n on its perpendicular complement, so its determinant is $n^p(p+1)$. Taking determinants in the unchanged matrices therefore gives

$$\det H(u, 0) = |\det C_F|^2 n^p (p+1) e^{2p \text{Re}(c/u)} \prod_{b=r}^{m-1} |d_b|^2 |u|^{2\beta_b}. \quad (\text{SC.u7})$$

No frame determinant, contour phase or Taylor coefficient in the original source Gram $G_{N,F} = I^* G_N I$ is removed. The determinant of the actual comparison is exactly the right side divided by $\det G_{N,F}$; its complete matrix is $G_{N,F}^{-1}$ times (SC.u6). For $p = 0$ all products and determinants are empty and equal one, and for $p = m$ this is the full determinant formula.

In particular $Y(F) = \text{im } W_r$ is constant as u varies on $u \neq 0$: D_r and C_F are invertible. Thus its normal u derivative and its u curvature vanish on this exact primary slice. Equivalently, $PB_hP^{-1} = \text{diag}(\beta_b)$ preserves $PJ_r(F)$ in (SC.u3). This evaluates that previously retained B_h/u normal term on the actual primary ideals. The t -normal map and its full acceleration are still (SC12)–(SC13); the tensor boundary requires the ordered character products calculated in TP/TPF rather than propagating the one-factor invariant-space dimension.

4.1.8 The exact derived base change at a collision

The trace and the positive-degree diagonal groups in the delivered SGA note have a further explicit specialization map. Retain the whole one-parameter algebra over $T = \mathbb{C}[t]$:

$$\mathcal{E} = T[S]/(\chi(S) - t) \simeq \mathbb{C}[S], \quad t \mapsto \chi(S), \quad M = \mathcal{E}/\chi' \mathcal{E}. \quad (\text{SC26})$$

The isomorphism sends $[P(t, S)]$ to $P(\chi(S), S)$; its inverse sends S to its original class. Monic division proves that $\mathcal{E}$ is free of rank q over T . Since χ' is a nonzero polynomial over $\mathbb{C}$, multiplication by χ' is injective on $\mathcal{E} \simeq \mathbb{C}[S]$. Consequently

$$0 \longrightarrow \mathcal{E} \xrightarrow{\chi'} \mathcal{E} \longrightarrow M \longrightarrow 0 \quad (\text{SC27})$$

is a free T -resolution of M . The diagonal resolution gives $HH_0(\mathcal{E}/T) = \mathcal{E}$, $HH_{2a+1}(\mathcal{E}/T) = M$ and $HH_{2a+2}(\mathcal{E}/T) = 0$ for $a \geq 0$. For clarity, that resolution is free over $\mathcal{B} = \mathcal{E} \otimes_T \mathcal{E}$ with alternating maps $X - Y$ and $\mathcal{C}_H = (H(X) - H(Y))/(X - Y)$, followed by the multiplication augmentation. Over $(T[Y]/H(Y))[X]$ both factors are monic and therefore non-zero-divisors; their product is $H(X)$ modulo $H(Y)$. Cancellation proves each alternating kernel equals the preceding image. Tensoring by the diagonal sets $X = Y$ and gives the maps $0, \chi', 0, \chi', \dots$, proving the stated groups.

Fix $t_0 \in \mathbb{C}$, put $\kappa = T/(t - t_0)$ and $E_{t_0} = \mathbb{C}[S]/(\chi - t_0)$. Tensoring (SC27) gives the complete kernel-and-cokernel sequence

$$\begin{aligned} 0 \longrightarrow \text{Tor}_1^T(M, \kappa) \longrightarrow E_{t_0} \xrightarrow{\chi'} E_{t_0} \\ \longrightarrow M \otimes_T \kappa \longrightarrow 0. \end{aligned} \quad (\text{SC28})$$

It computes the actual specialized diagonal groups:

$$\begin{aligned} HH_{2a+1}(E_{t_0}/\mathbb{C}) &= M \otimes_T \kappa = E_{t_0}/(\chi'), \\ HH_{2a+2}(E_{t_0}/\mathbb{C}) &= \text{Tor}_1^T(M, \kappa) = \text{ann}_{E_{t_0}}(\chi'). \end{aligned} \quad a \geq 0. \quad (\text{SC29})$$

Every arrow is induced by the same polynomial multiplication, so these are equivariant maps for the original S -action.

There is also a representative-level description of the Tor map. Resolve κ by $[T \xrightarrow{t-t_0} T]$ to identify $\text{Tor}_1^T(M, \kappa)$ with the kernel of $t - t_0$ on M . If $[v]_{\chi'}$ belongs to this kernel, write

$$(\chi - t_0)v = \chi'w. \quad [v]_{\chi'} \mapsto [w]_{\chi-t_0}. \quad (\text{SC30})$$

Replacing v by $v + \chi'a$ replaces w by $w + (\chi - t_0)a$, so the map is well-defined. Its image is annihilated by χ' . If its image is zero, $w = (\chi - t_0)a$ and cancellation of the nonzero polynomial $\chi - t_0$ gives $v = \chi'a$, proving injectivity. Conversely an annihilated class $[w]_{\chi-t_0}$ satisfies $\chi'w = (\chi - t_0)v$ for a polynomial v , giving its preimage. These operations commute with multiplication by S .

At a root λ of multiplicity m in $\chi - t_0$, its local algebra is $\mathbb{C}[x]/(x^m)$, $x = S - \lambda$, and $\chi' = mx^{m-1}a(x)$ with the original invertible local factor a (this equality is in the local algebra: before reduction, the derivative of $x^ma(x)$ also contains the term $x^ma'(x)$). The odd group has local module $\mathbb{C}[x]/(x^{m-1})$ and the even group has local module $(x)/(x^m)$; both have dimension

$m - 1$, with their full induced $S = \lambda + x$ action. The map $[b]_{x^{m-1}} \mapsto [xb]_{x^m}$ is an explicit module isomorphism between these local modules, using the stated local coordinate. For $m = 1$ both groups vanish. Thus the zero even group of the family and the nonzero even group of a collision fibre are related by the exact Tor term (SC28), not by omitting specialization data. The coefficient maps in the residue construction commute with base change; this calculation retains the additional derived term required when their kernels are specialized. Lifting every arrow at its original Split-Zero support keeps its receiving supported zero and external absence distinct as in (SC14).

4.1.9 The exact neighboring-minor identity and its theta factors

The ideal basis in (SC9) supplies three actual polynomial inclusions. Write their increasing-power coefficient matrices as

$$J_g = [g, Sg, \dots, S^{p-1}g], \quad J_- = [g, Sg, \dots, S^{p-2}g], \quad J_+ = [J_g, e_0]. \quad (\text{SC31})$$

For $p = 1$, J_- has zero columns. All three are injective on their stated coefficient spaces; the first two assertions follow from their distinct monic leading degrees, and the last from $e_0 \notin I(F)$. Let $C = (I^*I)^{-1}I^*J_g$. Since the columns of J_g lie in $\text{im } I$, this satisfies $IC = J_g$; it is invertible because both matrices are bases of that same subspace. The exact frame equations are

$$H_g(t) := (\Pi J_g)^* \Pi J_g = C^* H(t) C, \quad LC = (0, \dots, 0, 1), \quad h_g(t) = |\det C|^2 \det H(t), \quad (\text{SC32})$$

where $h_J(t) = \det((\Pi J)^* \Pi J)$ for each of the three inclusions. The scalar $|\det C|^2$ has been retained; it is independent of t , so it has zero logarithmic derivatives.

The Gram of ΠJ_+ has block form

$$\begin{pmatrix} H_g & (\Pi J_g)^* \Pi e_0 \\ (\Pi e_0)^* \Pi J_g & \|\Pi e_0\|^2 \end{pmatrix}. \quad (\text{SC33})$$

Eliminate its off-diagonal blocks using triangular matrices of determinant one. Its lower-right Schur complement is the squared norm of the orthogonal projection of Πe_0 away from $\text{im } \Pi J_g$, which is $\|z(t)\|^2$ by (SC11). Thus $h_+ = h_g \|z\|^2$. The last diagonal entry of H_g^{-1} , computed by the adjugate formula, is its upper-left principal minor divided by h_g , namely h_-/h_g . By (SC32) that entry is exactly $LH^{-1}L^*$. Formula (SC16) becomes

$$\boxed{\partial_{\bar{t}} \partial_t \log h_g(t) = \frac{|t|^2}{|u|^2} \frac{h_+(t) h_-(t)}{h_g(t)^2}}. \quad (\text{SC34})$$

This identity retains all three original polynomial maps. It is the precise neighboring-minor equation joining the period calculation to the source-volume and Toda calculations; it does not identify differently ordered or differently weighted families.

For each same inclusion $J = J_-, J_g, J_+$, define its original theta volume and its positive comparison determinant by

$$v_{N,J} = \det(J^* G_N J), \quad b_{N,J}(t) = \det((J^* G_N J)^{-1} J^* \Pi(t)^* \Pi(t) J). \quad (\text{SC35})$$

Injectivity and positivity show that both numbers are positive. For the empty inclusion both are 1. Taking the determinant of the literal product gives $h_J = v_{N,J} b_{N,J}$, and therefore

$$c_F(t) = \frac{1}{|u|^2} \frac{v_{N,J_+} v_{N,J_-}}{v_{N,J_g}^2} \frac{b_{N,J_+}(t) b_{N,J_-}(t)}{b_{N,J_g}(t)^2}. \quad (\text{SC36})$$

Every original theta mass and coefficient inclusion remains in the v factors; every period comparison remains in the b factors. This proves the metric dictionary required to use (SC34) within that original source programme.

For $p = q - 1$, the divisor g is monic linear. The square matrix $J_+ = [g, Sg, \dots, S^{q-2}g, e_0]$ has determinant $(-1)^{q-1}$: expand along its last column, and the remaining minor on rows $1, \dots, q-1$ is upper triangular with every diagonal entry 1. Thus, with that sign retained,

$$\det(\Pi J_+) = (-1)^{q-1} \det \Pi, \quad h_+(t) = (2\pi|u|)^q \exp\left(2 \operatorname{Re} \frac{\operatorname{Tr} \Phi_t(A + t\mathcal{R})}{u}\right). \quad (\text{SC37})$$

The top minor in (SC34) is therefore the explicitly evaluated full determinant in this codimension-one case. The remaining two minors and their exact theta factors are still those in (SC31)–(SC36).

5 The period curvature of the retained critical-comparison kernel

5.1 The period curvature of the retained critical-comparison kernel

This section joins the original theta representatives, the critical-line comparison of the Hochschild note (H16)–(H25), and the period curvature (SC1)–(SC37). The single-packet identification is proved first, including its full arithmetic unit. We then calculate the factorwise critical comparison on every cyclic tensor packet. All tensor targets below use the completed projective tensor product of the programme’s locally convex spaces; its finite surviving image is proved injective by a continuous retraction, without an assertion of tensor exactness for arbitrary spaces.

The proof-bearing comparison source is *The original theta complex inside the Hochschild trace realization*, (H1)–(H25), at the retained PR27 revision [a4a87844c257485ae6d669f5b3eb6d96b9e8b8d8](#); its original analytic constructions are retained in the cumulative source. The arithmetic section and unit maps are (A7)–(A13), (CT.1)–(CT.5), and (TVB.1)–(TVB.3). The calculations here prove the additional maps between those specified objects.

5.1.1 The single-packet polynomial and its complete arithmetic unit

Retain the original nonempty packet, with every selected zero’s complete order,

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad g(s) = 2\xi(s), \quad v_h(s) = g(s)/h(s), \quad d = \deg h. \quad (\text{PC1})$$

The packet is the same one used by the original source, including its reflection symmetry when that symmetry is used in its metric. Put $E_h = \mathbb{C}[s]/(h)$, and let j_h mean the full Hermite remainder. Its local coordinate at ρ is $x = s - \rho$, with basis $1, x, \dots, x^{m_\rho-1}$. Thus

$$v_{h,\rho} = \sum_{a=0}^{m_\rho-1} \frac{v_h^{(a)}(\rho)}{a!} x^a, \quad v_h = j_h(v_h), \quad \varepsilon_h = v_h^{-1}. \quad (\text{PC2})$$

This inverse exists with all its coefficients retained. Indeed write $h(s) = (s - \rho)^{m_\rho} h_\rho(s)$ and $g(s) = (s - \rho)^{m_\rho} g_\rho(s)$. Both remaining factors are nonzero at ρ , so the constant coefficient $a_0 = g_\rho(\rho)/h_\rho(\rho)$ is nonzero. If the coefficients of $v_{h,\rho}$ are a_a , its inverse coefficients satisfy

$$b_0 = a_0^{-1}, \quad b_n = -a_0^{-1} \sum_{a=1}^n a_a b_{n-a} \quad (1 \leq n < m_\rho). \quad (\text{PC3})$$

Multiplication makes the constant coefficient one and each other coefficient through degree $m_\rho - 1$ zero. The coprime polynomial factors of h give the Chinese-remainder isomorphism; assembling these inverses proves (PC2). The notation $j_h(h/g)$ for ε_h uses its holomorphic germ at every selected zero; no global entire reciprocal is asserted.

For $k = 1$, the sum variable is literally $S = s_1$. The isomorphism $\mathbb{C}[S] \rightarrow \mathbb{C}[s_1]$, $S \mapsto s_1$, pulls $(h(s_1))$ back to $(h(S))$. The monic annihilator defining the cyclic packet is consequently

$$\begin{aligned} \chi_{h,1}(S) &= h(S), \quad E = \mathbb{C}[S]/(h(S)), \quad A = M_S, \quad q = d, \\ \iota_1 : E &\xrightarrow{\sim} E_h, \quad [P(S)] \mapsto [P(s)], \quad \eta_h = M_{v_h} \iota_1. \end{aligned} \quad (\text{PC4})$$

The equality includes every exponent m_ρ . The inverse of η_h is $\iota_1^{-1}M_{\varepsilon_h}$. It intertwines the two original multiplication actions because multiplication by the unit commutes with s . Its actual multiplicative rule is $\eta_h(x)\eta_h(y) = v_h\eta_h(xy)$ and $\eta_h(1) = v_h$. Thus its source and target polynomial algebras and its module-map type remain explicit.

Use the original spaces and maps

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0\}, \quad Q = \mathcal{B}/\Theta V, \quad \mathbf{q} : \mathcal{B} \rightarrow Q, \\ \mathcal{M}F(s) &= \int_0^\infty F(x)x^{s-1} dx, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}F_h = v_h, \quad h(D)F_h = \Theta\phi_*. \end{aligned} \tag{PC5}$$

The source construction of F_h is the iterated original Euler inverse $S_\rho F = x^{-\rho} \int_x^\infty F(y)y^{\rho-1} dy$ on the vanishing Mellin fibre, as proved in (A5)–(A7) and (TVB.1). Let $\mathcal{T}_h P = P(D)F_h$, $\pi_h P = [P]_h$, and $J_h = j_h \mathcal{M} : \mathcal{B} \rightarrow E_h$, with the retained $\sigma_h : E_h \rightarrow Q$. The exact source relations are

$$\mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h = \eta_h \pi_h, \quad \mathbf{q} \mathcal{T}_h = \sigma_h \eta_h \pi_h. \tag{PC6}$$

Here and below π_h uses the variable identification (PC4). The first identity follows from $h(D)F_h = \Theta\phi_*$ and commutation with D . Integration by parts gives $\mathcal{M}\mathcal{T}_h P = v_h P$, proving the second identity. For the third, the retained section in (A10) is $\sigma_h = \mathbf{q} R_{\text{ref}}$, with $R_{\text{ref}} u = \mathcal{T}_h \text{rem}_h(\varepsilon_h u)$. Write $P = P_0 + hT$, $\deg P_0 < d$. Then $R_{\text{ref}} \eta_h[P] = \mathcal{T}_h P_0$, and the difference from $\mathcal{T}_h P$ is the displayed theta boundary $\Theta(T(D)\phi_*)$. This proves the precise equality with that section, rather than identifying two sections merely from equal jets. Since $J_h \Theta = 0$, its descended map is $\bar{J}_h : Q \rightarrow E_h$, defined by $\bar{J}_h \mathbf{q} = J_h$; the section identity is $\bar{J}_h \sigma_h = I_{E_h}$. Also $D\mathcal{T}_h P = \mathcal{T}_h(SP)$ and (PC6) imply $D_Q \sigma_h \eta_h = \sigma_h \eta_h A$. Surjectivity of η_h therefore proves $D_Q \sigma_h = \sigma_h M_s$ with its exact domains.

For each admitted original polynomial degree N , retain its actual remainder matrix B_N , source Gram O_N , and least representative

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad r_N = \mathcal{T}_h R_N. \tag{PC7}$$

The source Gram is for the original density $|v_h(1/2 + iy)|^2 dy / (2\pi)$ and the original monic coordinates. Since $B_N R_N = I$, (PC6) gives

$$j_E := \eta_h^{-1} J_h : \mathcal{B} \rightarrow E, \quad j_E r_N = I, \quad J_h r_N = \eta_h, \quad \mathbf{q} r_N = \sigma_h \eta_h, \quad j_E D = A j_E. \tag{PC8}$$

The last identity uses $J_h D = M_s J_h$ and the intertwining in (PC4). In particular the coefficient observer in (SC7) is j_E . In full Mellin-jet coordinates the same Gram is $\eta_h^{-*} G_N \eta_h^{-1}$; multiplying back by η_h^* and η_h recovers exactly G_N . The unit is present in both the source map and the coordinate congruence.

5.1.2 The weighted critical comparison and every recovered jet

Reserve $y \in \mathbb{R}$ for the critical-line coordinate, distinct from the complex period parameter t . The retained continuous map is

$$Q_{\text{crit}} = \mathcal{S}(\mathbb{R}) / \overline{\zeta(1/2 + iy) \mathcal{S}(\mathbb{R})}, \quad \Gamma[F] = [\mathcal{M}F(1/2 + iy)], \quad M = M_{1/2+iy}, \quad \Gamma D_Q = M\Gamma. \tag{PC9}$$

It is induced by the plus-sign Mellin restriction: the theta image restricts to $2\zeta(1/2 + iy)\mathcal{M}\phi(1/2 + iy)$. The continuity and strong-Schwartz domains are those proved in (H16)–(H20); continuity also makes it factor through the closed theta quotient. Let $Z_c = \{\rho \in Z : \Re \rho = 1/2\}$ and $Z_o = Z \setminus Z_c$. Retain the complete factors

$$h_c = \prod_{\rho \in Z_c} (S - \rho)^{m_\rho}, \quad h_o = \prod_{\rho \in Z_o} (S - \rho)^{m_\rho}, \quad h = h_c h_o. \tag{PC10}$$

An empty product is 1. Chinese remainders identify $E = E_c \oplus E_o$, where E_c is the ideal $(h_o)/(h)$ and E_o is the ideal $(h_c)/(h)$, with projections π_c, π_o . The notation also retains the isomorphisms to the products of their local primary algebras, including their nilpotent coordinates.

Define the actual weighted comparison

$$\tilde{\sigma}_h = \sigma_h \eta_h : E \rightarrow Q, \quad \mathcal{C}_h = \Gamma \tilde{\sigma}_h : E \rightarrow Q_{\text{crit}}. \quad (\text{PC11})$$

Then

$$\boxed{\ker \mathcal{C}_h = E_o, \quad \mathcal{C}_h A = M \mathcal{C}_h.} \quad (\text{PC12})$$

To prove the kernel, first take an off-critical root ρ . The reciprocal $(1/2 + iy - \rho)^{-1}$ and all its derivatives have polynomial bounds on the real axis. Multiplication by it preserves Schwartz space, the zeta ideal and its closure, and is inverse to $M - \rho$ on Q_{crit} . On a primary block $(A - \rho)^{m_\rho} = 0$, intertwining therefore forces the whole block to zero. Multiplication by v_h preserves each CRT block and does not change this conclusion.

For a critical root $\rho = 1/2 + i\gamma$ retain the evaluation

$$\Xi_\rho[f] = \sum_{j=0}^{m_\rho-1} \frac{i^{-j}}{j!} f^{(j)}(\gamma) x^j \in \mathbb{C}[x]/(x^{m_\rho}), \quad x = S - \rho. \quad (\text{PC13})$$

Every derivative used here vanishes on the zeta ideal and, by continuity, on its closure. The chain rule supplies i^j when differentiating $\mathcal{M}F(1/2 + iy)$, so (PC6) and $\bar{J}_h \sigma_h = I$ give the full formula

$$\Xi_\rho \mathcal{C}_h[P] = v_{h,\rho}[P]_\rho = \sum_{j < m_\rho} \sum_{a=0}^j \frac{v_h^{(a)}(\rho)}{a!} \frac{P^{(j-a)}(\rho)}{(j-a)!} x^j. \quad (\text{PC14})$$

In particular the critical observer recovers the actual unit on $e_0 = [1]$, rather than the constant one. Assemble the Ξ_ρ by the retained CRT map and then ι_1^{-1} into $\Xi_c : Q_{\text{crit}} \rightarrow E_c$. Put $v_c = \pi_c \iota_1^{-1}(v_h)$ and $\varepsilon_c = \pi_c \iota_1^{-1}(\varepsilon_h)$; these are inverse units in the ideal algebra E_c , whose identity is $e_c = \pi_c(e_0)$. Multiplication by this inverse defines the continuous map

$$\mathcal{R}_c := M_{\varepsilon_c} \Xi_c, \quad \mathcal{R}_c \mathcal{C}_h = \pi_c. \quad (\text{PC15})$$

This is a left inverse on the entire critical block, proving (PC12). For an empty critical packet both maps have zero-dimensional target. When $E_c \neq 0$, (PC14) proves $\mathcal{C}_h(e_0) \neq 0$. The finite image $W_h = \mathcal{C}_h(E_c)$ is preserved by M , and $\mathcal{C}_h|_{E_c}$ and $\mathcal{R}_c|_{W_h}$ are inverse maps.

5.1.3 The comparison kernel is a holomorphic period subbundle

Use exactly the polynomial $\chi = h$ in (SC1)–(SC6), its potential

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS, \quad h(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad u \in \mathbb{C}^*. \quad (\text{PC16})$$

Fix the original argument of u and rays $\theta_j = (\arg u + \pi + 2\pi j)/(q+1)$, with $\Gamma_j = -\ell_0^{\text{ray}} + \ell_j^{\text{ray}}$ and $\Pi_{jb}(t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS$. The original monomial factors are

$$(\Pi_{\text{mon}})_{j,b} = \frac{((q+1)|u|)^{(b+1)/(q+1)}}{q+1} \Gamma\left(\frac{b+1}{q+1}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/(q+1)} - 1). \quad (\text{PC17})$$

They have nonzero determinant: a column dependence of their last matrix gives a polynomial of degree at most q vanishing at all $q+1$ roots of unity. Its coefficients must be zero. Along any compact coefficient path the negative leading ray term dominates the lower-degree terms by an integrable bound $Cr^L \exp(-r^{q+1}/(2(q+1)|u|))$. Differentiation under the integral is therefore

valid. The original polynomial reduction $uP' + (h-t)P$ has leading degree $\deg P + q$ and the same leading coefficient, so it has a unique remainder below degree q . Its integral vanishes by the exact derivative and cancellation of the two origin values. Coefficient variation thus gives $\Pi' = \Pi B$ along a path, and the adjugate determinant identity gives $\det \Pi(s) = \det \Pi(0) \exp(\int \text{Tr } B)$. This proves invertibility at every t , including all collisions.

In the unchanged coefficient basis set

$$\ell([P]) = [S^{q-1}] \text{rem}_h P, \quad \mathcal{R} = e_0 \ell, \quad A_t = A + t\mathcal{R}, \quad \Pi'(t) = -\Pi(t)A_t/u. \quad (\text{PC18})$$

The last equality follows by differentiating in t : for the final column the sole reduction is $[h-t] = 0$, obtained from the differential of the constant polynomial one. Define maps on the period target, with range still in the original critical quotient,

$$\mathcal{C}_{h,t} = \mathcal{C}_h \Pi(t)^{-1}, \quad A_t^{\text{per}} = \Pi(t)A_t \Pi(t)^{-1}, \quad \mathcal{K}_{h,t} = \ker \mathcal{C}_{h,t} = \Pi(t)E_o. \quad (\text{PC19})$$

The equality of kernels follows from invertibility and (PC12). The inverse matrix is holomorphic, so this is a holomorphic kernel bundle of constant rank, with the indicated global frame from E_o . Its quotient is the fixed finite image W_h ; more explicitly $\mathcal{C}_{h,t} \Pi(t)|E_c = \mathcal{C}_h|E_c$ and $\Pi(t)|E_c \mathcal{R}_c$ is a right inverse on W_h .

The exact action defect is

$$\boxed{\mathcal{C}_{h,t} A_t^{\text{per}} - M \mathcal{C}_{h,t} = t \mathcal{C}_h(e_0) \ell \Pi(t)^{-1}.} \quad (\text{PC20})$$

Indeed multiplying the first term gives $\mathcal{C}_h(A+t\mathcal{R})\Pi^{-1}$, and the part containing A cancels $M\mathcal{C}_h\Pi^{-1}$ by (PC12). This proves the formula before restricting its domain.

Put $F = E_o$, $p = \dim F$. If $0 < p < q$, it is the ideal $(h_c)/(h)$, with ordered basis $[h_c], [Sh_c], \dots, [S^{p-1}h_c]$. The last member is monic of degree $q-1$, so $\ell|F$ is onto with kernel dimension $p-1$. An ideal containing e_0 would be all of E , hence $e_0 \notin F$. It follows from (PC14) and (PC20) that, for $t \neq 0$, the restriction of this action defect to $\mathcal{K}_{h,t}$ has

$$\text{im} = \mathbb{C}\mathcal{C}_h(e_0), \quad \ker = \Pi(t)(F \cap \ker \ell), \quad \text{rank} = 1. \quad (\text{PC21})$$

At $t = 0$ it is zero. If $F = 0$ its restricted domain is zero. If $F = E$, the comparison $\mathcal{C}_h$ is zero and every displayed defect is zero. These statements cover all actual sector possibilities without asserting that an off-critical zero exists.

5.1.4 The full curvature and both original metric observations

Take any specified coefficient inclusion $I : F \hookrightarrow E$; put $L = \ell I$, and for $0 < p < q$ define

$$Y = \Pi I, \quad H = Y^* Y, \quad P = Y H^{-1} Y^*, \quad N = I_q - P, \quad z = N \Pi e_0. \quad (\text{PC22})$$

This is exactly the kernel bundle (PC19), measured in the fixed period-coordinate Gram I_q . The Gram is positive because Y is injective. Multiplication proves that P is its orthogonal projection. Since $e_0 \notin F$ and Π is invertible, $z(t) \neq 0$ for every t . Original invariance $AI = I A_F$ gives

$$Y' = -Y A_F/u - (t/u) \Pi e_0 L, \quad N Y' = -(t/u) z L, \quad N(0) Y''(0) = -z(0) L/u. \quad (\text{PC23})$$

The final equation follows by differentiating (PC18); the term $\Pi(0)A^2 I/u^2$ lies inside the kernel frame and is removed by $N(0)$. Thus the first normal derivative vanishes at zero and the normal acceleration has rank one, with its original u factor.

For $t = x + iy_t$ use $\partial_t = (\partial_x - i\partial_{y_t})/2$ and its conjugate. Holomorphicity gives $\partial_t H = Y^* Y'$, $\partial_{\bar{t}} H = Y'^* Y$, and $\partial_{\bar{t}} \partial_t H = Y'^* Y'$. Differentiate $H^{-1} H = I$ and the determinant to obtain

$$\begin{aligned} \partial_{\bar{t}} \partial_t \log \det H &= \text{Tr}(H^{-1} Y'^* Y' - H^{-1} Y'^* Y H^{-1} Y^* Y') \\ &= \text{Tr}(H^{-1} Y'^* N Y') = \frac{|t|^2}{|u|^2} \|z\|^2 L H^{-1} L^*. \end{aligned} \quad (\text{PC24})$$

Every factor in the last coefficient is strictly positive except the explicit $|t|^2$: $z \neq 0$, $L \neq 0$ and $H^{-1} > 0$. The coefficient is real analytic, so the curvature vanishes at zero to exactly quadratic order. In the determinant-dual metric $(\det H)^{-1}$ its first Chern form is $\frac{i}{2\pi} \frac{|t|^2}{|u|^2} \|z\|^2 L H^{-1} L^* dt \wedge d\bar{t}$, with $i dt \wedge d\bar{t} = 2dx \wedge dy_t$. This proves that the SC curvature is the curvature of this actual critical-comparison kernel in the specified fixed period metric.

The original theta Gram remains separately and explicitly present. Let $G_{N,F} = I^* G_N I$, and set

$$B_{N,F} = G_{N,F}^{-1} H, \quad \tilde{G}_N = \Pi^{-*} G_N \Pi^{-1}, \quad P_N^\theta = \Pi I G_{N,F}^{-1} I^* G_N \Pi^{-1}. \quad (\text{PC25})$$

Thus $H = G_{N,F} B_{N,F}$ and $Y^* \tilde{G}_N Y = G_{N,F}$. Both identities follow by direct multiplication. The squared normal norm of Πe_0 for the moving target metric is exactly

$$e_0^* G_N e_0 - e_0^* G_N I G_{N,F}^{-1} I^* G_N e_0 > 0. \quad (\text{PC26})$$

It is positive because it is the squared norm of the nonzero G_N -orthogonal component of e_0 outside F . From (PC18), $\partial_t \Pi^{-1} = A_t \Pi^{-1} / u$, so

$$\partial_t \tilde{G}_N = \Pi^{-*} G_N A_t \Pi^{-1} / u, \quad \partial_t \tilde{G}_N = \Pi^{-*} A_t^* G_N \Pi^{-1} / \bar{u}. \quad (\text{PC27})$$

In the product rule for $Y^* \tilde{G}_N Y$ these terms cancel the derivatives of Y and Y^* exactly. Hence its Gram and determinant are constant in the same holomorphic frame. Formula (PC24) and this cancellation are related by the full displayed metric congruence, including all original theta masses and unit factors. For $F = 0$ the determinant is 1; for $F = E$, $N = 0$. Both endpoint curvatures are therefore zero.

5.1.5 The singular connection on the retained critical kernel

The fixed- u formulas above now extend on each original branch of $u \neq 0$ using (XD.u1)–(XD.u5), with $\chi = h$ at $k = 1$. Their coefficient identification is exactly (PC4), and the full v_h in (PC6)–(PC8) remains unchanged. Put $\Omega_u = -C_h(t)/u^2 + B_h/u$ in these original coordinates. The observer and its full kernel obey

$$\mathcal{C}_{h,u,t} = \mathcal{C}_h \Pi(u, t)^{-1}, \quad \ker \mathcal{C}_{h,u,t} = \Pi(u, t) E_o, \quad \partial_u \mathcal{C}_{h,u,t} = \mathcal{C}_h (C_h(t)/u^2 - B_h/u) \Pi^{-1}. \quad (\text{PC.u1})$$

Invertibility proves the first two identities; differentiating $\Pi^{-1} \Pi = I$ with (XD.u5) proves the third. The displayed critical target, all local inverse units in (PC15), and the right inverse $\Pi|_{E_c} \mathcal{R}_c$ are unchanged.

For $Y = \Pi I$, $H = Y^* Y$ and $N = I - Y H^{-1} Y^*$ from (PC22), the normal u map is

$$N \partial_u Y = N \Pi (-C_h(t)/u^2 + B_h/u) I. \quad (\text{PC.u2})$$

The full curvature and both metric derivatives are exactly (SC.u2)–(SC.u4), now on this particular kernel inclusion. To check the marked boundary parameter $t = 0$ directly, $C_h(0) = \Phi_h(A)$ preserves the ideal E_o , so the normal image of its order-two term is zero; its retained first-order term is $u^{-1} N \Pi B_h I$. Thus

$$\partial_{\bar{u}} \partial_u \log \det H(u, 0) = |u|^{-2} \text{Tr}(H^{-1} I^* B_h^* \Pi^* N \Pi B_h I), \quad \partial_{\bar{u}} \partial_t \log \det H(u, 0) = 0. \quad (\text{PC.u3})$$

The second equality follows because the normal t map is zero at $t = 0$, as already proved in (PC23). Empty and full kernels have zero normal maps and use the same empty determinant convention. At other t the complete two terms of (PC.u2), their cross pairing and the mixed u, t coefficient remain.

For each cyclic tensor packet (PC29), apply this calculation to the literal polynomial $\chi_{h,k}$ with its degree q_k , not to a product of one-factor period matrices. Replace h, E_o, I, C_h by $\chi_{h,k}, F_k, I_k, C_k$

in (PC.u1)–(PC.u3). Their proofs use precisely the already proved ideal and intertwining identities (PC34)–(PC38), so this gives the full u derivative of the same critical observation and the two-parameter curvature of every existing kernel minor. The marked-product period map instead has domain $E_h^{\otimes k}$ and restricts to E_k through the exact weighted injection η_k in (PC29). Its retained rectangular period form is $\eta_k^* \Pi_{\text{prod}}^* \Pi_{\text{prod}} \eta_k$. These typed maps and the parameter cyclic relation in (XD.u7)’s following paragraph carry the two families together; their different coefficient domains are never silently identified.

5.1.6 The exact completed tensor comparison and its collision jets

We now retain an arbitrary original integer $k \geq 1$. Set

$$B_k = E_h^{\otimes k} = \mathbb{C}[s_1, \dots, s_k] / (h(s_1), \dots, h(s_k)), \quad S = s_1 + \dots + s_k, \quad U_k = v_h^{\otimes k}, \quad (\text{PC28})$$

and retain the original cyclic algebra and its two distinct maps. The variable isomorphism between its tensor coefficient carriers is $\iota_k = \iota_1^{\otimes k} : E^{\otimes k} \xrightarrow{\sim} B_k$. Thus

$$E_k = \mathbb{C}[S] / (\chi_{h,k}), \quad A_k = M_S, \quad \alpha_k[P] = [P(S)] \in B_k^{\mathfrak{S}_k}, \quad \eta_k = M_{U_k} \alpha_k. \quad (\text{PC29})$$

The map α_k is unital; η_k is the original unit-weighted module injection. Define the fully typed comparison $\bar{\alpha}_k = \iota_k^{-1} \alpha_k : E_k \rightarrow E^{\otimes k}$. Then $\eta_k = \eta_h^{\otimes k} \bar{\alpha}_k$, since $\eta_h = M_{v_h} \iota_1$. Their source polynomial is exactly

$$\chi_{h,k}(S) = \prod_{\lambda} (S - \lambda)^{m_{\lambda}}, \quad m_{\lambda} = \max_{\substack{\rho_1 + \dots + \rho_k = \lambda \\ \rho_i \in \mathbb{Z}}} \left(1 + \sum_{i=1}^k (m_{\rho_i} - 1) \right). \quad (\text{PC30})$$

All tuples are ordered in the full tensor algebra; equal sums are grouped by their literal equality. To verify the exponent, the local factor at a tuple has coordinates $x_i = s_i - \rho_i$ and quotient by $(x_i^{m_{\rho_i}})$. The nilpotent sum $x_1 + \dots + x_k$ has highest nonzero power $D = \sum_i (m_{\rho_i} - 1)$: that power contains the monomial $\prod_i x_i^{m_{\rho_i} - 1}$ with the nonzero coefficient $D! / \prod_i (m_{\rho_i} - 1)!$. Higher powers vanish. Taking the maximum over equal sums proves (PC30) and the injectivity of α_k . The invariant unit vector already has that same annihilator, so passage to symmetric invariants does not alter it.

The coefficient map of the same collision-jet calculation. The nonzero factorial in the preceding proof also determines the entire coefficient-specialization map. Fix an ordered tuple $\rho = (\rho_1, \dots, \rho_k)$ occurring in (PC30), and retain $m_i = m_{\rho_i}$, $\lambda = \sum_i \rho_i$, $D = \sum_i (m_i - 1)$, and the original coordinates $x_i = s_i - \rho_i$. On the actual retained coefficient ring $A \subset \mathbb{C}$ carrying these coordinates and the full Taylor unit and inverse, the local map of (PC29) is

$$\begin{aligned} \mathcal{E}_A &= A[x_1, \dots, x_k] / (x_i^{m_i})_{i=1}^k, & J &= \sum_i x_i, & S &= \lambda + J, \\ \mathcal{C}_A &= A[S] / ((S - \lambda)^{D+1}), & \eta_{\rho}[P] &= U_{\rho} P(\lambda + J), & U_{\rho} &= \prod_i j_{\rho_i}(v_h). \end{aligned} \quad (\text{PC30a})$$

Every original coefficient of $j_{\rho_i}(v_h)$, including its inverse, is retained. This is a module map for the displayed original sum coordinate, since multiplication by S on the source is multiplication by $\lambda + J$ on the target. Its underlying unweighted ring map has value one at one, and its weighted map has the original value U_{ρ} at one.

For the specified coefficient map $\alpha : A \rightarrow k_0$ where it is defined, with finite target of characteristic $p > \max_i m_i$, put $\mathcal{R} = A_{\ker \alpha}$. The image of α is a finite field, so its kernel is maximal, and the map extends by $a/b \mapsto \alpha(a)/\alpha(b)$. The characteristic-zero domain $\mathcal{R}$ has every integer prime to p as a unit and p as a nonzero nonunit. Thus all the following maps are over this specified coefficient localization, with no extension to a larger CRT stage being presumed.

For every $0 \leq r \leq D$ retain the complete vector

$$V_r = \sum_{\substack{0 \leq b_i < m_i \\ \sum_i b_i = r}} \frac{x^{\mathbf{b}}}{\prod_i b_i!}, \quad W_r = U_\rho V_r, \quad J^r = r! V_r, \quad JW_r = (r+1)W_{r+1}. \quad (\text{PC30b})$$

Here $W_{D+1} = 0$. Each denominator is a unit in $\mathcal{R}$. To verify the first multiplication identity, expand $(\sum_i x_i)^r$: the surviving monomial $x^{\mathbf{b}}$ has exactly $r! / \prod_i b_i!$ ordered assignments. For the second, its degree- $r+1$ coefficient is $\sum_{i: b_i > 0} b_i / \prod_i b_i! = (r+1) / \prod_i b_i!$. Multiplication by the original unit proves the formula for W_r . A tuple of every degree $r \leq D$ exists by successively filling the capacities $m_i - 1$; choose one pivot $\mathbf{b}^*(r)$ for each such degree. The functionals $\lambda_r(x) = \mathbf{b}^*(r)! [x^{\mathbf{b}^*(r)}]x$ satisfy $\lambda_r(V_t) = \delta_{rt}$. Consequently $\pi(x) = \sum_r \lambda_r(x) V_r$ is a projection onto $\Lambda = \bigoplus_{r=0}^D \mathcal{R} V_r$. Subtracting this projection leaves exactly the span F of the nonpivot monomials. Conjugating by the entire unit therefore proves

$$\begin{aligned} \mathcal{E}_{\mathcal{R}} &= \Lambda_U \oplus U_\rho F, & \Lambda_U &= \bigoplus_{r=0}^D \mathcal{R} W_r, \\ L_U &:= \eta_\rho(\mathcal{C}_{\mathcal{R}}) = \bigoplus_{r=0}^D r! \mathcal{R} W_r, & \eta_\rho((S - \lambda)^r) &= r! W_r, \\ \Lambda_U / L_U &\simeq \bigoplus_{r=0}^D \mathcal{R} / (p^{d_r}), & d_r &= \sum_{a \geq 1} \lfloor r/p^a \rfloor, \quad [\sum_r c_r W_r] \mapsto (c_r \bmod p^{d_r})_r. \end{aligned} \quad (\text{PC30c})$$

Counting the factors of p in each integer up to r gives $r! = p^{d_r} \varepsilon_r$ with $\varepsilon_r \in \mathcal{R}^\times$, proving the quotient formula. Its kernel and surjectivity follow in the displayed basis. In particular the cokernel into the full local tensor also retains the free summand $U_\rho F$. The containing lattice is the exact saturation: $(L_U \otimes \text{Frac } \mathcal{R}) \cap \mathcal{E}_{\mathcal{R}} = \Lambda_U$. Indeed every factorial becomes invertible in the fraction field, and the free complement in the displayed splitting has zero component for an element in that rational span. No original vector has been divided by its factorial inside $\mathcal{E}_{\mathcal{R}}$.

Apply the actual map to $B = k_0$, or the quotient map to $B = \mathcal{R}/p\mathcal{R}$, and mark base changes by bars. All pivot maps and the full inverse unit base change, so the $\overline{W}_r$ remain independent. The exact result is

$$\begin{aligned} \overline{\eta}_\rho((S - \overline{\lambda})^r) &= \overline{r!} \overline{W}_r, & \ker \overline{\eta}_\rho &= ((S - \overline{\lambda})^p), \\ \text{im } \overline{\eta}_\rho &= \bigoplus_{r=0}^{\min(D, p-1)} B \overline{W}_r, & \text{ind}(M_{S-\overline{\lambda}} | \overline{\mathcal{C}}_{\mathcal{R}}) &= D+1, & \text{ind}(M_{\overline{J}} | \text{im } \overline{\eta}_\rho) &= \min(p, D+1). \end{aligned} \quad (\text{PC30d})$$

The kernel ideal is zero if $D < p$. The factorial is a unit below p and zero from p onwards, proving the image and kernel. In characteristic p , $\overline{J}^p = \sum_i \overline{x}_i^p = 0$, whereas $\overline{J}^{\min(p-1, D)} \overline{U}_\rho = \overline{\min(p-1, D)!} \overline{W}_{\min(p-1, D)} \neq 0$. These identities prove the two different indices with their exact module targets. They are joined by the explicit free resolution and its connecting map

$$\begin{aligned} 0 \longrightarrow \mathcal{C}_{\mathcal{R}} &\xrightarrow{\eta_\rho} \Lambda_U \longrightarrow \Lambda_U / L_U \longrightarrow 0, \\ \text{Tor}_1^{\mathcal{R}}(\Lambda_U / L_U, B) &= \bigoplus_{r=p}^D B \varepsilon_r, & \partial \varepsilon_r &= (S - \overline{\lambda})^r \in \ker \overline{\eta}_\rho. \end{aligned} \quad (\text{PC30e})$$

In the retained bases its differential is $\text{diag}(0!, \dots, D!)$. Its base change has exactly the displayed kernel, proving this Tor group and its connecting map, with differential sign positive.

There is also an exact formula for collisions after a coefficient map actually defined on the full one-factor CRT data in (PC28)–(PC30), with $p > m_\rho$ for every $\rho \in Z$. Retain those idempotents

and their original inverses, so their base changes still give the product of the ordered local factors. Let $\mathcal{Z}_{k,p} = \{\alpha(\sum_i \rho_i) : \boldsymbol{\rho} \in Z^k\}$ be the set of attained residual sums. Define

$$\psi_{h,k}(X) = \prod_{\mu \in \mathcal{Z}_{k,p}} (X - \mu)^{n_\mu}, \quad n_\mu = \max_{\substack{\boldsymbol{\rho} \in Z^k \\ \alpha(\sum_i \rho_i) = \mu}} \min \left(p, 1 + \sum_i (m_{\rho_i} - 1) \right). \quad (\text{PC30f})$$

Every maximum in this display is over a nonempty set. When Z is empty, $k \geq 1$ makes the tuple set empty, so the product is $\psi_{h,k} = 1$; both the original cyclic source $k_0[X]/(1)$ and its tensor image are zero. The global inequality on the orders is vacuous in this empty case. The multiplication matrix on each factor is $\mu + \bar{J}$, whose least annihilating power is (PC30d). More explicitly, write a nonzero polynomial as $P(X) = (X - \mu)^e Q(X)$ with $Q(\mu) \neq 0$. The matrix $Q(\mu + \bar{J})$ is the nonzero scalar $Q(\mu)$ times $I + H$, with H a polynomial in $\bar{J}$ having zero constant term and hence nilpotent. Its inverse is the finite geometric series $Q(\mu)^{-1} \sum_{j \geq 0} (-H)^j$, stopped after the nilpotence index. Thus $P(\mu + \bar{J}) = 0$ exactly when $\bar{J}^e = 0$. This proves that the least monic annihilating polynomial is the indicated power of $X - \mu$. On a product, a polynomial annihilates the operator exactly when it annihilates every factor; its monic annihilator is their least common multiple. Taking the maximum exponent at each equal residual sum proves (PC30f). The original reduced polynomial $\bar{\chi}_{h,k}$ still annihilates all factors, so $\psi_{h,k} \mid \bar{\chi}_{h,k}$. The original identity vector generates the polynomial image, and the full reduced unit is invertible; hence the precise specialized map of (PC29) has

$$\ker \bar{\eta}_k = (\psi_{h,k})/(\bar{\chi}_{h,k}), \quad \text{im } \bar{\eta}_k \simeq k_0[X]/(\psi_{h,k}), \quad [P] \mapsto \bar{U}_k P(\bar{S}). \quad (\text{PC30g})$$

The type of this isomorphism is a module isomorphism; the unweighted map has the corresponding algebra isomorphism. Its kernel is the annihilator ideal just proved, and its image is the definition of that polynomial image, proving both statements. This retains new residual collisions instead of reducing the characteristic-zero maximum without its map. It makes no claim that a local coefficient map extends to all the sum-CRT coefficients: the actual separating idempotent at degree $k \geq p$ in the original quartet corner gives p^{-1} by its divided difference (BTF.9), and forbids such an extension. The characteristic-zero formula (PC30), its critical tuple quotient below, and the completed complex comparison retain their original domains throughout.

Let $\chi_{c,k}$ be the same polynomial formed using only tuples from Z_c , with exponents r_λ given by the same maximum; an absent critical tuple at λ has $r_\lambda = 0$. When $Z_c = \emptyset$, define $\chi_{c,k} = 1$. Put $q_k = \deg \chi_{h,k}$ and $r_k = \deg \chi_{c,k}$. The tuple inclusion proves $r_\lambda \leq m_\lambda$ and hence $\chi_{c,k} \mid \chi_{h,k}$. The same nilpotence proof gives the exact injection

$$\alpha_{c,k} : \mathbb{C}[S]/(\chi_{c,k}) \hookrightarrow E_c^{\otimes k}, \quad [P] \mapsto P(A_c^{[k]})e_c^{\otimes k}, \quad \pi_c^{\otimes k} \bar{\alpha}_k = \alpha_{c,k} \pi_{\chi_{c,k}}. \quad (\text{PC31})$$

Here $A_c = A|_{E_c}$ and $A_c^{[k]}$ is its sum on the ordered tensor factors. The identity vector is the displayed $e_c^{\otimes k}$, so the codomain is precisely the original CRT ideal tensor. For an empty critical factor both the source and target of $\alpha_{c,k}$ are zero. This identity keeps the full source polynomial $\chi_{h,k}$ and the actual quotient map to its critical tuple polynomial.

The target is the completed projective tensor space $Q_{\text{crit}}^{\widehat{\otimes} \pi^k}$. There is no loss of the finite critical image on completion. In fact define $i_c = \mathcal{C}_h|_{E_c}$ and use the continuous map $\mathcal{R}_c$ in (PC15). Their tensor maps extend continuously to the projective completions, and on pure tensors their composition is identity. Pure tensors span the finite-dimensional $E_c^{\otimes k}$, which is already complete, so

$$\mathcal{R}_c^{\widehat{\otimes} k} i_c^{\widehat{\otimes} k} = I_{E_c^{\otimes k}}. \quad (\text{PC32})$$

This proves injectivity and a continuous retraction onto the finite image inside the completed target. Its associated continuous idempotent has that image as its kernel-complement, so the image is closed in the Hausdorff completed space. This argument uses the displayed maps and does not assume preservation of arbitrary exact sequences by completion.

The actual tensor comparison, with its original weighted inclusion, is

$$\mathcal{C}_k = \Gamma^{\widehat{\otimes} k} \sigma_h^{\widehat{\otimes} k} \eta_k = \mathcal{C}_h^{\widehat{\otimes} k} \overline{\alpha}_k : E_k \rightarrow Q_{\text{crit}}^{\widehat{\otimes} \pi^k}. \quad (\text{PC33})$$

Here $\sigma_h^{\widehat{\otimes} k} \eta_k$ is the existing arithmetic cohomology inclusion. Applying the raw critical evaluations gives $\Xi_c^{\widehat{\otimes} k} \mathcal{C}_k = (M_{v_c})^{\otimes k} \alpha_{c,k} \pi_{\chi_{c,k}}$. Applying their inverses (PC32) instead gives $\alpha_{c,k} \pi_{\chi_{c,k}}$. Every derivative coordinate has its product phase $i^{-\sum j_i} / \prod j_i!$ and the full tensor of the local units (PC14). The injectivity in (PC31)–(PC32) therefore proves the exact kernel and image dimension

$$\boxed{F_k := \ker \mathcal{C}_k = (\chi_{c,k})/(\chi_{h,k}), \quad \dim F_k = q_k - r_k, \quad \dim \text{im } \mathcal{C}_k = r_k.} \quad (\text{PC34})$$

To display all collision data, at an ambient sum λ write $x = S - \lambda$ and $\chi_{c,k}(S) = x^{r_\lambda} a_\lambda(x)$ with $a_\lambda(0) \neq 0$. The kernel in that original primary factor is the image of the explicit map

$$\mathbb{C}[x]/(x^{m_\lambda - r_\lambda}) \longrightarrow \mathbb{C}[x]/(x^{m_\lambda}), \quad [b] \longmapsto [x^{r_\lambda} a_\lambda(x)b]. \quad (\text{PC35})$$

It is injective: multiply by the recursively computed inverse of a_λ and cancel the displayed power of x . It is onto the local ideal by its definition. Thus all $m_\lambda - r_\lambda$ kernel jets remain, with action $S = \lambda + x$ and the local unit a_λ retained. If $\Re \lambda \neq k/2$, no all-critical tuple has that sum, so $r_\lambda = 0$ and the entire block belongs to the kernel. At a sum with real part $k/2$, the same exact formula still applies: an absent critical tuple kills the whole block, and a coinciding critical tuple retains precisely the quotient of length r_λ while the additional jets lie in (PC35). This is the complete relationship between factorwise critical selection and selection by the real part of the sum.

The original sum action also descends with its exact type. Multiplication by $1/2 + iy$ is continuous on the one-factor quotient and preserves its relation space. On the completed tensor target define the continuous operator

$$M^{[k]} = \sum_{a=1}^k I^{\widehat{\otimes}(a-1)} \widehat{\otimes} M \widehat{\otimes} I^{\widehat{\otimes}(k-a)}. \quad (\text{PC36})$$

On pure tensors this is the sum of the original k multipliers, and continuity extends it to the completed target. Since each η_h commutes with the original one-factor action, (PC12) gives $\mathcal{C}_k A_k = M^{[k]} \mathcal{C}_k$ on the cyclic image. At source level, the finite multinomial expansion of $P(D_1 + \dots + D_k) F_h^{\otimes k}$ and (PC6) give $\mathfrak{q}^{\widehat{\otimes} k} \mathcal{V}_k P = \sigma_h^{\widehat{\otimes} k} \eta_k [P]$. Thus both the completed comparison and its sum action are induced by the original weighted source, with no replacement of U_k .

5.1.7 The same kernel curvature in every original cyclic tensor packet

Form the original period matrix $\Pi_k(t)$ of (SC1)–(SC6) with the literal polynomial $\chi_{h,k}$, its full increasing-power basis, the same specified $u \neq 0$, and its own degree q_k in the ray formula. Write $e_{0,k} = [1]$, $\ell_k[P] = [S^{q_k-1}] \text{rem}_{\chi_{h,k}} P$ and $A_{k,t} = A_k + t e_{0,k} \ell_k$. The preceding convergence, remainder and determinant argument applies to this polynomial with all its coefficients; hence $\Pi'_k = -\Pi_k A_{k,t}/u$ and it is invertible at every t . Define

$$\mathcal{C}_{k,t} = \mathcal{C}_k \Pi_k^{-1}, \quad A_{k,t}^{\text{per}} = \Pi_k A_{k,t} \Pi_k^{-1}, \quad \mathcal{K}_{k,t} = \Pi_k F_k. \quad (\text{PC37})$$

Equations (PC34) and (PC36) prove

$$\ker \mathcal{C}_{k,t} = \mathcal{K}_{k,t}, \quad \mathcal{C}_{k,t} A_{k,t}^{\text{per}} - M^{[k]} \mathcal{C}_{k,t} = t \mathcal{C}_k (e_{0,k}) \ell_k \Pi_k^{-1}. \quad (\text{PC38})$$

If $0 < r_k < q_k$, then $\mathcal{C}_k(e_{0,k}) \neq 0$ by (PC31)–(PC33). Put $p_k = q_k - r_k$. The kernel ideal has basis $[\chi_{c,k}], \dots, [S^{p_k-1} \chi_{c,k}]$, with the last member monic of degree $q_k - 1$. Consequently $\ell_k|F_k$ is onto, its kernel has dimension $p_k - 1$, and $e_{0,k} \notin F_k$. For every $t \neq 0$, the restriction of (PC38) to $\mathcal{K}_{k,t}$ has rank one, image $\mathbb{C} \mathcal{C}_k(e_{0,k})$, and full kernel $\Pi_k(F_k \cap \ker \ell_k)$. At zero it vanishes.

With any specified inclusion $I_k : F_k \hookrightarrow E_k$, put $Y_k = \Pi_k I_k$, $H_k = Y_k^* Y_k$, $N_k = I - Y_k H_k^{-1} Y_k^*$, $z_k = N_k \Pi_k e_{0,k}$ and $L_k = \ell_k I_k$. The calculations (PC23)–(PC24) now give

$$\partial_{\bar{t}} \partial_t \log \det H_k = \frac{|t|^2}{|u|^2} \|z_k\|^2 L_k H_k^{-1} L_k^*. \quad (\text{PC39})$$

Its coefficient is strictly positive for $0 < r_k < q_k$, and the normal acceleration at zero is $-z_k(0)L_k/u$ with rank one. This calculation includes the collision-kernel jets of (PC35). The complete neighboring-minor form is obtained with

$$\begin{aligned} J_g &= [\chi_{c,k}, S\chi_{c,k}, \dots, S^{p_k-1}\chi_{c,k}], \quad J_- = [\chi_{c,k}, \dots, S^{p_k-2}\chi_{c,k}], \\ J_+ &= [J_g, e_{0,k}], \quad h_J = \det((\Pi_k J)^* \Pi_k J). \end{aligned} \quad (\text{PC40})$$

For $p_k = 1$, J_- is empty and $h_- = 1$. The Schur complement of the last column gives $h_+ = h_g \|z_k\|^2$; the adjugate formula for the last inverse-Gram entry gives $L_k H_k^{-1} L_k^* = h_- / h_g$ after the exact constant frame congruence of (SC32). Therefore

$$\partial_{\bar{t}} \partial_t \log h_g = \frac{|t|^2}{|u|^2} \frac{h_+ h_-}{h_g^2}. \quad (\text{PC41})$$

For the original tensor-source Gram $G_{N,k}$, each of these same inclusions has factors $v_{N,J} = \det(J^* G_{N,k} J)$ and $b_{N,J} = \det((J^* G_{N,k} J)^{-1} J^* \Pi_k^* \Pi_k J)$. The literal product identity $h_J = v_{N,J} b_{N,J}$ retains every theta mass, polynomial map and period factor in (PC41). The moving metric $\Pi_k^* G_{N,k} \Pi_k^{-1}$ has constant kernel Gram $I_k^* G_{N,k} I_k$, with the full derivative cancellation (PC27), replacing A_t by $A_{k,t}$. These are the two exact metric observations on the same critical-comparison kernel.

If $Z_c = \emptyset$, then $r_k = 0$, $F_k = E_k$ and $\mathcal{C}_k = 0$; the full kernel bundle has zero curvature. If all packet zeros are critical, $\chi_{c,k} = \chi_{h,k}$, $F_k = 0$, and the empty-kernel curvature is zero. If both sectors occur, an all-critical tuple proves $r_k > 0$, while a repeated off-critical root has sum $k\rho$ of real part different from $k/2$, proving $r_k < q_k$. Thus the strict case in (PC39) has an exact sector criterion at every original k .

Finally each coefficient arrow in this section has its original Split-Zero lift: $v^\bullet \mapsto (Tv)^\bullet$ and $\tau \mapsto \tau$, with the retained support label. The complete kernel (PC34), its smaller action-defect kernel in (PC38), and its cohomological source image remain separately represented. A vector killed by one of these comparisons lands at that receiving fibre's supported zero. The amplitude projection maps both that zero and external absence to ordinary zero, and the displayed split maps commute with that projection. This preserves the original support and the full arithmetic unit through the critical quotient, the completed tensor target, and the period curvature calculation.

6 The original Laplacian control along the exact period deformation

6.1 The original Laplacian control along the exact period deformation

We calculate the control parameter for the actual companion deformation, and then transport its complete numerical range through the periods. All metrics and coefficient vectors below are the original ones.

Fix the original cyclic algebra, admitted degree and positive Gram

$$\begin{aligned} E &= \mathbb{C}[S]/(\chi), \quad q = \deg \chi \geq 1, \quad A = M_S, \quad G = G_N > 0, \quad k \in \mathbb{R}, \\ e_0 &= [1], \quad \ell[P] = [S^{q-1}] \text{rem}_\chi P. \end{aligned} \quad (\text{LC1})$$

Here χ is the original monic polynomial, $N \in \mathbb{Z}$ satisfies $N \geq q-1$, and the deformation parameter is $t \in \mathbb{C}$. Put $R = e_0 \ell$, $A_t = A + tR$, and retain the actual control form and relative, Laplacian and shifted endomorphisms

$$W_t = A_t^* G + G A_t - kG, \quad X_t = G^{-1} W_t, \quad L_t = A_t^2 - kA_t, \quad B_t = A_t - \frac{k}{2} I. \quad (\text{LC2})$$

The control form has value $v^* W_t w = \langle v, X_t w \rangle_G$; thus its exact form-to-operator map is $X_t = G^{-1} W_t$. Here X_t is self-adjoint for $\langle v, w \rangle_G = v^* G w$. Its finite operator norm is defined by the supremum over the original nonzero vectors. Write $\epsilon_N = \|X_0\|_G$. This is an actual finite quantity, so $-\epsilon_N G \preceq W_0 \preceq \epsilon_N G$ follows from the spectral theorem for the displayed positive metric.

For the original reflection-stable source, its complete rank-two calculation (TVB.22)–(TVB.30) evaluates this same quantity. In the unchanged source-orthogonal columns b_j and norms ω_j , put

$$a = b_N^* G b_N, \quad d_+ = b_{N+1}^* G b_{N+1}, \quad z_N = b_N^* G b_{N+1}, \quad D_N = \det K_N.$$

For every integer $j \geq -1$ retain $K_j = \sum_{r=0}^j b_r b_r^* / \omega_r$, with the empty sum $K_{-1} = 0$ on E , and put $D_j = \det K_j$. The omitted-column kernel and determinant are $\tilde{K}_N = K_{N-1} + b_{N+1} b_{N+1}^* / \omega_{N+1}$ and $\tilde{D}_N = \det \tilde{K}_N$. At the first admitted degree $N = q-1$, the first $q-1$ monic remainder columns are independent, so K_{q-2} has rank $q-1$ and $D_{q-2} = 0$. No inverse of this boundary kernel is used. The retained identities give

$$\epsilon_N^2 = \frac{ad_+ - |z_N|^2}{\omega_N^2} = \alpha_{N+1} \frac{\Delta_N}{D_N}, \quad \alpha_{N+1} = \frac{\omega_{N+1}}{\omega_N}, \quad \Delta_N = D_{N+1} + D_{N-1} - D_N - \tilde{D}_N. \quad (\text{LC3})$$

The omitted-column determinant $\tilde{D}_N$, every boundary case and its positive source-minor sum are defined and proved in that full chapter. To check the radius here directly, the original control is $W_0 = G(b_{N+1} b_N^* + b_N b_{N+1}^*) G / \omega_N$. Its relative self-adjoint operator has rank at most two and trace zero; z_N is purely imaginary by the retained reflection phase. Expansion of its squared trace gives $2(ad_+ - |z_N|^2) / \omega_N^2$, so its two possible nonzero eigenvalues are exactly $\epsilon_N, -\epsilon_N$. The two-column determinant identity proves the second equality in (LC3). No estimate or limiting assumption is needed to evaluate it.

6.1.1 The exact additional radius of the companion term

The adjoint of R for the original metric is $R^{\dagger G} = G^{-1} \ell^* e_0^* G$. Thus

$$X_t - X_0 = tR + \bar{t} R^{\dagger G}. \quad (\text{LC4})$$

For $q > 1$, $\ell e_0 = 0$, hence $R^2 = 0$. Put

$$f = G^{-1} \ell^*, \quad a_0 = e_0^* G e_0 > 0, \quad b_0 = \ell G^{-1} \ell^* > 0, \quad \sigma_N = \sqrt{a_0 b_0}. \quad (\text{LC5})$$

The columns e_0, f are G -orthogonal because $e_0^* G f = \bar{\ell} e_0 = 0$. They are independent and their Gram is exactly $\text{diag}(a_0, b_0)$. In these original unscaled columns the matrix of (LC4) is

$$\begin{pmatrix} 0 & tb_0 \\ \bar{t}a_0 & 0 \end{pmatrix}. \quad (\text{LC6})$$

Indeed $Rf = b_0 e_0$, $Re_0 = 0$, $R^{\dagger G} e_0 = a_0 f$ and $R^{\dagger G} f = 0$. On their G -orthogonal complement both terms vanish. The characteristic polynomial on the displayed two-dimensional space is $x^2 - |t|^2 a_0 b_0$. Therefore the exact additional operator norm is $|t| \sigma_N$; all remaining eigenvalues are zero. This retains the two original vector lengths, with no choice of unit vectors.

For $q = 1$, instead $R = I_E$, so (LC4) equals $2 \operatorname{Re}(t) I_E$ and its exact norm is $2 |\operatorname{Re} t|$. Consequently the entirely specified quantity

$$E_N(t) = \epsilon_N + \begin{cases} |t| \sigma_N, & q > 1, \\ 2 |\operatorname{Re} t|, & q = 1 \end{cases} \quad \text{satisfies} \quad \|X_t\|_G \leq E_N(t), \quad -E_N(t)G \preceq W_t \preceq E_N(t)G. \quad (\text{LC7})$$

The inequality is the triangle inequality for the same operator norm. It is not an assertion that equality always holds. The two summands in (LC7) are the actual norms just computed, and the original ϵ_N remains available through (LC3). The full Laplacian change is also exact:

$$L_t - L_0 = t(AR + RA - kR) + t^2 R^2. \quad (\text{LC8})$$

Thus the quadratic term vanishes for $q > 1$, while for $q = 1$ it is the retained scalar $t^2 I_E$.

6.1.2 An explicit numerical-range enclosure with no missing input

Expansion of (LC2), without a commutation assumption, proves

$$L_t^* G - GL_t = A_t^* W_t - W_t A_t, \quad GL_t = W_t B_t - B_t^* G B_t - \frac{k^2}{4} G. \quad (\text{LC9})$$

For the second equality insert $W_t = B_t^* G + G B_t$, so the right side is $GB_t^2 - k^2 G/4 = GL_t$. The first equality follows by expansion; the two middle terms $A_t^* G A_t$ cancel.

For each unchanged nonzero $v \in E$, define the homogeneous observations

$$z = \frac{v^* GL_t v}{v^* G v}, \quad r = \frac{\|B_t v\|_G}{\|v\|_G}, \quad a + ib = \frac{\langle v, X_t B_t v \rangle_G}{\|v\|_G^2}. \quad (\text{LC10})$$

Cauchy-Schwarz and (LC7) give $a^2 + b^2 \leq E_N(t)^2 r^2$. The second identity in (LC9) gives $\text{Re } z = a - r^2 - k^2/4$ and $\text{Im } z = b$. Write $e = E_N(t)$. First, $a \leq er$ implies $\text{Re } z \leq (e^2 - k^2)/4 - (r - e/2)^2$. Second, direct multiplication gives

$$e^2 \left(\frac{e^2 - k^2}{4} - \text{Re } z \right) - (\text{Im } z)^2 = (a - e^2/2)^2 + e^2 r^2 - a^2 - b^2 \geq 0.$$

We have proved the actual enclosure

$$\boxed{\text{Re } z \leq \frac{E_N(t)^2 - k^2}{4}, \quad (\text{Im } z)^2 \leq E_N(t)^2 \left(\frac{E_N(t)^2 - k^2}{4} - \text{Re } z \right).} \quad (\text{LC11})$$

It includes every eigenvalue by taking its nonzero eigenvector; no diagonalizability is needed. It includes the case $E_N(t) = 0$. Every quantity in (LC11) is supplied by the original finite metric, the monic coefficient functional and the actual deformation parameter.

6.1.3 The complete period transport

For fixed $u \neq 0$, the full determinant proof makes the original period matrix $\Pi(u, t)$ invertible at every parameter. Put

$$\tilde{G} = \Pi^{-*} G \Pi^{-1}, \quad \tilde{A}_t = \Pi A_t \Pi^{-1}, \quad \tilde{L}_t = \Pi L_t \Pi^{-1}, \quad \tilde{W}_t = \Pi^{-*} W_t \Pi^{-1}. \quad (\text{LC12})$$

Direct multiplication gives $\tilde{W}_t = \tilde{A}_t^* \tilde{G} + \tilde{G} \tilde{A}_t - k \tilde{G}$ and $\tilde{L}_t = \tilde{A}_t^2 - k \tilde{A}_t$. For $x = \Pi v$, its norm and the complete numerical-range quotient are exactly those in (LC10). Its control quotient and adjoint-defect identity are likewise transported by the literal congruences in (LC12). Thus (LC11) applies unchanged in that period realization.

The transported fixed action is $\Pi A \Pi^{-1}$; the displayed deformed action differs from it by $t \Pi R \Pi^{-1}$, whose control contribution was computed in (LC4)–(LC7). The arithmetic source realizes precisely this change through $D_t^{(k)} = D^{(k)} + t r_N R j_E$, with $j_E r_N = I$ and the original boundary primitive retained, as proved in (SC7)–(SC8). These equalities supply the source, coefficient and period maps for the same Laplacian control. To retain its complete primitive, let $\mathcal{C}_{\text{prim}}$ be the original primitive term of the tensor complex and write $D_0 : \mathcal{C}_{\text{prim}} \rightarrow \mathcal{C}_{\text{prim}}$ for its original Euler action of cochain degree zero. Thus $K_N^{\text{prim}} : E \rightarrow \mathcal{C}_{\text{prim}}$ and $D^{(k)} d = d D_0$ on

this term. The original boundary observation $j_E d = 0$ makes the added finite-rank action vanish there, so $D_t^{(k)} d = dD_0$ as well. Put $K = K_N^{\text{prim}}$. Using $D_t^{(k)} r_N = r_N A_t + dK$, expand both terms of the square:

$$\begin{aligned} (D_t^{(k)})^2 r_N &= D_t^{(k)}(r_N A_t) + D_t^{(k)} dK \\ &= (r_N A_t + dK)A_t + dD_0 K = r_N A_t^2 + d(KA_t + D_0 K). \end{aligned}$$

Subtract $kD_t^{(k)} r_N = kr_N A_t + kdK$ with its original sign. This proves the complete quadratic source comparison

$$\boxed{((D_t^{(k)})^2 - kD_t^{(k)})r_N - r_N L_t = d(D_0 K_N^{\text{prim}} + K_N^{\text{prim}} A_t - kK_N^{\text{prim}})}. \quad (\text{LC12a})$$

The term $K_N^{\text{prim}} A_t = K_N^{\text{prim}} A + tK_N^{\text{prim}} R$ retains the full deformation contribution. All terms inside the displayed primitive have the same target $\mathcal{C}_{\text{prim}}$; the original differential, source action and boundary class remain explicit. No smallness uniform in packet size is claimed for the fully calculated ϵ_N or σ_N .

The same complete transport now varies in $u \neq 0$ as well as t . With $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$ from (XD.u1)–(XD.u5), the precise derivative equations are

$$\partial_u \tilde{G} = -\Pi^* G \Omega_u \Pi^{-1}, \quad \partial_{\bar{u}} \tilde{G} = -\Pi^* \Omega_u^* G \Pi^{-1}, \quad \partial_u \tilde{L}_t = \Pi[\Omega_u, L_t] \Pi^{-1}, \quad \partial_u \tilde{A}_t = \Pi[\Omega_u, A_t] \Pi^{-1}. \quad (\text{LC.u1})$$

Here G, A_t, L_t are independent of u . Each formula follows by differentiating its literal definition (LC12), inserting $\Pi_u = \Pi \Omega_u$ and $(\Pi^{-1})_u = -\Omega_u \Pi^{-1}$, and retaining both product terms. Thus the full numerical-range identity (LC10)–(LC12) holds at every such u with exactly the same original bound $E_N(t)$, while its moving metric has the displayed derivatives. This statement makes no bound on the unweighted target metric. For the original source lift the exact u commutator is $r_N[A, \Omega_u]j_E + dK_N^{\text{prim}} \Omega_u j_E$ by (DS74); the primitive is in the original cochain degree.

The actual comparison numbers used in (LC14)–(LC16) are now the eigenvalues $m(u, t), M(u, t)$ of $G^{-1} \Pi(u, t)^* \Pi(u, t)$. Their original proof by two quadratic-form inequalities applies pointwise without alteration. The full u and mixed curvatures of this same constituent are (SC.u1)–(SC.u4), with its actual inclusion I and Gram $I^* G I$. In particular the exact order-two and order-one normal terms must both be used at $t \neq 0$; the original t -curvature comparison (LC15) remains its own calculated matrix entry.

6.1.4 The exact finite comparison with constituent curvature

Let $I : F \hookrightarrow E$ be an actual injective coefficient map with $AI = IA_F$ and $0 < p := \dim F < q$. In particular $q > 1$, so the radius σ_N in (LC5) applies. Retain the original orthogonal projection and its two scalar observations:

$$\begin{aligned} G_F &= I^* G I, & P_F &= I G_F^{-1} I^* G, & L &= \ell I, & f &= G^{-1} \ell^*, \\ a_\perp &= e_0^* G (1 - P_F) e_0, & b_F &= L G_F^{-1} L^* = \ell I G_F^{-1} I^* \ell^*. \end{aligned} \quad (\text{LC13})$$

The injection and positive G give $G_F > 0$. Multiplication proves $P_F^2 = P_F$ and $P_F^* G = G P_F$, and its image is exactly $I(F)$. Consequently $a_\perp = \|(1 - P_F)e_0\|_G^2$ and $b_F = \|P_F f\|_G^2$, the second identity following from $I^* G f = I^* \ell^*$, with the same column f as in (LC5).

Both numbers are strictly positive on this actual constituent. Indeed A -invariance implies invariance under every polynomial in A , so $I(F)$ is an ideal of E . Its inverse image in $\mathbb{C}[S]$ is (g) for a unique monic divisor g of χ . Writing $\chi = gh$, the map $[a]_h \mapsto [ga]_\chi$ is an isomorphism onto $I(F)$: it is onto by the ideal description, and $gh \mid ga$ implies $h \mid a$ by cancellation in the polynomial domain. Thus $\deg h = p$ and the complete original ideal has basis $[g], [Sg], \dots, [S^{p-1}g]$. The last vector is monic of degree $q-1$, proving $L \neq 0$; positivity of G_F^{-1} gives $b_F > 0$. An ideal containing

$e_0 = [1]$ is all of E . Properness therefore gives $e_0 \notin I(F)$ and $a_\perp > 0$. Every repeated factor remains in this ideal calculation.

At the same $u \neq 0$ and parameter t , let $m(t)$ and $M(t)$ be the actual smallest and largest eigenvalues of $B(t) = G^{-1}\Pi(t)^*\Pi(t)$. This operator is G -self-adjoint and positive because $v^*GB(t)v = \|\Pi(t)v\|_{I_q}^2 > 0$ for each nonzero v . Its finite spectrum therefore gives $0 < m(t) \leq M(t) < \infty$ and the quadratic-form inequalities $m(t)G \preceq \Pi(t)^*\Pi(t) \preceq M(t)G$. These are evaluated finite comparison quantities, with no bound uniform in the arithmetic inputs imposed on them.

Put $Y = \Pi I$, $H_F = Y^*Y$, and $P^{\text{per}} = YH_F^{-1}Y^*$, the orthogonal projection in the specified period-coordinate metric I_q . Its residual is $z = (1 - P^{\text{per}})\Pi e_0$. The exact comparison is

$$\boxed{m(t)a_\perp \leq \|z(t)\|_{I_q}^2 \leq M(t)a_\perp, \quad \frac{b_F}{M(t)} \leq LH_F(t)^{-1}L^* \leq \frac{b_F}{m(t)}. \quad (\text{LC14})}$$

To prove the first pair, orthogonal projection gives $\|z\|_{I_q}^2 = \min_{x \in F} \|\Pi(e_0 - Ix)\|_{I_q}^2$. The same minimization in the original metric has value $a_\perp$, attained at $x = G_F^{-1}I^*Ge_0$. The lower quadratic-form bound holds for every x , so its minimum is at least $m(t)a_\perp$. Evaluating the upper bound at the displayed original minimizer gives the upper inequality. Compression by I gives $m(t)G_F \preceq H_F \preceq M(t)G_F$. Inversion of positive forms, followed by the unchanged row L , proves the second pair. Explicitly, for any positive matrix T and column w , completing the square proves $w^*T^{-1}w = \max_x (2\operatorname{Re}(w^*x) - x^*Tx)$. Applying this formula to the three positive matrices verifies the inverse order and both scalar factors without a commutation assumption.

The coefficient of the actual constituent curvature in (SC16) is $c_F(t) = \|z(t)\|_{I_q}^2 LH_F(t)^{-1}L^*/|u|^2$. For completeness, the period equation and $AI = IA_F$ give $Y' = -YA_F/u - t\Pi e_0 L/u$, hence $(1 - P^{\text{per}})Y' = -tzL/u$. Holomorphy of Y gives $\partial_t H_F = Y^*Y'$ and $\partial_{\bar{t}} H_F = Y'^*Y$; differentiating its inverse in the determinant derivative proves $\partial_{\bar{t}}\partial_t \log \det H_F = \operatorname{Tr}(H_F^{-1}Y'^*(1 - P^{\text{per}})Y')$. Substitution yields the full equality and the bounds

$$\partial_{\bar{t}}\partial_t \log \det H_F(t) = |t|^2 c_F(t), \quad \boxed{\frac{m(t)}{M(t)} \frac{a_\perp b_F}{|u|^2} \leq c_F(t) \leq \frac{M(t)}{m(t)} \frac{a_\perp b_F}{|u|^2}. \quad (\text{LC15})}$$

Every factor in (LC14) is positive, so multiplying its appropriate two lower or two upper bounds proves (LC15). In particular the coefficient is positive even at $t = 0$, while the actual curvature at that parameter is zero through its retained factor $|t|^2$.

The comparison to the Laplacian radius has an exact two-term loss. Orthogonality of P_F gives $a_0 = \|P_F e_0\|_G^2 + a_\perp$ and $b_0 = b_F + \|(1 - P_F)f\|_G^2$. Multiplying with every term retained therefore proves

$$\boxed{\sigma_N^2 - a_\perp b_F = \|P_F e_0\|_G^2 b_0 + a_\perp \|(1 - P_F)G^{-1}\ell^*\|_G^2 \geq 0. \quad (\text{LC16})}$$

Combining (LC15)–(LC16) and the exact additional norm (LC4)–(LC6) gives, on this same original constituent,

$$c_F(t) \leq \frac{M(t)}{m(t)} \frac{\sigma_N^2}{|u|^2}, \quad \partial_{\bar{t}}\partial_t \log \det H_F(t) \leq \frac{M(t)}{m(t)} \frac{\|X_t - X_0\|_G^2}{|u|^2}.$$

Thus the companion control, the original constituent projections, and its period curvature have an explicit finite comparison with the complete loss and both metric comparison eigenvalues retained. The proof assigns no uniform estimate to $M(t)/m(t)$ or σ_N .

7 Complete circle and critical-observation diamond

7.1 The exact diamond of the circle and tensor-critical observations

This calculation uses the original packet h , the original cyclic polynomial $\chi = \chi_{h,k}$, and both actual observation maps. Its proof dependencies are the complete *Periodized theta-source recovery*

and arithmetic endpoint control, (P1)–(P64), in `sources/web_periodized_source_delivery/Tau_Periodized_Source_Control/NOTE.tex`, and the complete period/critical-kernel bridge (PC1)–(PC41), in `work/period_critical_kernel_bridge_20260913.tex`. The former is the delivered source of SHA256

acf63a56d5f50cfaa23f57ee52a66df261ec241ce13fd81ccac39a3bbbdd3ff;

the latter's final typeset source has SHA256

7a7733e0d6910398b6de716604a82ef5287a55002e1874327490a70ccc3dd414.

Both complete proofs are retained beside this continuation. The analytic theta section and critical quotient retain the full Hochschild comparison (H1)–(H42) used there. All tensor completions below are the same completed projective tensor products as in (PC32).

7.1.1 The original source, two actual targets, and their inverses

Retain the original nonempty packet h , $k \geq 2$, and $L > 0$ for the positive-weight all-integer circle realization. The final paragraph of this subsection gives the exact $k = 1$ support modification. Write Z for the original selected zero set, with its complete multiplicities m_ρ . Put

$$R = \mathbb{C}[S], \quad E = R/(\chi), \quad q = \deg \chi, \quad A = M_S, \quad e_0 = [1]_\chi, \quad (DC1)$$

$$\chi(S) = \prod_{\lambda} (S - \lambda)^{m_\lambda}, \quad m_\lambda = \max_{\substack{\rho_1 + \dots + \rho_k = \lambda \\ \rho_i \in Z}} \left(1 + \sum_{i=1}^k (m_{\rho_i} - 1) \right).$$

Every tuple is ordered. Equal sums are combined only by their literal equality in $\mathbb{C}$. In the tuple-local algebra $\mathbb{C}[x_1, \dots, x_k]/(x_i^{m_{\rho_i}})$, the highest nonzero power of $x_1 + \dots + x_k$ has degree $D = \sum_i (m_{\rho_i} - 1)$ and coefficient $D! / \prod_i (m_{\rho_i} - 1)!$ on $\prod_i x_i^{m_{\rho_i} - 1}$. All higher powers vanish. Taking the maximum over equal sums proves (DC1), including every exponent. The nonempty original packet gives $q \geq 1$.

Use only tuples in $Z_c = \{\rho \in Z : \operatorname{Re} \rho = 1/2\}$ to define $a = \chi_{c,k}$. At an ambient root λ let r_λ be the same maximum over those all-critical tuples, and set $r_\lambda = 0$ if there is none. Thus

$$a(S) = \prod_{\lambda} (S - \lambda)^{r_\lambda}, \quad 0 \leq r_\lambda \leq m_\lambda, \quad a \mid \chi. \quad (DC2)$$

If Z_c is empty, $a = 1$. This definition retains the full nilpotent length of the all-critical cyclic image.

The actual tensor-critical map is

$$C = \Gamma^{\widehat{\otimes} k} \sigma_h^{\widehat{\otimes} k} \eta_k : E \longrightarrow Q_{\text{crit}}^{\widehat{\otimes} k}, \quad \eta_k = M_{v_h^{\otimes k}} \alpha_k, \quad v_h = j_h((2\xi)/h). \quad (DC3)$$

Its full map type includes $\iota_1 : \mathbb{C}[S]/h(S) \rightarrow E_h = \mathbb{C}[s]/h(s)$, $[P(S)] \mapsto [P(s)]$, and $\bar{\alpha}_k = (\iota_1^{-1})^{\otimes k} \alpha_k$. Consequently $\eta_k = \eta_h^{\otimes k} \bar{\alpha}_k$ with $\eta_h = M_{v_h} \iota_1$. Write $q_{\mathcal{B}} : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V$ for the original quotient map. The original source comparison is still $q_{\mathcal{B}}^{\widehat{\otimes} k} \mathcal{V}_{h,k} = \sigma_h^{\widehat{\otimes} k} \eta_k \pi_\chi$. No arithmetic unit is replaced by 1.

By (PC31)–(PC35), $\ker C = (a)/(\chi)$. Set $W_a = \operatorname{im} C$. The induced map and its actual inverse are

$$j_a : R/(a) \xrightarrow{\sim} W_a, \quad j_a[P]_a = C[P]_\chi, \quad (DC4)$$

$$j_a^{-1} = \alpha_{c,k}^{-1} \mathcal{R}_c^{\widehat{\otimes} k} \Big|_{W_a}, \quad C = j_a \pi_a.$$

Here $\pi_a : E \rightarrow R/(a)$ is polynomial reduction, and $\alpha_{c,k} : R/(a) \hookrightarrow E_c^{\otimes k}$ is the injection $[P] \mapsto P(A_c^{[k]})e_c^{\otimes k}$, with e_c the identity of the original critical CRT ideal. The inverse in (DC4) has domain the image of this injection, because (PC33) proves $\mathcal{R}_c^{\widehat{\otimes} k} C = \alpha_{c,k} \pi_a$. In each one-factor critical coordinate,

$$\Xi_\rho[f] = \sum_{j=0}^{m_\rho-1} \frac{i^{-j}}{j!} f^{(j)}(\operatorname{Im} \rho) x^j, \quad \mathcal{R}_c = M_{\varepsilon_c} \Xi_c, \quad \varepsilon_c v_c = e_c. \quad (DC5)$$

Thus (DC4) retains the full tensor product of the factors $i^{-j}/j!$ and the recursively inverted original local units. It is an isomorphism of finite-dimensional R -modules. The action on W_a is the restriction M_a of the original sum multiplier $M^{[k]}$, and $M_a j_a = j_a M_S$. No algebra multiplication on the ambient critical quotient is needed.

For the circle retain the original frequencies, scalar measure, and vector amplitude:

$$\begin{aligned} S_n &= k/2 + 2\pi in/L, & \nu_n &= (2\pi/L)m_{h,k}(2\pi n/L) > 0, \\ b &= p_L = \prod_{n \in \mathcal{Z}_L} (S - S_n), & \mathcal{Z}_L &= \{n \in \mathbb{Z} : \chi(S_n) = 0\}, \\ \beta_n &= \widehat{\mathcal{U}_k F_h^{\otimes k}}(2\pi n/L), & \|\beta_n\|_{\mathcal{K}}^2 &= 2\pi m_{h,k}(2\pi n/L). \end{aligned} \quad (\text{DC6})$$

The frequencies are distinct, so b is squarefree and divides χ ; an empty product is 1. In the actual circle Fourier basis $L^{-1/2}e^{-2\pi inr/L}$, the polynomial source has coefficient $L^{-1/2}\beta_n P(S_n)$. The scalar-to-vector map $c \mapsto (L^{-1/2}\beta_n c_n)_n$ has squared norm $\sum_n \nu_n |c_n|^2$, proving the isometry with the weighted cyclic space without rescaling any β_n or ν_n .

For completeness, the Hilbert quotient in (P56) is exactly

$$W_b = \ell^2(\mathbb{Z}, \nu) / \overline{\{(\chi(S_n)P(S_n))_n : P \in R\}} \xrightarrow{\sim} \bigoplus_{n \in \mathcal{Z}_L} \mathbb{C}, \quad \|v\|_{W_b}^2 = \sum_{n \in \mathcal{Z}_L} \nu_n |v_n|^2. \quad (\text{DC7})$$

Indeed the relation closure vanishes on $\mathcal{Z}_L$. For the reverse inclusion divide a vector vanishing there by $\chi(S_n)$ off that finite set; the result belongs to $\ell^2(|\chi|^2 \nu)$. Polynomial density in this space follows from the original exponential moment (P54): an orthogonal vector defines a finite complex measure with analytic Fourier transform in a strip, whose derivatives at zero all vanish. The power series, the identity theorem, and Fourier uniqueness force the vector to vanish. Multiplying the approximating polynomials by χ proves the reverse inclusion. This also proves the stated isometric quotient. All finite vectors in (DC7) are attained by polynomial interpolation. The actual observation, its injection, and inverse are therefore

$$\begin{aligned} \theta &= \theta_L : E \rightarrow W_b, & \theta[P]_\chi &= (P(S_n))_{n \in \mathcal{Z}_L}, & \ker \theta &= (b)/(\chi), \\ j_b &: R/(b) \xrightarrow{\sim} W_b, & j_b[P]_b &= (P(S_n))_{n \in \mathcal{Z}_L}, & \theta &= j_b \pi_b, \\ j_b^{-1}(v) &= \left[\sum_{n \in \mathcal{Z}_L} v_n \prod_{\substack{m \in \mathcal{Z}_L \\ m \neq n}} \frac{S - S_m}{S_n - S_m} \right]_b. \end{aligned} \quad (\text{DC8})$$

Every denominator is nonzero. The action M_b on W_b is $(v_n) \mapsto (S_n v_n)$, so $M_b j_b = j_b M_S$ and $\theta A = M_b \theta$. The class $j_b[1]_b$ has squared norm $\sum_{n \in \mathcal{Z}_L} \nu_n$, which is positive exactly when $\mathcal{Z}_L$ is nonempty and is zero otherwise. This actual number has not been assigned the value one. For $k = 1$ the identical proof uses only the actual support $I_L = \{n : w_h(2\pi n/L) > 0\}$ and $\mathcal{Z}_L = \{n \in I_L : \chi(S_n) = 0\}$ in (DC6)–(DC8). All formulas below use that support-specific b and remain exact. For the actual full-order packet, this restriction omits no sampled root of h . At such a root ρ write $g(s) = (s - \rho)^{m_\rho} g_\rho(s)$ and $h(s) = (s - \rho)^{m_\rho} h_\rho(s)$, where $g_\rho(\rho) h_\rho(\rho) \neq 0$ by the original complete-order hypothesis. Hence $v_h(\rho) = g_\rho(\rho)/h_\rho(\rho) \neq 0$. For $\rho = 1/2 + 2\pi in/L$ the actual weight is $\nu_n = |v_h(\rho)|^2/L > 0$, by $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$. Thus for $k = 1$ the polynomial b is also exactly the product over all sampled roots of $\chi = h$, even if some other lattice atoms have zero weight. No positivity of those other atoms is used.

7.1.2 The common quotient and the actual joint image

For any monic divisor $s \mid \chi$, denote the reduction by $\pi_s : E \rightarrow E_s = R/(s)$ and its kernel by $K_s = (s)/(\chi)$. In particular $E_1 = 0$, $K_1 = E$, $E_\chi = E$, and $K_\chi = 0$. Put

$$d = \gcd(a, b), \quad l = \text{lcm}(a, b) = ab/d, \quad d \mid a, b \mid l \mid \chi. \quad (\text{DC9})$$

All gcds and lcms are monic. Define the maps into the common quotient E_d by their full types:

$$f_a : W_a \rightarrow E_d, \quad f_a = \pi_d^a j_a^{-1}, \quad f_b : W_b \rightarrow E_d, \quad f_b = \pi_d^b j_b^{-1}, \quad f_a C = f_b \theta = \pi_d. \quad (\text{DC10})$$

Here $\pi_d^a : E_a \rightarrow E_d$ and $\pi_d^b : E_b \rightarrow E_d$ are reduction. Both maps are onto, commute with S , and are continuous on these finite-dimensional spaces. They give the actual fibre product

$$W_\vee = \{(x, y) \in W_a \oplus W_b : f_a x = f_b y\}, \quad J = (C, \theta) : E \rightarrow W_\vee. \quad (\text{DC11})$$

Thus the compatibility condition is expressed through the exact weighted inverse (DC4) and original-circle inverse (DC8).

There is an explicit isomorphism

$$\begin{aligned} j_l : E_l &\xrightarrow{\sim} W_\vee, & j_l[P]_l &= (j_a[P]_a, j_b[P]_b), & J &= j_l \pi_l, \\ \ker J &= K_l, & \dim W_\vee &= \deg l = \deg a + \deg b - \deg d. \end{aligned} \quad (\text{DC12})$$

Well-definedness follows from $a, b \mid l$. If both components vanish, a and b divide P ; their prime-power factorizations show that l divides P . This proves injectivity. To prove surjectivity with an explicit inverse, write $a = da'$, $b = db'$ and compute polynomials u, v with $ua' + vb' = 1$. Given compatible representatives X modulo a and Y modulo b , $Y - X = dT$ for some $T \in R$. Then

$$P = X + auT, \quad P - X = auT, \quad P - Y = -bvT. \quad (\text{DC13})$$

It has the required two residues. A different choice of representatives, Bezout coefficients, or T changes P by a multiple of both a and b , hence of l . This proves the well-defined inverse and the entire isomorphism. The Euclidean algorithm terminates because its successive nonzero remainders have strictly decreasing degrees, and its back substitution gives u, v . The same proof covers $a = 1$, $b = 1$, or $d = 1$.

The complete exact quotient sequence is

$$0 \longrightarrow E_l \xrightarrow{[P] \mapsto ([P]_a, [P]_b)} E_a \oplus E_b \xrightarrow{(x, y) \mapsto \pi_d^a x - \pi_d^b y} E_d \longrightarrow 0. \quad (\text{DC14})$$

The difference map is onto because its first component is onto; its kernel and first-map injectivity were just proved. Transport by j_a, j_b, j_l proves the identical sequence for the actual images. In parallel, the kernel diamond has

$$\begin{aligned} K_a \cap K_b &= K_l, & K_a + K_b &= K_d, \\ 0 \longrightarrow K_l &\xrightarrow{z \mapsto (z, -z)} K_a \oplus K_b \xrightarrow{(x, y) \mapsto x + y} K_d \longrightarrow 0. \end{aligned} \quad (\text{DC15})$$

The intersection follows from the same divisibility argument. Bezout applied to a, b gives $d = ua + vb$, which proves the equality of the sum with K_d . The displayed signs then prove exactness. Every map in (DC14)–(DC15) commutes with multiplication by S .

In particular $E_d = E/(K_a + K_b)$ is the largest common quotient of the original two observations in the following exact sense. For any vector space T and linear maps $g_a : W_a \rightarrow T$, $g_b : W_b \rightarrow T$ with $g_a C = g_b \theta$, their common composite vanishes on $K_a + K_b$. It therefore has a unique factor $g_d : E_d \rightarrow T$, defined by $g_d[P]_d = g_a C[P]_\chi$. It is well defined by that vanishing and satisfies $g_a = g_d f_a$, $g_b = g_d f_b$ by surjectivity. If the given maps intertwine specified actions, so does g_d , by applying the surjective map π_d . This is the common-quotient assertion with its domain and universality fully specified.

7.1.3 Every local length and both exact factorization criteria

At a root λ of χ , use $x = S - \lambda$ and retain

$$\begin{aligned} \chi &= x^{m_\lambda} c_\lambda(x), & a &= x^{r_\lambda} a_\lambda(x), & b &= x^{s_\lambda} b_\lambda(x), \\ c_\lambda(0) a_\lambda(0) b_\lambda(0) &\neq 0, & s_\lambda &\in \{0, 1\}, \\ d &= x^{\min(r_\lambda, s_\lambda)} d_\lambda(x), & l &= x^{\max(r_\lambda, s_\lambda)} l_\lambda(x), & d_\lambda(0) l_\lambda(0) &\neq 0. \end{aligned} \quad (\text{DC16})$$

These are the literal polynomial cofactors at λ , with no chosen constant rescaling. The source local ring is $R_\lambda = \mathbb{C}[x]/(x^{m_\lambda})$. For $s \in \{a, b, d, l\}$, write its exponent as $e_\lambda(s)$ and its cofactor as $s_\lambda^{\text{unit}}(x)$. The exact kernel inclusion is

$$\mathbb{C}[x]/(x^{m_\lambda - e_\lambda(s)}) \longrightarrow R_\lambda, \quad [B] \longmapsto [x^{e_\lambda(s)} s_\lambda^{\text{unit}}(x) B]. \quad (\text{DC17})$$

It is injective and its image is precisely the local component of K_s . Indeed the unit is invertible by recursive coefficient division; multiplying by that inverse and cancelling the stated power of x proves injectivity. Its image is the principal ideal by definition. The quotient is the reduction $R_\lambda \rightarrow \mathbb{C}[x]/(x^{e_\lambda(s)})$. All these maps intertwine the actual action $\lambda + x$. Thus the four quotient lengths are respectively $r_\lambda, s_\lambda, \min(r_\lambda, s_\lambda)$, and $\max(r_\lambda, s_\lambda)$, and the four kernel lengths are m_λ minus those lengths. No primary component or nilpotent jet is discarded from the source.

There is a map of the actual images $T_{ab} : W_a \rightarrow W_b$ with $\theta = T_{ab}C$ if and only if $b \mid a$. In that case it is unique and its complete formula is

$$T_{ab} = j_b \pi_b^a j_a^{-1}, \quad M_b T_{ab} = T_{ab} M_a. \quad (\text{DC18})$$

Necessity follows by applying the assumed factorization to K_a : it gives $K_a \subset K_b$. Since the class of a is in K_a , this means $a = bP + \chi Q$ for some P, Q , and $b \mid \chi$ then gives $b \mid a$. Sufficiency is the displayed reduction map; uniqueness follows because C is onto W_a . Intertwining follows from the corresponding property of every factor. The reverse factorization exists if and only if $a \mid b$, and is exactly

$$T_{ba} = j_a \pi_a^b j_b^{-1}, \quad C = T_{ba} \theta. \quad (\text{DC19})$$

The same proof establishes it even if an image is zero. Consequently the two observations have mutually inverse factorizations over E if and only if $a = b$. Since b is squarefree, (DC19) requires every critical cyclic exponent to be at most one and every such root to be actually sampled. Formula (DC18) instead requires each sampled root to occur in a ; it places no upper bound on its critical multiplicity.

Their extra observations are calculated by the original maps:

$$\begin{aligned} \theta(K_a) &\simeq K_a/K_l \simeq [a]_b E_b, & \dim \theta(K_a) &= \deg b - \deg d, \\ C(K_b) &\simeq K_b/K_l \simeq [b]_a E_a, & \dim C(K_b) &= \deg a - \deg d. \end{aligned} \quad (\text{DC20})$$

Here $[a]_b E_b$ is the ideal generated by the residue of a in $R/(b)$, and $[b]_a E_a$ is the ideal generated by the residue of b in $R/(a)$. Explicitly the first isomorphism sends the class of aP modulo K_l to $j_b[aP]_b$. Its kernel is exactly $K_a \cap K_b$, and all values in that ideal are attained. The ideal in $R/(b)$ is also the ideal generated by d , by Bezout and reduction modulo b , so its dimension is $\deg b - \deg d$. Interchanging a, b proves the second line. Alternatively (DC17) calculates the local lengths as $s_\lambda - \min(r_\lambda, s_\lambda)$ and $r_\lambda - \min(r_\lambda, s_\lambda)$.

This gives a concrete criterion for a circle value of a class killed by C . It occurs exactly at a sampled λ with $s_\lambda = 1$ and $r_\lambda = 0$. Let $e_\lambda \in E$ be the CRT idempotent equal to 1 in the λ -primary ring and 0 in every other primary ring. Polynomial Bezout produces this idempotent in the original E . Then

$$C e_\lambda = 0, \quad \theta e_\lambda = (\delta_{\lambda, S_n})_{n \in \mathbb{Z}_L}, \quad \|\theta e_\lambda\|^2 = \nu_n \quad (S_n = \lambda). \quad (\text{DC21})$$

Such a λ has real part $k/2$ because it is sampled. Its absence from a means precisely that no all-critical tuple has this sum; the ambient tuples realizing it have at least one off-critical member. The statement is a criterion and explicit map for actual roots of the retained packet; it asserts no existence of an off-critical zero. When $r_\lambda \geq 1$ and $s_\lambda = 1$, the circle sees the constant critical jet and all $r_\lambda - 1$ higher critical jets lie in $C(K_b)$. If mixed tuples raise m_λ above r_λ , the additional $m_\lambda - r_\lambda$ source jets remain in K_a and K_l . For an unsampled root every surviving critical jet lies in $C(K_b)$. These are all local possibilities, directly from (DC16)–(DC20).

7.1.4 The same period transport and all four action defects

Keep the period matrix $\Pi(t) = \Pi_k(t)$ of the original polynomial χ from (PC37), with its original ordered rays, $u \in \mathbb{C}^*$, and increasing-power basis. In particular its entries are the integrals of $S^j dS$ against $\exp((S^{q+1}/(q+1) + \sum_{\ell=0}^{q-1} c_\ell S^{\ell+1}/(\ell+1) - tS)/u)$ on those rays. It is invertible for every t , including repeated-root parameters, by the complete fixed-ray and determinant proof (PC16)–(PC18). Retain

$$\ell[P]_\chi = [S^{q-1}] \text{rem}_\chi P, \quad R_0 = e_0 \ell, \quad A_t = A + tR_0, \quad \Pi' = -\Pi A_t/u, \quad A_t^{\text{per}} = \Pi A_t \Pi^{-1}. \quad (\text{DC22})$$

Here R_0 is an endomorphism and is distinct from the polynomial ring R used above. The following indexed families use four formal roles $\mathcal{I} = \{\text{critical, circle, common, joint}\}$ with associated polynomials $(s_i)_{i \in \mathcal{I}} = (a, b, d, l)$. Subscripts a, b, d, l on the actual targets and observations below are shorthand for these four roles. In K_s , $\deg s$, and polynomial reduction, s means the associated polynomial. Thus even if two or all four polynomials coincide, their four actual target realizations and maps remain separately specified. Set $W_d = E_d$, $j_d = I$, $T_d = \pi_d$, $T_a = C$, $T_b = \theta$, $T_l = J$, and $W_l = W_\vee$. Let M_s be multiplication by S on W_s through j_s . For $s = l$ it is explicitly $(M_a, M_b)|_{W_\vee}$; this restriction is well defined by (DC10). For all four indices, $T_s = j_s \pi_s$, $T_s A = M_s T_s$, and $\ker T_s = K_s$.

Define the four actual period observations on the same target coordinate space $\mathbb{C}^q$ by

$$T_{s,t} = T_s \Pi^{-1}, \quad \mathcal{K}_{s,t} = \ker T_{s,t} = \Pi K_s, \quad s \in \{a, b, d, l\}. \quad (\text{DC23})$$

They are onto their fixed images, with holomorphic kernels and constant ranks $\deg s$. If j_s^{-1} is followed by any polynomial representative of degree less than $\deg s$, its composition with Π is an explicit holomorphic right inverse of $T_{s,t}$. The diagram remains exactly the same:

$$f_a T_{a,t} = f_b T_{b,t} = T_{d,t}, \quad T_{l,t} = (T_{a,t}, T_{b,t}), \quad \mathcal{K}_{a,t} \cap \mathcal{K}_{b,t} = \mathcal{K}_{l,t}, \quad \mathcal{K}_{a,t} + \mathcal{K}_{b,t} = \mathcal{K}_{d,t}. \quad (\text{DC24})$$

Every equality follows by composing its previously proved source equality with the invertible map Π^{-1} . For the transported original action $B_t = \Pi A \Pi^{-1}$ this is an exact diagram of R -modules: $T_{s,t} B_t = M_s T_{s,t}$. Indeed multiplication cancels the two middle factors $\Pi^{-1} \Pi$ and gives the original identity $T_s A = M_s T_s$. The deformed action A_t^{per} instead has the following fully calculated defects.

Before restricting any domain, the complete action defect is

$$\boxed{T_{s,t} A_t^{\text{per}} - M_s T_{s,t} = t T_s(e_0) \ell \Pi^{-1}}. \quad (\text{DC25})$$

Indeed the first term is $T_s(A + t e_0 \ell) \Pi^{-1}$; the A part is $M_s T_s \Pi^{-1}$ by the original intertwining. This proves the formula without asserting invariance of a deformed kernel. The weights in its four target vectors are explicit:

$$T_a(e_0) = C(e_0), \quad T_b(e_0) = (1)_{n \in \mathcal{Z}_L}, \quad T_d(e_0) = [1]_d, \quad T_l(e_0) = (C(e_0), (1)_{n \in \mathcal{Z}_L}). \quad (\text{DC26})$$

The first vector has the raw jet recovery $(M_{v_c})^{\otimes k} \alpha_{c,k}[1]_a$ before applying the inverse units; the second has its original norm from (DC8). If $\deg s > 0$, $T_s(e_0) \neq 0$ because j_s is injective. If $\deg s = 0$, its target and every displayed defect are zero.

For any monic $v \mid \chi$ with $\deg v < q$, the ideal K_v has the exact basis $[v], [Sv], \dots, [S^{q-\deg v-1}v]$. Monicity gives different leading degrees, proving independence; division by χ/v proves spanning. Its last member has degree $q-1$ and leading coefficient 1, so $\ell|_{K_v}$ is onto. Consequently, for $t \neq 0$, $\deg s > 0$, and $\deg v < q$, restriction of the s -defect to $\mathcal{K}_{v,t}$ has

$$\begin{aligned} \text{im } (T_{s,t} A_t^{\text{per}} - M_s T_{s,t})|_{\mathcal{K}_{v,t}} &= \mathbb{C} T_s(e_0), \\ \ker (T_{s,t} A_t^{\text{per}} - M_s T_{s,t})|_{\mathcal{K}_{v,t}} &= \Pi(K_v \cap \ker \ell), \quad \text{rank} = 1. \end{aligned} \quad (\text{DC27})$$

For $v = \chi$ the restricted domain is zero; for $t = 0$ or $s = 1$ the whole restricted defect is zero. In particular the self-kernel defects have rank one precisely when $t \neq 0$ and $0 < \deg s < q$. Formula (DC27) also calculates the critical defect on the circle kernel and the circle defect on the critical kernel, with their full domains. The observations themselves on those crossed kernels retain (DC20): $T_{b,t}(\mathcal{K}_{a,t}) = \theta(K_a)$ and $T_{a,t}(\mathcal{K}_{b,t}) = C(K_b)$.

The factorizations (DC18)–(DC19), whenever their proved divisibility criterion holds, remain unchanged after period transport. Conversely a factorization at even one parameter t can be composed with $\Pi(t)$ and gives the original factorization, so its same divisibility criterion is necessary. The maps f_a, f_b carry the first two defects to the common defect, and the joint defect has the two original defects as components, because $f_a C(e_0) = f_b \theta(e_0) = [1]_d$. Thus both the observations and their entire failure of deformed-action intertwining commute with the exact diamond.

7.1.5 The exact extra circle summand

Squarefreeness of the actual circle polynomial gives a more explicit joint-image formula. Put $b_{\text{new}} = b/d$. Its roots are exactly the sampled roots absent from a , so $\gcd(a, b_{\text{new}}) = 1$ and $l = ab_{\text{new}}$. Denote by W_{new} the actual circle coordinate subspace on these roots, with its inherited metric $\sum_{S_n: b_{\text{new}}(S_n)=0} \nu_n |v_n|^2$ and diagonal action S_n . There is an exact isomorphism

$$\begin{aligned} W_V &\xrightarrow{\sim} W_a \oplus W_{\text{new}}, & (x, y) &\longmapsto (x, y_{\text{new}}), \\ (x, z) &\longmapsto (x, y), & y_n &= \begin{cases} (j_a^{-1}x)(S_n), & d(S_n) = 0, \\ z_n, & b_{\text{new}}(S_n) = 0. \end{cases} \end{aligned} \quad (\text{DC28})$$

Evaluation in the first case is independent of a polynomial representative because $a(S_n) = 0$. It gives exactly the shared reduction in (DC10); the two cases exhaust the sampled roots. Thus the displayed maps are mutually inverse. They intertwine the original actions, since evaluation of SP gives $S_n P(S_n)$. This is the actual-image transport of the CRT isomorphism $R/(l) \simeq R/(a) \oplus R/(b_{\text{new}})$. Its first factor retains the original unit-weighted critical map and all of its nilpotent jets; its second factor retains the actual new circle amplitudes and weights. It asserts an isomorphism of finite-dimensional R -modules. If one assigns the actual circle norm to the second component of $W_V \subset W_a \oplus W_b$, that norm under (DC28) is exactly $\sum_{d(S_n)=0} \nu_n |(j_a^{-1}x)(S_n)|^2 + \sum_{b_{\text{new}}(S_n)=0} \nu_n |z_n|^2$. The shared first sum remains present, so the module isomorphism has not silently been asserted to preserve a chosen direct-sum metric on $W_a \oplus W_{\text{new}}$.

The new circle coordinates have an explicit original polynomial source lift. For a prescribed vector $z = (z_n)_{b_{\text{new}}(S_n)=0}$, set

$$P_z(S) = a(S) \sum_{b_{\text{new}}(S_n)=0} \frac{z_n}{a(S_n)} \prod_{\substack{b_{\text{new}}(S_m)=0 \\ m \neq n}} \frac{S - S_m}{S_n - S_m}. \quad (\text{DC29})$$

Each $a(S_n)$ and each $S_n - S_m$ in a denominator is nonzero, by the proved coprimality and distinct sampled frequencies. The factor a puts $[P_z]_\chi$ in K_a . At a new sample the interpolation product gives exactly z_n ; at a shared sample the factor a gives zero. Consequently $C[P_z]_\chi = 0$ and $\theta[P_z]_\chi = (0, z)$, with squared circle norm $\sum \nu_n |z_n|^2$ on precisely the new samples. At source level the lift is the unchanged $\mathcal{V}_{h,k} P_z$, whose cohomology class is $\sigma_h^{\otimes k} \eta_k [P_z]_\chi$. Thus the circle summand in (DC28) has an actual unit-weighted arithmetic source, including every coefficient $a(S_n)^{-1}$. When there are no new samples, the sum and its lift are zero. For $k = 1$, every sampled root of $\chi = h$ is itself critical, so $b \mid a$ and this extra circle summand is always zero.

The differential equation of all four transported observations retains the same forcing as (DC25):

$$\boxed{u \partial_t T_{s,t} - M_s T_{s,t} = t T_s(e_0) \ell \Pi^{-1}, \quad s \in \{a, b, d, l\}.} \quad (\text{DC30})$$

Differentiating $\Pi^{-1}\Pi = I$ and using (DC22) gives $(\Pi^{-1})' = A_t\Pi^{-1}/u$. Multiplication by uT_s and the original relation $T_sA = M_sT_s$ prove (DC30). The fixed maps f_a, f_b carry these differential equations to the common one by (DC10); the joint equation is their paired equation. The original factors u, t , and the full four vectors in (DC26) remain in both formulas.

The entire diamond now also has its exact singular u equation. Use the division (XD.u1) for the same cyclic χ , and write $C_\chi(t), B_\chi$ in its unchanged increasing-power basis. On every retained contour branch let $T_{s,u,t} = T_s\Pi(u,t)^{-1}$. For all four formal roles,

$$\boxed{\partial_u T_{s,u,t} = T_s(C_\chi(t)/u^2 - B_\chi/u)\Pi(u,t)^{-1}, \quad s \in \{a, b, d, l\}.} \quad (\text{DC31})$$

This follows from $\partial_u \Pi^{-1} = -\Omega_u \Pi^{-1}$ and the original constant maps T_s , with $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$. The same composition gives kernels $\Pi(u,t)K_s$ and right inverses from (DC23), so all intersections, sums and exact quotients in (DC24) remain valid in both parameters. Multiplying (DC31) by f_a, f_b gives the common equation because $f_a T_a = f_b T_b = T_d$; the joint equation has the same two components. The factorization maps (DC18)–(DC19) are parameter-independent and commute with this derivative for the same reason. This proves the full differential diamond.

The actual circle weights of (DC6)–(DC8) are unchanged. Let $D_\nu = \text{diag}(\nu_n)_{n \in \mathcal{Z}_L}$ in the stated circle coordinate basis, including an empty matrix for an empty sample set. Its exact pulled-back positive semidefinite form on the period coordinates is

$$H_b^{\text{obs}} = \Pi^{-*} T_b^* D_\nu T_b \Pi^{-1}, \quad \ker H_b^{\text{obs}} = \Pi K_b, \quad \partial_u H_b^{\text{obs}} = -\Pi^{-*} T_b^* D_\nu T_b \Omega_u \Pi^{-1}, \quad \partial_{\bar{u}} H_b^{\text{obs}} = -\Pi^{-*} \Omega_u^* T_b^* D_\nu T_b \Pi^{-1} \quad (\text{DC32})$$

The kernel identity follows since all retained ν_n are strictly positive, so zero norm is precisely $T_b \Pi^{-1} x = 0$. The derivatives follow by the same inverse-matrix product rule, keeping the original weights in every term. Under (DC28), the shared and new circle sums therefore still use exactly those same ν_n ; the map itself is constant in u, t . The actual polynomial lift (DC29), all inverse-unit factors and the receiving support labels remain unchanged.

All coefficient arrows above have their original supported Split-Zero lifts: $v^\bullet \mapsto (Tv)^\bullet$ and $\tau \mapsto \tau$, with the specified support retained. The kernel inclusions (DC15), local maps (DC17), and period kernels (DC23) identify exactly the represented classes sent to the receiving supported zero. Projection to ordinary coefficients commutes with every displayed arrow. The common quotient and joint image therefore preserve the same support data while recording these different, explicitly calculated coefficient kernels.

8 Complete residue constituent curvature extension

8.1 Residue endomorphisms, metric variation, and the forced period Laplacian

The delivered residue–constituent note proves (RC1)–(RC53). Its transverse-rank and period-curvature calculation is the same exact calculation as (SC9)–(SC25), and its comparison with the original positive form is already (LC13)–(LC15). We retain these shared results and prove the additional endomorphism, variation, differential, and translation maps below. The complete preceding determinant and constituent proofs are included in the standalone edition. No original arithmetic source norm, coefficient, multiplicity, or contour factor is changed.

8.1.1 A universal inverse to the residue endomorphism basis

Let B be an arbitrary commutative ring with identity. Fix the monic polynomial and its original power-basis quotient

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a \in B[S], \quad q \geq 1, \quad E_B = B[S]/(\chi), \quad e_i = [S^i] \quad (0 \leq i < q). \quad (\text{RCX1})$$

Polynomial division by the monic polynomial proves that these e_i are a basis: subtract leading multiples until the degree is less than q ; a nonzero multiple of a monic degree- q polynomial cannot have degree less than q , even when B has zero divisors. Let $A = M_S$, $e = e_0$, and let $\ell : E_B \rightarrow B$ be the coefficient of S^{q-1} in this remainder. Set

$$R = e\ell, \quad \mathcal{B}_{ij} = \ell(e_i e_j), \quad 0 \leq i, j < q. \quad (\text{RCX2})$$

The matrix $\mathcal{B}$ is symmetric. Its entries with $i + j < q - 1$ are zero, and its antidiagonal entries are one. Reversing its columns therefore gives a triangular matrix with diagonal entries one. The column-reversal permutation has $q(q - 1)/2$ inversions, so

$$\det \mathcal{B} = (-1)^{q(q-1)/2}, \quad \mathcal{B}^{-1} = (-1)^{q(q-1)/2} \text{adj}(\mathcal{B}). \quad (\text{RCX3})$$

In particular, this inverse uses no discriminant, root separation, or division by a coefficient. The pairing $\beta(x, y) = \ell(xy)$ induces an isomorphism $E_B \rightarrow \text{Hom}_B(E_B, B)$ with matrix $\mathcal{B}$.

For a matrix T in the original power basis, put $K(T) = T\mathcal{B}^{-1}$. The complete reconstruction is

$$\boxed{T = \sum_{i,j=0}^{q-1} K(T)_{ij} A^i R A^j, \quad K(T) = T\mathcal{B}^{-1}.} \quad (\text{RCX4})$$

Indeed $A^i e = e_i$, and the row ℓA^j has entries $\ell(e_j e_b) = \mathcal{B}_{jb}$. Thus the matrix of the displayed sum is $K(T)\mathcal{B} = T$. Conversely, a zero sum has $K\mathcal{B} = 0$, so $K = 0$. This proves, with explicit inverse, the isomorphism of free B -modules

$$B^{q \times q} \longrightarrow \text{End}_B(E_B), \quad (K_{ij}) \longmapsto \sum_{i,j} K_{ij} A^i R A^j. \quad (\text{RCX5})$$

Its image is contained in the unital B -algebra generated by A, R ; hence that algebra is all of $\text{End}_B(E_B)$. This includes $q = 1$, where $R = I$. For a ring map $B \rightarrow B'$, monic division, $\mathcal{B}$, its displayed inverse, and (RCX4) all map to their corresponding formulas over B' . Thus this is a universal coefficient and base-change identity, including every nonreduced fibre.

The residue dual rows give literal matrix units:

$$d^j = \sum_{b=0}^{q-1} (\mathcal{B}^{-1})_{jb} \ell A^b, \quad U_{ij} = e_i d^j = \sum_b (\mathcal{B}^{-1})_{jb} A^i R A^b. \quad (\text{RCX6})$$

The equality $d^j(e_k) = \delta_{jk}$ follows by multiplication of $\mathcal{B}^{-1}\mathcal{B}$. Consequently $U_{ij}U_{kl} = \delta_{jk}U_{il}$ and $\sum_i U_{ii} = I$. These equations specify the multiplication law, not merely a dimension count.

For the original complex arithmetic source, retain exactly

$$j_E r_N = I, \quad j_E D^{(k)} = A j_E, \quad j_E d = 0, \quad G_N = r_N^* r_N > 0. \quad (\text{RCX7})$$

The maps have domains $r_N : E \rightarrow \mathcal{X}$ and $j_E : \mathcal{X} \rightarrow E$ on the admitted strong-Schwartz source $\mathcal{X}$; d is its original boundary map. Every $T \in \text{End}_{\mathbb{C}}(E)$ has the canonical finite-rank source lift

$$\widehat{T} = r_N T j_E, \quad j_E \widehat{T} = T j_E, \quad \widehat{T} r_N = r_N T, \quad \widehat{T} d = 0, \quad \widehat{T} \widehat{U} = \widehat{T} \widehat{U}. \quad (\text{RCX8})$$

Each identity follows by inserting $j_E r_N = I$ or $j_E d = 0$. In particular $\widehat{T} = r_N j_E$, an idempotent on the full source with image $r_N E$. Thus the lift is a unital representation in the corner with identity $r_N j_E$, and is injective because $j_E \widehat{T} r_N = T$. No assertion that j_E is bounded on an unweighted completion is required or made. Taking $T = U_{ij}$ in (RCX8) gives every matrix unit at the original source, with the same boundary kernel. Taking $T = R$ gives the residue perturbation $r_N R j_E$ already used in (SC7)–(SC8).

8.1.2 The original quotient-unit and residue metric product

Now $B = \mathbb{C}$, and $I : F \hookrightarrow E$ is a fixed nonzero proper A -invariant inclusion of dimension p . Keep the original unit e and row $L = \ell I$. Both $e \notin I(F)$ and $L \neq 0$ follow directly from (RCX3): if $e \in I(F)$, all $A^i e$ belong to $I(F)$; if $L = 0$, invariance gives $\ell A^j I = 0$ for all j , and the invertible residue matrix forces $I = 0$.

For every positive Hermitian form M on the original E , define

$$\begin{aligned} M_F &= I^* M I, & P_M &= I M_F^{-1} I^* M, & r_M &= (1 - P_M)e, & w_M &= I M_F^{-1} L^*, \\ q_M &= r_M^* M r_M, & d_M &= L M_F^{-1} L^*, & \mathbf{c}_F(M) &= q_M d_M. \end{aligned} \quad (\text{RCX9})$$

The equations $P_M^2 = P_M$, $P_M^* M = M P_M$, and $\text{im } P_M = I(F)$ follow by multiplication. Thus $I^* M r_M = 0$. Completing the square gives

$$q_M = \min_{v \in F} (e - Iv)^* M (e - Iv), \quad d_M = w_M^* M w_M = \sup_{v \neq 0} \frac{|Lv|^2}{v^* M_F v}. \quad (\text{RCX10})$$

For the supremum, write $x = M_F^{-1} L^*$; then $Lv = x^* M_F v$, and Cauchy–Schwarz in M_F proves the upper bound $x^* M_F x = d_M$, attained at $v = x \neq 0$. The minimum is attained at $v = M_F^{-1} I^* M e$, with residual $r_M \neq 0$. Consequently $q_M, d_M > 0$, and $r_M^* M w_M = 0$. At $M = G_N$, (RCX10) has exactly the original source expression

$$\mathbf{c}_F(G_N) = \left(\min_{v \in F} \|r_N(e - Iv)\|^2 \right) \left(\sup_{v \neq 0} \frac{|Lv|^2}{\|r_N Iv\|^2} \right). \quad (\text{RCX11})$$

In particular every original arithmetic Taylor unit and source mass encoded in r_N remains in the two displayed factors.

For any invertible coefficient map $C : E' \rightarrow E$, retain the transported data $I' = C^{-1} I$, $e' = C^{-1} e$, $\ell' = \ell C$, and $M' = C^* M C$. Direct substitution gives

$$M'_F = M_F, \quad P_{M'} = C^{-1} P_M C, \quad r_{M'} = C^{-1} r_M, \quad w_{M'} = C^{-1} w_M, \quad \mathbf{c}_{F'}(M') = \mathbf{c}_F(M). \quad (\text{RCX12})$$

For a positive real scalar s , the formulas separately give $q_{sM} = s q_M$, $d_{sM} = s^{-1} d_M$, and $\mathbf{c}_F(sM) = \mathbf{c}_F(M)$. These are proved homogeneity identities with both factors retained; no form is replaced by a rescaled form in the arithmetic construction.

8.1.3 An exact rank-two differential for metric variation

Let $M(s) > 0$ be a C^1 path on a closed real interval; a dot denotes its actual derivative. All inverses in (RCX9) are C^1 . Differentiating $q = r^* M r$, the two terms with $\dot{r}$ vanish: $\dot{r} \in I(F)$, whereas r is M -orthogonal to $I(F)$. Differentiating M_F^{-1} gives $(M_F^{-1})^\cdot = -M_F^{-1} \dot{M}_F M_F^{-1}$. Hence

$$\dot{q} = r^* \dot{M} r, \quad \dot{d} = -w^* \dot{M} w, \quad \boxed{\frac{d}{ds} \log \mathbf{c}_F(M(s)) = \frac{r^* \dot{M} r}{q} - \frac{w^* \dot{M} w}{d}}. \quad (\text{RCX13})$$

Define endomorphisms on the original E , with its original form,

$$Q_r = \frac{r r^* M}{q}, \quad Q_w = \frac{w w^* M}{d}, \quad Z_M = Q_r - Q_w, \quad X = M^{-1} \dot{M}. \quad (\text{RCX14})$$

Each Q is its indicated rank-one M -orthogonal projection. Their products in both orders vanish because $r^* M w = 0$. It follows that Z_M has eigenvalues 1, -1 , and zero on the remaining $q - 2$ dimensions. The complete differential is

$$\boxed{\frac{d}{ds} \log \mathbf{c}_F(M(s)) = \text{Tr}(Z_M X)}. \quad (\text{RCX15})$$

Indeed the trace of a column times a row is their row-column contraction, giving exactly (RCX13). This also proves that scalar metric directions cancel: $X = \alpha I$ gives $\text{Tr } Z_M = 0$, while the original scalar α and both rank-one terms remain specified.

The operator X is self-adjoint in M , since $MX = \dot{M}$ is Hermitian. Let $\lambda_-(s)$ and $\lambda_+(s)$ be its least and greatest real eigenvalues. Each quotient in (RCX13) is a Rayleigh quotient of this same X , so

$$\left| \frac{d}{ds} \log \mathfrak{c}_F(M(s)) \right| \leq \lambda_+(s) - \lambda_-(s). \quad (\text{RCX16})$$

The positive equality holds exactly when r lies in the greatest eigenspace and w in the least; negative equality reverses them. To verify the equality statement, diagonalize the M -self-adjoint operator: a Rayleigh quotient is a weighted mean of its eigenvalues, with nonnegative weights and positive total weight, and attains an extreme precisely on the corresponding eigenspace. Integrating the actual continuous differential yields

$$\log \frac{\mathfrak{c}_F(M(1))}{\mathfrak{c}_F(M(0))} = \int_0^1 \text{Tr}(Z_{M(s)} M(s)^{-1} \dot{M}(s)) ds. \quad (\text{RCX17})$$

This gives an exact observation on the metric path as well as the bound obtained by integrating (RCX16).

For the nested original forms $G_i \succeq G_j > 0$, put

$$T = G_i^{-1} G_j, \quad 0 < \theta = \lambda_{\min}(T) \leq \eta = \lambda_{\max}(T) \leq 1, \quad M(s) = G_i + s(G_j - G_i). \quad (\text{RCX18})$$

No diagonalization of an arithmetic source is performed here: the spectral theorem is used to evaluate the indicated relative operator. Algebraic multiplication gives

$$M(s) = G_i(I + s(T - I)), \quad M(s)^{-1} \dot{M}(s) = (I + s(T - I))^{-1}(T - I). \quad (\text{RCX19})$$

Thus its eigenvalues are $(\lambda - 1)/(1 + s(\lambda - 1))$. The derivative of this expression in λ is $1/(1 + s(\lambda - 1))^2 > 0$. The spread therefore integrates exactly to

$$\int_0^1 (\lambda_+(s) - \lambda_-(s)) ds = \log \eta - \log \theta. \quad (\text{RCX20})$$

Combining (RCX16)–(RCX20), and keeping $V_n = \det G_n$, proves the sharpened endpoint estimate

$$\left| \log \frac{\mathfrak{c}_F(G_j)}{\mathfrak{c}_F(G_i)} \right| \leq \log \frac{\eta}{\theta} \leq -\log \theta \leq \log \frac{V_i}{V_j}. \quad (\text{RCX21})$$

For the last inequality, all q eigenvalues of T lie in $(0, 1]$ and one is θ , so $\det T \leq \theta$. Taking negative logarithms and $\det T = V_j/V_i$ proves it. This improves (RC36) by retaining the greatest eigenvalue η as well as the least. It assigns no unproved uniform bound to these actual finite quantities.

8.1.4 All original volume factors in the coupling

The ideal classification gives $I(F) = (g)/(\chi)$, with g monic, $\chi = gh$, and $\deg h = p$. Indeed A -invariance is invariance under every polynomial in A , so $I(F)$ is an ideal; its preimage in $\mathbb{C}[S]$ is generated by a monic divisor g of χ . The map $[a]_h \mapsto [ga]_\chi$ is bijective onto this ideal: surjectivity is its definition and injectivity follows from $gh \mid ga \Rightarrow h \mid a$. Keep its columns and frame map

$$J_g = [g, Sg, \dots, S^{p-1}g] = IC, \quad J_- = [g, \dots, S^{p-2}g], \quad J_+ = [J_g, e], \quad C = (I^* I)^{-1} I^* J_g. \quad (\text{RCX22})$$

Here C is invertible, $LC = e_{p-1}^T$, and the empty determinant for $p = 1$ is one. The last equality follows because $S^{p-1}g$ is monic of degree $q - 1$, while the preceding columns have smaller degree.

All displayed columns are independent, including J_+ because $e \notin I(F)$. For $v_J(M) = \det(J^*MJ)$, the Schur complement for J_+ and the last diagonal inverse cofactor for $J_g^*MJ_g$ give

$$q_M = \frac{v_{J_+}(M)}{v_{J_g}(M)}, \quad d_M = \frac{v_{J_-}(M)}{v_{J_g}(M)}, \quad \mathbf{c}_F(M) = \frac{v_{J_+}(M)v_{J_-}(M)}{v_{J_g}(M)^2}. \quad (\text{RCX23})$$

For the first identity, the Schur complement is $e^*Me - e^*MJ_g(J_g^*MJ_g)^{-1}J_g^*Me = q_M$. For the second, use $LC = e_{p-1}^T$ and $J_g^*MJ_g = C^*M_F C$; the inverse last cofactor is precisely the determinant of the first $p-1$ Gram columns divided by the full determinant. These proofs retain the frame determinant $\det(J_g^*MJ_g) = |\det C|^2 \det M_F$.

Write $v_{n,J} = v_J(G_n)$. The exact endpoint observation is

$$\log \frac{\mathbf{c}_F(G_j)}{\mathbf{c}_F(G_i)} = \log \frac{v_{j,J_+}}{v_{i,J_+}} + \log \frac{v_{j,J_-}}{v_{i,J_-}} - 2 \log \frac{v_{j,J_g}}{v_{i,J_g}}. \quad (\text{RCX24})$$

This supplies the original Toda-volume expression for the rank-two differential (RCX15). It follows by taking logarithms of the three positive factors in (RCX23); none is discarded. In particular a scalar form change by s contributes $(p+1)\log s + (p-1)\log s - 2p\log s = 0$ to (RCX24), exactly matching (RCX12), with every dimension retained.

8.1.5 The mixed fourth jet and the real-axis derivatives

Fix the original $u \in \mathbb{C}^\times$, the full period matrix $\Pi(t)$, and its equation $\Pi' = -\Pi(A + tR)/u$. The full determinant and contour convergence are proved in (XD1)–(XD26) and (SC1)–(SC6b), included before this section. Put $Y = \Pi I$, $H = \Pi^* \Pi$, $H_F = Y^* Y$, and $\psi(t) = \log \det H_F(t)$. With $P = Y H_F^{-1} Y^*$ and $z = (1 - P)\Pi e$, the exact equations are

$$(1 - P)Y' = -\frac{t}{u}zL, \quad \partial_t \partial_{\bar{t}} \psi = \frac{|t|^2}{|u|^2} \mathbf{c}_F(H(t)). \quad (\text{RCX25})$$

For completeness, $Y' = -Y A_F/u - t\Pi e L/u$ gives the first. Holomorphy gives $\partial_t H_F = Y^* Y'$; differentiate the inverse in $\partial_t \log \det H_F = \text{Tr}(H_F^{-1} Y^* Y')$. The result is $\text{Tr}(H_F^{-1} Y'^*(1 - P)Y')$. Substitution of the first equation, (RCX10), and the period quotient-norm minimum gives the second.

All entries are real analytic and $H_F > 0$, so ψ has a convergent expansion in $t, \bar{t}$ near zero. This can also be seen by treating the two variables independently in $\det(I^* \Pi(\bar{s})^* \Pi(t) I)$; its value at $(0, 0)$ is positive, so it has a holomorphic logarithm on a sufficiently small polydisc. Equation (RCX25) then gives

$$\begin{aligned} \psi_{t\bar{t}}(0) &= \psi_{t\bar{t}\bar{t}}(0) = \psi_{t\bar{t}\bar{t}\bar{t}}(0) = 0, & \psi_{t\bar{t}\bar{t}\bar{t}}(0) &= \psi_{t\bar{t}\bar{t}\bar{t}}(0) = 0, \\ \psi_{t\bar{t}\bar{t}\bar{t}}(0) &= \mathbf{c}_F(H(0))/|u|^2 > 0, & [t^2 \bar{t}^2] \psi &= \mathbf{c}_F(H(0))/(4|u|^2). \end{aligned} \quad (\text{RCX26})$$

The first row is obtained by applying the indicated derivatives to $t\bar{t} \mathbf{c}_F(H(t))/|u|^2$ and evaluating at zero. The factorial in the second row is $2!2! = 4$.

Here is the exact map to real-axis derivatives, including the terms a mixed derivative does not determine. Let

$$\phi(t) = \log \det(I^* \Pi(0)^* \Pi(t) I), \quad \phi(0) = \log \det H_F(0) \quad (\text{RCX27})$$

on its local nonzero determinant neighbourhood. Its derivatives of positive order are the pure t -derivatives of ψ . For $x \in \mathbb{R}$, $d/dx = \partial_t + \partial_{\bar{t}}$. Reality of ψ , the binomial theorem for commuting derivatives, and (RCX26) prove

$$\begin{aligned} \left. \frac{d^2}{dx^2} \psi(x) \right|_0 &= 2 \text{Re } \phi''(0), \\ \left. \frac{d^4}{dx^4} \psi(x) \right|_0 &= 2 \text{Re } \phi^{(4)}(0) + 6 \frac{\mathbf{c}_F(H(0))}{|u|^2}. \end{aligned} \quad (\text{RCX28})$$

Thus the mixed observation and the real derivatives are related by these exact equations, with the pure holomorphic jets retained. In particular the first equation and $\Pi''(0) = \Pi(0)A^2/u^2 - \Pi(0)R/u$ give

$$\phi'(0) = -\text{Tr } A_F/u, \quad \phi''(0) = -\frac{1}{u}LH_F(0)^{-1}I^*H(0)e. \quad (\text{RCX29})$$

To check the second identity, the second logarithmic derivative is the trace of the second matrix derivative minus the square of the first logarithmic matrix derivative. The A_F^2/u^2 terms cancel exactly, leaving the displayed rank-one trace.

8.1.6 Every period derivative and the forced Laplacian

All higher pure jets in (RCX28) are given by an explicit finite noncommutative recursion in the original A, R, u . Set

$$Q_0(t) = I, \quad Q_{n+1}(t) = Q'_n(t) - \frac{1}{u}(A + tR)Q_n(t). \quad (\text{RCX30})$$

Induction using the product rule and the period equation proves $\Pi^{(n)}(t) = \Pi(t)Q_n(t)$ for every $n \geq 0$. Every Q_n is a polynomial in t with coefficients in the algebra over $\mathbb{C}[u, u^{-1}]$ generated by A, R . In particular, with $A_t = A + tR$,

$$\begin{aligned} Q_1 &= -A_t/u, & Q_2 &= A_t^2/u^2 - R/u, \\ Q_3 &= -A_t^3/u^3 + (RA_t + 2A_tR)/u^2, \\ Q_4 &= A_t^4/u^4 - (RA_t^2 + 2A_tRA_t + 3A_t^2R)/u^3 + 3R^2/u^2. \end{aligned} \quad (\text{RCX31})$$

The order of RA_t and A_tR is retained: differentiating A_t^2 gives $RA_t + A_tR$, and the product $-A_tQ_2/u$ contributes one additional A_tR/u^2 . Differentiating Q_3 gives $-(RA_t^2 + A_tRA_t + A_t^2R)/u^3 + 3R^2/u^2$; the additional $-A_tQ_3/u$ contributes $A_t^4/u^4 - (A_tRA_t + 2A_t^2R)/u^3$. Adding them proves the displayed Q_4 , including the $3R^2$ term. For $q > 1$, $\ell e = 0$ and $R^2 = 0$; for $q = 1$, $R = I$ and that term remains $3I/u^2$.

To evaluate the remaining fourth jet entirely from the displayed data, put

$$D_n = H_F(0)^{-1}I^*H(0)Q_n(0)I \quad (1 \leq n \leq 4).$$

Then

$$\phi^{(4)}(0) = \text{Tr}(D_4 - 4D_1D_3 - 3D_2^2 + 12D_1^2D_2 - 6D_1^4). \quad (\text{RCX32})$$

Indeed the local matrix in (RCX27), after left multiplication by $H_F(0)^{-1}$, is $I + \sum_{n=1}^4 D_n t^n/n! + O(t^5)$. The convergent trace-log expansion $\text{Tr} \log(I + Z) = \sum_{m \geq 1} (-1)^{m+1} \text{Tr}(Z^m)/m$ follows by differentiating both sides near zero and using their common value zero at $Z = 0$. Collecting the t^4 coefficient gives respectively D_4 , $-4D_1D_3 - 3D_2^2$, $12D_1^2D_2$, and $-6D_1^4$ after multiplication by $4!$. Cyclic trace, rather than a commutation assumption, combines the three cubic terms. This proves (RCX32) and completely evaluates the real fourth derivative in (RCX28).

For the original tensor degree $k \geq 1$, let $L_t = A_t^2 - kA_t$. The following polynomial matrix identity also holds for every real k , as a separate algebraic parameter. Equation (RCX31) proves the exact differential realization

$$\boxed{u^2\Pi'' + ku\Pi' + u\Pi R = \Pi L_t.} \quad (\text{RCX33})$$

The added term is the same $R = e\ell$ whose complete algebra and normal action were computed above. On the admitted source domain stable under $D^{(k)}$ and its square, let $\hat{R} = r_N R j_E$ and $Y_x(t) = \Pi(t)j_E x$ for a fixed source vector x . Applying (RCX33) at zero and twice using $j_E D^{(k)} = A j_E$ gives

$$(u^2\partial_t^2 + ku\partial_t)Y_x|_0 + u\Pi(0)j_E\hat{R}x = \Pi(0)j_E((D^{(k)})^2 - kD^{(k)})x. \quad (\text{RCX34})$$

Every term has target $\mathbb{C}^q$ with its specified period coordinates; the source vector remains in the same admitted domain.

The primitive also has a complete formula. Let $\mathcal{C}_{\text{prim}}$ be the original primitive source, $D_0 : \mathcal{C}_{\text{prim}} \rightarrow \mathcal{C}_{\text{prim}}$ its Euler action, and retain

$$D^{(k)}d = dD_0, \quad D^{(k)}r_N - r_N A = dK, \quad K : E \rightarrow \mathcal{C}_{\text{prim}}.$$

Put $D_t = D^{(k)} + t\hat{R}$. Equations (RCX8) give $D_t d = dD_0$ and $D_t r_N = r_N A_t + dK$. Therefore direct expansion gives

$$\boxed{(D_t^2 - kD_t)r_N - r_N L_t = d(D_0 K + K A_t - kK)}. \quad (\text{RCX35})$$

Indeed $D_t(r_N A_t) = (r_N A_t + dK)A_t$ and $D_t dK = dD_0 K$; subtract $k(r_N A_t + dK)$. This reproduces the complete primitive (LC12a) and attaches it to the new forced period equation. In particular the term tKR in KA_t is present with its original sign.

8.1.7 Translation of the marked curvature and first derivative

For a fixed complex a , keep the actual substitution data

$$\begin{aligned} \chi_a(S) &= \chi(S - a), \quad \Phi_a(S) = \Phi(S - a) - \Phi(-a), \\ (C_a)_{rb} &= \binom{b}{r} a^{b-r} \quad (r \leq b), \quad f_a(t) = \exp((- \Phi(-a) - ta)/u). \end{aligned} \quad (\text{RCX36})$$

The zero entries for $r > b$ are retained. The literal substitution $P(S) \mapsto P(X + a)$ gives $C_a^{-1} = C_{-a}$, $\det C_a = 1$, $C_a e = e$, and $\ell C_a = \ell$, because translation preserves the highest coefficient in degree below q . It follows in particular that $C_a R C_a^{-1} = R$. Also $C_a A_a(t) C_a^{-1} = A_t + aI$.

The original oriented exponential contour integral gives

$$\Pi_a(t) = f_a(t) \Pi(t) C_a. \quad (\text{RCX37})$$

To retain its analytic justification, substitute $S = X + a$. The exponent becomes $(\Phi(X) - tX - \Phi(-a) - ta)/u$, and the polynomial factor becomes $(X + a)^b$. The substituted contour is the original ray contour translated by $-a$. Truncate at radius R_0 and connect the corresponding endpoints by segments of length $|a|$. Along each segment the real part of the leading exponent is $-R_0^{q+1}/((q+1)|u|) + O(R_0^q)$, uniformly on the segment. Its integral tends to zero even with the polynomial factor. Cauchy's theorem applies because the integrand is entire; the finite connectors cancel with the opposite ray orientations. This proves equality with the original contour and yields (RCX37) with every scalar and column coefficient present.

Transport the same constituent by $I_a = C_a^{-1}I$. The exact Gram and coupling relations are

$$\begin{aligned} Y_a &= f_a Y, \quad H_a = |f_a|^2 C_a^* H C_a, \quad H_{F,a} = |f_a|^2 H_F, \\ \mathbf{c}_{F_a}(H_a) &= \mathbf{c}_F(H), \quad \log \det H_{F,a} = \log \det H_F + 2p \operatorname{Re} \frac{-\Phi(-a) - ta}{u}. \end{aligned} \quad (\text{RCX38})$$

The coupling equality follows from the complete pullback (RCX12) and its proved scalar law. In the determinant identity the added function is the real part of an affine holomorphic function of t at fixed u , so its $\partial_t \partial_{\bar{t}}$ derivative and every pure or mixed $t, \bar{t}$ derivative of total order at least two vanish. All mixed curvatures, (RCX26), and the real derivatives in (RCX28) are therefore unchanged under this specified translation.

For real $u > 0$, the first real-axis derivative instead gives

$$-u \frac{d}{dx} \log \frac{\det H_{F,a}(x)}{\det(I_a^* C_a^* G_N C_a I_a)} \Big|_0 - kp = 2 \operatorname{Re} \operatorname{Tr} A_F - kp + 2p \operatorname{Re} a. \quad (\text{RCX39})$$

The denominator is exactly $\det(I^* G_N I)$, independent of x , and the numerator derivative follows from (RCX29) and (RCX38). This specifies the complete relation between translation, curvature, and the original trace defect. The original arithmetic object is the marked fibre $a = 0$; the comparison creates no claim that an auxiliary translated polynomial is an arithmetic zero packet.

8.1.8 Every u, t jet and the retained translation derivatives

The recursion (RCX30) now has its full two-parameter version on each original branch of $u \neq 0$. Use the exact matrices $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$ and $\Omega_t = -A_t/u$ from (XD.u1)–(XD.u6). Starting at $Q_{0,0} = I$, define

$$Q_{r+1,s} = \partial_u Q_{r,s} + \Omega_u Q_{r,s}, \quad Q_{r,s+1} = \partial_t Q_{r,s} + \Omega_t Q_{r,s}. \quad (\text{RCX.u1})$$

These prescriptions are compatible. The commutator of their two left-acting differential operators is multiplication by $\partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t] = 0$. Starting at I they therefore give the same matrix for every ordering with r u derivatives and s t derivatives. The product rule and $\partial_a \Pi = \Pi \Omega_a$ prove by induction the exact identity $\partial_u^r \partial_t^s \Pi = \Pi Q_{r,s}$. All coefficient matrices remain in the algebra on the same remainder frame. In particular, retaining the order of every product,

$$\begin{aligned} Q_{1,0} &= -C_\chi/u^2 + B_\chi/u, \\ Q_{2,0} &= C_\chi^2/u^4 + (2C_\chi - C_\chi B_\chi - B_\chi C_\chi)/u^3 + (B_\chi^2 - B_\chi)/u^2, \\ Q_{1,1} &= C_\chi A_t/u^3 + (A_t - B_\chi A_t)/u^2. \end{aligned} \quad (\text{RCX.u2})$$

Indeed $\partial_u C_\chi = \partial_u B_\chi = 0$ and $\partial_u(-A_t/u) = A_t/u^2$. Expanding $\partial_u \Omega_u + \Omega_u^2$ and $\partial_u \Omega_t + \Omega_u \Omega_t$ gives exactly these three formulas. Setting $r = 0$ gives every earlier matrix (RCX30)–(RCX32), including its $3R^2/u^2$ term.

For $H = \Pi^* \Pi$ the exact derivative is $\partial_u H = H \Omega_u$ and $\partial_{\bar{u}} H = \Omega_u^* H$. Thus along an arbitrary specified differentiable curve $(u(s), t(s))$ the metric derivative in (RCX13)–(RCX17) is the actual Hermitian matrix

$$\dot{H} = \Omega_{\text{path}}^* H + H \Omega_{\text{path}}, \quad \Omega_{\text{path}} = \dot{u} \Omega_u + \dot{t} \Omega_t. \quad (\text{RCX.u3})$$

Substitution into (RCX13) supplies both original Rayleigh quotients and their exact signed difference, including the full singular u terms. The constituent Hermitian Hessian is (SC.u2) on the same I, H_F , so the pure t fourth-jet identity (RCX26)–(RCX28) is retained as that exact slice. Differentiating the already proved forced equation (RCX33) also gives its complete u descendant:

$$2u \Pi_{tt} + u^2 \Pi_{utt} + k \Pi_t + ku \Pi_{ut} + \Pi R + u \Pi_u R = \Pi \Omega_u L_t. \quad (\text{RCX.u4})$$

The coefficient factors and every derivative term follow from the product rule; L_t and R are independent of u . Composing with the fixed original j_E gives the exact same identity on its admitted theta-source domain. Its original Euler-action connection defect is precisely (DS74), so no boundary primitive is discarded by this extension.

For the translated phase retain $h_a(t) = -\Phi(-a) - ta$. The exact identity (RCX38) holds in both parameters:

$$\psi_a(u, t) - \psi(u, t) = 2p \operatorname{Re}(h_a(t)/u). \quad (\text{RCX.u5})$$

This is pluriharmonic on $u \neq 0$, so all mixed holomorphic–antiholomorphic derivatives of this difference vanish. Its pure u derivatives are explicitly retained:

$$\partial_u(\psi_a - \psi) = -p h_a(t)/u^2, \quad \partial_u^2(\psi_a - \psi) = 2p h_a(t)/u^3, \quad \partial_u \partial_t(\psi_a - \psi) = p a/u^2. \quad (\text{RCX.u6})$$

They follow by differentiating $p(h_a/u + \overline{h_a/u})$ in Wirtinger coordinates, with $\partial_t h_a = -a$. Consequently the earlier vanishing of derivatives of order at least two concerns the fixed- u t derivatives only. The full translated singular connection is the proved map (DT.14)–(DT.16), and these additional pure derivatives are carried into (RCX.u1)–(RCX.u4) through that exact map.

8.1.9 Source chronology and exact comparison scope

The delivered note’s historical statement about translation source hashes is resolved by the pre-existing sealed translation manifest. The reviewed and final source SHA-256 values, respectively, are

39163624c54704c203a69481ff4d7df4262afa108ac1e0989171d5ba8d700c42
bd1831cda196602d34afc1fa5039d9d80041711fba72bc8b45c782f67c25bade

The exact one-field change is the geometry margin from 25 to 23 millimetres; reversing this one byte-string substitution reproduces the reviewed hash. The original dated seal records that verification, its four-page visual review, and the later README delivery. The raw delivered note remains unchanged.

The new finite suite checks actual examples of the residue basis, the transverse map, the mixed jet with factorial, the metric inequalities, the forced Laplacian, and translation. These examples supplement the full proofs above. The universal endomorphism identity, exact differential (RCX15), endpoint bound (RCX21), and source primitive (RCX35) supply explicit relations to the original coefficient, metric, and cochain objects. They do not provide an upper estimate uniform over the remaining arithmetic endpoint family; the original determinant quantity remains explicit in (RCX21) and (RCX24).

The concurrent finite-sampling calculation (FC1)–(FC23) acts on these same original forms. Its exact quotient Gram is $G_N = (J_N M_N^{-1} J_N^*)^{-1}$, and its finite-circle Gram is $H_N = (J_N M_N(L, J)^{-1} J_N^*)^{-1}$. The supplied source-error calculation gives the actual constants $\alpha = 1 - \eta_P - \eta_T > 0$, $\beta = 1 + \eta_P$, with $\alpha G_N \preceq H_N \preceq \beta G_N$. Here is the complete finite-dimensional map to our observation: minimizing both forms on $e + I(F)$ gives $\alpha q_{G_N} \leq q_{H_N} \leq \beta q_{G_N}$; restricting to $I(F)$ and reversing inverse order gives $\beta^{-1} d_{G_N} \leq d_{H_N} \leq \alpha^{-1} d_{G_N}$. Thus

$$\frac{\alpha}{\beta} \mathbf{c}_F(G_N) \leq \mathbf{c}_F(H_N) \leq \frac{\beta}{\alpha} \mathbf{c}_F(G_N). \quad (\text{RCX40})$$

This is exactly (FC9) in our retained unit/residue notation. Equations (RCX13)–(RCX17) further evaluate the full differential along the straight path between these particular forms; no commutation between the form and the generator is inserted. The original full-volume observer and all constituent minors remain those of (RCX21)–(RCX24).

9 Complete analytic pole classes, connection, periods and endpoint transport

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 6e8bba584c5788e0d1ea29127ace2d89d9a65a2e111da01df5870d9c9b57d1e7

Complete source. `sources/next_edition/analytic_pole_conversions_v2/sources/analytic_pole/ANALYTIC_POLE_RESIDUE_TRANSPORT.md`

Source SHA-256. 6e8bba584c5788e0d1ea29127ace2d89d9a65a2e111da01df5870d9c9b57d1e7

Current mathematical continuation. APU1–23 below are inserted into the complete original proof at the AP13–AP14 transition. Their source is `sources/current_boundary/fragments/analytic_pole_u_connection.tex`, SHA-256 1b42c530ccee6fd019b8536170277e3909ba43bfe4e3b2fefab3f7b9fc666c05. The preceding source record identifies the preserved historical body; the exact current body and reversible wrapper mapping are pinned by the boundary patch manifest.

9.0.1 Actual Taylor-unit gauge: analytic pole classes and their connection

13 September 2026. This continues UG1–19 on the original arithmetic packet. The accompanying derivation supplied the exact weighted-residue sequence and the connection on simple-pole lifts. This note proves those constructions in full, calculates their entire finite scaling transport and dual, and compares them with the already proved formal contraction. It does not substitute a new source, a new metric, a simple-root packet, or an assumed arithmetic bound.

9.0.2 1. Original input and the fixed nonzero-parameter complex

Keep the monic polynomial h of degree $d > 0$ with every selected actual zero order, $g = 2\xi$, and the unique polynomial ν of degree less than d representing the complete Taylor unit $v_h = j_h(g/h)$. Thus $\gcd(h, \nu) = 1$, and no factor of h or Taylor coefficient of its unit has been removed. Let P_ν be the set of distinct zeros of ν , and r its cardinality; $r = 0$ when ν is constant. The multiplicities of ν are retained in ν itself throughout the gauge.

Fix the original primitive Φ with $\Phi' = h$ and its specified additive constant. For $u \in \mathbb{C}^*$ and $t \in \mathbb{C}$ define

$$\begin{aligned} D &= u \frac{d}{ds} + h(s) - t, \\ w(s; u, t) &= \exp((\Phi(s) - ts)/u), \\ B &= \mathbb{C}[s], \quad B_{\text{loc}} = \mathbb{C}[s, \nu^{-1}], \quad M = B_{\text{loc}}/B. \end{aligned} \tag{AP1}$$

Use complexes $[B \xrightarrow{D} B ds]$ and $[B_{\text{loc}} \xrightarrow{D} B_{\text{loc}} ds]$ in degrees 0, 1. The equation $wDF = u(wF)'$ is literal, with the constant in Φ retained. Although w is entire and nowhere zero in s , it is not asserted rational. The quotient M consists exactly of finite sums of principal parts at P_ν : partial fractions gives existence and uniqueness, with the polynomial part discarded only in this explicitly specified quotient.

9.0.3 2. The entire additional pole cohomology is calculated

Define, for $[F] \in M$,

$$\text{Res}_w[F] = (\text{Res}_{s=p} w(s)F(s) ds)_{p \in P_\nu}. \tag{AP2}$$

Adding a polynomial changes none of these residues. Also $\text{Res}_w D = 0$ by AP1, since a derivative of a meromorphic Laurent series has zero residue.

The exact sequence is

$$0 \longrightarrow M \xrightarrow{D} M \xrightarrow{\text{Res}_w} \mathbb{C}^{P_\nu} \longrightarrow 0. \tag{AP3}$$

Here is a proof of every assertion. If F has pole order $n \geq 1$ at p and leading term $c(s-p)^{-n}$, DF has leading pole term $-unc(s-p)^{-n-1}$. Multiplication by $h-t$ has pole order at most n , so it cannot cancel this term. Thus $D[F] = 0$ in M implies that F has no poles, and $[F] = 0$. This proves injectivity.

For each p , the rational function $1/(w(p)(s-p))$ has residue vector equal to the p -th coordinate vector. It is analytic at every other point of P_ν . This proves surjectivity without suppressing the factor $w(p)$.

Suppose $\text{Res}_w[F] = 0$. In a disk at p containing no other point of P_ν , the Laurent series of wF has finitely many negative terms and no $(s-p)^{-1}$ term. Integrating term by term therefore gives a meromorphic primitive H_p , with no logarithm. Put $Q_p = H_p/(uw)$. Then $DQ_p = F$ locally. Let Q be the rational function obtained by summing the finite principal parts of the Q_p at all p . The difference $Q - Q_p$ is holomorphic at p , so $DQ - F$ is holomorphic there. This rational function has no other possible finite poles, hence is a polynomial. Therefore $D[Q] = [F]$ in M , which proves exactness in the middle. Changing any integration constant changes only a holomorphic local function and does not change this construction of the principal part.

All higher pole orders are explicitly observed, rather than ignored:

$$\begin{aligned} \text{Res}_p \frac{w(s)}{(s-p)^n} ds &= \frac{w^{(n-1)}(p)}{(n-1)!}, \\ D(s-p)^{-n} &= -un(s-p)^{-n-1} \\ &\quad + \sum_{j=0}^d \frac{h^{(j)}(p)}{j!} (s-p)^{j-n} - t(s-p)^{-n}. \end{aligned} \tag{AP4}$$

These formulas give the triangular reduction of each higher pole while retaining every derivative of h and every derivative of w that contributes. They explain why there is one quotient coordinate per distinct pole point; they do not replace h by its radical or discard its nilpotent zero jets.

9.0.4 3. The original arithmetic classes inject into the analytic extension

The polynomial complex has $H^0 = 0$ and H^1 of dimension d , with basis $1, s, \dots, s^{d-1}$. Indeed, for a nonzero polynomial of degree n , the highest term of DF has degree $n + d$, with unchanged leading coefficient. Successive subtraction of such leading terms reduces every polynomial uniquely to degree less than d . Write this unchanged remainder module as H_{pol} .

The same highest-pole argument shows H^0 of the localized complex is zero: a rational solution of $DF = 0$ has no finite pole, and then the preceding polynomial argument applies. The short exact sequence of the three complexes, together with AP3, gives

$$0 \longrightarrow H_{\text{pol}} \xrightarrow{i} H_{\text{loc}} \xrightarrow{\text{Res}_w} \mathbb{C}^{P_\nu} \longrightarrow 0. \quad (\text{AP5})$$

One can check injectivity without a long exact sequence. If a polynomial F equals DQ for rational Q , the leading-pole calculation forces Q to be a polynomial. Thus no original polynomial cohomology class becomes a boundary. The kernel and surjectivity assertions follow from AP3 by subtracting DQ and retaining the resulting polynomial.

An explicit basis of H_{loc} is

$$1, s, \dots, s^{d-1}, \quad r_p = \frac{1}{s-p} \quad \text{for } p \in P_\nu. \quad (\text{AP6})$$

For existence, subtract $\sum_p (\text{Res}_w F)_p r_p / w(p)$. The remainder has zero weighted residues, and AP3 reduces it modulo D to a polynomial, then to degree less than d . For independence, applying AP2 to a linear combination forces every pole coefficient to vanish because $w(p) \neq 0$; injectivity in AP5 then gives independence of the polynomial coefficients. Hence $\dim H_{\text{loc}} = d + r$. The original d -dimensional arithmetic fibre, with all repeated roots of h , is present as the explicitly injected summand of this vector-space basis.

9.0.5 4. Actual parameter connection and both explicit residue frames

The differential operator $L = u d/dt - s$ commutes with D :

$$\left[u \frac{d}{dt}, D \right] = -u, \quad \left[-s, u \frac{d}{ds} \right] = u, \quad [-s, h - t] = 0. \quad (\text{AP7})$$

It therefore descends as a connection operator on the analytic family over $\mathbb{C}^* \times \mathbb{C}$, not as an unsupported multiplication endomorphism on the fixed de Rham quotient. Let $A(t)$ be the original companion matrix of $h - t$ in the polynomial frame, and let e_0 be its constant-polynomial column. Set

$$P = \text{diag}(p)_{p \in P_\nu}, \quad B_0 = (e_0 \quad \dots \quad e_0), \quad A_{\text{ext}}(t) = \begin{bmatrix} A(t) & B_0 \\ 0 & P \end{bmatrix}. \quad (\text{AP8})$$

Here each pole basis vector satisfies the exact identity

$$Lr_p = -\frac{s}{s-p} = -pr_p - 1. \quad (\text{AP9})$$

For the last polynomial vector, reduction of s^d uses $D(1) = h - t$, so the polynomial block is exactly the already specified $A(t)$, without a derivative correction. Thus in the full frame AP6 the connection is

$$L = u \frac{d}{dt} - A_{\text{ext}}(t). \quad (\text{AP10})$$

All coefficients depend holomorphically on (u, t) for $u \neq 0$. In fact A_{ext} itself has no singular u coefficient; this statement about the matrix does not identify its rank- $(d + r)$ frame with the formal cohomology at $u = 0$.

Put $W = \text{diag}(w(p; u, t))$. The residue map in AP6 is the rectangular matrix

$$\text{Res}_w = \begin{bmatrix} 0 & W \end{bmatrix}, \quad u \frac{dW}{dt} = -PW. \quad (\text{AP11})$$

For parameter-dependent representatives direct differentiation proves

$$u \frac{d}{dt} \text{Res}_w[F] = \text{Res}_w \left[\left(u \frac{d}{dt} - s \right) F \right]. \quad (\text{AP12})$$

Thus the quotient connection in weighted-residue coordinates is $u d/dt$. The ordinary simple-pole coefficients instead have $u d/dt - P$. Their exact transition is multiplication by W , not deletion of an exponential phase.

For completeness use the second frame $r_p/w(p)$. Its residue coordinates are the standard coordinate vectors, and

$$L\left(\frac{r_p}{w(p)}\right) = -\frac{1}{w(p)}. \quad (\text{AP13})$$

The pole diagonal becomes zero while its coupling to the original polynomial class is $-e_0/w(p)$. This is the same connection under the explicit diagonal frame map $\text{diag}(I_d, W^{-1})$; no factor has been set to one in the original frame.

9.0.6 The complete two-parameter connection on the original analytic pole extension

This calculation continues AP1–26 with its original monic h of degree $d > 0$, primitive $\Phi' = h$, full Taylor representative ν of $j_h(g/h)$, and localization $\mathbb{C}[s, \nu^{-1}]$. The polynomial ν and its complete multiplicities are fixed; its distinct pole points $P_\nu = \{p : \nu(p) = 0\}$ do not vary with u or t . Write $r = \#P_\nu$, $n = d + 1$, and

$$f(s, t) = \Phi(s) - ts, \quad D = u\partial_s + h(s) - t, \quad w(s; u, t) = e^{f(s, t)/u}. \quad (\text{APU1})$$

All formulas below are on $S = \mathbb{C}_u^\times \times \mathbb{C}_t$. The original coefficient frame in AP6 is

$$(1, s, \dots, s^{d-1}, (r_p)_{p \in P_\nu}), \quad r_p = (s - p)^{-1}. \quad (\text{APU2})$$

The free polynomial submodule, the localized module, and their quotient are the actual cohomology modules in AP5. In particular there is no new source frame that discards their extension.

Degree-correct chain connections. On both the polynomial and localized complexes, in degree $j = 0, 1$, define

$$\nabla_u^j = \partial_u - \frac{f}{u^2} + \frac{1-j}{u}, \quad \nabla_t^j = \partial_t - \frac{s}{u}. \quad (\text{APU3})$$

Multiplication by f and differentiation in the parameters preserve the stated localization, since all poles are fixed. The two connections are actual cochain maps:

$$\nabla_u^1 D = D \nabla_u^0, \quad \nabla_t^1 D = D \nabla_t^0. \quad (\text{APU4})$$

Indeed $\partial_u D - D \partial_u = \partial_s$, $[-f/u^2, D] = f_s/u = (h - t)/u$, and $D/u = \partial_s + (h - t)/u$; their three terms cancel in the first identity. For the second, $[\partial_t, D] = -1$ and $[-s/u, D] = 1$. The original degree-zero $1/u$ term is essential in the first cancellation. Their curvature vanishes in each degree:

$$[\nabla_u^j, \nabla_t^j] = 0. \quad (\text{APU5})$$

The only nonzero terms are $[\partial_u, -s/u] = s/u^2$ and $[-f/u^2, \partial_t] = f_t/u^2 = -s/u^2$. The scalar $(1 - j)/u$ commutes with ∇_t^j . Thus the descended connection is flat on the original two-parameter family, with no commutation assertion inferred from pointwise matrix coincidence.

Every coefficient of the u -connection. For $0 \leq b < d$, perform the original monic polynomial division

$$f(s, t)s^b = (h(s) - t)q_b(s, t) + c_b(s, t), \quad \deg c_b < d.$$

Here $\deg q_b \leq b + 1$, so $\deg q'_b \leq b < d$. Let $C(t)$ have coefficient column c_b and $B(t)$ have coefficient column q'_b , both in the original ascending monomial frame. Subtraction of Dq_b gives the exact differential remainder $[fs^b] = [c_b - uq'_b]$. Therefore the polynomial block is

$$\Omega_{\text{pol}} = -\frac{C(t)}{u^2} + \frac{B(t)}{u}. \quad (\text{APU6})$$

This is the full polynomial connection from the marked-family calculation, with its derivative correction retained.

For a fixed $p \in P_\nu$, set

$$f_p(t) = f(p, t), \quad q_p(s, t) = \frac{f(s, t) - f(p, t)}{s - p}, \quad Q_p(s, t) = q_p(s, t) - \frac{h(s) - t}{n}. \quad (\text{APU7})$$

Since h is monic of degree d , the leading coefficient of Φ is $1/n$. Thus q_p has degree d and leading coefficient $1/n$; its ordinary quotient by $h - t$ is precisely the constant $1/n$, and Q_p has degree less than d . The derivative of this quotient is zero. Subtracting $D(1/n)$ therefore proves the full differential-remainder equality $[q_p] = [Q_p]$, at every $u \neq 0$ and every t . Now the literal rational identity $fr_p = q_p + f_p r_p$ gives

$$\nabla_u^1[r_p] = -\frac{[Q_p]}{u^2} - \frac{f_p[r_p]}{u^2}. \quad (\text{APU8})$$

No higher-pole or derivative term is omitted: fixed p implies $\partial_u r_p = \partial_t r_p = 0$, and the only polynomial quotient just computed is constant.

Let $Q(t)$ be the d -by- r matrix of these exact Q_p columns and $F(t) = \text{diag}(f_p(t))$. Retain $A(t)$, $B_0 = (e_0, \dots, e_0)$, and $P = \text{diag}(p)$ from AP8. The complete coefficient connections in the frame APU2 are

$$\nabla_u = \partial_u + \Omega_u, \quad \Omega_u = \begin{bmatrix} -C(t)/u^2 + B(t)/u & -Q(t)/u^2 \\ 0 & -F(t)/u^2 \end{bmatrix}, \quad \nabla_t = \partial_t + \Omega_t, \quad \Omega_t = -\frac{1}{u} \begin{bmatrix} A(t) & B_0 \\ 0 & P \end{bmatrix}. \quad (\text{APU9})$$

The second formula is exactly AP10 divided by u , not a change of its connection convention.

The off-diagonal flatness can also be checked directly in these original coefficient matrices. First,

$$\partial_t Q_p = -\frac{d}{n} e_0, \quad B(t) e_0 = \frac{1}{n} e_0.$$

The first identity differentiates APU7, using $\partial_t q_p = -1$; the second uses the constant derivative of the leading linear quotient of f by $h - t$. Also reducing

$$(s - p)Q_p = f - f_p - \frac{(s - p)(h - t)}{n}$$

modulo the original polynomial $h - t$ proves

$$(A(t) - pI_d)Q_p = C(t)e_0 - f_p e_0. \quad (\text{APU10})$$

Consequently the upper-right curvature block equals

$$\frac{B_0 + \partial_t Q - BB_0}{u^2} + \frac{CB_0 + QP - AQ - B_0 F}{u^3} = 0.$$

The lower-right curvature is zero because $F_t = -P$; the polynomial block is flat by APU4–5. This proves the matrix curvature identity $\partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t] = 0$ with every extension column retained.

Weighted residues, oriented loops, and their exact horizontal maps. Preserve separately the two actual matrices

$$W_{\text{res}}(u, t) = \text{diag}(e^{f_p(t)/u}), \quad W_{\text{loop}}(u, t) = (2\pi i)W_{\text{res}}(u, t). \quad (\text{APU11})$$

The first is the matrix W in AP11: its rectangular map $\mathcal{R} = [0 \ W_{\text{res}}]$ is the weighted residue, with no $2\pi i$ in its definition. The second gives the positively oriented small-loop period map $\mathcal{L} = [0 \ W_{\text{loop}}] = (2\pi i)\mathcal{R}$. Direct differentiation gives

$$\partial_u W_{\text{res}} = -FW_{\text{res}}/u^2, \quad \partial_t W_{\text{res}} = -PW_{\text{res}}/u, \quad \partial_u \mathcal{R} = \mathcal{R}\Omega_u, \quad \partial_t \mathcal{R} = \mathcal{R}\Omega_t. \quad (\text{APU12})$$

The same equations hold for $\mathcal{L}$. Equivalently, for every parameter-dependent rational representative $a(s; u, t)$,

$$\partial_u \text{Res}_w[a] = \text{Res}_w[\nabla_u^1 a], \quad \partial_t \text{Res}_w[a] = \text{Res}_w[\nabla_t^1 a].$$

To prove these equalities, use $\partial_u w = -(f/u^2)w$ and $\partial_t w = -(s/u)w$ in its convergent Laurent expansion on a circle containing that fixed pole and no other pole. Parameter differentiation and the finite contour integral commute. These identities descend because both residue maps kill D -boundaries by AP1. Thus

AP5 is an exact sequence of the actual flat connections in both parameters: its polynomial inclusion is horizontal and its weighted-residue quotient has the ordinary trivial connection.

The explicit AP13 frame comparison remains available: replace only the pole columns by $r_p/w(p)$, through the displayed matrix $\text{diag}(I_d, W_{\text{res}}^{-1})$. In that frame their exact derivatives are

$$\nabla_u[r_p/w(p)] = -[Q_p]/(u^2 w(p)), \quad \nabla_t[r_p/w(p)] = -e_0/(uw(p)). \quad (\text{APU13})$$

The pole diagonal terms cancel against derivatives of the displayed inverse weights. The off-diagonal polynomial extension remains, with its full inverse exponential in both directions.

For the unchanged rapid-decay contours and positively oriented small loops of AP19–19b, their complete period matrix satisfies

$$\partial_u \Pi_{\text{ext}} = \Pi_{\text{ext}} \Omega_u, \quad \partial_t \Pi_{\text{ext}} = \Pi_{\text{ext}} \Omega_t. \quad (\text{APU14})$$

Here the contours can be held fixed locally in an admitted u -sector: smaller compact subsectors retain the same decaying ends; their finite detours avoid the fixed poles. The negative leading exponential dominates its polynomial derivatives and simple-pole factors uniformly on compact parameter subsets. This proves differentiation under the integral. The differential relations integrate to zero at the decaying ends or around the closed loops, so APU6–8 give the stated matrices. The lower period block is exactly W_{loop} , and the determinant remains the full AP19a expression

$$\det \Pi_{\text{ext}} = \det \Pi_{\text{pol}} (2\pi i)^r \exp \left(\frac{\sum_{p \in P_\nu} f_p(t)}{u} \right). \quad (\text{APU15})$$

In particular its u -derivative is fixed by $\partial_u \det \Pi_{\text{ext}} = (\text{Tr } \Omega_u) \det \Pi_{\text{ext}}$; the polynomial trace, every pole phase, and the $2\pi i$ factors retain their original values.

The full Taylor gauge and its dual commute with the new connection. The original cochain isomorphism multiplies both degrees by the complete polynomial ν , with inverse ν^{-1} , and has

$$D^\nu = \nu D \nu^{-1} = u \partial_s + h - t - u \nu' / \nu.$$

Because ν is fixed in both parameters, it commutes with ∇_u^j and ∇_t^j in APU3. Conjugating APU4–5 therefore proves the same chain and flatness identities for D^ν . In the actual transported frame

$$(\nu, \nu s, \dots, \nu s^{d-1}, (\nu/(s-p))_{p \in P_\nu})$$

the matrices are exactly APU9. The target weight is w/ν , and

$$(w/\nu)(\nu a) = wa, \quad (w/\nu) D^\nu a = u((w/\nu)a)'. \quad (\text{APU16})$$

These prove preservation of every old contour and every loop period, including all multiplicities of ν . If $\nu/(s-p)$ has no rational pole, its weighted form still equals $w/(s-p)$. The loop class and its original weighted pole are thereby retained.

For the ordinary linear dual, the two connections are $\partial_u - \Omega_u^T$ and $\partial_t - \Omega_t^T$. The injection of the weighted-residue dual is $\mathcal{R}^T$. Transposing APU12 gives

$$(\partial_u - \Omega_u^T)(\mathcal{R}^T a) = \mathcal{R}^T \partial_u a, \quad (\partial_t - \Omega_t^T)(\mathcal{R}^T a) = \mathcal{R}^T \partial_t a. \quad (\text{APU17})$$

Thus its full two-parameter horizontality is proved, with no positive pairing inferred from an algebraic dual. The dual of the Taylor gauge is the inverse transpose of its actual multiplication map, preserving the inverse unit.

The exact endpoint arrow to the original arithmetic source. The singular u -connection is specified on $u \neq 0$. Its comparison with the original marked endpoint uses the actual arrow already calculated in AP23–26. Since $\gcd(h, \nu) = 1$, each retained pole satisfies $h(p) \neq 0$. Set

$$R_p(s) = -\frac{h(s) - h(p)}{(s-p)h(p)}, \quad Q_0 = (R_p)_{p \in P_\nu}, \quad J_0 = [I_d \ Q_0]. \quad (\text{APU18})$$

The polynomial identity $(s-p)R_p = 1 - h(s)/h(p)$ proves that R_p is the exact image of r_p in $E_h = \mathbb{C}[s]/h$. Thus J_0 sends every pole coordinate and every polynomial coordinate to its original marked class, with all jets of h retained. It has the exact kernel $\{(-Q_0 c, c) : c \in \mathbb{C}^r\}$ and satisfies

$$J_0 A_{\text{ext}}(0) = A(0) J_0. \quad (\text{APU19})$$

These identities follow by multiplying the preceding polynomial inverse by s , exactly as in AP25. No value of the singular matrix Ω_u at $u = 0$ is assigned by this endpoint arrow.

For completeness its original theta-source representative is also explicit. Retain the original even Schwartz source V , rapid-decay target $\mathcal{B}$, theta map $\Theta\phi(x) = 2\sum_{a \geq 1} \phi(ax)$, Euler operator $\mathcal{D} = -x\partial_x$, and seed

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g.$$

The original Euler inverse $F_h \in \mathcal{B}$ satisfies $h(\mathcal{D})F_h = \Theta\phi_*$ and $\mathcal{M}F_h = g/h$. The polynomial chain map is the original

$$P \longmapsto P(\mathcal{D})\phi_*, \quad P ds \longmapsto P(\mathcal{D})F_h, \quad \Theta P(\mathcal{D})\phi_* = (hP)(\mathcal{D})F_h. \quad (\text{APU20})$$

Consequently the coefficient map

$$\Psi_h : E_h \longrightarrow Q_{\text{ar}} = \mathcal{B}/\Theta V, \quad [P] \longmapsto [P(\mathcal{D})F_h] \quad (\text{APU21})$$

is well defined: replacing P by $P + hT$ changes its source representative by $\Theta T(\mathcal{D})\phi_*$. The original Mellin-jet map sends it to $j_h(g/h)[P] = [\nu][P]$. Since $[\nu]$ is the full invertible Taylor class, this proves injectivity on the original packet. Equivalently, with the retained corrected section $\sigma_h = \Psi_h[\nu]^{-1}$, it is $\Psi_h = \sigma_h[\nu]$. The complete endpoint-source arrow is therefore

$$\mathbb{C}^{d+r} \xrightarrow{J_0} E_h \xrightarrow{[\nu]} E_h \xrightarrow{\sigma_h} Q_{\text{ar}}, \quad v \longmapsto [(J_0 v)(\mathcal{D})F_h]. \quad (\text{APU22})$$

All inverse Taylor coefficients are those of the original $[\nu]^{-1}$; they have not been replaced by a constant. Since multiplication by $[\nu]$ commutes with the original $A(0)$, APU19 intertwines the displayed endpoint action with this actual theta-source action. On the transported gauge fibre the coefficient is $[\nu]J_0 v$, whose image under σ_h is the same source vector. This is the precise unit and endpoint comparison for the original rational pole extension.

Backward propagation through the entire finite scaling transport. Retain AP15–19b’s exact transport, for every complex a ,

$$C_{\text{ext},a}(u, t) = \Pi_{\text{ext}}(u, t)^{-1} \Pi_{\text{ext}}(u, t + ua), \quad \partial_a C_{\text{ext},a} = -C_{\text{ext},a} A_{\text{ext}}(t + ua).$$

Its invertible diagonal blocks and full upper-right integral are the ones in AP15; this paragraph propagates APU9 through that already existing finite map. Put $t_a = t + ua$. At fixed a , the exact two parameter derivatives are

$$\boxed{\begin{aligned} \partial_u C_{\text{ext},a} &= -\Omega_u(u, t) C_{\text{ext},a} \\ &\quad + C_{\text{ext},a} (\Omega_u(u, t_a) + a\Omega_t(u, t_a)), \\ \partial_t C_{\text{ext},a} &= -\Omega_t(u, t) C_{\text{ext},a} + C_{\text{ext},a} \Omega_t(u, t_a). \end{aligned}} \quad (\text{APU23})$$

To prove every term, differentiation of $\Pi_{\text{ext}}^{-1} \Pi_{\text{ext}} = I$ and APU14 gives

$$\partial_u \Pi_{\text{ext}}^{-1} = -\Omega_u \Pi_{\text{ext}}^{-1}, \quad \partial_t \Pi_{\text{ext}}^{-1} = -\Omega_t \Pi_{\text{ext}}^{-1}.$$

The derivative of the shifted factor is, by the ordinary chain rule and APU14,

$$\partial_u (\Pi_{\text{ext}}(u, t + ua)) = \Pi_{\text{ext}}(u, t + ua) (\Omega_u(u, t + ua) + a\Omega_t(u, t + ua)),$$

whereas its t -derivative is $\Pi_{\text{ext}}(u, t + ua) \Omega_t(u, t + ua)$. Multiplying these inverse and shifted derivatives proves APU23. In particular the $a\Omega_t$ term is the derivative of the actual shifted parameter and cannot be omitted. Equivalently, under the precise base map $F_a : S \rightarrow S$, $F_a(u, t) = (u, t + ua)$, the pulled-back connection has coefficients $\Omega_u(u, t_a) + a\Omega_t(u, t_a)$ and $\Omega_t(u, t_a)$; the matrix $C_{\text{ext},a}$ is the horizontal map from that pullback to the unshifted coefficient bundle, by APU23.

The original weighted-residue identity AP18 uses the rectangular matrix $\mathcal{R} = [0 \ W_{\text{res}}]$:

$$\mathcal{R}(u, t) C_{\text{ext},a}(u, t) = \mathcal{R}(u, t + ua).$$

It remains horizontal in both parameters. In fact its left-hand u -derivative is, by APU12 and APU23,

$$\begin{aligned} \mathcal{R}(u, t) \Omega_u(u, t) C_{\text{ext},a} + \mathcal{R}(u, t) \partial_u C_{\text{ext},a} \\ = \mathcal{R}(u, t + ua) (\Omega_u(u, t + ua) + a\Omega_t(u, t + ua)), \end{aligned}$$

which is the exact u -derivative of the right side by APU12 and the same chain rule. The t -derivative is $\mathcal{R}(u, t + ua)\Omega_t(u, t + ua)$, by the two cancelling unshifted terms in the identical calculation. Multiplication by the constant $2\pi i$ proves these same compatibilities for the positively oriented loop-period map $\mathcal{L}$. Thus the finite scaling, its full extension integral, both parameter derivatives, and both residue conventions all remain in the same original commuting diagrams.

For constant ν , the set P_ν is empty and every additional block, residue coordinate and loop row in APU7–23 is the unique empty map; the polynomial connection remains unchanged. For a nonempty pole set, all coefficient-face maps act on these matrices slot by slot. Constant zero insertion commutes with both parameter derivatives and every displayed coefficient map; hence the independent original support labels and their represented zeros are retained by the entire two-parameter calculation.

9.0.7 5. Entire finite scaling and its retained extension term

Let $C_a(u, t)$ be the original polynomial scaling transport, satisfying

$$\frac{dC_a}{da} = -C_a A(t + ua), \quad C_0 = I_d. \quad (\text{AP14})$$

It is the previously proved entire solution, including its full higher terms. Define an integral along the straight segment from 0 to a in the complex a -plane (equivalently the primitive of its entire integrand):

$$\begin{aligned} K_a &= - \int_0^a C_b(u, t) B_0 \exp(-(a - b)P) db, \\ C_{\text{ext}, a} &= \begin{bmatrix} C_a & K_a \\ 0 & \exp(-aP) \end{bmatrix}. \end{aligned} \quad (\text{AP15})$$

Differentiating the integral gives $K'_a = -C_a B_0 - K_a P$ and $K_0 = 0$. Consequently

$$\frac{dC_{\text{ext}, a}}{da} = -C_{\text{ext}, a} A_{\text{ext}}(t + ua), \quad C_{\text{ext}, 0} = I_{d+r}. \quad (\text{AP16})$$

Existence and uniqueness for this finite polynomial-coefficient matrix ODE prove that these are the actual entire finite connection transports. The block integral retains the coupling into the original polynomial class. Both diagonal blocks are invertible, and the inverse has upper-right block $-C_a^{-1} K_a \exp(aP)$. The groupoid law is

$$C_{\text{ext}, a}(u, t) C_{\text{ext}, b}(u, t + ua) = C_{\text{ext}, a+b}(u, t). \quad (\text{AP17})$$

It follows either by the shifted ODE and uniqueness, or from the commuting operator's formal flow $\exp(aL)F = e^{-as}F(u, t + ua, s)$. The latter is coefficientwise in a when F is rational; no claim that $e^{-as}F$ is itself a rational function at a nonzero complex a is needed. The finite matrix ODE supplies its entire cohomological continuation.

The residue quotient has the exact transport identity

$$\begin{bmatrix} 0 & W(u, t) \end{bmatrix} C_{\text{ext}, a}(u, t) = \begin{bmatrix} 0 & W(u, t + ua) \end{bmatrix}, \quad (\text{AP18})$$

because $w(p; u, t + ua) = w(p; u, t)e^{-ap}$. In weighted residues the entire action is parameter translation; in the unchanged simple-pole frame the factor e^{-ap} remains. The exponentials and their polynomial extension K_a are both displayed, including the case when p is not real.

9.0.8 6. Small-loop periods, the actual unit gauge and dual maps

For a positively oriented small circle γ_p enclosing only p , the exact additional period is

$$\int_{\gamma_p} wF ds = 2\pi i (\text{Res}_w F)_p. \quad (\text{AP19})$$

Thus the extra periods are concrete loops, with their orientation and $2\pi i$ factor retained. Every original polynomial image has zero such loop period. For any original rapid-decay contour avoiding P_ν , the incoming polynomial period is still its old integral, with its original ray orientation and phase.

Choose the finite parts of the original d rapid-decay contours to avoid every point of P_ν , including zero when zero belongs to that set. Such detours leave their polynomial integrals unchanged, since those integrands are entire. Different detours for a rational integrand can add the small loops AP19; those are

retained rows, not discarded contour ambiguities. At infinity w has the same decay on the specified rays as in the original polynomial period proof, and the simple-pole functions are $O(1/s)$, so all new integrals converge. On compact parameter subsets in an admitted u -sector and bounded t , the negative leading exponential dominates the lower terms and justifies differentiation.

In the unchanged AP6 frame these d contours followed by the r small loops give the full period matrix

$$\begin{aligned}\Pi_{\text{ext}} &= \begin{bmatrix} \Pi_{\text{pol}} & V_{\text{pole}} \\ 0 & (2\pi i)W \end{bmatrix}, \\ \det \Pi_{\text{ext}} &= \det \Pi_{\text{pol}} (2\pi i)^r \prod_p w(p).\end{aligned}\tag{AP19a}$$

Π_{pol} is the original full polynomial period matrix, whose nonzero determinant was proved by the monomial ray matrix and coefficient-connection transport. Hence AP19a is invertible: it realizes all $d+r$ analytic classes and all their loop periods. Integrals of D -exact forms vanish at the decaying endpoints or on closed loops. Differentiating with t therefore gives

$$\begin{aligned}\frac{d\Pi_{\text{ext}}}{dt} &= -\Pi_{\text{ext}} A_{\text{ext}}(t)/u, \\ C_{\text{ext},a} &= \Pi_{\text{ext}}(u,t)^{-1} \Pi_{\text{ext}}(u,t+ua).\end{aligned}\tag{AP19b}$$

The second identity follows from the same ODE and initial value as AP16, for fixed nonzero u in the admitted sector. No period matrix is evaluated at $u=0$.

On the same localized complex define the actual gauge differential

$$D^\nu = \nu D \nu^{-1} = u \frac{d}{ds} + h - t - u \frac{\nu'}{\nu}.\tag{AP20}$$

The cochain map is multiplication by ν in both degrees. Its inverse is multiplication by ν^{-1} in B_{loc} . The target weight is w/ν , and the identities

$$\begin{aligned}(w/\nu) D^\nu F &= u((w/\nu)F)', \\ (w/\nu)(\nu F) &= wF\end{aligned}\tag{AP21}$$

show exact equality of every small-loop and every old rapid-decay period under this map. The target extra class $\nu/(s-p)$ need not have a pole as a rational function; its weighted form $(w/\nu)\nu/(s-p) = w/(s-p)$ retains the pole. The original image is ν times H_{pol} , not the unmodified polynomial frame.

The ordinary algebraic dual of AP5 is the exact sequence

$$0 \longrightarrow (\mathbb{C}^{P_\nu})^* \xrightarrow{\text{Res}_w^T} H_{\text{loc}}^* \xrightarrow{i^T} H_{\text{pol}}^* \longrightarrow 0.\tag{AP22}$$

In AP6 the injected column for α is $(0, W^T \alpha)$. The dual connection is $\frac{d}{dt} + A_{\text{ext}}(t)^T/u$. Its action on that column is $(0, W^T \alpha')$: the derivative $-PW/u$ cancels PW/u , and the upper block is zero. Thus the loop-functional injection is explicitly horizontal for the trivial weighted-residue dual. Under the gauge the dual is the inverse transpose of multiplication by ν . This statement concerns the ordinary linear dual; the previous τ -base dagger and moment-line dual retain their separate, already calculated conjugation and weight-line maps. No positive pairing is inferred from AP22, and no original inverse Taylor unit is replaced by ν on a dual.

9.0.9 7. Exact relation to the formal contraction, including the endpoint

UG uses $\mathbb{C}[s][[u, t]]$ and $\mathbb{C}[s, \nu^{-1}][[u, t]]$, with coefficientwise completion. On the added pole module h is invertible because $\gcd(h, \nu) = 1$. Writing H for this inverse and $E = u \frac{d}{ds} - t$, its exact formal inverse is the ordered sum $\sum_{n \geq 0} (-HE)^n H$. That is the contraction proved in UG. AP3 instead fixes $u \neq 0$ first and calculates a cokernel of dimension r , nonzero when $r > 0$. For constant ν , $r = 0$ and the additional pole complex and its residue map are zero; localization adds no analytic or formal classes in that case.

The two operations are related by the original rational complexes and the localization arrow, but the formal completion has no evaluation homomorphism at an arbitrary nonzero complex u . A formal coefficient sequence may diverge there. Thus the previously displayed formal contraction is not evaluated to claim that AP3's residue classes vanish. AP3 gives exactly the missing cokernel.

There is a concrete endpoint calculation. For p in P_ν , $\gcd(h, \nu) = 1$ gives $h(p) \neq 0$. At $u = t = 0$ the class of $1/(s-p)$ modulo h is the original polynomial class

$$\left[\frac{1}{s-p} \right] = - \left[\frac{h(s) - h(p)}{(s-p)h(p)} \right] \quad \text{in } \mathbb{C}[s]/(h). \quad (\text{AP23})$$

Indeed multiplying the displayed polynomial by $s-p$ gives $1 - h(s)/h(p)$. This is the literal inverse of $s-p$ in the original algebra, with every jet of h retained. For formal u, t before the endpoint, $D(1/(s-p))$ gives

$$(h(p) - t) \left[\frac{1}{s-p} \right] = u \left[\frac{1}{(s-p)^2} \right] - \left[\frac{h(s) - h(p)}{s-p} \right]. \quad (\text{AP24})$$

The successive higher-pole relations AP4 and ordered formal inverse supply the remaining u, t coefficients. AP23 is not a rank- $(d+r)$ basis at $u=0$. It explicitly describes how these rational lifts are represented in the original rank- d formal fibre.

The weighted residue factors also retain their parameter behavior: $w(p) = \exp((\Phi(p) - tp)/u)$. At fixed t with $\Phi(p) - tp \neq 0$ this has an essential singularity at $u=0$. If that numerator vanishes, that particular value is one; its exact derivative is always $u d_t w(p) = -p w(p)$. For $p=0$, $w(0)$ is independent of t : it is identically one when $\Phi(0)=0$, and is $\exp(\Phi(0)/u)$ otherwise. Thus no universal singularity or t -dependence is asserted for every residue coordinate. Even when $w(p)=1$, AP4 on higher pole orders uses $w^{(n)}(p)$, whose highest inverse- u term is $w(p)(h(p) - t)^n/u^n$. Because $h(p) \neq 0$, these contain unbounded inverse- u powers as the pole order increases near $t=0$. Consequently when $r > 0$, AP2 does not define a coefficientwise continuous residue map on UG's completed pole module. For $r=0$ its zero map is of course continuous. The obstruction to evaluating the nonzero-pole formal contraction is not replaced by a false assertion that every individual loop factor is singular.

There is also an exact map for the entire displayed endpoint matrix. Let Q_0 have p -th column the polynomial in AP23, expressed in the original degree- d frame, and put $J_0 = [I_d \ Q_0]$. Direct multiplication of AP23 gives

$$\begin{aligned} A Q_0 &= B_0 + Q_0 P, \\ J_0 A_{\text{ext}}(0) &= A J_0, \\ \ker J_0 &= \{(-Q_0 c, c) : c \in \mathbb{C}^r\}. \end{aligned} \quad (\text{AP25})$$

The map $c \mapsto (-Q_0 c, c)$ carries P to $A_{\text{ext}}(0)$, by the first identity. Thus the endpoint has an explicit invariant r -dimensional kernel and the original arithmetic quotient, rather than an unexplained rank comparison. Entire matrix exponentiation yields

$$J_0 C_{\text{ext},a}(0,0) = \exp(-aA) J_0. \quad (\text{AP26})$$

Here $C_{\text{ext},a}(0,0) = \exp(-aA_{\text{ext}}(0))$ is the matrix specialization of AP16; J_0 then maps it to actual formal-fibre cohomology. Multiplication by the full unit v_h after J_0 gives the gauge endpoint. It commutes with A since that unit is an element of the original algebra, so the same action diagram holds. The endpoint comparison therefore retains the full unit as well as all jets.

Completed scope: the original arithmetic polynomial cohomology injects into the analytic localization. The complete two-parameter connection, weighted-residue and oriented-loop maps, finite-scaling pullback including its shifted endpoint derivative, full Taylor gauge, dual and original theta-source endpoint have now been proved in APU1–23. All original additional pole classes and their finite-scaling extension integral remain in the same maps. The formal and analytic constructions retain their exact common maps and different coefficient operations. This does not supply a uniform θ -norm bound or an arithmetic Frobenius comparison not constructed in these equations.

9.0.10 Provenance and verification scope

Input proof: ACTUAL_TAYLOR_UNIT_FORMAL_GAUGE.md, SHA256 9ded2486160b76bf65f92d44b73cdd0c59a3ca7e5334f88682e10592d5a45b2f. Original scaling proof after the accepted integer-power-map clarification: ../f1_scaling_frobenius/ACTUAL_TAU_SCALING_AND_EXPONENTIAL_FLOW.md, SHA256 4e4cb66e3cbbff6379ae389ddaaad65045ac0ed380f17a87ac7c4241c45e6c08. The accompanying derivation supplied AP3's local-primitive construction and AP8–AP13's exact connection in the preceding calculation. This note supplies the full written continuation, AP15–AP18's entire extension transport, AP22's explicit dual and AP23–AP24's endpoint comparison. The independent reviewer supplied the determinant extension AP19a and the endpoint intertwiner AP25–AP26; the text incorporates their full proofs and includes the review's corrected $p=0$ residue-factor exception explicitly. No new Lean execution or certified arithmetic-zero computation is claimed.

10 Original cyclic coefficients and the tensor boundary maps

10.1 The cyclic exponent with its retained coefficient map

The characteristic-zero nilpotent exponent in the cyclic-sum calculation has the following full coefficient form. This gives the original map before specialization, its saturation, its precise cokernel, and the specialized map. In particular the characteristic-zero injection and the specialized injection are no longer treated as the same assertion.

Let ρ be the retained primary centre, let $m \geq 1$ be its complete order, let $k \geq 1$, and put $K = k(m-1)$. Retain a coefficient subring $A \subset \mathbb{C}$ containing ρ , its already specified unital map $\alpha : A \rightarrow k_0$ to a finite field of characteristic $p > m$, and every coefficient of the original full unit and its inverse. The finite-field phase used below requires the stronger $p > m+1$. Since the image of α is a finite subring of a field, multiplication by any nonzero element of that image is an injective self-map of a finite set, hence a surjective map. The image is therefore a field and $\mathfrak{q} = \ker \alpha$ is maximal. Put $\mathcal{S} = A_{\mathfrak{q}}$. The extension $\tilde{\alpha} : \mathcal{S} \rightarrow k_0$ sends a/b to $\alpha(a)/\alpha(b)$; cross multiplication proves independence of the chosen fraction. The ring $\mathcal{S}$ is a characteristic-zero domain, p is a nonzero nonunit, and every integer prime to p is a unit. These statements use this actual coefficient map; they make no discrete-valuation or principal-ideal assumption on $\mathcal{S}$.

The coefficient datum and the phase datum have separate explicit domains. If A_{coef} carries the original centre and the full unit and inverse, the coefficient maps below are defined over A_{coef} . Adjoining the original phase requires the map $A_{\text{coef}} \rightarrow A_{\text{coef}}[1/(m+1)]$; a source presentation which also retains $1/(m+1)!$ uses that larger, explicitly given localization. A coefficient map extends to either localization exactly when its inverted elements have nonzero images: the extension must send each inverse to the inverse of its image, and that formula proves existence and uniqueness in the nonzero case. Thus the factorial calculation with $p > m$ does not itself construct a phase specialization. For $m=1, p=2$, its coefficient map, when given on A_{coef} , sends the one-dimensional cyclic image by multiplication by the retained unit $\bar{a}_0^k$, which is nonzero; $K=0$ and the nilpotent has index one. This map has no extension to $A_{\text{coef}}[1/2]$, since the identity $2(1/2) = 1$ would acquire images $0 = 1$. Every finite-field phase statement below uses its actual stronger datum $p > m+1$, with all other retained denominator exclusions.

The original objects and the two explicit coordinate substitutions are

$$\begin{aligned} E_{\mathcal{S}} &= \mathcal{S}[s_1, \dots, s_k] / ((s_i - \rho)^m)_{i=1}^k \simeq \mathcal{S}[y_1, \dots, y_k] / (y_i^m)_{i=1}^k, & y_i &= s_i - \rho, \\ J &= \sum_i y_i, & M &= \sum_i s_i = k\rho + J, \\ \mathcal{C}_{\mathcal{S}}^{\text{orig}} &= \mathcal{S}[Z] / ((Z - k\rho)^{K+1}) \simeq \mathcal{C}_{\mathcal{S}} = \mathcal{S}[T] / (T^{K+1}), & T &= Z - k\rho. \end{aligned} \tag{BPC.1}$$

The inverse substitutions are $s_i = y_i + \rho$ and $Z = T + k\rho$; both respect the displayed ideals and their composites fix every generator. Successive division by the monic relations y_i^m proves that the ordered monomials $y^{\mathbf{b}}$, $0 \leq b_i < m$, are an $\mathcal{S}$ -basis.

Write $h = (s - \rho)^m$, $g = 2\xi$, and retain

$$\begin{aligned} v(y) &= \sum_{j=0}^{m-1} a_j y^j, & a_j &= \frac{(g/h)^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!}, \\ w(y) &= \sum_{j=0}^{m-1} b_j y^j, & b_0 &= a_0^{-1}, & b_r &= -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}, \\ U &= \prod_i v(y_i), & U^{-1} &= \prod_i w(y_i). \end{aligned} \tag{BPC.2}$$

All these coefficients belong to the specified A . Expanding $g = (s - \rho)^m (g/h)$ proves the derivative identity. The recurrence makes the constant coefficient of vw equal to one and all its positive coefficients of degree below m zero; thus it proves the stated inverse. Multiplication by U is an

actual $\mathcal{S}$ -linear automorphism of $E_{\mathcal{S}}$, commuting with J and M . The same proof applies to the full unit induced by a retained quartet packet on this primary factor, using that unit's original coefficients throughout.

The unchanged cyclic ring map and the actual unit-weighted module map are

$$f : \mathcal{C}_{\mathcal{S}} \longrightarrow E_{\mathcal{S}}, \quad T^r \longmapsto J^r, \quad f_U : \mathcal{C}_{\mathcal{S}} \longrightarrow E_{\mathcal{S}}, \quad T^r \longmapsto UJ^r, \quad f_U^{\text{orig}}(Z^j) = UM^j. \quad (\text{BPC.3})$$

All surviving degrees in $E_{\mathcal{S}}$ are at most K , so $J^{K+1} = 0$ and these maps are well defined. The weighted map is a map of modules for $T \mapsto J$, or for $Z \mapsto M$ in the original coordinates: $f_U(Tc) = Jf_U(c)$ and $f_U^{\text{orig}}(Zc) = Mf_U^{\text{orig}}(c)$. Its value at one is the actual U . These formulas also give its type when $U \neq 1$; they do not require it to be a unital ring homomorphism.

For $0 \leq r \leq K$ set

$$V_r = \sum_{\substack{0 \leq b_i < m \\ \sum_i b_i = r}} \frac{y^{\mathbf{b}}}{\prod_i b_i!}, \quad W_r = UV_r, \quad V_{K+1} = W_{K+1} = 0. \quad (\text{BPC.4})$$

Every denominator here is a unit because $b_i < m < p$. For each $r \leq K$, a tuple of this degree exists: fill the slots successively up to capacity $m - 1$. Select that tuple $\mathbf{b}^*(r)$ and put $\lambda_r(x) = \mathbf{b}^*(r)! [y^{\mathbf{b}^*(r)}]x$. Then $\lambda_r(V_t) = \delta_{rt}$. The coefficient projection $\pi(x) = \sum_r \lambda_r(x) V_r$ deletes every selected pivot coordinate from $x - \pi(x)$. Consequently, with F the span of all unselected monomials, the maps just given prove

$$E_{\mathcal{S}} = \Lambda \oplus F, \quad \Lambda = \bigoplus_{r=0}^K \mathcal{S} V_r, \quad E_{\mathcal{S}} = \Lambda_U \oplus UF, \quad \Lambda_U = U\Lambda = \bigoplus_{r=0}^K \mathcal{S} W_r, \quad \pi_U = U\pi U^{-1}. \quad (\text{BPC.5})$$

The sum defining each V_r is retained in full; its selected pivot only supplies this explicit splitting.

The multinomial expansion and a direct coefficient calculation give

$$J^r = r! V_r, \quad JV_r = (r+1) V_{r+1}, \quad f_U(T^r) = r! W_r, \quad JW_r = (r+1) W_{r+1}. \quad (\text{BPC.6})$$

For the first formula the coefficient of $y^{\mathbf{b}}$ is the number $r!/\mathbf{b}!$ of its ordered assignments among the r factors. Terms exceeding a primary order vanish in the defining quotient. For the second formula the coefficient at degree $r+1$ is $\sum_{i: b_i > 0} b_i/\mathbf{b}! = (r+1)/\mathbf{b}!$. The last two formulas follow by multiplying by the retained U . At degree K there is exactly one tuple, so

$$J^K = \frac{K!}{((m-1)!)^k} \prod_i y_i^{m-1} \neq 0, \quad J^{K+1} = 0, \quad f_U(T^K) = \frac{K!}{((m-1)!)^k} a_0^k \prod_i y_i^{m-1}. \quad (\text{BPC.7})$$

Its coefficient is nonzero in the characteristic-zero domain $\mathcal{S}$. Every positive Taylor term of U kills the top monomial, which proves the last equality with its full original scalar. Thus the old characteristic-zero exponent is exactly $K+1 = 1 + k(m-1)$, and f_U is injective: its columns are the independent W_r multiplied by the nonzero $r!$. This includes $m=1$: then $K=0$, the zeroth power is the identity and the zero nilpotent has index one.

In the original coordinates the same map has matrix

$$[f_U^{\text{orig}}]_{(s^{\mathbf{b}}), (Z^j)} = P_k^{-1} [U]_{(y^{\mathbf{b}})} \mathbf{V} \Delta B, \quad \Delta = \text{diag}(0!, 1!, \dots, K!), \quad (\text{BPC.8})$$

where $P_k = P^{\otimes k}$, $P_{ab} = \binom{b}{a} \rho^{b-a}$ for $a \leq b$ and zero otherwise, $B_{rj} = \binom{j}{r} (k\rho)^{j-r}$ for $r \leq j$ and zero otherwise, and $\mathbf{V}_{\mathbf{b}r} = 1/\mathbf{b}!$ for $|\mathbf{b}| = r$, zero otherwise. The inverses of P and B replace ρ and $k\rho$ by their negatives, as follows by the inverse substitutions in (BPC.1). The j -th column on the right is the original polynomial $U \sum_r \binom{j}{r} (k\rho)^{j-r} r! V_r = UM^j$, proving the formula and retaining every original coordinate.

Write $L_U = f_U(\mathcal{C}_{\mathcal{S}})$ and $d_r = v_p(r!) = \sum_{a \geq 1} \lfloor r/p^a \rfloor$. Counting, for every integer from one to r , each of its factors of p proves the sum formula. The complementary factor $\varepsilon_r = r!/p^{d_r}$ is prime to p , so is a unit in $\mathcal{S}$. We obtain the actual inclusion and quotient

$$L_U = \bigoplus_{r=0}^K r! \mathcal{S} W_r \subset \Lambda_U, \quad C_U := \Lambda_U / L_U \simeq \bigoplus_{r=0}^K \mathcal{S} / (p^{d_r}), \quad \left[\sum_r c_r W_r \right] \mapsto (c_r \bmod p^{d_r})_r. \quad (\text{BPC.9})$$

The kernel and surjectivity follow coordinate by coordinate from the displayed basis; the convention $\mathcal{S}/(p^0) = 0$ is used. The extension of these quotient coordinates to $E_{\mathcal{S}}$ is $x \mapsto \lambda_r(U^{-1}x) \bmod p^{d_r}$, and its kernel also retains the complementary summand UF . In particular $E_{\mathcal{S}}/L_U \simeq UF \oplus C_U$. Over $Q = \text{Frac } \mathcal{S}$ every $r!$ is invertible. The splitting (BPC.5) therefore proves the precise saturation

$$(L_U \otimes_{\mathcal{S}} Q) \cap E_{\mathcal{S}} = \Lambda_U \quad \text{inside } E_{\mathcal{S}} \otimes_{\mathcal{S}} Q. \quad (\text{BPC.10})$$

Indeed an integral element in the rational span of Λ_U has zero UF -component, since that free module injects into its scalar extension. This proves both inclusions without dividing an original vector by its factorial in $E_{\mathcal{S}}$.

Now take either $B = \mathcal{S}/p\mathcal{S}$ or the specified finite-field target $B = k_0$, using the actual coefficient map, and put a bar on each base change. The splitting and its inverse in (BPC.5) base change because all their coefficients lie in $\mathcal{S}$. Thus every $\overline{W}_r$ remains a distinct basis vector of a direct summand. The exact specialized map is

$$\overline{f}_U(\overline{T}^r) = \begin{cases} r! \overline{W}_r, & r < p, \\ 0, & r \geq p, \end{cases} \quad \ker \overline{f}_U = (\overline{T}^p) = \bigoplus_{r=p}^K B \overline{T}^r, \quad \text{im } \overline{f}_U = \bigoplus_{r=0}^{\min(K, p-1)} B \overline{W}_r. \quad (\text{BPC.11})$$

The direct sum is empty when $K < p$. The factorial is a unit for $r < p$, and zero for $r \geq p$; the independent basis proves every claim. The unchanged abstract source $B[T]/T^{K+1}$ consequently has nilpotency index $K+1$, whereas multiplication by $\overline{J}$ on its actual image, on $\overline{\Lambda}_U$, and on $\overline{E}_{\mathcal{S}}$ has index

$$\min(p, K+1). \quad (\text{BPC.12})$$

For the upper bound, in characteristic p the intermediate binomial coefficients vanish, so $\overline{J}^p = \sum_i \overline{y}_i^p = 0$, while $\overline{J}^{K+1} = 0$ still holds. For the lower bound put $d = \min(p-1, K)$. Then $\overline{J}^d \overline{W}_0 = \overline{d!} \overline{W}_d \neq 0$. This vector is in the image, which proves all three lower bounds. All additional chains of the containing lattice are retained by

$$e_{j,t} = \left(\prod_{a=1}^t \overline{j p + a} \right) \overline{W}_{j p + t}, \quad 0 \leq j p \leq K, \quad 0 \leq t \leq \min(p-1, K - j p). \quad (\text{BPC.13})$$

Each coefficient here is a unit. Equation (BPC.6) makes $\overline{J} e_{j,t} = e_{j,t+1}$ until the last position; the next coefficient is zero modulo p , or the next degree is $K+1$. These are all the reduced lattice chains.

The coefficient loss has the exact homological map

$$\begin{aligned} 0 &\longrightarrow \mathcal{C}_{\mathcal{S}} \xrightarrow{f_U} \Lambda_U \longrightarrow C_U \longrightarrow 0, \\ 0 &\longrightarrow \text{Tor}_1^{\mathcal{S}}(C_U, B) \xrightarrow{\partial} \overline{\mathcal{C}}_{\mathcal{S}} \xrightarrow{\overline{f}_U} \overline{\Lambda}_U \longrightarrow C_U \otimes_{\mathcal{S}} B \longrightarrow 0, \\ \text{Tor}_1^{\mathcal{S}}(C_U, B) &= \bigoplus_{r=p}^K B \epsilon_r, \quad \partial \epsilon_r = \overline{T}^r, \quad C_U \otimes_{\mathcal{S}} B = \bigoplus_{r=p}^K B[\overline{W}_r]. \end{aligned} \quad (\text{BPC.14})$$

The first line is an explicit free resolution whose matrix is Δ in (BPC.8). Tensoring that complex with B gives the diagonal matrix $\overline{\Delta}$. Its kernel and cokernel are exactly (BPC.11), proving the last

two lines and the sign of ∂ , which is the inclusion into the degree-one free module for differential $+f_U$. When $K \geq p$, multiplication by $\bar{T}$ on this kernel has index $K - p + 1$: its $(K - p)$ -th power sends $\bar{T}^p$ to the nonzero $\bar{T}^K$. The original source, actual image, containing lattice, and connecting image are thus all present with their proved maps.

10.2 The same cyclic map carries the tensor boundary projector

Put $N = m + 1$ and retain the primitive fixed at the original point $s = 0$:

$$\Phi(s) = \frac{(s - \rho)^N - (-\rho)^N}{N}, \quad c = -\frac{(-\rho)^N}{N}, \quad \sum_i \Phi(s_i) = \sum_i \frac{y_i^N}{N} + kc. \quad (\text{BTP.1})$$

The ordered tensor complex has the commuting differentials $u\partial_{s_i} + h(s_i)$ and the original form $ds_1 \wedge \cdots \wedge ds_k$. The one-factor monic remainder decomposition, tensored with its explicit contracting homotopy, concentrates its cohomology in degree k , with the free basis $s^{\mathbf{b}} ds_1 \wedge \cdots \wedge ds_k$, $0 \leq b_i < m$. No permutation quotient is taken.

For $\zeta = e^{2\pi i/N}$ and the fixed branch of $u^{1/N}$, keep

$$\begin{aligned} \beta_b &= (b + 1)/N, & d_b &= N^{\beta_b - 1} e^{\pi i \beta_b} \Gamma(\beta_b), \\ W_{jb} &= \zeta^{j(b+1)} - 1, & D(u) &= \text{diag}(d_b u^{\beta_b}), \\ \Pi_k(u) &= e^{kc/u} W^{\otimes k} D(u)^{\otimes k} P_k. \end{aligned} \quad (\text{BTP.2})$$

Here $1 \leq j \leq m$, $0 \leq b < m$, and the rows are the ordered products of the original relative contours $\Gamma_j = -\ell_0 + \ell_j$, whose outward rays have angles $(\arg u + \pi)/N + 2\pi j/N$. The one-variable substitution $t = r^N/(N|u|)$ gives precisely $d_b u^{\beta_b}$ on the zeroth ray and its factor $\zeta^{j(b+1)}$ on the j -th ray. Their oriented difference gives W_{jb} . Translation to $y = s - \rho$ is the original one-factor contour equality: the remote connector integral tends to zero by the bound $\exp(-R^N/(N|u|) + O(R^{N-1}))$ times a polynomial. It leaves the factor $e^{c/u}$ intact. Absolute integrability permits Fubini on the ordered product contours, and the binomial substitution gives P_k , proving (BTP.2). The polynomial $\sum_{j=1}^{N-1} a_j(z^j - 1)$ cannot vanish at all N -th roots unless every coefficient is zero; this proves W invertible. Also $d_b \neq 0$, so every $\Pi_k(u)$ is invertible for $u \neq 0$.

In y -coordinates let

$$E_r y^{\mathbf{b}} = \mathbf{1}_{\{|\mathbf{b}|+k \equiv r \pmod{N}\}} y^{\mathbf{b}}, \quad Q_r = P_k^{-1} E_r P_k, \quad \hat{Q}_r = U_k Q_r U_k^{-1}, \quad U_k = [U]_{(s^{\mathbf{b}})}. \quad (\text{BTP.3})$$

These diagonal zero-one masks exist over the exact retained coefficient ring. The period monodromy and its projectors are

$$M_k = W^{\otimes k} \text{diag}(\zeta^{|\mathbf{b}|+k})(W^{\otimes k})^{-1}, \quad \mathcal{P}_r = \frac{1}{N} \sum_{j=0}^{N-1} \zeta^{-rj} M_k^j, \quad \mathcal{P}_r \Pi_k = \Pi_k Q_r. \quad (\text{BTP.4})$$

One positive circuit multiplies the $\mathbf{b}$ -th diagonal entry in (BTP.2) by $\zeta^{|\mathbf{b}|+k}$; its scalar exponential is single-valued. This proves the monodromy formula. The finite geometric sum is one at matching character and zero otherwise, proving the projectors and their intertwining identity. The horizontal coefficient convention has the inverse monodromy, with the same invariant index set.

The source-period map is exactly $\Pi_k U_k^{-1}$. Indeed the original source has Mellin transform $f(s_1, \dots, s_k) \prod_i (g/h)(s_i)$, whose full jet is $U_k[f]_s$; integration by parts for $D_i = -x_i \partial_{x_i}$ gives multiplication by s_i with vanishing endpoint terms in the original source space. It follows that

$$\mathcal{P}_r \Pi_k U_k^{-1} = \Pi_k U_k^{-1} \hat{Q}_r. \quad (\text{BTP.5})$$

This is the exact source comparison, including the entire unit and its inverse. It does not replace the source norm by the auxiliary period coordinate norm.

The invariant dimension is

$$d_{0,k} = \frac{m^k + m(-1)^k}{m+1}. \quad (\text{BTP.6})$$

To prove it, filter the ordered tuples by their total character: $N^{-1} \sum_{j=0}^{N-1} (\sum_{a=1}^{N-1} \zeta^{ja})^k$. The inner sum is m for $j=0$ and -1 otherwise, giving the formula. Thus the one-factor vanishing remains true at $k=1$; tensor invariants appear for every $m \geq 2, k \geq 2$, and for $m=1$ precisely when k is even. These are the ordinary local-system invariants at the original puncture.

The original arithmetic matrix is $A_k = k\rho I + R_k$, where $R_k = P_k^{-1}[J]P_k$. Each surviving multiplication by y_i increases the boundary character by one. Therefore, on every monomial,

$$\begin{aligned} Q_r R_k^j &= R_k^j Q_{r-j}, & [Q_0, A_k] &= R_k(Q_{-1} - Q_0), \\ Q_0 A_k Q_0 &= k\rho Q_0, & (I - Q_0) A_k Q_0 &= R_k Q_0. \end{aligned} \quad (\text{BTP.7})$$

Terms that exceed a primary order are zero on both sides. Conjugation by the commuting U_k gives the identical formulas on the actual source invariant plane. In particular the scalar compression retains the explicit outgoing map.

For $\text{im } Q_0 \neq 0$, set $d_{\min} = N \lceil k/N \rceil - k$ and $q_* = \lfloor (K - d_{\min})/N \rfloor$. Put $R_0 = Q_0 R_k^N Q_0|_{\text{im } Q_0}$. Then the complete dilation on this plane is

$$Q_0 a^{A_k} Q_0|_{\text{im } Q_0} = a^{k\rho} \sum_{j=0}^{q_*} \frac{(\log a)^{jN}}{(jN)!} R_0^j, \quad R_0^{q_*} \neq 0, \quad R_0^{q_*+1} = 0. \quad (\text{BTP.8})$$

Expand the original finite exponential through degree K . Equation (BTP.7) removes exactly the powers whose degree is not divisible by N . Every invariant input has degree at least $d_{\min}$, proving the upper cutoff. At that minimum degree, distribute any increment $jN \leq K - d_{\min}$ among the remaining slot capacities. Its multinomial coefficient is positive and distinct monomials cannot cancel, proving the stated nonzero powers. The source compression is the conjugate of (BTP.8) by the full U_k .

The original terminal monomial $\tau_k = \prod_i y_i^{m-1}$ satisfies

$$A_k \tau_k = k\rho \tau_k, \quad U_k \tau_k = a_0^k \tau_k, \quad Q_0 \tau_k = \hat{Q}_0 \tau_k = \mathbf{1}_{\{N|k\}} \tau_k. \quad (\text{BTP.9})$$

Multiplication by every y_i kills this top monomial, proving the first two formulas. Its character is $km \equiv -k \pmod{N}$, proving the third, including the unit conjugation. Its exact period is

$$\Pi_k \tau_k = e^{kc/u} \left(N^{-1/N} e^{\pi i m/N} \Gamma(m/N) \right)^k u^{km/N} (\zeta^{-j} - 1)_{j=1}^m \otimes k. \quad (\text{BTP.10})$$

This is its column in (BTP.2); all its factors are nonzero. The corresponding actual corrected-section class has observation $a_0^{-k} \Pi_k \tau_k$. For the original amplified quartet vector the retained coefficient is $C_{h_0,k} = K!((m-1)!)^{-k}((g/h_0)(\rho))^k$, so its observation is exactly $C_{h_0,k} a_0^{-k} \Pi_k \tau_k$. Multiplication by (g/h_0) on the terminal monomial has this scalar because all its positive Taylor terms vanish; this is also (BPC.7) for that retained unit. Therefore, on the infinite subsequence $N \mid k$, the same original terminal arithmetic eigenclass lies in the boundary invariant plane. Its original dilation still satisfies

$$\log \left(\frac{|a^{k\rho}|^2}{a^k} \right) = 2k(\text{Re } \rho - 1/2) \log a. \quad (\text{BTP.11})$$

Taking the real part of $k\rho \log a$ proves the identity. No invariant-plane projection has altered this arithmetic scalar.

10.3 Finite-field boundary and the actual coefficient-specialization restriction

Use the already specified finite-field map with $p > N$, and write $k_0 = \mathbb{F}_{Q_0}$, $L = \mathbb{F}_{Q_0^d}$, $d = \text{ord}_{(\mathbb{Z}/N\mathbb{Z})^\times}(Q_0)$. Thus L contains μ_N . Retain $\psi = \psi_0 \circ \text{Tr}_{L/k_0}$, geometric Frobenius, the original compact-support one-factor sheaf, and every one-factor Gauss constant. The already proved SPF decomposition, distributed over the ordered tensor factors on the common u -line, gives the exact isomorphism

$$\mathcal{F}_L^{\otimes k} \simeq \mathcal{L}_\psi(kc/u) \otimes \bigoplus_{\substack{\chi = (\chi_1, \dots, \chi_k) \\ \chi_i^N = 1, \chi_i \neq 1}} \mathcal{A}_{A_\chi} \otimes \mathcal{K}_{\prod_i \chi_i}(u), \quad A_\chi = \prod_i \left(-\chi_i(N) \sum_{z \in L^\times} \chi_i(z) \psi(z) \right). \quad (\text{BTF.1})$$

The torsor tensor laws add the Artin–Schreier phases and multiply the Kummer characters. Distribution of tensor over direct sum retains every ordered tuple, even if different tuples have equal product characters. These facts prove the displayed map from the established one-factor map. The compact-support Kunneth map places it in degree k on the original ordered product source, since each one-factor complex is concentrated in degree one. No factor permutation changes its sign.

For $c \neq 0$, choose the original additive character $\psi_0(x) = \vartheta(\text{Tr}_{k_0/\mathbb{F}_p}(bx))$. The cover $u = v^N$, $z^p - z = bc/v^N$ has deck group $\mu_N \times \mathbb{F}_p$. Its degree is Np : the second right side has pole order N prime to p , whereas a pole of $f^p - f$ has order divisible by p , so it is a nontrivial degree- p extension. The displayed commuting deck maps $(v, z) \mapsto (\zeta v, z + a)$ then exhaust that degree. On the tuple line the original action is

$$(\zeta, a) \mapsto \lambda_{\prod_i \chi_i}(\zeta)^{-1} \vartheta(-ka), \quad (\text{BTF.2})$$

where $\lambda_\chi(x^{(Q-1)/N}) = \chi(x)$ and $Q = Q_0^d$. These are the original SPF inverse deck-action conventions, tensored without changing them. Averaging over the complete finite inertia image in the characteristic-zero ℓ -adic coefficient field proves

$$P_{I_0, k} = \begin{cases} P_{0, k}^{\text{tame}}, & kc = 0, \\ 0, & kc \neq 0, \end{cases} \quad \dim V_k^{I_0} = \begin{cases} \frac{m^k + m(-1)^k}{m+1}, & kc = 0, \\ 0, & kc \neq 0. \end{cases} \quad (\text{BTF.3})$$

The tame average is one exactly when $\prod_i \chi_i = 1$. The wild average $p^{-1} \sum_a \vartheta(-ka)$ is one for $p \mid k$ and zero otherwise. For $c = 0$ the original additive torsor is trivial, so only the tame average is present. Since $kc = 0$ is equivalent to $c = 0$ or $p \mid k$, these computations prove the full case distinction. The tuple count is the same geometric sum proved in (BTP.6).

For $kc \neq 0$, the Artin–Schreier factor of $z^p - z = bkc/u$ has a single break one: z^{-1} is a uniformizer and its difference under $z \mapsto z + a$, $a \neq 0$, has valuation two. Every tuple line has that same wild factor, while Kummer twists are tame. Therefore

$$\text{Swan}_0(\mathcal{F}_L^{\otimes k}) = \begin{cases} 0, & kc = 0, \\ m^k, & kc \neq 0, \end{cases} \quad N_{\text{mon}, k} = 0 \text{ after } u = v^N, \quad z_k^p - z_k = bkc/v^N. \quad (\text{BTF.4})$$

The cover trivializes every geometric character, so its monodromy is the identity and its logarithm is zero. This leaves the original-inertia invariant space in (BTF.3) unchanged as a separately specified stalk over the original base.

The exact boundary Frobenius for $kc = 0$ over L is the direct sum of the scalars A_χ with $\prod_i \chi_i = 1$. Over k_0 , geometric Frobenius sends χ to χ^{Q_0} : conjugating the SPF deck projector by Frobenius replaces its deck argument by $\zeta^{Q_0^{-1}}$, and reindexing yields that character power. For an orbit $\mathcal{O}$ of length e , each χ_i descends uniquely by norm to $\kappa_{i,e}$ over $L_e = \mathbb{F}_{Q_0^e}$. Indeed its order

divides $Q_0^e - 1$, so it kills the norm kernel; norm surjectivity proves uniqueness. Retaining the original denominator N , the one-factor SPF calculation over this orbit field gives

$$\Lambda_{\mathcal{O}} = \prod_i (-\kappa_{i,e}(N) G_{L_e}(\kappa_{i,e}, \psi_0 \circ \text{Tr}_{L_e/k_0})) , \quad \det(1 - tF_0 \mid V_k^{I_0}) = \prod_{\mathcal{O}} (1 - \Lambda_{\mathcal{O}} t^e). \quad (\text{BTF.5})$$

For the descent map when L_e does not contain all N -th roots, factor the source power map as $y \mapsto x = y^{N/r_i} \mapsto z = x^{r_i} = y^N$, where $r_i = \text{ord } \chi_i$. Functions constant on the μ_{N/r_i} -orbits are exactly functions on the intermediate root set; the same map is the identity at the zero fibre. This identifies the descended $\kappa_{i,e}$ -summand of the original sheaf, whose phase is still $z/(Nu) + c/u$. Thus the displayed scalar is the actual F_0^e -action. A basis $v, F_0 v, \dots, F_0^{e-1} v$ of distinct tuple lines gives the factor $1 - \Lambda_{\mathcal{O}} t^e$. The unscaled Gauss absolute-square calculation $|G_{L_e}|^2 = Q_0^e$ therefore gives $|\Lambda_{\mathcal{O}}| = Q_0^{ek/2}$ and weight k for every nonzero boundary orbit block.

To specify the comparison of characteristic types, put $B_0 = \mathbb{Z}[\zeta_N, 1/N]$ and

$$e_j = \frac{1}{N} \sum_{a=0}^{N-1} \zeta_N^{-ja} \gamma^a \in B_0[C_N], \quad M_{B_0} = B_0[y]dy/y^m B_0[y]dy, \quad e_j \longmapsto y^{j-1} dy \quad (1 \leq j < N). \quad (\text{BTF.6})$$

Give γ the action $\gamma(y^b dy) = \zeta_N^{b+1} y^b dy$. The finite geometric sum proves that the e_j are complete orthogonal idempotents, and the displayed basis map is an isomorphism $\bigoplus_{j=1}^{N-1} B_0 e_j \simeq M_{B_0}$. The actual maps $B_0 \rightarrow L$ and $B_0 \rightarrow \overline{\mathbb{Q}}_\ell$ send ζ_N to the respective chosen primitive roots. They give, respectively, the coefficient monomial-form projector and the actual SPF Gauss-line projector: the latter is $N^{-1} \sum_a \zeta_\ell^{-ja} D_\zeta^{-a} = N^{-1} \sum_a \zeta_\ell^{ja} D_\zeta^a$. If H_j is that complete Gauss line, retain its Frobenius and reconstruct its isotypic component by $(ae_j) \otimes h \mapsto ah$ from $(B_0 e_j \otimes \overline{\mathbb{Q}}_\ell) \otimes H_j^{\text{triv}}$, where only the deck action on the second factor is made trivial. The scalar map is bijective on each summand and preserves both deck action and the retained splitting-field Frobenius, proving the comparison. Ordered tensor products of these maps retain every tuple. This is the common integral representation and its two proved base changes; it introduces no field map from characteristic p to $\overline{\mathbb{Q}}_\ell$.

The terminal form has deck-pullback character $\zeta^m = \zeta^{-1}$ in each factor; let its primitive Kummer character be χ_{ter} . Its ordered sheaf line is consequently

$$\mathcal{L}_\psi(kc/u) \otimes \mathcal{A}_{[-\chi_{\text{ter}}(N)G_L(\chi_{\text{ter}}, \psi)]^k} \otimes \mathcal{K}_{\chi_{\text{ter}}}^k(u). \quad (\text{BTF.7})$$

It has invariants exactly for $N \mid k$ and $kc = 0$; this means $N \mid k$ if $c = 0$, and $pN \mid k$ if $c \neq 0$. Its original base orbit has length d , because its character is primitive, with polynomial $1 - [-\chi_{\text{ter}}(N)G_L(\chi_{\text{ter}}, \psi)]^k t^d$. These claims follow from (BTF.2)–(BTF.5) without assigning a finite-field Frobenius to an original complex arithmetic scalar.

On a coefficient stage admitting the specified map, the original amplified coefficient instead has the literal coefficient image

$$\overline{C}_{h_0,k} = \overline{K}! ((m-1)!)^{-k} \overline{u}_0^k, \quad u_0 = (g/h_0)(\rho), \quad \overline{C}_{h_0,k} \neq 0 \iff K < p \text{ and } \overline{u}_0 \neq 0. \quad (\text{BTF.8})$$

The denominator is a unit since $m-1 < p$, while $K!$ vanishes exactly for $K \geq p$. Retaining the inverse of the original full unit ensures $\overline{u}_0 \neq 0$. The target is the coefficient algebra $k_0[y_1, \dots, y_k]/(y_i^m)$, with coefficient map obtained by applying α to every coefficient in its unchanged monomial basis. Its terminal basis vector remains nonzero; the original multiple vanishes precisely as in (BTF.8). This is the map (BPC.11). It does not multiply a characteristic- p scalar into the characteristic-zero Gauss line (BTF.7); their common projector maps are instead (BTF.6).

For $m \geq 2$, the wild-cancellation orders $p \mid k$ have $K \geq p$, so the specialized original terminal multiple vanishes on every stage admitting this local coefficient map. The sheaf invariant lines calculated above remain as specified by (BTF.7). For $m = 1$, $K = 0$ and $C_{h_0,k} = u_0^k$; there is

no factorial loss. Here $N = 2$ and the quadratic Gauss constant $\gamma_\chi = -\chi(2)G_L(\chi, \psi)$ satisfies $\gamma_\chi^2 = \chi(-1)Q$. Indeed complex conjugation gives $\overline{G} = \chi(-1)G$, while $G\overline{G} = Q$, which proves the algebraic equality in its cyclotomic number field and hence in its ℓ -adic realization. Thus at $k = 2pr$ with $c \neq 0$, the exact invariant eigenvalue is $(\chi(-1)Q)^{pr}$, and the coefficient image is the retained nonzero $\overline{w}_0^k$.

Finally, the local coefficient map does not silently extend through new CRT denominators. In the original off-critical quartet branch $\rho = 1/2 + \delta + i\gamma$, $\delta > 0$, the complete degree- k split corner has centres $\lambda_0 = k\rho$ and $\lambda_1 = (k-p)\rho + p(1-\bar{\rho})$ whenever $k \geq p$. Retain its actual terminal CRT polynomial $e(Z) = \sum_r e_r Z^r$ with $e(\lambda_0) = 1$, $e(\lambda_1) = 0$. In its own coefficient ring set

$$\mathcal{D} = \sum_{r \geq 1} e_r \sum_{j=0}^{r-1} \lambda_0^{r-1-j} \lambda_1^j, \quad 1 = p(\rho + \bar{\rho} - 1)\mathcal{D}, \quad p^{-1} = (\rho + \bar{\rho} - 1)\mathcal{D}. \quad (\text{BTF.9})$$

The divided difference telescopes term by term: $(\lambda_0 - \lambda_1)\mathcal{D} = e(\lambda_0) - e(\lambda_1) = 1$. Since $\lambda_0 - \lambda_1 = p(\rho + \bar{\rho} - 1)$, this proves all the formulas. Hence that full stage inverts p and has no unital characteristic- p specialization. Replacing p by any rational prime $q \leq k$ proves that the degree- k split corner inverts every such prime. The directed coefficient union of all these retained stages therefore contains $\mathbb{Q}$: each reciprocal prime occurs at its corresponding stage. The explicit local maps (BPC.3), (BPC.8), (BPC.11) and (BPC.14) continue to relate the local characteristic-zero source, its lattice and its valid coefficient specialization. The absence of an extension of that specialization is not the zero image of a vector under a nonexistent map.

The depth and boundary character can be carried together on those same local maps. Define $D_c T^r = \mathbf{1}_{\{r+k \equiv c \pmod{N}\}} T^r$ and use the full-unit character masks $\widehat{Q}_c$ in (BTP.3). Every term in V_r has degree r , so

$$\widehat{Q}_c f_U = f_U D_c, \quad \widehat{Q}_c W_r = \mathbf{1}_{\{r+k \equiv c\}} W_r, \quad \text{Ann}_{\mathcal{S}}([W_r]) = p^{dr} \mathcal{S}. \quad (\text{BTF.10})$$

These masks preserve both lattices and their free resolution, so the Tor connecting map in (BPC.14) has the same character: $\overline{D}_c \partial \epsilon_r = \mathbf{1}_{\{r+k \equiv c\}} \overline{T}^r$. For $k = p\ell$, $m \geq 2$, its surviving reduced image vector $\overline{W}_{p-1}$ has the exact calculation

$$\begin{aligned} \overline{J} \overline{W}_{p-1} &= 0, \quad \overline{W}_{p-1} \neq 0, \quad J W_{p-1} = p W_p \neq 0, \\ \widetilde{\overline{Q}}_0 \overline{W}_{p-1} &= \overline{W}_{p-1} \iff p(\ell+1) \equiv 1 \pmod{N}. \end{aligned} \quad (\text{BTF.11})$$

Equation (BPC.6) and the primitive pivot prove the first three assertions. Its degree is $p-1$, so (BTF.10) gives the congruence. Since $p > N$, it selects an infinite positive residue class of ℓ . The precursor $p W_p$ is retained, so this new reduced kernel direction has not been identified with the original complex terminal eigenclass. Every coefficient map above acts at each existing nonbottom split-support label without changing that label; a killed coefficient remains its represented zero, and the external absent symbol is fixed separately. Thus this update propagates the arithmetic exponent, tensor boundary projection and coefficient depth on the same original maps.

Abstract

For the original monic packet polynomial, retaining all selected zero multiplicities, we calculate the missing u -connection of its polynomial exponential complex in the original monomial lattice. The connection has an explicit second-order coefficient, a first-order coefficient of trace $d/2$, and a degree-correct chain realization. We calculate its full tensor version and the exact period determinant, including its gamma factors and ordered contour phase. We also apply Deligne's lissity and local-monodromy theorems to the actual finite-field family on $u \neq 0$, obtaining a genuine mixed boundary extension there. The original theta source, its Taylor unit, and its metric enter through explicit maps. The exact single-primary branch now includes its full period matrix, boundary monodromy, finite cover, source-unit maps and finite-field character decomposition through every ordered tensor degree, together with the exact original coefficient lattice and its specialization maps. The general multi-primary sector transitions and original arithmetic uniform four-volume estimate remain continuing calculations on the retained objects.

11 Fixed objects and original source maps

Let Z be the original nonempty finite set of selected zeros of $g = 2\xi$, with the complete selected orders m_ρ , and put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho} = s^d + \sum_{b=0}^{d-1} h_b s^b, \quad d = \sum_{\rho \in Z} m_\rho \geq 1. \quad (1)$$

The primitive and its constant are fixed as

$$\Phi_h(s) = \frac{s^{d+1}}{d+1} + \sum_{b=0}^{d-1} \frac{h_b s^{b+1}}{b+1}, \quad \Phi_h(0) = 0, \quad f_t(s) = \Phi_h(s) - ts. \quad (2)$$

Retain the original source spaces, Fourier convention, and operators

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^m F(x)| < \infty \text{ for all } b \in \mathbb{Z}, m \geq 0\}, \\ D &= -x\partial_x, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \\ \widehat{\phi}(\xi) &= \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx. \end{aligned} \quad (3)$$

The seed is

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g. \quad (4)$$

The inherited Euler inverses supply the original $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h = g/h$. Define

$$E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = j_h v_h \in E_h^\times, \quad \alpha_h(P) = P(D)\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h. \quad (5)$$

The original identities are

$$\Theta\alpha_h(P) = \mathcal{T}_h(hP), \quad J_h \mathcal{T}_h(P) = v_h[P]_h. \quad (6)$$

Thus $[\mathbb{C}[s] \xrightarrow{h} \mathbb{C}[s]ds]$ maps to $[V \xrightarrow{\Theta} \mathcal{B}]$ by $(\alpha_h, \mathcal{T}_h)$. Its degree-one map is exactly $\sigma_h M_{v_h}$ into $Q = \mathcal{B}/\Theta V$. The linear source section is not asserted to be a ring homomorphism. Tensoring these chain maps preserves the full factor $v_h^{\otimes k}$ and the original ordered exterior signs.

12 The full lattice and its marked fibre

For $R_k = \mathbb{C}[u, t_1, \dots, t_k]$ set

$$\mathcal{C}_k^j = R_k[s_1, \dots, s_k] \otimes \bigwedge^j (ds_1, \dots, ds_k), \quad \mathcal{D} = \sum_{i=1}^k L_i ds_i \wedge (-), \quad L_i = u\partial_{s_i} + h(s_i) - t_i. \quad (7)$$

The L_i commute, so the exterior signs give $\mathcal{D}^2 = 0$.

Proposition 12.1 (The original complete lattice). *The complex (7) has cohomology only in degree k , with*

$$\mathcal{H}_k := H^k(\mathcal{C}_k) = \bigoplus_{0 \leq a_i < d} R_k[s_1^{a_1} \cdots s_k^{a_k} ds_1 \wedge \cdots \wedge ds_k]. \quad (8)$$

Specialization at $u = t_1 = \cdots = t_k = 0$ identifies this displayed module with $E_h^{\otimes k}$, retaining every primary jet.

Proof. In one factor $L = u\partial_s + h - t$ increases the degree of a nonzero polynomial by exactly d and retains its leading coefficient. Consequently L is injective. Subtracting $L(as^j)$ at each highest degree gives a remainder of degree below d . A polynomial of degree below d in the image of L must be zero by the same degree identity. This proves the direct module decomposition

$$R[s] = L(R[s]) \oplus R[s]_{<d}, \quad R = \mathbb{C}[u, t].$$

The inverse of L on its image is R -linear and contracts the image two-term complex. Tensor the direct decompositions in the specified factor order. Every tensor summand with a contractible factor contracts by that factor's homotopy with the sign contributed by preceding degrees; the cross terms cancel by the tensor differential rule. Only the top remainder summand remains. The splitting is over R_k , so it remains a splitting after the stated coefficient specialization. $\square$

The family is a literal deformation of the original finite polynomial source images: with $V_* = \alpha_h(\mathbb{C}[s])$ and $\mathcal{B}_h = \mathcal{T}_h(\mathbb{C}[s])$, its one-factor differential is

$$d_{u,t}^\theta = \Theta|_{V_*} + u\mathcal{T}_h\partial_s\alpha_h^{-1} - t\mathcal{T}_h\alpha_h^{-1}. \quad (9)$$

The inverse maps here are defined on the displayed injective images. Equation (6) gives $d_{u,t}^\theta\alpha_h = \mathcal{T}_hL$. The derivative term is also the original function-space commutator

$$\mathcal{T}_h(P') = (\log x)P(D)F_h - P(D)((\log x)F_h),$$

because $[\log x, D] = 1$. In Mellin coordinates this subtraction retains and cancels the two equal v'_hP terms, leaving v_hP' .

If χ is the minimal polynomial of $S = \sum_i s_i$ on $E_h^{\otimes k}$, the weighted cyclic injection is

$$\eta : \mathbb{C}[S]/\chi \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k}P\left(\sum_i s_i\right)1. \quad (10)$$

For fixed-order monic division $\chi(S)P(S) = \sum_i h(s_i)Q_i(s)$, the ordered primitive $\sum_i (-1)^{i-1}Q_i ds_1 \wedge \cdots \widehat{ds_i} \cdots \wedge ds_k$ has image $\sum_i (h_iQ_i - t_iQ_i + u\partial_iQ_i)$. It follows that

$$[\chi(S)P(S)]_{\mathcal{H}_k} = \sum_i [t_iQ_i - u\partial_{s_i}Q_i]_{\mathcal{H}_k}. \quad (11)$$

The quotient by the cyclic image and this parameter defect remain in the full family.

13 A genuine finite-field mixed extension

Choose the same finitely generated coefficient ring $R_0 \subset \mathbb{C}$ containing the coefficients of h , the matrices of the complete v_h and its inverse, and $1/(d+1)!$. Fix a specified finite-field point $R_0 \rightarrow \kappa = \mathbb{F}_Q$ of characteristic $p > d+1$. For the rest of this section polynomial coefficients mean their images under this specified map. Let

$$\begin{aligned} S &= \mathbb{G}_{mu} \times \mathbb{A}_{\mathbf{t}}^k, & \pi : \mathbb{A}_{\mathbf{S}}^k \times S &\longrightarrow S, \\ \mathcal{L} &= \mathcal{L}_\psi \left(\frac{\sum_i \Phi_h(s_i) - \sum_i t_i s_i}{u} \right), & \mathcal{F} &= R^k \pi_! \mathcal{L}. \end{aligned} \quad (12)$$

Here ψ is a fixed nontrivial additive character with values in the chosen ℓ -adic coefficient field, $\ell \neq p$.

Proposition 13.1 (The actual open family and boundary stalk). *The sheaf $\mathcal{F}$ in (12) is lisse, of rank d^k , and pointwise pure of weight k ; all $R^j \pi_! \mathcal{L}$ with $j \neq k$ vanish. Fix any finite-field*

parameter value $\mathbf{t} = \mathbf{t}_0$, after a residue-field extension when necessary, and write $\mathcal{F}_{\mathbf{t}_0}$ for its pullback to $\mathbb{G}_m$. For $j : \mathbb{G}_m \hookrightarrow \mathbb{A}_u^1$ and $i : \{0\} \hookrightarrow \mathbb{A}_u^1$ there is an exact sequence

$$0 \longrightarrow j_! \mathcal{F}_{\mathbf{t}_0} \longrightarrow j_* \mathcal{F}_{\mathbf{t}_0} \longrightarrow i_*(V^I) \longrightarrow 0, \quad (13)$$

where V is the local geometric generic representation and I its inertia group. The boundary stalk V^I is mixed of weights at most k .

Proof. The highest homogeneous phase is $\sum_i s_i^{d+1}/((d+1)u)$. Its partial derivatives s_i^d/u cannot vanish simultaneously at a projective point. Therefore the explicit coefficient map from S to the universal degree- $(d+1)$ polynomial parameter space lands entirely in Deligne's good open. Weil II, Lemma 3.7.3, proves lissity of the universal proper-support cohomology sheaves there. Proper-support base change pulls these sheaves to (12). Statement 3.7.2.3, with polynomial degree $d+1$ and affine dimension k , proves the asserted concentration, rank, and weight. This check uses no simplicity assumption on finite critical roots.

The stalk of $j_* \mathcal{F}_{\mathbf{t}_0}$ at zero is V^I ; the stalk of $j_! \mathcal{F}_{\mathbf{t}_0}$ there is zero. On the open the map is the identity, proving (13) on every geometric stalk. The extension by zero is mixed of weight k .

On an open finite-index inertia subgroup with unipotent action, let N be the actual logarithmic monodromy operator and M its monodromy filtration. Weil II 1.8.4 applies to the lisse pointwise pure $\mathcal{F}_{\mathbf{t}_0}$ and proves that $\mathrm{Gr}_r^M V$ is pure of weight $k+r$. By 1.8.8, V^I is obtained from $\ker N$ by finite-group invariants; the filtration it inherits has no graded pieces with $r > 0$. Its nonzero graded pieces therefore have weights at most k . No graded rank or numerical matrix for N is inserted in this argument. $\square$

This supplies an actual use of Deligne's mixed-boundary machinery on the finite-field branch. The coefficient specialization of (8), and the theta source map (6), are separately explicit. A comparison from their marked complex fibre to the local representation in (13) has not been proved by these constructions. The complex boundary data relevant to such a comparison are calculated next, with all original coefficients retained.

14 Exact division and the full boundary connection

For $0 \leq b < d$ perform monic division in the original variable s :

$$f_t(s)s^b = (h(s) - t)q_b(s, t) + r_b(s, t), \quad \deg_s r_b < d, \quad \deg_s q_b \leq b+1 \leq d. \quad (14)$$

Let $C_h(t)$ and $B_h(t)$ be the matrices whose b th columns are the coefficients of r_b and $\partial_s q_b$, respectively, in the ascending monomial basis $1, s, \dots, s^{d-1}$.

Lemma 14.1 (Exact differential remainder). *In the one-factor top lattice,*

$$[f_t s^b] = [r_b - u \partial_s q_b]. \quad (15)$$

There is no further reduction of the displayed quotient derivative.

Proof. Subtract $Lq_b = (h - t)q_b + u \partial_s q_b$ from (14). The derivative has degree below d , so the right side is already in the retained remainder basis. $\square$

Theorem 14.2 (Degree-correct flat connection). *On the one-factor complex localized at u , the degree-one and degree-zero connections in the u direction are*

$$\nabla_u^1 = \partial_u - \frac{f_t}{u^2}, \quad \nabla_u^0 = \partial_u - \frac{f_t}{u^2} + \frac{1}{u}. \quad (16)$$

The induced top connection in the original monomial lattice is exactly

$$\boxed{\nabla_u = \partial_u - \frac{C_h(t)}{u^2} + \frac{B_h(t)}{u}}. \quad (17)$$

On the ordered k -factor complex the degree- j operator and its top cohomology operator are

$$\nabla_u^{(j)} = \partial_u - \frac{\sum_i f_{t_i}(s_i)}{u^2} + \frac{k-j}{u}, \quad (18)$$

$$\boxed{\nabla_u} = \boxed{\partial_u - \frac{\sum_i C_{h,i}(t_i)}{u^2} + \frac{\sum_i B_{h,i}(t_i)}{u}}. \quad (19)$$

These commute with the existing parameter connections $\nabla_{t_i} = \partial_{t_i} - s_i/u$ and are compatible with the original complex differential.

Proof. The commutators on polynomial coefficients are

$$[L, \partial_u] = -\partial_s, \quad [L, -f_t/u^2] = -(h-t)/u,$$

so $[L, \nabla_u^1] = -L/u$ and $\nabla_u^1 L = L(\nabla_u^1 + 1/u) = L\nabla_u^0$. This proves the chain identity, including the degree-zero correction. The differential remainder (15) proves (17).

For (18), the commutator with each L_i is $-L_i/u$, and the scalar difference between adjacent exterior degrees contributes $+L_i/u$. Their cancellation proves chain compatibility in the original order. Top cohomology and (15) give (19). Finally, differentiating $-s_i/u$ in u gives $+s_i/u^2$, whereas the commutator of $-\sum f_{t_i}/u^2$ with ∂_{t_i} is $-s_i/u^2$. All other terms commute. This proves the asserted flatness. $\square$

Proposition 14.3 (The exact first-order trace). *$B_h(t)$ is independent of t . In the unchanged ascending monomial basis it is upper triangular and satisfies*

$$(B_h)_{bb} = \frac{b+1}{d+1}, \quad \boxed{\text{Tr } B_h = \frac{d}{2}}. \quad (20)$$

Proof. In (14), compare coefficients of degrees greater than d . The perturbation $-ts^{b+1}$ and the product $-tq_b$ both have degree at most d . All coefficients of q_b except possibly its constant coefficient are therefore determined independently of t . Differentiation removes that constant. The leading term of q_b is $s^{b+1}/(d+1)$, so its derivative has degree b and leading coefficient $(b+1)/(d+1)$. Summing those diagonal entries proves (20). $\square$

The matrix B_h alone is not asserted to be the local monodromy residue: the order-two coefficient $C_h(t)$ is present and generally does not commute with B_h . Equations (17) and (19) retain their exact relationship.

15 The full primary jets at the boundary

Proposition 15.1 (Leading coefficient on every original primary block). *Retain the Chinese-remainder summands $E_\rho = \mathbb{C}[s]/(s-\rho)^{m_\rho}$ and their projectors. At $t=0$,*

$$C_h(0)|_{E_\rho} = \Phi_h(\rho)I_{E_\rho}, \quad A_h|_{E_\rho} = \rho I_{E_\rho} + N_\rho, \quad \dim E_\rho = m_\rho. \quad (21)$$

For the ordered tuple $(\rho_1, \dots, \rho_k)$ the leading matrix in (19) is scalar $\sum_i \Phi_h(\rho_i)$ on a block of dimension $\prod_i m_{\rho_i}$.

Proof. Write $h(s) = (s-\rho)^{m_\rho} a_\rho(s)$, where $a_\rho(\rho) \neq 0$. Integration of its Taylor polynomial, with the original constant retained, gives the exact identity

$$\Phi_h(s) - \Phi_h(\rho) = \sum_{j \geq 0} \frac{a_\rho^{(j)}(\rho)}{j!} \frac{(s-\rho)^{m_\rho+j+1}}{m_\rho+j+1}. \quad (22)$$

The sum is finite. Every term is divisible by $(s-\rho)^{m_\rho+1}$. The matrix $C_h(0)$ is multiplication by $[\Phi_h]_h$, by its definition in (14); hence its restriction is the first expression in (21). Multiplication by s remains $\rho + N_\rho$ on that same full module. The tensor statement follows by applying the sum of these matrices to the ordered tensor projectors. $\square$

Equal sums of critical values retain their separate original projectors and source labels. The scalar exponential factors for two tuples have the exact modulus ratio

$$\exp\left(\operatorname{Re} \frac{\sum_i \Phi_h(\rho_i) - \sum_i \Phi_h(\rho'_i)}{u}\right). \quad (23)$$

For a nonzero difference, equal modulus occurs on $\arg u = \arg(\text{difference}) + \pi/2 \pmod{\pi}$. These are candidate Stokes directions of the displayed exponential factors. Equation (23) supplies no assertion that a particular Stokes entry is nonzero, and does not replace the calculation of the matrices for (19).

16 Exact ordered period determinant

Keep the original rays and contours

$$\vartheta_j = \frac{\arg u + \pi}{d+1} + \frac{2\pi j}{d+1}, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq d), \quad (\Pi_h)_{jb} = \int_{\Gamma_j} e^{f_t(s)/u} s^b ds. \quad (24)$$

The leading radial exponent is $-r^{d+1}/((d+1)|u|)$; it proves absolute convergence and coefficient differentiation on compact parameter sets. Contours on nearby parameters deform within the same decaying sectors, with their original orientations. Every formula below uses a fixed retained branch for u and the ordered rays.

Theorem 16.1 (Full period determinant, including its constant). *Let $c_h(t) = \operatorname{Tr} C_h(t)$, and put*

$$\begin{aligned} \zeta &= e^{2\pi i/(d+1)}, & \beta_b &= \frac{b+1}{d+1}, \\ W_{jb} &= \zeta^{j(b+1)} - 1 \quad (1 \leq j \leq d, \ 0 \leq b < d), \\ K_d &= (d+1)^{-d/2} e^{\pi i d/2} \left(\prod_{b=0}^{d-1} \Gamma(\beta_b) \right) \det W. \end{aligned} \quad (25)$$

Then $K_d \neq 0$ and

$$\boxed{\det \Pi_h(u, t) = K_d u^{d/2} e^{c_h(t)/u}.} \quad (26)$$

For the ordered external product $\Pi_k = \bigotimes_i \Pi_h(u, t_i)$,

$$\boxed{\det \Pi_k = K_d^{kd^{k-1}} u^{kd^k/2} \exp\left(\frac{d^{k-1} \sum_i c_h(t_i)}{u}\right).} \quad (27)$$

Both identities retain all lower coefficients of h and extend across critical-root collisions.

Proof. Differentiating (24) in u inserts $-f_t/u^2$. The exact reduction (15) gives

$$\partial_u \Pi_h = \Pi_h \left(-\frac{C_h(t)}{u^2} + \frac{B_h}{u} \right).$$

The determinant equation, valid as the exterior-power equation even before invertibility is known, is

$$\partial_u \det \Pi_h = \left(-\frac{c_h(t)}{u^2} + \frac{d}{2u} \right) \det \Pi_h. \quad (28)$$

Thus the remaining factor after division by $u^{d/2} e^{c_h(t)/u}$ is independent of u on a contour branch.

We prove its independence of t and all lower coefficients without deleting their effects. On the dense open of simple roots of $h - t$,

$$c_h(t) = \sum_{h(\rho)=t} f_t(\rho), \quad \partial_t c_h(t) = - \sum_{h(\rho)=t} \rho = -\operatorname{Tr} A_h(t).$$

The terms containing derivatives of the roots vanish because $f'_t(\rho) = 0$. Both sides are polynomial expressions in the coefficients and t , so this identity holds across collisions too. The period equation $\partial_t \Pi_h = \Pi_h(-A_h(t)/u)$ cancels the derivative of $e^{c_h(t)/u}$, proving that the remaining factor is independent of t .

Differentiation in h_b inserts $s^{b+1}/((b+1)u)$. For a column s^j , its numerator has degree $j + b + 1 \leq 2d - 1$. In division by $h - t$ the quotient derivative has degree at most $j + b - d < j$. Consequently the differential-reduction correction has zero trace. The determinant logarithmic derivative in h_b is $\text{Tr } M_{s^{b+1}}/((b+1)u)$. On the simple-root open this equals $(\partial_{h_b} c_h(t))/u$, since again the root derivative is multiplied by $f'_t(\rho) = 0$. Polynomial continuation proves the identity for all coefficients. Therefore the remaining factor has zero derivative in every lower coefficient.

It remains to compute that factor at $h = s^d, t = 0$, with the original contours. Substitution $v = r^{d+1}/((d+1)|u|)$ on each ray gives

$$(\Pi_h)_{jb} = [(d+1)u]^{\beta_b} e^{\pi i \beta_b} \frac{\Gamma(\beta_b)}{d+1} (\zeta^{j(b+1)} - 1). \quad (29)$$

The sum of the β_b is $d/2$, so taking determinants gives exactly (25) and (26). For invertibility of W , suppose $\sum_{r=1}^d a_r(z^r - 1)$ vanishes at the d nontrivial roots of $z^{d+1} = 1$. It also vanishes at $z = 1$. Its degree is at most d , so it is identically zero and every $a_r = 0$. All gamma factors are nonzero, proving $K_d \neq 0$.

The periods and their parameter derivatives are entire in the lower coefficients on compact sets by the fixed leading radial decay. This justifies the coefficient continuation used above and proves the identities also at collisions. Finally, $\det(A \otimes B) = (\det A)^{\dim B} (\det B)^{\dim A}$ in the retained ordered product bases gives (27). $\square$

17 Exact comparison with the original source metric

The period Gram determinant following from (27) is

$$\det(\Pi_k^* \Pi_k) = |K_d|^{2kd^{k-1}} |u|^{kd^k} \exp\left(2d^{k-1} \text{Re} \frac{\sum_i c_h(t_i)}{u}\right). \quad (30)$$

Retain, independently, the actual one-factor source Gram

$$(G_\theta)_{ab} = \frac{1}{2\pi} \int_{\mathbb{R}} \overline{(1/2 + iy)^a} (1/2 + iy)^b |v_h(1/2 + iy)|^2 dy, \quad 0 \leq a, b < d. \quad (31)$$

This is the Gram of $\mathcal{T}_h(s^b) = D^b F_h$ in the original $L^2(dx)$ source norm by Mellin–Plancherel. Its complete multiplier is $v_h = g/h$. It is positive definite: a zero norm forces the continuous source function $\mathcal{T}_h(P)$ to vanish; its Mellin transform Pv_h is then zero, and the nonzero entire v_h forces $P = 0$. The original ordered tensor source Gram is $G_\theta^{\otimes k}$.

The exact comparison of the two finite forms is the endomorphism

$$T_{\text{per}} = (G_\theta^{\otimes k})^{-1} (\Pi_k^* \Pi_k). \quad (32)$$

It is self-adjoint and positive in the original source metric, since

$$\langle v, T_{\text{per}} w \rangle_{G_\theta^{\otimes k}} = v^* \Pi_k^* \Pi_k w, \quad \langle v, T_{\text{per}} v \rangle_{G_\theta^{\otimes k}} = \|\Pi_k v\|^2 > 0 \quad (v \neq 0).$$

Its determinant is exactly (30) divided by $(\det G_\theta)^{kd^{k-1}}$. This preserves both metrics rather than identifying them.

On the weighted cyclic image (10), the period form is $\eta^* \Pi_k^* \Pi_k \eta$. Its relation to the original cyclic source metric retains the rectangular map $\Pi_k \eta$, the complement of η , and the transverse map

$$E \xrightarrow{\eta} E_h^{\otimes k} \xrightarrow{\sum_i \mathcal{R}_{h,i}} E_h^{\otimes k} \longrightarrow E_h^{\otimes k} / \eta E.$$

The full determinant (30) does not determine the singular values of this rectangular restriction. The displayed maps are the exact relationship; no original relation norm has been set to zero.

18 Compatibility with independently retained mixed faces

For an original coefficient mask A , retain its A -indexed coefficient complex and the original outer theta-leg label. Let j_A^B be the specified constant zero-insertion linear map for $A \subseteq B$. It inserts represented zero in each added slot. The preceding complex differential and connection act with the same coefficients in each slot. Therefore

$$j_A^B \mathcal{D}_A = \mathcal{D}_B j_A^B, \quad j_A^B \nabla_A = \nabla_B j_A^B. \quad (33)$$

These equations hold componentwise on retained coordinates; on the inserted coordinates each derivative and matrix coefficient sends zero to zero. Since j_A^B is constant in all parameters, differentiation commutes with it as well.

On the carrier retaining its mask, a vector killed by an observation therefore remains the receiving represented zero with that mask. The empty coefficient face maps to the zero vector in the receiving face under zero insertion; this face-group map is not a τ -preserving pointed lift. A nonempty outer theta-leg label remains present when the coefficient mask is empty. No colimit over the entire mask poset is taken, since its terminal full face would discard the independent face labels.

The scalar structural specialization is still

$$\mathbb{F}_{1,\tau} \longrightarrow G(\mathbb{C}[u, \mathbf{t}]) \xrightarrow{G(\text{ev}_0)} G(\mathbb{C}).$$

It sends represented u, t_i to represented zero e , and τ to τ . The original absolute point, its source, and the coefficient specialization are retained. Equation (33) is an explicit compatibility with the mixed face maps; it supplies no claim that a Boolean mask itself is a Deligne weight.

19 Completed single-primary boundary calculation and the continuing mixed-support calculation

The full single-primary calculation SP.1–36 and SPF.1–29, included in this edition as *The actual single-primary marked family and its complete boundary monodromy* and its finite-field companion, sharpens the preceding connection and determinant calculations on their exact primary subcase. The formulas in this section are reused from that complete proof, with the original source, constant, unit and basis retained. They replace the earlier statement that this subcase's local inertia and periods still required calculation.

19.1 The exact primary specialization, connection and period matrix

Choose an actual nontrivial zero ρ of $g = 2\xi$, with its complete order $m \geq 1$. In this subsection fix $t = 0$ and put

$$n = m+1, \quad h(s) = (s-\rho)^m, \quad y = s-\rho, \quad \Phi_h(s) = \frac{(s-\rho)^n - (-\rho)^n}{n} = \frac{y^n}{n} + c, \quad c = -\frac{(-\rho)^n}{n}.$$

Thus $\Phi_h(0) = 0$ and $\Phi'_h = h$; the constant c has its original value. In the ordered rows $e_s = (1, s, \dots, s^{m-1})$ and $e_y = (1, y, \dots, y^{m-1})$, the exact coefficient map is

$$e_s = e_y P, \quad P_{ab} = \binom{b}{a} \rho^{b-a} \quad (a \leq b), \quad (P^{-1})_{ab} = \binom{b}{a} (-\rho)^{b-a} \quad (a \leq b),$$

with entries zero when $a > b$. Substitution in the binomial theorem proves both maps and their inverse relation; $\det P = 1$.

The original complex has differential $L_u = u\partial_s + (s - \rho)^m$. Its leading-degree term gives the free degree-one remainder module $\bigoplus_{b=0}^{m-1} \mathbb{C}[u][s^b ds]$, whose marked fibre is $\mathbb{C}[s]/h ds$, with the full primary order. In these unchanged coefficient columns the exact division and its derivative are

$$\Phi_h(s)s^b = h(s)\frac{(s - \rho)s^b}{n} + cs^b, \quad \left(\frac{(s - \rho)s^b}{n}\right)' = \frac{(b + 1)s^b - b\rho s^{b-1}}{n}.$$

Subtracting $L_u((s - \rho)s^b/n)$ proves

$$C_s = cI_m, \quad (B_s)_{bb} = \frac{b + 1}{n}, \quad (B_s)_{b-1,b} = -\frac{b\rho}{n}, \quad \nabla_u = \partial_u - \frac{cI_m}{u^2} + \frac{B_s}{u}.$$

All other entries of B_s vanish. The degree-zero lift is still $\partial_u - \Phi_h/u^2 + 1/u$; it intertwines L_u with the degree-one lift $\partial_u - \Phi_h/u^2$. Writing $\beta_a = (a + 1)/n$ gives the exact conjugation

$$PB_sP^{-1} = \text{diag}(\beta_0, \dots, \beta_{m-1}), \quad \text{Tr } B_s = m/2.$$

These are SP.1–9, including the original coefficient and chain maps.

On a fixed branch of $\log u$, retain the ordered outward rays of angles $(\arg u + \pi)/n + 2\pi j/n$ and their differences $\Gamma_j = -\ell_0 + \ell_j$, $1 \leq j \leq m$. Set

$$\zeta = e^{2\pi i/n}, \quad W_{ja} = \zeta^{j(a+1)} - 1, \quad d_a = n^{\beta_a-1} e^{\pi i \beta_a} \Gamma(\beta_a), \quad D(u) = \text{diag}(d_a u^{\beta_a}).$$

The complete original period matrix, with original monomial columns, is

$$\Pi_s(u) = \left(\int_{\Gamma_j} e^{\Phi_h(s)/u} s^b ds \right)_{jb} = e^{c/u} W D(u) P.$$

SP.10–14 prove this equality on the original contours: translating their junction by ρ leaves connector integrals tending to zero by $\text{Re}(\Phi_h/u) = -R^n/(n|u|) + O(R^{n-1}) + O(1)$; the finite junction paths cancel with their original orientations. The radial substitution then gives exactly the d_a above. The matrix W is invertible by the degree- $(n - 1)$ polynomial root argument in SP.14. In particular,

$$\det \Pi_s = K_m u^{m/2} e^{mc/u}, \quad K_m = n^{-m/2} e^{\pi i m/2} \left(\prod_{a=0}^{m-1} \Gamma(\beta_a) \right) \det W \neq 0.$$

Thus the preceding determinant has its complete matrix refinement on this precise subcase, with all Gamma factors, contour phases, Pascal coefficients and the original scalar retained.

19.2 The computed boundary and its exact marked-source maps

The transported cycles satisfy $\Gamma_j \mapsto \Gamma_{j+1} - \Gamma_1$ for $j < m$ and $\Gamma_m \mapsto -\Gamma_1$. Consequently SP.16–21 give

$$M_{\text{cyc}} = W \text{diag}(\zeta, \zeta^2, \dots, \zeta^m) W^{-1}, \quad M_{\text{cyc}}^n = I, \quad \ker(M_{\text{cyc}} - I) = 0.$$

The horizontal coefficient convention is determined by the exact fundamental matrix

$$H_s(u) = P^{-1} e^{-c/u} \text{diag}(u^{-\beta_a}).$$

It has inverse eigencharacters, with the same zero invariant space. Under $u = v^n$, the displayed connection becomes

$$\partial_v - \frac{nc}{v^{n+1}} I_m + \frac{1}{v} \text{diag}(1, \dots, m).$$

The meromorphic change $x_y = \text{diag}(v^{-1}, \dots, v^{-m})z$ removes precisely these integer residues and retains the common irregular factor e^{-c/v^n} . This exact fundamental solution has identity monodromy after the cover; its unipotent logarithm is $N_{\text{boundary}} = 0$. Its sector restrictions agree on

overlaps with the same branch, so there are no additional Stokes factors in this one-critical-value calculation. Both ordinary invariant stalks and the derived local invariant and coinvariant groups vanish before the cover, since $M_{\text{cyc}} - I$ is invertible.

On the same marked fibre let J_m be multiplication by y , with $(J_m)_{a+1,a} = 1$. The original multiplication and its precise relation to the boundary connection are

$$A_s = P^{-1}(\rho I_m + J_m)P, \quad N_{\text{ar}} = P^{-1}J_m P, \quad [B_s, A_s] = \frac{1}{n}N_{\text{ar}}, \quad [\nabla_u, A_s] = \frac{N_{\text{ar}}}{nu}.$$

With $D_0 = \text{diag}(d_a)$ and $K_0 = W D_0 J_m D_0^{-1} W^{-1}$, the complete period transport is

$$\Pi_s A_s \Pi_s^{-1} = \rho I_m + u^{1/n} K_0, \quad M_{\text{cyc}} K_0 M_{\text{cyc}}^{-1} = \zeta K_0.$$

The coefficient calculation on each monomial proves these commutators and conjugations, including the last nilpotent column. They retain the original arithmetic nilpotent through its exact nonhorizontal map; see SP.22–24.

For the entire quotient $v_h = g/h$, retain every coefficient

$$a_j = \frac{v_h^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!}, \quad 0 \leq j < m, \quad a_0 \neq 0.$$

Its marked multiplication and inverse are

$$U_s = P^{-1} \left(\sum_{j=0}^{m-1} a_j J_m^j \right) P, \quad U_s^{-1} = P^{-1} \left(\sum_{j=0}^{m-1} b_j J_m^j \right) P,$$

$$b_0 = a_0^{-1}, \quad b_r = -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}.$$

Multiplication of these finite series proves their inverse relation, using $J_m^m = 0$. Its complete period and connection images are

$$\Pi_s U_s \Pi_s^{-1} = \sum_{j=0}^{m-1} a_j u^{j/n} K_0^j, \quad [\nabla_u, U_s] = \frac{1}{nu} P^{-1} \left(\sum_{j=1}^{m-1} j a_j J_m^j \right) P.$$

These retain the full Taylor unit, rather than only its leading coefficient.

The original Euler/theta square remains, with $D = -x\partial_x$ and $\mathcal{M}F_h = v_h$,

$$\alpha_h(P) = P(D)\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h, \quad \Theta\alpha_h(P) = \mathcal{T}_h(hP), \quad J_h\mathcal{T}_h(P) = U_s[P]_s.$$

Here $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, $\Theta\phi = 2 \sum_{r \geq 1} \phi(rx)$, and $\mathcal{M}\Theta\phi_* = g$, with the original function spaces and Mellin convention. The degree-one map is the retained $\sigma_h U_s : E_h \hookrightarrow Q_{\text{ar}}$. The exact translated source vector is $\mathcal{T}_h(y^b) = (D - \rho)^b F_h$, and at general u the source differential is the specified $\Theta + u\mathcal{T}_h\partial_s\alpha_h^{-1}$ on $\alpha_h(\mathbb{C}[s])$. For $v_{\text{pol}} = \sum_{j < m} a_j(s - \rho)^j$, the original differential gives the full representative defect

$$v_{\text{pol}} L_u Q = L_u(v_{\text{pol}} Q) - u v'_{\text{pol}} Q.$$

If $v_{\text{pol}} p = hq + r$ for an original remainder p , its differential remainder is $r - uq'$, whereas the constant coefficient extension U_s returns r . SP.25–31 prove these identities and retain precisely the derivative term needed away from $u = 0$.

The exact finite-cover lattice map is also available:

$$e^{-c/v^n} \Pi_s(v^n) = W D_0 \text{diag}(v, v^2, \dots, v^m) P.$$

After the two displayed constant invertible changes, its cokernel is $\bigoplus_{a=0}^{m-1} \mathbb{C}\{v\}/(v^{a+1})$, of length $m(m+1)/2$. Its image lattice has rank- m fibre modulo v , while its inclusion in the ambient constant lattice induces zero modulo v . This is the complete lattice map between the retained marked fibre and the computed local boundary; the factored exponential remains an irregular factor. See SP.32–33.

In the unchanged source metric of the preceding calculation, the full matrix refinement is

$$T_{\text{per}}(u) = G_{\theta}^{-1} \Pi_s(u)^* \Pi_s(u) = e^{2\text{Re}(c/u)} G_{\theta}^{-1} P^* D(u)^* W^* W D(u) P.$$

It is positive and self-adjoint in G_{θ} , with the complete multiplier g/h in that Gram. For $N \geq m-1$, the enlarged polynomial source has the exact period factorization and kernel

$$\text{Per}_N = \Pi_s J_{u,N}, \quad \ker J_{u,N} = L_u(\mathbb{C}[s]_{\leq N-m}).$$

Thus its period Gram has rank m ; on the displayed remainder frame it is positive definite. SP.34–36 prove this by the original Mellin source and the endpoint calculation for $u d(e^{\Phi_h/u} Q)$. The original relation space and source norm therefore remain in subsequent restriction calculations.

19.3 The exact finite-field boundary on the same primary phase

For the finite-field statement retain the specified coefficient homomorphism $R \rightarrow k_0 = \mathbb{F}_{Q_0}$, with $p > n$, including ρ , $1/n!$, and every coefficient of the full Taylor unit and its inverse. Use the explicit constant extension

$$k = \mathbb{F}_Q, \quad Q = Q_0^d, \quad d = \text{ord}_{(\mathbb{Z}/n\mathbb{Z})^\times}(Q_0),$$

so $n \mid Q - 1$. Fix $\ell \neq p$, geometric Frobenius, and the nontrivial additive character ψ obtained from the specified one on k_0 by the field trace. SPF.1–29 compute the actual sheaf $\mathcal{F} = R^1 \pi_{s!} \mathcal{L}_{\psi}(\Phi_h(s)/u)$. With

$$X_n = \{\chi : k^\times \rightarrow \overline{\mathbb{Q}}_{\ell}^\times : \chi^n = 1\}, \quad G_k(\chi, \psi) = \sum_{z \in k^\times} \chi(z) \psi(z),$$

and $\mathcal{A}_\gamma$ denoting the geometrically constant line with geometric Frobenius eigenvalue γ , the exact parameter-sheaf isomorphism is

$$\mathcal{F} \simeq \mathcal{L}_{\psi}(c/u) \otimes \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} \mathcal{A}_{-\chi(n)G_k(\chi, \psi)} \otimes \mathcal{K}_{\chi}(u).$$

The Kummer convention has trace χ . SPF.6–19 prove the isomorphism using the original translation, the finite power map $y \mapsto y^n$, its character projectors including the zero stalk, and compact-support base change. The minus sign is the degree-one cohomological sign; $\chi(n)$ is the original coefficient factor. Both it and the common Artin–Schreier factor remain.

If the specialized $c \neq 0$, the common wild character has break one, so

$$V^{I_0} = 0, \quad \text{Swan}_0(\mathcal{F}) = m.$$

If $c = 0$, the nontrivial tame Kummer characters instead give

$$V^{I_0} = 0, \quad \text{Swan}_0(\mathcal{F}) = 0.$$

Writing $\psi(x) = \vartheta(\text{Tr}_{k/\mathbb{F}_p}(bx))$, the explicit cover $u = v^n$, followed when needed by $z^p - z = bc/v^n$, kills all geometric inertia. Its restricted unipotent logarithm is zero, while its restricted invariant space is the full rank m . SPF.22–27 prove the conductor, original invariant space and finite-cover statements by the exact local ramification calculation. Finally SPF.28–29 prove $|G_k(\chi, \psi)|^2 = Q$ directly for every nontrivial χ ; all local eigenvalues consequently have the original weight one. These are completed local and weight calculations for the specified sheaf and coefficient specialization.

19.4 Continuation through the full mixed-support family

The preceding complete primary calculation now supplies actual boundary matrices, period maps, conductors, unit defects and source forms to the earlier family. The original independent coefficient faces, outer theta-leg labels, proper-source kernels and structural specialization remain those already defined above; every constant coefficient map retains the corresponding represented zero and its label.

For several distinct primary critical values, the continuing calculation includes the sector transitions and local extensions between the computed blocks, with their exact maps through the retained mixed faces. The single-primary tensor calculation now gives the following complete continuation on the original parameter line.

The one-factor calculation gives the original finite-field sheaf's exact decomposition, weight one, invariant-space vanishing and zero finite-cover logarithm. The ordered tensor calculation now carries that result through every positive tensor order on the same parameter line. Keeping $n = m + 1$, the original phase has constant kc , and the character on an ordered monomial form is $\zeta^{\sum_i(b_i+1)}$, where $0 \leq b_i < m$ and $\zeta = e^{2\pi i/n}$. The complete sum of these characters gives the exact ordinary complex invariant dimension

$$d_{0,k} = \frac{1}{n} \sum_{j=0}^{n-1} \left(\sum_{a=1}^{n-1} \zeta^{ja} \right)^k = \frac{m^k + m(-1)^k}{m+1}. \quad (\text{SPBT.1})$$

The inner sum is m for $j = 0$ and -1 otherwise, proving the equality. It is zero at $k = 1$; for $m \geq 2$ it is nonzero at every $k \geq 2$, and for $m = 1$ it is one at even k and zero at odd k .

More precisely, let E_0 be the diagonal mask selecting $\sum_i(b_i + 1) \equiv 0 \pmod{n}$, let $P_k = P^{\otimes k}$ be the original Pascal coordinate map, and let U_k be the complete original source-unit matrix. The retained tensor period matrix Π_k , with all its Gamma constants and original contours, gives

$$Q_0 = P_k^{-1} E_0 P_k, \quad \widehat{Q}_0 = U_k Q_0 U_k^{-1}, \quad \mathcal{P}_0 \Pi_k U_k^{-1} = \Pi_k U_k^{-1} \widehat{Q}_0. \quad (\text{SPBT.2})$$

The diagonal power matrix in the exact tensor period formula commutes with E_0 , so multiplication of those matrices proves this identity. Thus $\widehat{Q}_0$ is the projector on the actual original-source jet, and its period observation is $\Pi_k U_k^{-1}$; neither unit nor inverse is removed. The terminal monomial $\prod_i (s_i - \rho)^{m-1}$ has character ζ^{-k} , and hence survives exactly when $n \mid k$. The original sum multiplication still acts on that same vector by $k\rho$, since multiplication by each $s_i - \rho$ kills it. Its dilation eigenvalue remains $a^{k\rho}$.

On the actual finite-field coefficient stage with $p > m + 1$, the entire ordered sheaf is the direct sum of the original Gauss lines for the ordered character tuples, tensored with the unchanged additive factor $\mathcal{L}_\psi(kc/u)$. The exact original-inertia projector is the tame average multiplied by its wild average; the latter is one when $kc = 0$, zero otherwise. Consequently

$$\dim V_k^{I_0} = \begin{cases} d_{0,k}, & kc = 0, \\ 0, & kc \neq 0, \end{cases} \quad \text{Swan}_0(V_k) = \begin{cases} 0, & kc = 0, \\ m^k, & kc \neq 0. \end{cases} \quad (\text{SPBT.3})$$

The wild average is $p^{-1} \sum_{a \in \mathbb{F}_p} \vartheta(-ka)$ when the original $c \neq 0$, which proves the two cases; for $c = 0$ the original additive torsor is trivial. The Swan value follows because every nontrivial $1/u$ Artin-Schreier character has break one, shared by all m^k ordered summands. Each nonzero invariant orbit block has its full product of the original negative Gauss constants and has weight k . The terminal character survives precisely for $n \mid k$ and $kc = 0$, hence for $pn \mid k$ in the branch $c \neq 0$.

The coefficient lattice now records the same original cyclic map as $T^r \mapsto r!UV_r$. In the actual coefficient localization $\mathcal{S} = A_{\ker \alpha}$ of the given coefficient datum, its image L_U lies in its saturation Λ_U , with the exact quotient $\Lambda_U/L_U \simeq \bigoplus_r \mathcal{S}/(p^{v_p(r!)}).$ The quotient of the full tensor target also retains its free complementary summand, as proved in (BPC.5) and (BPC.9). The complete

proof in (BPC.1)–(BPC.14) identifies its specialized kernel (T^p), including all the original unit coefficients and the Tor connecting map. The unequal-order and residual-collision forms are (PC30a)–(PC30g). These statements apply only on their actual coefficient-map domains; the phase denominators and the additional CRT denominators have the explicit extension tests in (BTF.9) and the preceding coefficient-domain calculation. The common cyclotomic projector maps in (BTF.6) relate the coefficient characters to the actual Gauss-line representation through their two base changes. They preserve the distinct coefficient and sheaf scalar targets. Thus one-factor vanishing is retained as its exact $k = 1$ case, and the tensor invariants, original arithmetic action and coefficient specialization are carried forward with their proved maps.

The continuing induced relation-form calculation on the weighted cyclic image retains the parameter defect (11), the transverse map, its rectangular period restriction, and all original source cross-pairings. The explicit primary formula for T_{per} above can now enter that calculation, while its complete coefficient Gram and restriction maps remain. An arithmetic Frobenius comparison and the uniform four-volume estimate still require their own calculations on those original objects; the completed primary boundary result has not performed those additional steps.

Source and verification record

The original packet-family inputs are those of *Tau Marked Product Four Endpoint*, 13 September 2026, NOTE.tex, read in full, with SHA256

40749e2a9d599a322e7b125eb43ad0878
a066a51b61da52cc3bb2ffb94cbd841.

The relevant primary source is Pierre Deligne, *La conjecture de Weil II*, Publications mathématiques de l’IHÉS 52 (1980), 137–252: 1.8.4, 1.8.8, 3.3.1–3.3.6, 3.7.2.3, 3.7.3, 6.1.13 and 6.2.1–6.2.6. The local typed text of those statements and surrounding proofs was read for this audit.

The companion exact symbolic checker verifies differential reduction, flatness, $\partial_t B_h = 0$, the trace and diagonal of B_h , the critical-value trace derivative, and complete jet divisibility for four polynomial fixtures of degrees 1 through 4, including repeated roots. All passed. These fixtures are explicitly not asserted to be arithmetic zero packets; the proofs here apply to the original h .

20 The actual single-primary marked family and its complete boundary monodromy

This calculation specializes the original marked polynomial family to a single selected zero with its complete order. The parameter t is fixed at its original value zero. The variable s , its ascending monomial basis, the primitive with value zero at $s = 0$, and the original theta source are retained. A translation below is a displayed coordinate isomorphism on that same family; its constant term and contour transport are part of the calculation.

20.1 The actual packet and the exact Pascal map

Let ρ be a selected actual nontrivial zero of $g = 2\xi$, of complete order $m \geq 1$. Put

$$n = m + 1, \quad h(s) = (s - \rho)^m, \quad E_h = \mathbb{C}[s]/((s - \rho)^m), \quad A_s = M_s : E_h \rightarrow E_h. \quad (\text{SP.1})$$

The original polynomial primitive is exactly

$$\Phi_h(s) = \frac{(s - \rho)^n - (-\rho)^n}{n} = \frac{y^n}{n} + c, \quad y = s - \rho, \quad c = -\frac{(-\rho)^n}{n}. \quad (\text{SP.2})$$

Differentiation gives $\Phi'_h = h$, and substitution $s = 0$ gives $\Phi_h(0) = 0$. Thus its constant has not been reassigned. Since a nontrivial zero ρ is nonzero, this complex number c is nonzero; later coefficient specializations may send it to zero and are treated separately.

Write $e_s = (1, s, \dots, s^{m-1})$ and $e_y = (1, y, \dots, y^{m-1})$ as ordered rows. The coefficient map from the former basis to the latter is the explicit Pascal matrix

$$e_s = e_y P, \quad P_{ab} = \begin{cases} \binom{b}{a} \rho^{b-a}, & 0 \leq a \leq b < m, \\ 0, & a > b, \end{cases} \quad (P^{-1})_{ab} = \begin{cases} \binom{b}{a} (-\rho)^{b-a}, & a \leq b, \\ 0, & a > b. \end{cases} \quad (\text{SP.3})$$

The binomial theorem proves the first identity. Substituting $s = y + \rho$ and $y = s - \rho$ in succession gives the identity on every monomial, proving the inverse formula. Both matrices have diagonal entries one and determinant one. If x_s is an original coefficient column, its new coordinates are exactly $x_y = P x_s$.

The polynomial-exponential complex at $t = 0$ is

$$\mathbb{C}[u, s] \xrightarrow{L_u} \mathbb{C}[u, s] ds, \quad L_u = u \partial_s + (s - \rho)^m. \quad (\text{SP.4})$$

Under the same translation $ds = dy$ and $L_u = u \partial_y + y^m$. For a nonzero polynomial Q , the degree of $L_u Q$ in s is exactly $\deg Q + m$, with the same leading coefficient. Successive subtraction of its leading monomials therefore gives a unique remainder of degree below m . The complex is injective in degree zero and its degree-one cohomology is the free module

$$\mathcal{H} = \bigoplus_{b=0}^{m-1} \mathbb{C}[u] [s^b ds], \quad \mathcal{H}/u\mathcal{H} = E_h ds. \quad (\text{SP.5})$$

The unique-remainder proof also verifies the equality after specialization: at $u = 0$ the differential is multiplication by the original h , retaining its complete primary order.

20.2 The two boundary coefficients in the unchanged basis

The BC connection is defined by monic division of $\Phi_h(s)s^b$ by $h(s)$. Here that division is explicit:

$$\Phi_h(s)s^b = h(s) \frac{(s - \rho)s^b}{n} + cs^b, \quad q_b(s) = \frac{(s - \rho)s^b}{n}, \quad r_b(s) = cs^b. \quad (\text{SP.6})$$

For $0 \leq b < m$, its quotient derivative is already a remainder:

$$q'_b(s) = \frac{(b+1)s^b - b\rho s^{b-1}}{n}, \quad (\text{SP.7})$$

where the second term is zero for $b = 0$. Subtracting $L_u q_b$ in (SP.6) proves $[\Phi_h s^b] = [cs^b - u q'_b]$ in (SP.5). Thus the exact coefficient matrices and connection are

$$C_s = cI_m, \quad (B_s)_{bb} = \frac{b+1}{n}, \quad (B_s)_{b-1,b} = -\frac{b\rho}{n} \quad (1 \leq b < m), \quad \nabla_u = \partial_u - \frac{cI_m}{u^2} + \frac{B_s}{u}, \quad (\text{SP.8})$$

and all other entries of B_s are zero. The induced degree-one formula comes from the original chain connection $\nabla_u^1 = \partial_u - \Phi_h/u^2$ and $\nabla_u^0 = \partial_u - \Phi_h/u^2 + 1/u$. Indeed $[L_u, \partial_u] = -\partial_s$ and $[L_u, -\Phi_h/u^2] = -h/u$, so $\nabla_u^1 L_u = L_u \nabla_u^0$. This proves the connection on the stated complex, with the degree-zero correction and the marked scalar retained.

In the y -coordinates, the same division of $(y^n/n+c)y^a$ gives the quotient y^{a+1}/n . Consequently

$$PB_s P^{-1} = \text{diag}(\beta_0, \dots, \beta_{m-1}), \quad \beta_a = \frac{a+1}{n}, \quad PC_s P^{-1} = cI_m, \quad \text{Tr } B_s = \frac{m}{2}. \quad (\text{SP.9})$$

This also proves the matrix conjugation directly, without replacing the original matrix by one having the same eigenvalues.

20.3 The original ordered contours and their exact translation

Fix a branch $\log u = \log |u| + i \arg u$ on a simply connected sector of $\mathbb{C}^\times$. Put

$$\vartheta_j = \frac{\arg u + \pi}{n} + \frac{2\pi j}{n}, \quad \ell_j = \{re^{i\vartheta_j} : 0 \leq r < \infty\}, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq m), \quad (\text{SP.10})$$

with each ℓ_j oriented toward infinity. These are the original ordered contours in the s -plane. Their periods, in the original monomial columns, are

$$(\Pi_s(u))_{jb} = \int_{\Gamma_j} e^{\Phi_h(s)/u} s^b ds, \quad 1 \leq j \leq m, \quad 0 \leq b < m. \quad (\text{SP.11})$$

We prove the translation needed to evaluate these exact contours. For $a \in [0, 1]$, translate the broken contour by $a\rho$. Its two ends remain in the same ordered decay sectors, while its junction moves from 0 to $a\rho$. Truncate its two rays at radius R . Cauchy's theorem applies to the quadrilateral swept by each finite ray because the integrand is entire in s . The two junction paths have opposite orientations in $-\ell_0 + \ell_j$ and cancel. Each remaining connector has length at most $|\rho|$ and lies at $s = Re^{i\vartheta_l} + a\rho$, $l = 0, j$. Uniformly for $a \in [0, 1]$,

$$\operatorname{Re} \frac{\Phi_h(s)}{u} = -\frac{R^n}{n|u|} + O(R^{n-1}) + O(1). \quad (\text{SP.12})$$

This follows by expanding $(Re^{i\vartheta_l} + (a-1)\rho)^n$, whose leading term divided by nu has the displayed negative real value. The polynomial factor is $O((1+R)^b)$; the connector integrals therefore tend to zero as $R \rightarrow \infty$. The same estimate proves absolute convergence of all rays. It follows that each period over its original Γ_j equals the period over $\rho + \Gamma_j$. This is the exact sector-preserving contour deformation, with the original orientation and junction cancellation.

On the latter contour set $s = \rho + y$. The binomial expansion is exactly (SP.3), while (SP.2) retains the factor $e^{c/u}$. Let

$$\zeta = e^{2\pi i/n}, \quad W_{ja} = \zeta^{j(a+1)} - 1, \quad d_a = n^{\beta_a-1} e^{\pi i \beta_a} \Gamma(\beta_a), \quad D(u) = \operatorname{diag}(d_a u^{\beta_a})_{a=0}^{m-1}. \quad (\text{SP.13})$$

On the j -th y -ray, the substitution $v = r^n/(n|u|)$, with its specified positive real radial variable, gives

$$\int_{\ell_j} e^{y^n/(nu)} y^a dy = n^{\beta_a-1} e^{\pi i \beta_a} \Gamma(\beta_a) u^{\beta_a} \zeta^{j(a+1)}.$$

Subtracting the $j = 0$ ray proves the complete ordered period matrix in the original columns:

$$\boxed{\Pi_s(u) = e^{c/u} W D(u) P.} \quad (\text{SP.14})$$

Every factor in this equality has been specified, including the original primitive constant, the gamma constants, all powers of ρ , and the ordered contour phase.

For completeness, W is invertible. If its columns have a linear relation with coefficients $a_1, \dots, a_m$, then $\sum_{r=1}^m a_r (z^r - 1)$ vanishes at every nontrivial n -th root of unity. It also vanishes at 1. Its degree is at most $m = n-1$; division by the degree- n polynomial $z^n - 1$, or the elementary bound on the number of roots of a nonzero polynomial, makes it the zero polynomial. All its coefficients a_r are zero. The gamma factors in (SP.13) are positive-real gamma values times nonzero constants, since $0 < \beta_a < 1$. Thus (SP.14) is invertible for every $u \neq 0$. Its determinant is exactly

$$\det \Pi_s(u) = K_m u^{m/2} e^{mc/u}, \quad K_m = n^{-m/2} e^{\pi i m/2} \left(\prod_{a=0}^{m-1} \Gamma(\beta_a) \right) \det W \neq 0. \quad (\text{SP.15})$$

Here $\sum_a \beta_a = m/2$ and $\det P = 1$ prove every factor. This is the single-primary value of the original BC determinant, with $\operatorname{Tr} C_s = mc$.

20.4 Complete monodromy, its finite cover, and the boundary invariants

Under one positive circuit of $u = 0$, the ordered decay rays move ℓ_j to ℓ_{j+1} , with indices modulo n . Thus their transported relative cycles satisfy

$$\Gamma_j \mapsto \begin{cases} \Gamma_{j+1} - \Gamma_1, & 1 \leq j < m, \\ -\Gamma_1, & j = m. \end{cases} \quad (\text{SP.16})$$

The equality follows by writing each as $-\ell_1 + \ell_{j+1}$ and subtracting the defining $-\ell_0 + \ell_1$. It remains valid for $m = 1$, when the second case is the single map $\Gamma_1 \mapsto -\Gamma_1$. Let M_{cyc} be the matrix whose rows record (SP.16). The constant exponential is single-valued on $\mathbb{C}^\times$. Equation (SP.14) therefore gives

$$\Pi_s^{\text{cont}} = M_{\text{cyc}} \Pi_s, \quad M_{\text{cyc}} = W \text{diag}(\zeta, \zeta^2, \dots, \zeta^m) W^{-1}, \quad M_{\text{cyc}}^n = I, \quad \ker(M_{\text{cyc}} - I) = 0. \quad (\text{SP.17})$$

For example the row difference in (SP.16), applied to the column $W_{\bullet a}$, is $\zeta^{a+1}(\zeta^{j(a+1)} - 1)$, proving the stated diagonalization directly. None of $\zeta, \dots, \zeta^m$ equals one.

The signs for horizontal coefficient solutions can also be specified without an implicit duality convention. The connection (SP.8) acts on coefficient columns. Its horizontal equation is $x'_s = (cI/u^2 - B_s/u)x_s$, whose fundamental matrix is

$$H_s(u) = P^{-1} e^{-c/u} \text{diag}(u^{-\beta_a})_{a=0}^{m-1}. \quad (\text{SP.18})$$

Differentiate it and use (SP.9) to verify the equation. Its coefficient monodromy is conjugate to $\text{diag}(\zeta^{-1}, \dots, \zeta^{-m})$. This is the inverse convention to the period-cycle action in (SP.17). Its eigenvalue set is the same set of all nontrivial n -th roots, so it too has zero invariants.

On the finite ramified cover $u = v^n$, the pulled-back connection in the y -basis is exactly

$$\partial_v - \frac{nc}{v^{n+1}} I_m + \frac{1}{v} \text{diag}(1, 2, \dots, m). \quad (\text{SP.19})$$

The meromorphic basis change $x_y = \text{diag}(v^{-1}, \dots, v^{-m})z$ cancels precisely the integer diagonal term and leaves $\partial_v - nc v^{-n-1} I_m$. Its horizontal factor remains e^{-c/v^n} . It is not a meromorphic gauge removing that irregular factor when $c \neq 0$. Analytic continuation of the displayed fundamental matrix around $v = 0$ is, nevertheless, the identity. Thus the actual finite-cover unipotent monodromy logarithm is exactly

$$N_{\text{boundary}} = 0. \quad (\text{SP.20})$$

There is no uncomputed sector transition in this one-critical-value calculation: (SP.18) is an exact diagonal fundamental solution on every branch, and after (SP.19) all columns have the same explicit scalar exponential and integer powers. On two overlapping sectors with the same branch they are identical. This proves the absence of additional Stokes factors for this specified family; it is not a claim about a packet with several distinct critical values.

Let $j : \Delta^\times \hookrightarrow \Delta$ be a small complex disc punctured at zero, and let $\mathcal{V}_{\text{an}}$ denote either the period local system or its horizontal dual, with the convention just specified. Its stalk at zero under ordinary j_* is the invariant subspace: a local section is a vector unchanged by a circuit of the puncture. Equation (SP.17) or (SP.18) proves

$$(j_* \mathcal{V}_{\text{an}})_0 = 0, \quad (j! \mathcal{V}_{\text{an}})_0 = 0. \quad (\text{SP.21})$$

Moreover the circle cochain complex is $[\mathbb{C}^m \xrightarrow{M-I} \mathbb{C}^m]$; cutting the circle at one point and identifying its endpoints by M gives exactly this cellular differential. Since $M - I$ is invertible, both its kernel and cokernel are zero. Thus the derived local invariant and coinvariant groups also vanish. This statement concerns the local system specified above and does not replace the marked algebraic fibre $\mathcal{H}/u\mathcal{H}$ in (SP.5).

20.5 The same full marked fibre retains the arithmetic nilpotent

In the y -basis, let J_m be the matrix of multiplication by y on $\mathbb{C}[y]/y^m$; explicitly $(J_m)_{a+1,a} = 1$ for $0 \leq a < m-1$, with all other entries zero. Then

$$A_s = P^{-1}(\rho I_m + J_m)P, \quad N_{\text{ar}} := A_s - \rho I_m = P^{-1}J_mP, \quad J_m^m = 0, \quad J_m^{m-1} \neq 0. \quad (\text{SP.22})$$

For $m = 1$, the last statement means $J_1^0 = I_1 \neq 0$ and $J_1 = 0$; its nilpotency index is one. These identities follow by applying multiplication to each of the ordered monomials; no monodromy identification is used. They reconstruct the full original companion multiplication matrix, since the polynomial identity $(s - \rho)^m = 0$ determines its last column and (SP.3) its complete change of coordinates.

The relation between this arithmetic action and the boundary connection is also explicit. From $\beta_{a+1} - \beta_a = 1/n$,

$$[B_s, A_s] = \frac{1}{n}N_{\text{ar}}, \quad [\nabla_u, A_s] = \frac{N_{\text{ar}}}{nu}. \quad (\text{SP.23})$$

The first identity is verified on every y^a , including the last one where $J_m y^{m-1} = 0$, and then conjugated by P . The scalar $C_s = cI$ commutes with A_s , and A_s is constant in u , proving the second. Thus for $m > 1$ this marked arithmetic action is not a horizontal endomorphism of the displayed boundary connection. The exact nonhorizontal term has been computed, rather than discarding the action.

It can be transported through the actual period isomorphism. Set $D_0 = \text{diag}(d_a)$, with the constants in (SP.13), and

$$K_0 = W D_0 J_m D_0^{-1} W^{-1}.$$

The scalar exponential cancels by its actual inverse in conjugation, and (SP.14) gives

$$\Pi_s A_s \Pi_s^{-1} = \rho I_m + u^{1/n} K_0, \quad K_0^m = 0, \quad K_0^{m-1} \neq 0, \quad M_{\text{cyc}} K_0 M_{\text{cyc}}^{-1} = \zeta K_0. \quad (\text{SP.24})$$

Indeed the $(a+1, a)$ entry of $\text{diag}(u^\beta) J_m \text{diag}(u^{-\beta})$ is $u^{1/n}$. The same calculation with ζ^{a+1} proves the last identity. Differentiating (SP.24) retains the exact $1/(nu)$ commutator in (SP.23). This is the concrete connection between the nonzero arithmetic nilpotent and the finite boundary monodromy, on the same rank- m space at each $u \neq 0$.

20.6 The complete Taylor unit and the original theta source

The entire quotient and its full Taylor coefficients are

$$v_h(s) = \frac{g(s)}{(s - \rho)^m}, \quad a_j = \frac{v_h^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!} \quad (0 \leq j < m), \quad a_0 \neq 0. \quad (\text{SP.25})$$

The equality follows by dividing the convergent Taylor series of g , whose first nonzero term is of order m . In the unchanged marked fibre its multiplication matrix is

$$U_s = P^{-1} \left(\sum_{j=0}^{m-1} a_j J_m^j \right) P, \quad U_s^{-1} = P^{-1} \left(\sum_{j=0}^{m-1} b_j J_m^j \right) P, \quad b_0 = a_0^{-1}, \quad b_r = -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}. \quad (\text{SP.26})$$

Multiplying the two finite series proves the inverse: the constant coefficient is one and every coefficient of J_m^r , $1 \leq r < m$, is zero by the displayed recursion. Terms of degree at least m vanish by $J_m^m = 0$. All coefficients a_j , including their derivative constants, remain in the same matrix. When $u \neq 0$, the occurrence of this same U_s on the displayed remainder lattice means its constant coefficient extension in that specified basis. It is not an assertion of an unrecorded multiplication operation on twisted polynomial cohomology. The exact multiplication defect is calculated below.

Retain the original function spaces $V, \mathcal{B}$, Euler operator $D = -x\partial_x$, Fourier convention, theta map $\Theta\phi(x) = 2\sum_{r\geq 1}\phi(rx)$, and seed

$$\phi_* = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g, \quad \mathcal{M}F(s) = \int_0^\infty F(x)x^s \frac{dx}{x}.$$

Specifically V is the even Schwartz subspace with $\phi(0) = \int \phi = 0$, and $\mathcal{B}$ consists of smooth functions on $\mathbb{R}_{>0}$ with finite seminorms $\sup_{x>0} x^b |D^r F(x)|$ for every integer b and every nonnegative integer r . The original iterated Euler inverse gives $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$. The actual polynomial maps are

$$\alpha_h(P) = P(D)\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h, \quad \Theta\alpha_h(P) = \mathcal{T}_h(hP), \quad J_h\mathcal{T}_h(P) = U_s[P]_s. \quad (\text{SP.27})$$

Here $J_h = j_h\mathcal{M}$ and $[P]_s$ is the original remainder coordinate column. These are the inherited exact Euler/theta identities in original marked equations (18)–(19) and BC's source square. In particular $[\mathbb{C}[s] \xrightarrow{h} \mathbb{C}[s]ds]$ maps to $[V \xrightarrow{\Theta} \mathcal{B}]$ by $(\alpha_h, \mathcal{T}_h)$. Its degree-one injection is $\sigma_h U_s : E_h \hookrightarrow Q_{\text{ar}} = \mathcal{B}/\Theta V$, where σ_h is the original corrected section.

The coordinate change is realized on these exact source vectors:

$$\mathcal{T}_h(y^b) = (D - \rho)^b F_h, \quad (\mathcal{T}_h(s^b))_{b=0}^{m-1} = (\mathcal{T}_h(y^a))_{a=0}^{m-1} P. \quad (\text{SP.28})$$

The binomial theorem for the commuting scalar ρ and operator D proves both identities. Thus the Pascal map, full nilpotent and Taylor unit have explicit source representatives on the same theta complex. At general u , the original source deformation is still $\Theta + u\mathcal{T}_h\partial_s\alpha_h^{-1}$ on $\alpha_h(\mathbb{C}[s])$; its square with (SP.4) commutes. There is no assertion that the unit-valued coefficient map turns this deformed differential into the undeformed theta map.

Under the period map, (SP.26) has the exact image

$$\Pi_s U_s \Pi_s^{-1} = \sum_{j=0}^{m-1} a_j u^{j/n} K_0^j, \quad [\nabla_u, U_s] = \frac{1}{nu} P^{-1} \left(\sum_{j=1}^{m-1} j a_j J_m^j \right) P. \quad (\text{SP.29})$$

The first follows by taking powers in (SP.24); the second uses $[B_y, J_m^j] = (j/n)J_m^j$, proved by induction from (SP.23). These formulas specify exactly how every Taylor coefficient is transported and why its arithmetic multiplication is not silently identified with a constant flat monodromy operator.

More explicitly put $v_{\text{pol}}(s) = \sum_{j=0}^{m-1} a_j (s - \rho)^j$, the exact original Taylor remainder of v_h . For every polynomial $Q(s)$, the Leibniz rule in the original differential gives

$$v_{\text{pol}} L_u Q = L_u(v_{\text{pol}} Q) - u v'_{\text{pol}} Q. \quad (\text{SP.30})$$

Indeed the $h v_{\text{pol}} Q$ terms agree and the derivative of $v_{\text{pol}} Q$ contributes exactly the additional $u v'_{\text{pol}} Q$. At the marked fibre $u = 0$ this is ordinary multiplication of the full class $v_h \in E_h^\times$, with inverse (SP.26). Away from that fibre it records precisely the derivative term needed to compare multiplication on representatives with the specified coefficient extension U_s . For a remainder polynomial p of degree less than m , divide $v_{\text{pol}} p = hq + r$ monically. The degree of q is at most $m - 2$, with $q = 0$ when the product has degree less than m . Consequently q' already has degree below m , and

$$[v_{\text{pol}} p]_{\mathcal{H}_u} = [r - uq']_{\mathcal{H}_u}. \quad (\text{SP.31})$$

Subtracting $L_u q = hq + uq'$ proves the identity with its specified correction. The constant matrix U_s gives the coefficient r ; (SP.31) computes exactly its difference from multiplying these representatives and then taking the differential remainder. No term involving v'_h or its Taylor coefficients has been deleted.

20.7 The finite-cover lattice and the original source metric

The marked lattice also has an explicit image on the cover $u = v^n$. After retaining and factoring its common scalar e^{c/v^n} in the period expression, its columns are

$$e^{-c/v^n} \Pi_s(v^n) = W D_0 \operatorname{diag}(v, v^2, \dots, v^m) P. \quad (\text{SP.32})$$

This is an equality on the punctured disc that also defines a holomorphic matrix on the whole v -disc. The constant invertible matrices $W D_0, P$ identify its image lattice with $\bigoplus_{a=0}^{m-1} v^{a+1} \mathbb{C}\{v\}$. Its cokernel in the displayed ambient holomorphic free module is consequently

$$\bigoplus_{a=0}^{m-1} \mathbb{C}\{v\}/(v^{a+1}), \quad \text{length} = \sum_{a=0}^{m-1} (a+1) = \frac{m(m+1)}{2}. \quad (\text{SP.33})$$

Both assertions follow by multiplying by those constant inverses and computing the image and quotient in each one-dimensional coordinate. The image lattice itself has a rank- m quotient by multiplication by v , as does the pulled-back marked lattice. Its inclusion into the ambient constant lattice induces zero modulo v , since every column contains a positive power of v . Thus the vanishing boundary invariants and the nonzero marked fibre have their exact lattice map and cokernel recorded; neither is used to erase the other's coefficients. The scalar exponential factored in (SP.32) remains an irregular factor and is not being declared a meromorphic gauge.

In the original source norm, retain the positive matrix

$$(G_\theta)_{ab} = \frac{1}{2\pi} \int_{\mathbb{R}} \overline{(1/2 + it)^a} (1/2 + it)^b |v_h(1/2 + it)|^2 dt, \quad 0 \leq a, b < m. \quad (\text{SP.34})$$

Mellin–Plancherel identifies this with the Gram of $\mathcal{T}_h(s^b)$ in the actual $L^2(dx)$ source. The source moments are finite by the inherited arithmetic source estimates. A zero polynomial-vector norm would make its entire Mellin transform $P v_h$ zero; since $v_h \neq 0$, this forces $P = 0$. Hence G_θ is positive definite, with its original mass and full multiplier retained. The actual comparison endomorphism is

$$T_{\text{per}}(u) = G_\theta^{-1} \Pi_s(u)^* \Pi_s(u) = e^{2\operatorname{Re}(c/u)} G_\theta^{-1} P^* D(u)^* W^* W D(u) P. \quad (\text{SP.35})$$

Its positivity and self-adjointness are in G_θ : $x^* G_\theta T_{\text{per}} x = \|\Pi_s x\|^2 > 0$ for $x \neq 0$. The ratio of its largest and smallest eigenvalues is unchanged by its explicit positive scalar $e^{2\operatorname{Re}(c/u)}$. Every other factor in (SP.35), including the original Pascal matrix and G_θ , remains. In particular the scalar cancellation alone gives no estimate for that ratio.

Fix $u \neq 0$. For a larger polynomial source $\mathbb{C}[s]_{\leq N}$, $N \geq m-1$, let $J_{u,N}$ be the unique differential remainder for (SP.4). It is onto the same m -dimensional remainder module and has exact kernel

$$\ker J_{u,N} = L_u(\mathbb{C}[s]_{\leq N-m}), \quad \operatorname{Per}_N = \Pi_s(u) J_{u,N}. \quad (\text{SP.36})$$

When $N = m-1$ the relation space is zero. The degree argument for (SP.5) proves the kernel and surjectivity. For the period factorization, each polynomial differential relation has integral

$$\int_{\Gamma_j} e^{\Phi_h/u} L_u Q ds = u \int_{\Gamma_j} d(e^{\Phi_h/u} Q) = 0;$$

the endpoint values vanish by the same polynomial decay estimate (SP.12), and the junction endpoints cancel with the original orientations. Thus the period Gram at degree N has rank exactly m , with kernel (SP.36). Its restriction to the displayed m -dimensional remainder frame is positive definite. It is not a positive definite Gram on an enlarged source with more columns.

20.8 Support labels and the scope of the computed boundary

All constant coefficient maps above act at each original nonbottom support label by $(A, x) \mapsto (A, fx)$, with external $\tau \mapsto \tau$. The Pascal matrix, its inverse, the finite Taylor-unit matrix and its inverse preserve the labelled coefficient space. A killed coefficient is its represented zero at that label. The coefficient specialization at $u = 0, t = 0$ sends represented parameter values to the represented zero e , as in the original marked structural map $\mathbb{F}_{1,\tau} \rightarrow G(\mathbb{C}[u, t]) \rightarrow G(\mathbb{C})$. It does not send the marked primary fibre to external absence.

Equations (SP.17)–(SP.21) compute the actual one-factor boundary monodromy, invariant stalk and finite-cover nilpotent logarithm. Equations (SP.22)–(SP.33) reconstruct the original arithmetic action, full Taylor unit, theta representatives and marked lattice through explicit maps. They do not identify N_{boundary} with N_{ar} . These conclusions are restricted to this complete single-primary packet at $t = 0$; tensor products may multiply its characters to the trivial character, and no assertion of vanishing tensor invariants is made.

21 The finite-field sheaf of the single-primary phase

21.1 Specified coefficient specialization and all constants

Let the specified single primary polynomial be

$$h(s) = (s - \rho)^m, \quad m \geq 1, \quad n = m + 1. \quad (\text{SPF.1})$$

Retain the given coefficient ring R , including ρ , $1/n!$, and every coefficient of the full Taylor class $v_h = j_h(g/h)$ and its inverse in the retained m -dimensional remainder basis. The finite-field datum for this calculation is the specified ring homomorphism $\alpha : R \rightarrow k_0 = \mathbf{F}_{Q_0}$, of characteristic $p > n$. All coefficients, including the unit and its inverse, are mapped by this same homomorphism. We continue to write ρ for $\alpha(\rho)$. At $t = 0$, the specified primitive with value zero at $s = 0$ is exactly

$$\Phi(s) = \frac{(s - \rho)^n - (-\rho)^n}{n}, \quad \Phi'(s) = h(s), \quad c = -\frac{(-\rho)^n}{n}. \quad (\text{SPF.2})$$

In particular, the constant c is retained throughout. The only constant-field extension we require is

$$k = \mathbf{F}_Q, \quad Q = Q_0^d, \quad d = \text{ord}_{(\mathbf{Z}/n\mathbf{Z})^\times}(Q_0). \quad (\text{SPF.3})$$

Thus $n \mid Q - 1$. This is an explicit finite extension because $\gcd(Q_0, n) = 1$. All original coefficients are mapped by $R \xrightarrow{\alpha} k_0 \hookrightarrow k$.

Fix $\ell \neq p$, work over $\overline{\mathbf{Q}}_\ell$, and fix a nontrivial additive character $\psi_0 : k_0 \rightarrow \overline{\mathbf{Q}}_\ell^\times$. On k use $\psi = \psi_0 \circ \text{Tr}_{k/k_0}$. For a further extension $k_r = \mathbf{F}_{Q^r}$, write

$$\psi_r = \psi \circ \text{Tr}_{k_r/k}, \quad \chi_r = \chi \circ \text{N}_{k_r/k}. \quad (\text{SPF.4})$$

Every Frobenius below is geometric Frobenius, the inverse of $x \mapsto x^{Q^r}$ at a k_r -rational point. Define $\mathcal{L}_\psi(f)$ by the Artin–Schreier torsor convention whose trace there is $\psi_r(f)$. Define the multiplicative Kummer sheaf $\mathcal{K}_\chi$ by the convention whose trace there is $\chi_r(x)$. These conventions fix both inverse characters and cohomological signs.

Let

$$\begin{aligned} \pi_s : \mathbf{A}_s^1 \times \mathbf{G}_{m,u} &\longrightarrow \mathbf{G}_{m,u}, \\ \mathcal{F}_0 &= R^1\pi_{s!}\mathcal{L}_{\psi_0}(\Phi(s)/u) \quad \text{over } k_0, \\ \mathcal{F} &= (\mathcal{F}_0)_k = R^1\pi_{s!}\mathcal{L}_\psi(\Phi(s)/u) \quad \text{over } k. \end{aligned} \quad (\text{SPF.5})$$

The equality is compact-support base change. Our computation concerns this precise sheaf. It does not replace the specified phase by an expression involving g/h , and does not remove that unit from the original coefficient ring.

The source isomorphism

$$T : \mathbf{A}_y^1 \times \mathbf{G}_{m,u} \longrightarrow \mathbf{A}_s^1 \times \mathbf{G}_{m,u}, \quad T(y, u) = (y + \rho, u), \quad T^{-1}(s, u) = (s - \rho, u) \quad (\text{SPF.6})$$

fixes the coefficient ring and satisfies $\pi_s T = \pi_y$. It preserves compact supports because it is an isomorphism. For any original coefficient function $v(s)$, the transported function is exactly $v(y + \rho)$; for an invertible such function the inverse is exactly $v^{-1}(y + \rho)$. Direct substitution into (SPF.2), and the additive torsor tensor-product law, give

$$T^* \mathcal{L}_\psi(\Phi(s)/u) = \mathcal{L}_\psi(y^n/(nu)) \otimes \pi_y^* \mathcal{L}_\psi(c/u). \quad (\text{SPF.7})$$

The tensor-product law follows by adding two torsor coordinates: if $z_i^p - z_i = f_i$, their sum satisfies the equation for $f_1 + f_2$; the associated one-dimensional character representations multiply. Projection formula therefore yields

$$\mathcal{F} = \mathcal{L}_\psi(c/u) \otimes R^1 \pi_{y!} \mathcal{L}_\psi(y^n/(nu)). \quad (\text{SPF.8})$$

21.2 The actual power-map splitting, including its zero stalk

Put $X_n = \{\chi : k^\times \rightarrow \overline{\mathbf{Q}}_\ell^\times : \chi^n = 1\}$, so $\#X_n = n$, and let $q : \mathbf{A}_y^1 \rightarrow \mathbf{A}_z^1$ be $z = y^n$. Let $j : \mathbf{G}_{m,z} \hookrightarrow \mathbf{A}_z^1$. Then the following is an isomorphism of sheaves over k , not merely an equality of trace functions:

$$q_* \overline{\mathbf{Q}}_\ell \simeq \overline{\mathbf{Q}}_{\ell, \mathbf{A}_z^1} \oplus \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} j_! \mathcal{K}_\chi. \quad (\text{SPF.9})$$

Here is the explicit splitting. The deck group on $\mathbf{G}_m$ is $\mu_n(k)$. On the stalk with basis e_r , indexed by $r^n = z$, set $D_\zeta e_r = e_{\zeta r}$. For $\chi \in X_n$ there is a unique character $\lambda_\chi : \mu_n(k) \rightarrow \overline{\mathbf{Q}}_\ell^\times$ with

$$\lambda_\chi(x^{(Q-1)/n}) = \chi(x) \quad (x \in k^\times), \quad e_\chi = \frac{1}{n} \sum_{\zeta \in \mu_n(k)} \lambda_\chi(\zeta) D_\zeta. \quad (\text{SPF.10})$$

Uniqueness and existence follow because $x \mapsto x^{(Q-1)/n}$ surjects onto $\mu_n(k)$, with kernel the subgroup of n th powers, on which χ is trivial. Character orthogonality gives $e_\chi e_{\chi'} = \delta_{\chi, \chi'} e_\chi$ and $\sum_{\chi \in X_n} e_\chi = 1$. Its image is the λ_χ^{-1} -eigenline of the operators D_ζ . For $z = x \in k_r^\times$, a chosen root has $r^{Q^r} = x^{(Q^r-1)/n} r$; geometric Frobenius consequently acts on this eigenline by $\lambda_\chi(x^{(Q^r-1)/n}) = \chi_r(x)$. It is therefore the stated Kummer summand with the stated descent convention.

At $z = 0$, the geometric fibre of q has one point. Its constant-sheaf stalk has trivial deck action. Thus e_χ has zero image there for $\chi \neq 1$, and the trivial projector has a one-dimensional stalk. The unit map from the constant sheaf identifies the latter summand with the constant sheaf on $\mathbf{A}_z^1$. Each other summand is canonically $j_! \mathcal{K}_\chi$: the adjunction map is an isomorphism at every geometric stalk, on $\mathbf{G}_m$ and at zero. Finally, q is finite, so its higher direct images on this sheaf vanish; the argument has neither suppressed the ramified fibre nor added a term at it.

21.3 Compact-support cohomology and the negative Gauss line

For a nonzero scalar a , write

$$\mathcal{E}_{\chi, a} = \mathcal{K}_\chi(z) \otimes \mathcal{L}_\psi(az) \quad \text{on } \mathbf{G}_{m, z}, \quad H_\chi = H_c^1(\mathbf{G}_{m, \bar{k}}, \mathcal{E}_{\chi, 1}). \quad (\text{SPF.11})$$

For $\chi \neq 1$ one has

$$H_c^i(\mathbf{G}_{m, \bar{k}}, \mathcal{E}_{\chi, a}) = 0 \quad (i \neq 1), \quad \dim H_\chi = 1. \quad (\text{SPF.12})$$

We give the complete application of the curve formulas. A lisse sheaf on the connected, nonproper curve $\mathbf{G}_m$ has no nonzero compactly supported section: a nonzero locally constant section has support the whole curve. Thus $H_c^0 = 0$. Poincaré duality identifies H_c^2 with the Tate-twisted dual of the global invariants of the dual sheaf. Those invariants are zero, since the Artin–Schreier factor at infinity has nontrivial wild inertia and the Kummer factor is tame. The Swan conductors at zero and infinity are respectively 0 and 1. The second value also follows from the direct local calculation below, with the local parameter z^{-1} and nonzero numerator a . The Euler–Poincaré formula for this rank-one sheaf gives

$$\chi_c(\mathbf{G}_m, \mathcal{E}_{\chi,a}) = 1 \cdot \chi_c(\mathbf{G}_m, \overline{\mathbf{Q}}_\ell) - \text{Swan}_0 - \text{Swan}_\infty = 0 - 0 - 1 = -1. \quad (\text{SPF.13})$$

Together with the two vanishing groups and the cohomological dimension bound for a curve, this proves (SPF.12). The formulas used here, including their proofs and geometric Frobenius convention, are in Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, Sections 2.1–2.3, especially 2.3.1–2.3.3.

The trivial summand of (SPF.9) instead contributes

$$H_c^i(\mathbf{A}_k^1, \mathcal{L}_\psi(az)) = 0 \quad \text{for every } i, a \neq 0. \quad (\text{SPF.14})$$

Indeed $H_c^0 = 0$ for the same support reason, $H_c^2 = 0$ by its nontrivial wild inertia at infinity and duality, and Euler–Poincaré gives $1 - 1 = 0$, hence $H_c^1 = 0$. This proves why the final rank is $n - 1$, with no trivial character contribution.

Set

$$G_k(\chi, \psi) = \sum_{z \in k^\times} \chi(z) \psi(z). \quad (\text{SPF.15})$$

The trace formula, with (SPF.12), gives the exact one-dimensional geometric Frobenius action

$$\text{Fr}_Q | H_\chi = -G_k(\chi, \psi). \quad (\text{SPF.16})$$

The minus sign is the cohomological sign in degree one. There is no Tate twist in this identity.

This computation is an isomorphism of the entire parameter sheaf. On $\mathbf{G}_{m,z} \times \mathbf{G}_{m,a}$, use the source isomorphism $(z, a) \mapsto (w = az, a)$, whose inverse is $(w, a) \mapsto (w/a, a)$. The multiplicative Kummer torsor law gives

$$\mathcal{K}_\chi(w/a) \otimes \mathcal{L}_\psi(w) \simeq \text{pr}_w^*(\mathcal{K}_\chi(w) \otimes \mathcal{L}_\psi(w)) \otimes \text{pr}_a^* \mathcal{K}_{\chi^{-1}}(a). \quad (\text{SPF.17})$$

This law follows by multiplying or dividing the root coordinates of Kummer torsors; the associated character representations multiply or invert accordingly. The isomorphism preserves compact supports in every fibre, so projection formula and base change identify the parameter sheaf with $H_\chi \otimes \mathcal{K}_{\chi^{-1}}(a)$. Here H_χ is the pullback from $\text{Spec } k$ of the one-dimensional Galois module in (SPF.16); it is geometrically constant.

21.4 The exact decomposition on the original parameter

For a nonzero scalar γ , denote by $\mathcal{A}_\gamma$ the geometrically constant one-dimensional sheaf with geometric Fr_Q -eigenvalue γ . In its use below γ is the actual eigenvalue in (SPF.16), or that eigenvalue times a root of unity, so this notation denotes an existing continuous Galois representation. Combining the preceding source maps with $a = 1/(nu)$ gives

$$\boxed{\mathcal{F} \simeq \mathcal{L}_\psi(c/u) \otimes \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} \mathcal{A}_{-G_k(\chi, \psi)} \otimes \mathcal{K}_{\chi^{-1}}(1/(nu)).} \quad (\text{SPF.18})$$

In particular, $\mathcal{F}$ is lisse of rank $n - 1 = m$. The inverse Kummer character here is essential. Since inversion twice returns χ and multiplication by the constant n has trace $\chi(n)$, the same isomorphism on $\mathbf{G}_{m,u}$ is exactly

$$\boxed{\mathcal{F} \simeq \mathcal{L}_\psi(c/u) \otimes \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} \mathcal{A}_{-\chi(n)G_k(\chi, \psi)} \otimes \mathcal{K}_\chi(u).} \quad (\text{SPF.19})$$

Neither the factor n nor the additive factor c/u has been dropped.

For completeness this determines every finite-extension trace. Applying the trace formula to the base change of the one-dimensional group H_χ proves directly

$$-G_{k_r}(\chi_r, \psi_r) = (-G_k(\chi, \psi))^r. \quad (\text{SPF.20})$$

Thus the usual lifting sign is derived from this actual group, without an independent convention for a normalized Gauss sum. For $u \in k_r^\times$,

$$\begin{aligned} \text{Tr}(\text{Fr}_{Q^r, u} | \mathcal{F}_{\bar{u}}) &= - \sum_{s \in k_r} \psi_r(\Phi(s)/u) \\ &= -\psi_r(c/u) \sum_{\substack{\chi \in X_n \\ \chi \neq 1}} \chi_r(nu) G_{k_r}(\chi_r, \psi_r) \\ &= \psi_r(c/u) \sum_{\substack{\chi \in X_n \\ \chi \neq 1}} (-\chi(n)G_k(\chi, \psi))^r \chi_r(u). \end{aligned} \quad (\text{SPF.21})$$

To verify the middle equality also by finite sums, translate s to $y = s - \rho$. The number of n th roots of $z \in k_r^\times$ is $\sum_{\chi \in X_n} \chi_r(z)$. There is one root at zero. The trivial-character part is $1 + \sum_{z \neq 0} \psi_r(az) = 0$. For each other character the change of variable $w = az$ gives $\chi_r(a)^{-1} G_{k_r}(\chi_r, \psi_r)$; here $a^{-1} = nu$. In the last equality, $\chi_r(n) = \chi(n)^r$, since $n \in k^\times$ and its extension-field norm is n^r .

21.5 Inertia at zero, including a zero constant specialization

Let I_0 denote geometric inertia over $\bar{k}((u))$, let P_0 be its wild subgroup, and let $V = \mathcal{F}_{\bar{\eta}}$. Write $\psi(x) = \vartheta(\text{Tr}_{k/\mathbf{F}_p}(bx))$, where $\vartheta : \mathbf{F}_p \rightarrow \bar{\mathbf{Q}}_\ell^\times$ is a fixed nontrivial additive character and $b \in k^\times$. Such a unique b exists by nondegeneracy of the finite-field trace pairing. Then $\mathcal{L}_\psi(c/u) = \mathcal{L}_\vartheta(bc/u)$. Put $d_0 = bc$.

First suppose $c \neq 0$. The Artin–Schreier equation

$$z^p - z = d_0/u \quad (\text{SPF.22})$$

defines a degree- p extension of $\bar{k}((u))$. Indeed $f^p - f = d_0/u$ has no Laurent-series solution: if f has a pole of order $r > 0$, then $f^p - f$ has pole order pr , whereas the right side has order one; if f has no pole, neither does $f^p - f$. An Artin–Schreier polynomial of prime degree with no root is irreducible, since its roots form one translate of $\mathbf{F}_p$ and its Galois group is a subgroup of that group, hence has order either one or p .

Let v_E be its integer-valued normalized valuation, and write $v_E(u) = e$ for the ramification index. The equation forces $v_E(z) < 0$, hence $p v_E(z) = -e$. Thus $p \mid e$; since $1 \leq e \leq [E : \bar{k}((u))] = p$, one has $e = p$ and $v_E(z) = -1$. The extension is therefore totally ramified and z^{-1} is a uniformizer. For $\sigma_a(z) = z + a$, $a \in \mathbf{F}_p^\times$,

$$v_E(\sigma_a(z^{-1}) - z^{-1}) = v_E\left(\frac{-a}{z(z+a)}\right) = 2. \quad (\text{SPF.23})$$

The lower ramification groups are consequently $G_0 = G_1 = \mathbf{F}_p$, $G_2 = 1$; their unique positive upper break is one. Every nontrivial character of this group has Swan conductor one. In particular the character ε_{d_0} supplied by $\mathcal{L}_{\mathcal{O}}(d_0/u)$ is nontrivial on P_0 . All geometrically constant factors of (SPF.19) are trivial on I_0 , and all Kummer factors are trivial on P_0 . Thus

$$V|_{P_0} \simeq \varepsilon_{d_0}^{\oplus(n-1)}, \quad V^{P_0} = 0, \quad V^{I_0} = 0, \quad \text{Swan}_0(\mathcal{F}) = n - 1 \quad (c \neq 0). \quad (\text{SPF.24})$$

The first vanishing follows explicitly by choosing $\sigma \in P_0$ with $\varepsilon_{d_0}(\sigma) \neq 1$: an invariant vector v would satisfy $(\varepsilon_{d_0}(\sigma) - 1)v = 0$, hence $v = 0$. Swan conductors add under direct sum.

Now suppose $c = 0$. This occurs exactly when $\rho = 0$ in the specified residue field, since $n \neq 0$ there. The additive factor is the constant sheaf, and

$$V|_{I_0} \simeq \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} \mathcal{K}_{\chi}|_{I_0}, \quad V^{I_0} = 0, \quad \text{Swan}_0(\mathcal{F}) = 0 \quad (c = 0). \quad (\text{SPF.25})$$

Each displayed character is nontrivial on tame inertia: the totally ramified Kummer cover $u = v^n$ has deck group μ_n , and its summand is the nontrivial λ_{χ}^{-1} -character proved in (SPF.10). An invariant in a direct sum of these eigenlines has invariant components in each, hence is zero. Thus this specialization does not produce an invariant line.

Here is one finite cover that kills all geometric inertia in either case and keeps track of the additive factor:

$$u = v^n, \quad z^p - z = \frac{bc}{v^n}. \quad (\text{SPF.26})$$

Both maps are finite etale on the punctured u -line or its pullback: their derivatives there are $nv^{n-1} \neq 0$ and $-1 \neq 0$. The pullback of $\mathcal{K}_{\chi}(u)$ is $\mathcal{K}_{\chi^n}(v)$, hence constant. The pullback of the Artin–Schreier factor is trivialized by the displayed torsor itself. For $c = 0$, the first map alone suffices and the second is a disjoint trivial torsor. For $c \neq 0$, the second map remains a nontrivial degree- p extension after the first: its right side has pole order n , and $p \nmid n$, so the preceding pole-order argument applies. The composite has degree np . Constants $\mathcal{A}_{-\chi(n)G_k}$ can retain arithmetic Frobenius action; they have no geometric inertia action.

On the resulting finite-index subgroup of I_0 , every monodromy matrix is the identity. Its unipotent logarithm is therefore exactly

$$N = \log(1) = 0. \quad (\text{SPF.27})$$

Equivalently, the original inertia representation has finite image and is a direct sum of finite-order characters, so its unipotent part is already the identity. Formula (SPF.27) and the vanishing $V^{I_0} = 0$ hold simultaneously. Invariants refer to the original group I_0 ; after the cover its restricted representation is trivial, so the new inertia invariants are the whole rank- m vector space.

21.6 Purity proved by the exact Gauss calculation

For every complex embedding of the cyclotomic field of the character values, and every nontrivial χ ,

$$|G_k(\chi, \psi)|^2 = \sum_{t \in k^{\times}} \chi(t) \sum_{y \in k^{\times}} \psi((t-1)y) = (Q-1) - \sum_{\substack{t \in k^{\times} \\ t \neq 1}} \chi(t) = Q. \quad (\text{SPF.28})$$

The first equality expands the square of the sum, uses the complex conjugates $\overline{\chi(y)} = \chi(y)^{-1}$ and $\overline{\psi(y)} = \psi(-y)$, and sets $x = ty$. The inner sum is $Q-1$ at $t = 1$ and -1 otherwise, by nontrivial additive-character orthogonality. Multiplicative-character orthogonality gives $\sum_{t \neq 0} \chi(t) = 0$, so the sum over $t \neq 1$ is -1 . This proves every equality, with the original unscaled Gauss sum.

Every local eigenvalue in (SPF.21) is

$$\psi_r(c/u) \left(-\chi(n)G_k(\chi, \psi) \right)^r \chi_r(u), \quad \left| \psi_r(c/u) \left(-\chi(n)G_k(\chi, \psi) \right)^r \chi_r(u) \right| = Q^{r/2}. \quad (\text{SPF.29})$$

All character factors have absolute value one, and (SPF.28) gives the last equality. Thus the specified sheaf is pointwise pure of weight one. No division by $Q^{1/2}$, half Tate twist, or change of the original phase was made. Over k_0 the same purity follows as well: for a closed point of degree e , pass to a constant extension in which its Frobenius has a positive power covered by (SPF.29). If its original eigenvalue is β and that power is r , the computed equality $|\beta^r| = Q_0^{er/2}$ implies $|\beta| = Q_0^{e/2}$. The inertia calculation is geometric and is unchanged by constant-field extension.

This establishes the exact finite-field decomposition, purity, invariant-space vanishing, and finite-cover logarithm of the sheaf in (SPF.5). The finite-field parameter and its Frobenius eigenvalues have been constructed here. No complex off-critical zeta zero or identification of such a zero with one of these finite-field Frobenius eigenvalues is constructed by this calculation.

Primary sources for the cohomological foundations. The compact-support curve formulas and geometric Frobenius convention used above are Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups* (Annals of Mathematics Studies 116), Sections 2.1–2.3; the unscaled Gauss convention is Section 4.0. The Artin–Schreier torsor construction is the exact sequence in the Stacks Project, Section 59.63, tag 0A3J; the Kummer torsor construction is Section 59.28, tag 03PK. All specializations, source maps, power-map projectors, Gauss factors, and local ramification computations needed for this particular sheaf have been proved above.

22 Ordered tensor primary packets: invariant characters and the actual arithmetic action

This calculation takes the ordered tensor product of the complete single-primary marked family, with a common parameter u and with every original t -parameter fixed at zero. It does not pass to a symmetric quotient. Its source is the full packet attached to a selected actual zero when such a packet is given; it asserts the existence of no zero off the critical line. The constant in the original primitive, the original monomial coordinates, and the full source unit are retained.

22.1 The original ordered family and its period matrix

Let ρ be the selected zero of $g = 2\xi$ of complete order $m \geq 1$, put $N = m + 1$, and let $k \geq 1$ be an integer. For $1 \leq i \leq k$, put $y_i = s_i - \rho$. The precise phase and marked algebra are

$$\begin{aligned} h(s) &= (s - \rho)^m, & c &= -\frac{(-\rho)^N}{N}, \\ \Phi_k(s_1, \dots, s_k) &= \sum_{i=1}^k \frac{(s_i - \rho)^N - (-\rho)^N}{N} = \sum_{i=1}^k \frac{y_i^N}{N} + kc, \\ E_k &= \mathbb{C}[s_1, \dots, s_k] / (h(s_1), \dots, h(s_k)). \end{aligned} \tag{TP.1}$$

Each primitive is zero at its original base point $s_i = 0$. Index every ordered basis lexicographically by $\mathbf{b} = (b_1, \dots, b_k) \in \{0, \dots, m-1\}^k$. Write $s^{\mathbf{b}} = \prod_i s_i^{b_i}$, $y^{\mathbf{b}} = \prod_i y_i^{b_i}$, $|\mathbf{b}| = \sum_i b_i$, and $L(\mathbf{b}) = |\mathbf{b}| + k$. The unchanged top differential is $ds_1 \wedge \dots \wedge ds_k = dy_1 \wedge \dots \wedge dy_k$.

The one-factor Pascal matrix and its ordered tensor power are

$$P_{ab} = \begin{cases} \binom{b}{a} \rho^{b-a} & a \leq b, \\ 0 & a > b, \end{cases} \quad (P^{-1})_{ab} = \begin{cases} \binom{b}{a} (-\rho)^{b-a} & a \leq b, \\ 0 & a > b, \end{cases} \quad P_k = P^{\otimes k}. \tag{TP.2}$$

Here all one-factor indices run from zero to $m-1$. The binomial theorem gives $e_s = e_y P_k$, where e_s, e_y are the ordered rows of monomials. Substitution in both directions proves the inverse formula, and $\det P = 1$. Coefficient columns therefore satisfy $x_y = P_k x_s$, with no change of the polynomial object.

The ordered exponential complex is the tensor product over $\mathbb{C}[u]$ of the two-term complexes with differential

$$L_i = u\partial_{s_i} + h(s_i). \quad (\text{TP.3})$$

Equivalently its total differential on polynomial forms is $u d + d\Phi_k \wedge$, with the usual ordered wedge signs. The operators L_i commute for different indices, proving that the square of this differential is zero. In one variable the leading degree of $L_i Q$, for $Q \neq 0$, is $\deg Q + m$, with the same leading coefficient. Monic subtraction therefore gives a unique remainder of degree below m . This decomposes the two-term complex as a contractible complex mapping isomorphically onto its image, plus a free $\mathbb{C}[u]$ -module of rank m in degree one. To see contractibility explicitly, send an image element $L_i Q$ back to its unique Q , and send its remainder to zero. Tensoring a contracting homotopy with the identity contracts every summand having a contractible factor; the total-complex signs cancel the differentials in the other factors. Consequently all cohomology vanishes except in degree k , where

$$\mathcal{H}_k = \bigoplus_{\mathbf{b}} \mathbb{C}[u][s^{\mathbf{b}} ds_1 \wedge \cdots \wedge ds_k], \quad \mathcal{H}_k / u\mathcal{H}_k = E_k ds_1 \wedge \cdots \wedge ds_k. \quad (\text{TP.4})$$

In particular its rank is m^k , and every primary order is retained.

Fix a branch of $u^{1/N}$ on a sector of $u \neq 0$, let $\zeta = e^{2\pi i/N}$, and for $0 \leq a < m$ define

$$\beta_a = \frac{a+1}{N}, \quad d_a = N^{\beta_a-1} e^{\pi i \beta_a} \Gamma(\beta_a), \quad W_{ja} = \zeta^{j(a+1)} - 1 \quad (1 \leq j \leq m), \quad D(u) = \text{diag}(d_a u^{\beta_a}). \quad (\text{TP.5})$$

Use the original rays at angles $(\arg u + \pi)/N + 2\pi j/N$, oriented outwards, and the original relative contour $\Gamma_j = -\ell_0 + \ell_j$. The ordered product contour is $\Gamma_{j_1} \times \cdots \times \Gamma_{j_k}$, with its product orientation. There is no permutation of its factors or top form. The full original period matrix is

$$\Pi_k(u) = e^{kc/u} W_k D_k(u) P_k, \quad W_k = W^{\otimes k}, \quad D_k(u) = D(u)^{\otimes k}. \quad (\text{TP.6})$$

Here is a direct verification retaining the original contours. In each variable, translate its contour to $\rho + \Gamma_j$. Cauchy's theorem on the finite quadrilateral swept by each ray applies to the entire integrand. The junction connectors cancel between the negative and positive rays. At their remote endpoints $s = Re^{i\vartheta} + a\rho$, $0 \leq a \leq 1$, the exponent has real part $-R^N/(N|u|) + O(R^{N-1}) + O(1)$, uniformly in a . Each connector has length at most $|\rho|$; its polynomially weighted integral tends to zero. Absolute integrability on each ray follows from the same estimate. Translation is thus a proved contour equality, preserving the scalar $e^{c/u}$. On the translated ray the substitution $v = r^N/(N|u|)$ gives precisely $d_a u^{\beta_a} \zeta^{j(a+1)}$; subtracting the zeroth ray gives the entry of $WD(u)$. The Pascal expansion gives the factor P . Absolute convergence permits Fubini on every product contour and proves (TP.6), including its orientation and all constants.

The matrix W is invertible: a relation among its columns gives a polynomial $\sum_{r=1}^m a_r (z^r - 1)$ of degree at most $m = N - 1$, vanishing at every N -th root of unity, including one. A nonzero polynomial cannot have more roots than its degree, so each $a_r = 0$. Every $d_a \neq 0$, since $0 < \beta_a < 1$. Thus (TP.6) is invertible for every $u \neq 0$. For reference its determinant, without discarding its scalar, is

$$\det \Pi_k(u) = K_m^{km^{k-1}} u^{km^k/2} e^{km^k c/u}, \quad K_m = N^{-m/2} e^{\pi i m/2} \left(\prod_{a=0}^{m-1} \Gamma(\beta_a) \right) \det W. \quad (\text{TP.7})$$

Indeed $\sum_a \beta_a = m/2$, and $\det(X \otimes Y) = (\det X)^{\dim Y} (\det Y)^{\dim X}$. The latter identity follows first on triangular matrices by multiplying their diagonal entries and then on all complex matrices by triangularizing them separately. Applying it repeatedly proves the displayed exponent km^{k-1} , also when $m = 1$.

22.2 Full monodromy and the exact character projectors

Let B_s have column b equal to $((b+1)s^b - b\rho s^{b-1})/N$ in the original basis, and let $B_k = \sum_i I^{\otimes(i-1)} \otimes B_s \otimes I^{\otimes(k-i)}$. The exact connection on (TP.4) is

$$\nabla_u = \partial_u - \frac{kc}{u^2}I + \frac{B_k}{u}, \quad P_k B_k P_k^{-1} = \text{diag}(L(\mathbf{b})/N). \quad (\text{TP.8})$$

For one factor, divide $\Phi(s)s^b$ monically by $h(s)$: the quotient is $(s - \rho)s^b/N$ and the remainder is cs^b . Subtracting its L_u -image gives $[\Phi s^b] = [cs^b - u((s - \rho)s^b/N)']$. This is exactly $\partial_u - \Phi/u^2$ on cohomology and gives the stated column. The binomial coordinate map makes this column diagonal with entries β_a . Summing the tensor-factor connections proves (TP.8).

On one positive circuit of $u = 0$, each relative contour changes by $\Gamma_j \mapsto \Gamma_{j+1} - \Gamma_1$ for $j < m$, and $\Gamma_m \mapsto -\Gamma_1$. This follows directly by advancing each ray index by one modulo N . Applying this to the columns of W gives eigenvalue ζ^{a+1} . Consequently the full period-cycle monodromy is

$$\Pi_k^{\text{cont}} = M_k \Pi_k, \quad M_k = W_k \text{diag}(\zeta^{L(\mathbf{b})}) W_k^{-1}, \quad M_k^N = I. \quad (\text{TP.9})$$

The scalar $e^{kc/u}$ is single-valued and has not been removed. The horizontal coefficient fundamental matrix is $P_k^{-1} e^{-kc/u} \text{diag}(u^{-L(\mathbf{b})/N})$, as follows by differentiation using (TP.8). Its monodromy has the inverse eigenvalues. Both conventions have the same invariant indices, namely $N \mid L(\mathbf{b})$.

For a residue $r \in \mathbb{Z}/N\mathbb{Z}$, define the diagonal mask E_r on the y -basis by $E_r y^{\mathbf{b}} = \mathbf{1}_{L(\mathbf{b}) \equiv r} y^{\mathbf{b}}$. The period-space and marked-space projectors are, respectively,

$$\begin{aligned} \mathcal{P}_r &= \frac{1}{N} \sum_{j=0}^{N-1} \zeta^{-rj} M_k^j = W_k E_r W_k^{-1}, \\ Q_r &= P_k^{-1} E_r P_k, \quad \mathcal{P}_r \Pi_k = \Pi_k Q_r. \end{aligned} \quad (\text{TP.10})$$

The geometric sum of N -th roots is N when its exponent is divisible by N and zero otherwise; this proves the first equality and the identities $Q_r Q_t = \mathbf{1}_{r=t} Q_r$, $\sum_r Q_r = I$, and the same identities for $\mathcal{P}_r$. The diagonal matrix $D_k(u)$ commutes with every E_r , proving the last equality. In particular the pulled-back projector is constant in u . These are algebraic projectors; no orthogonality in the original theta-source metric is assumed.

Their dimensions are exactly

$$d_r = \text{rank } Q_r = \begin{cases} (m^k + m(-1)^k)/N, & r = 0, \\ (m^k - (-1)^k)/N, & r \neq 0. \end{cases} \quad (\text{TP.11})$$

In fact the character sum giving the number of ordered tuples is $N^{-1} \sum_{j=0}^{N-1} \zeta^{-rj} (\sum_{l=1}^{N-1} \zeta^{jl})^k$. Its inner sum is m at $j = 0$ and -1 otherwise. The sum of the remaining outer characters is m for $r = 0$ and -1 for $r \neq 0$. This proves (TP.11) as an actual integer count, including all small values of m, k .

The ordinary invariant stalk of this punctured-disc local system is $\ker(M_k - I) = \text{im } \mathcal{P}_0$, of dimension d_0 . Its coinvariant space is identified with that same summand by $\mathcal{P}_0$: on every other summand $M_k - I$ is invertible. Thus both circle cohomology groups have dimension d_0 . On the cover $u = v^N$, (TP.8) becomes

$$\partial_v - \frac{Nkc}{v^{N+1}}I + \frac{1}{v} \text{diag}(L(\mathbf{b})). \quad (\text{TP.12})$$

The basis change by $\text{diag}(v^{-L(\mathbf{b})})$ removes precisely the integer residue. The remaining exact horizontal scalar is e^{-kc/v^N} . It is not a meromorphic gauge removing the irregular term when $kc \neq 0$. Nevertheless all displayed solutions are single-valued on this cover, so its analytic monodromy is the identity and its unipotent logarithm is zero. The exact diagonal solution leaves no additional sector transition to determine for this tensor of the one-critical-value family. The invariant and coinvariant statements above refer to the ordinary local system; they do not impose an unmentioned growth condition at the puncture.

22.3 The arithmetic action moves the invariant character

Let J be the matrix of multiplication by y on $\mathbb{C}[y]/y^m$: $J_{a+1,a} = 1$, all other entries zero. Put $J_i = I^{\otimes(i-1)} \otimes J \otimes I^{\otimes(k-i)}$, $J_{\text{tot}} = \sum_i J_i$, and

$$R_k = P_k^{-1} J_{\text{tot}} P_k, \quad A_k = M_{s_1+\dots+s_k} = k\rho I + R_k. \quad (\text{TP.13})$$

This is multiplication in the original algebra (TP.1), since $s_i = \rho + y_i$. No geometric nilpotent replaces R_k . Put $K = k(m-1)$. The commuting nilpotents have the exact multinomial powers

$$J_{\text{tot}}^r y^{\mathbf{b}} = \sum_{\substack{\alpha \in \mathbb{Z}_{\geq 0}^k, |\alpha|=r \\ b_i + \alpha_i \leq m-1 \ (1 \leq i \leq k)}} \frac{r!}{\prod_i \alpha_i!} y^{\mathbf{b}+\alpha}, \quad R_k^{K+1} = 0, \quad R_k^K \neq 0. \quad (\text{TP.14})$$

Expansion of the commuting operators proves the sum. A term with an exponent at least m vanishes in (TP.1), giving its exact index restrictions. At $r = K$, the input one has nonzero top coefficient $K!/((m-1)!)^k$; at larger r no tuple survives. This also proves the conventions when $K = 0$: $R_0^0 = I \neq 0$ and $R_k = 0$.

Define $D_0 = \text{diag}(d_a)$ and $K_0 = W D_0 J D_0^{-1} W^{-1}$, and let K_{tot} be the sum of its ordered factor placements. The actual period transport of the arithmetic action is

$$\Pi_k A_k \Pi_k^{-1} = k\rho I + u^{1/N} K_{\text{tot}}, \quad M_k K_{\text{tot}} M_k^{-1} = \zeta K_{\text{tot}}. \quad (\text{TP.15})$$

Conjugating J by $\text{diag}(u^{\beta_a})$ multiplies each nonzero entry by $u^{\beta_{a+1}-\beta_a} = u^{1/N}$; the tensor placements give the first identity. Replacing these powers by ζ^{a+1} gives the second. The scalar $e^{kc/u}$ cancels by its displayed inverse in this conjugation. Equivalently the exact connection commutator is

$$[B_k, A_k] = R_k/N, \quad [\nabla_u, A_k] = R_k/(Nu). \quad (\text{TP.16})$$

These follow on each monomial from the difference of its two residue eigenvalues, including each truncated top term, and then from (TP.8). Thus the arithmetic action is explicitly related to the boundary connection, with its precise nonhorizontal term.

Multiplication by a surviving y_i increases L by one. For all residues and integers $r \geq 0$, this gives

$$\begin{aligned} Q_t R_k^r &= R_k^r Q_{t-r}, & [Q_0, A_k] &= R_k(Q_{-1} - Q_0), \\ Q_0 A_k Q_0 &= k\rho Q_0, & (I - Q_0) A_k Q_0 &= R_k Q_0 = Q_1 R_k Q_0. \end{aligned} \quad (\text{TP.17})$$

All subscripts are taken modulo N . The first identity holds on every basis monomial, including zero output terms. The other identities follow from it and from $N \geq 2$. They calculate both the compression and the outgoing map. In particular a scalar compression is not an assertion that the invariant plane is preserved.

For a precise nonvanishing statement put

$$d_{\min} = N \lceil k/N \rceil - k \in \{0, \dots, N-1\}. \quad (\text{TP.18})$$

The possible degrees $|\mathbf{b}|$ fill every integer from zero to K : distribute any such integer successively into the k slots, never exceeding their capacity $m-1$. Thus $Q_0 \neq 0$ exactly when $d_{\min} \leq K$. Moreover $R_k Q_0 \neq 0$ exactly when $d_{\min} < K$. If strict inequality holds, a tuple of that degree has an unfilled slot, and (TP.14) with $r = 1$ gives nonzero output with no cancellation, since its distinct monomials have positive integer coefficients. If equality holds, the only possible invariant tuple is the top one and every J_i kills it. This proves both directions.

For $k = 1$ we have $d_{\min} = m > K = m-1$, so there are no invariants. For $m = 1$, $R_k = 0$ and the unique tensor character is invariant exactly when k is even. For every $m \geq 2, k \geq 2$, instead,

$$Q_0 \neq 0, \quad R_k Q_0 \neq 0. \quad (\text{TP.19})$$

Indeed at $k = 2$, $d_{\min} = m-1 < 2m-2 = K$. For $k \geq 3$, $K = k(m-1) \geq m+1 = N$, while $d_{\min} \leq N-1$. This proves (TP.19), including $m = 2, k = 3$, the smallest equality case for the capacity bound.

22.4 The complete compressed dilation, including every return power

Work on $V_0 = \text{im } Q_0$. If this space is zero, all the following compressed maps are the unique zero maps. Suppose it is nonzero and put

$$q_* = \left\lfloor \frac{K - d_{\min}}{N} \right\rfloor = \lfloor km/N \rfloor - \lceil k/N \rceil, \quad T = Q_0 R_k^N Q_0|_{V_0}. \quad (\text{TP.20})$$

Since R_k^N commutes with Q_0 , $T^q = Q_0 R_k^{qN} Q_0|_{V_0}$. For every $0 \leq q \leq q_*$ this map is nonzero; for $q > q_*$ it is zero. To prove nonvanishing, choose an invariant tuple of minimum degree. Its remaining capacities sum to $K - d_{\min}$, so an increment tuple of size qN can be distributed within them. Its term in (TP.14) is nonzero and cannot cancel. For vanishing, every input has degree at least $d_{\min}$, so all output terms exceed capacity when $qN > K - d_{\min}$. Thus the nilpotency index of T is exactly $q_* + 1$, with the identity zeroth power when $q_* = 0$.

For $a > 0$, define the original ordered dilation on functions by simultaneous replacement $x_i \mapsto x_i/a$. Its generator is $\sum_i D_i$, with $D_i = -x_i \partial_{x_i}$, and hence its action on this marked packet is a^{A_k} . Write $z = \log a$. The complete compression is the finite expression

$$Q_0 a^{A_k} Q_0|_{V_0} = a^{k\rho} \sum_{q=0}^{q_*} \frac{z^{qN}}{(qN)!} T^q. \quad (\text{TP.21})$$

Indeed $a^{A_k} = a^{k\rho} \sum_{r=0}^K z^r R_k^r / r!$ by (TP.14). The first identity in (TP.17) kills its compression unless $N \mid r$, and (TP.20) determines exactly which surviving powers are nonzero. Equivalently, the sum without its scalar is the restriction of $N^{-1} \sum_{j=0}^{N-1} \exp(\zeta^j z R_k) Q_0$; the same root-of-unity sum proves this equality. In the original y -monomial coordinates its exact entry, from invariant $\mathbf{b}$ to invariant $\mathbf{d}$, is

$$\begin{cases} a^{k\rho} \frac{z^{|\mathbf{d}|-|\mathbf{b}|}}{\prod_i (d_i - b_i)!}, & d_i \geq b_i \text{ for all } i, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{TP.22})$$

The two invariant conditions make the exponent divisible by N . Formula (TP.14) cancels its total factorial against that of the exponential and proves (TP.22). Conjugation by the explicit columns of P_k^{-1} returns this same matrix to the original s -basis.

In particular $k = 2$ always has $q_* = 0$, so (TP.21) is scalar even though (TP.19) proves nonzero outgoing arithmetic action when $m \geq 2$. Conversely nonconstant returns do occur: for $m = 2, N = 3, k = 3$, V_0 is spanned in the y -basis by 1 and $y_1 y_2 y_3$, and $R_k^3(1) = 6y_1 y_2 y_3$. The compression is exactly $a^{3\rho}(I + z^3 T/6)$, with that nonzero T . When $q_* \geq 1$, the compressed maps are not a one-parameter representation: after removing the scalar, their derivative at $z = 0$ is zero but their coefficient of z^N is the nonzero $T/N!$. A differentiable matrix-valued function F with $F(0) = I$ and $F(z+w) = F(z)F(w)$ satisfies $F'(z) = F(z)F'(0)$ by differentiating in w ; zero derivative at zero would make it constant. This proves the asserted failure without identifying compression with a quotient action.

22.5 The full Taylor unit and its exact original-source projector

Retain the entire quotient $v_h = g/h$ and all coefficients of its marked Taylor class:

$$a_j = \frac{v_h^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!} \quad (0 \leq j < m), \quad a_0 \neq 0, \quad U_s = P^{-1} \left(\sum_{j=0}^{m-1} a_j J^j \right) P, \quad U_k = U_s^{\otimes k}. \quad (\text{TP.23})$$

The Taylor expansion at the complete-order zero proves the derivative equality. The inverse coefficients are $b_0 = a_0^{-1}$ and $b_r = -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}$; multiplying the finite series modulo J^m

proves the inverse. Thus (TP.23) is invertible and keeps every coefficient. It commutes with A_k , since all its factors are polynomials in the commuting multiplication matrices.

For a multi-index $\mathbf{j} \in \{0, \dots, m-1\}^k$, write $a_{\mathbf{j}} = \prod_i a_{j_i}$ and let $R_i = P_k^{-1} J_i P_k$, so $R_k = \sum_i R_i$. The full graded unit, its period transport and its connection defect are

$$\begin{aligned} U_k &= \sum_{\mathbf{j}} a_{\mathbf{j}} \prod_i R_i^{j_i}, \\ \Pi_k U_k \Pi_k^{-1} &= \sum_{\mathbf{j}} a_{\mathbf{j}} u^{|\mathbf{j}|/N} \prod_i K_i^{j_i}, \\ [\nabla_u, U_k] &= \frac{1}{Nu} \sum_{\mathbf{j}} |\mathbf{j}| a_{\mathbf{j}} \prod_i R_i^{j_i}. \end{aligned} \tag{TP.24}$$

Here K_i is the corresponding placement of K_0 . The first equality expands the tensor product, the second uses (TP.15) factor by factor, and the third follows from the residue difference $|\mathbf{j}|/N$ on each surviving monomial. Zero terms at the truncation boundary satisfy the identities as well.

The actual source maps can now be specified. On the original one-variable spaces retain $D = -x\partial_x$, $\Theta\phi(x) = 2\sum_{r \geq 1} \phi(rx)$, $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, and the original F_h with Mellin transform v_h . Here the Mellin convention is $\mathcal{M}F(s) = \int_0^\infty F(x)x^s dx/x$. The source V is the even Schwartz subspace with $\phi(0) = \int \phi = 0$; the target $\mathcal{B}$ has all seminorms $\sup_{x>0} x^b |D^r F(x)|$ finite for every integer b and nonnegative integer r . The inherited original source identity is $\mathcal{M}\Theta\phi_* = g$, and the exact Euler inverse constructs this $F_h \in \mathcal{B}$. Define the actual finite-packet maps on multivariate polynomials by

$$\begin{aligned} \mathcal{T}_k(f) &= f(D_1, \dots, D_k) \prod_{i=1}^k F_h(x_i), \\ \alpha_k(f) &= f(D_1, \dots, D_k) \prod_{i=1}^k \phi_*(x_i), \\ \mathcal{J}_k \mathcal{T}_k(f) &= U_k[f]_s. \end{aligned} \tag{TP.25}$$

Here $\mathcal{J}_k$ means the multivariable Mellin transform followed by reduction modulo all $h(s_i)$, and $[f]_s$ is the unchanged remainder column. Every displayed source lies in the original smooth product source; it is an explicit finite sum of tensor functions, not an assertion that the whole completed source equals its algebraic tensor product. Integration by parts gives $\mathcal{M}(D_i F) = s_i \mathcal{M}F$; all endpoint terms vanish by the stated seminorms. Fubini then gives $\mathcal{M}_k \mathcal{T}_k(f) = f(s) \prod_i v_h(s_i)$. Its full jet is exactly the last equality of (TP.25). The factorwise identities $\Theta\alpha_h(f) = \mathcal{T}_h(hf)$ give the ordered tensor source square with every usual total-complex sign. The corrected one-factor section σ_h therefore gives the actual degree- k injection $\sigma_h^{\otimes k} U_k$ on this finite tensor packet. This retains the original full unit; it is not set equal to the identity.

In particular, if x_s is a marked coefficient column and $q = U_k x_s$ is its actual theta-jet coordinate, the period observation of that source is $\Pi_k U_k^{-1} q$. Its exact invariant projector is

$$\widehat{Q}_0 = U_k Q_0 U_k^{-1}, \quad \mathcal{P}_0 \Pi_k U_k^{-1} = \Pi_k U_k^{-1} \widehat{Q}_0, \quad [\widehat{Q}_0, A_k] = U_k [Q_0, A_k] U_k^{-1}. \tag{TP.26}$$

The first two formulas follow by substituting $q = U_k x_s$ into (TP.10). The last uses the proved commutation of U_k with A_k . Thus the actual invariant source plane is $U_k \text{im } Q_0$, with the same exact arithmetic leakage conjugated through its original unit. On the cohomology image the corresponding projector is $\sigma_h^{\otimes k} \widehat{Q}_0 (\sigma_h^{\otimes k})^{-1}$, where the inverse is only on that injective image. Its maps are fully specified.

For comparison, when $V_0 \neq 0$, compressing the coefficient matrix U_k by Q_0 is another concrete

operation. Its full expression on V_0 is

$$Q_0 U_k Q_0|_{V_0} = a_0^k I_{V_0} + Z, \quad Z = \sum_{\substack{\mathbf{j} \neq 0 \\ N || \mathbf{j}}} a_{\mathbf{j}} Q_0 \prod_i R_i^{j_i} Q_0 \Big|_{V_0}, \quad Z^{q_*+1} = 0. \quad (\text{TP.27})$$

Every omitted term changes the character and so is killed by (TP.10); every term of Z raises degree by at least N . The degree bound defining q_* proves its stated vanishing power, without requiring any a_j other than a_0 to be nonzero. Its exact inverse is $a_0^{-k} \sum_{r=0}^{q_*} (-a_0^{-k} Z)^r$. Multiplying this finite geometric series proves the claim. This inverse is not substituted for the full U_k^{-1} in (TP.26).

Finally the difference between marked coefficient multiplication and multiplication of twisted representatives is also explicit. Put $v_{\text{pol}}(s) = \sum_{j=0}^{m-1} a_j (s - \rho)^j$ and $V_{\text{pol}} = \prod_i v_{\text{pol}}(s_i)$. For every polynomial f , the actual differential obeys

$$V_{\text{pol}} L_i f = L_i(V_{\text{pol}} f) - u(\partial_{s_i} V_{\text{pol}}) f. \quad (\text{TP.28})$$

This is the Leibniz rule, with the full derivative term. To compute its complete remainder, divide in one variable $v_{\text{pol}} s^b = h q_b + r_b$, with $\deg r_b < m$. Let H_s have column b the original coordinate column of q'_b . Since $\deg q_b \leq m-2$, q'_b is already a remainder. Subtracting $L_u q_b = h q_b + u q'_b$ gives $[v_{\text{pol}} s^b] = [r_b - u q'_b]$. For explicit coefficients in the y -basis,

$$(H_y)_{ab} = \begin{cases} (a+1)a_{a+m+1-b}, & 0 \leq a+m+1-b < m, \\ 0, & \text{otherwise,} \end{cases} \quad H_s = P^{-1} H_y P, \quad (\text{TP.29})$$

$$\text{Rem}_u(V_{\text{pol}} \cdot) = (U_s - u H_s)^{\otimes k}.$$

on the retained original remainder frame. Indeed the quotient of $v_{\text{pol}} y^b$ has terms $a_j y^{j+b-m}$ exactly when $j+b \geq m$; differentiating gives the stated matrix entry. The tensor formula follows by replacing the factors successively; their variables are distinct and their top-form order stays fixed. This proves the exact full representative correction on the same frame. It does not assert that multiplication descends on arbitrary twisted-cohomology representatives.

22.6 The original terminal class on its exact invariant subsequence

The complete tensor primary algebra contains its original terminal jet

$$\tau_k = \prod_{i=1}^k y_i^{m-1}, \quad R_k \tau_k = 0, \quad A_k \tau_k = k \rho \tau_k, \quad U_k \tau_k = a_0^k \tau_k. \quad (\text{TP.30})$$

Its displayed coefficient in the retained basis is one, so it is nonzero. Each multiplication by y_i kills it, proving the next two statements by (TP.13). Every positive Taylor term in (TP.23) also kills it, proving the last statement with its full scalar. Since $L(m-1, \dots, m-1) = km \equiv -k \pmod{N}$,

$$Q_0 \tau_k = \hat{Q}_0 \tau_k = \begin{cases} \tau_k, & N \mid k, \\ 0, & N \nmid k. \end{cases} \quad (\text{TP.31})$$

The equality for Q_0 is its diagonal-mask definition, and that for $\hat{Q}_0$ follows from $U_k^{-1} \tau_k = a_0^{-k} \tau_k$ in (TP.26). Thus on the infinite subsequence $k = N\ell$, $\ell \geq 1$, the same nonzero arithmetic terminal class lies in the actual boundary-invariant plane, in both marked and original jet coordinates.

Its complete period vector is

$$\Pi_k \tau_k = e^{kc/u} d_{m-1}^k u^{km/N} (W_{\bullet, m-1})^{\otimes k}, \quad d_{m-1} = N^{-1/N} e^{\pi i m/N} \Gamma(m/N), \quad W_{j, m-1} = \zeta^{-j} - 1. \quad (\text{TP.32})$$

This is column $(m-1, \dots, m-1)$ of (TP.6) in its specified y -coordinates. Every entry $W_{j,m-1}$ is nonzero, so the vector remains nonzero for $u \neq 0$. Its period-cycle monodromy is multiplication by $\zeta^{km} = \zeta^{-k}$, agreeing with (TP.31). For the original corrected-section class $u_\rho^{\otimes k} = \sigma_h^{\otimes k} \tau_k$, the period observation defined in (TP.26) is precisely $a_0^{-k} \Pi_k \tau_k$, not the vector with this scalar omitted.

For the comparison with the original MAI/PAM counterfactual branch, retain its selected $\rho = 1/2 + \delta + i\gamma$, with $0 < \delta < 1/2$ and $\gamma > 0$, reached by its original reflection and conjugation arrows. Let h_0 be its original reflected quartet packet containing the same order- m zero, put $w = h_0/h$, and retain its original CRT idempotent e_ρ . The map $E_h \rightarrow e_\rho E_{h_0}$, $[f] \mapsto e_\rho[f]$, is an isomorphism with inverse reduction modulo h : the idempotent is one on this primary factor and zero on all the other factors. This proves both composites directly on every CRT component. Moreover $w(\rho) \neq 0$, and the exact unit relation on this factor is $(g/h_0) = v_h/w$; the inverse of the class of w is its unique truncated Taylor inverse. Thus this local passage retains the whole original unit, not only its constant coefficient.

Write $K = k(m-1)$ as before. The original amplified terminal coefficient is

$$c_{h_0,k} = \frac{K!}{((m-1)!)^k} ((g/h_0)(\rho))^k \neq 0, \quad \eta_k v_k = c_{h_0,k} \tau_k, \quad \mathcal{I}_k v_k = c_{h_0,k} u_\rho^{\otimes k}. \quad (\text{TP.33})$$

Here $v_k, \eta_k, \mathcal{I}_k$ are the original cyclic terminal vector, coefficient map and cohomology map of MAI6a. The selected quartet corner $k\rho$ arises only by choosing ρ in every tensor factor, as in that construction. The first equality for that map is also its direct local calculation: (TP.14) at $r = K$ leaves precisely $K!/((m-1)!)^k \prod_i y_i^{m-1}$; the original complete unit acts on that terminal monomial by $((g/h_0)(\rho))^k$. All other Taylor terms vanish because they exceed a retained primary order. The original corrected-section transition $\sigma_{h_0} \iota_{h,h_0} = \sigma_h$ carries this identity to the second cohomology equality. Thus the original source period observation of this specified amplified vector is exactly

$$c_{h_0,k} a_0^{-k} \Pi_k \tau_k = \frac{K!}{((m-1)!)^k} \left(\frac{(g/h_0)(\rho)}{(g/h)(\rho)} \right)^k e^{kc/u} d_{m-1}^k u^{km/N} (W_{\bullet,m-1})^{\otimes k}. \quad (\text{TP.34})$$

Every factor is retained and nonzero over $\mathbb{C}$. The ratio of unit values is $w(\rho)^{-1}$, as follows from the full unit relation above. No rescaling of the original vector was used to get (TP.34).

For $a > 0$, (TP.30) gives $a^{A_k} \tau_k = a^{k\rho} \tau_k$; the original source action has the same equality after applying its established theta-boundary homotopy and passing to cohomology. If $\delta = \text{Re } \rho - 1/2$, its exact discrepancy is

$$\log \left(\frac{|a^{k\rho}|^2}{a^k} \right) = 2k\delta \log a. \quad (\text{TP.35})$$

This follows by taking the real part of $k\rho \log a$. On $k = N\ell$, it holds on the same nonzero original terminal class inside the boundary-invariant plane. The invariant projection therefore does not eliminate that discrepancy. This calculation supplies no bound on it by a constant independent of k .

22.7 The exact rank of arithmetic departure from the invariant plane

Let $E_{k,d}$ be the span of $y^{\mathbf{b}}$ with $|\mathbf{b}| = d$, and define $h_d = \dim E_{k,d}$, with $h_d = 0$ outside $0 \leq d \leq K$. The exact coefficient count is

$$\sum_{d=0}^K h_d t^d = (1 + t + \dots + t^{m-1})^k. \quad (\text{TP.36})$$

Expanding the product proves it by the ordered monomial basis. To compute the rank of J_{tot} on these levels, define on one factor $Fy^b = b(m-b)y^{b-1}$, with $F1 = 0$, and use the positive

diagonal form with $\|y^b\|^2 = b!(m-1)!/(m-1-b)!$. The ratio of consecutive diagonal entries is $(b+1)(m-1-b)$; hence the adjoint of J is exactly F . This form is used only to prove a basis-independent algebraic rank. It is not substituted for the original theta-source norm.

Use the tensor product of these forms and put $F_{\text{tot}} = \sum_i F_i$. Direct evaluation on each one-factor monomial gives $[F, J]y^b = (m-1-2b)y^b$. Operators in different factors commute, so on degree d this proves

$$\|J_{\text{tot}}x\|^2 - \|F_{\text{tot}}x\|^2 = (K-2d)\|x\|^2 \quad (x \in E_{k,d}). \quad (\text{TP.37})$$

If $d < K/2$, the identity makes J_{tot} injective on $E_{k,d}$. If $d+1 > K/2$, its reversed identity makes F_{tot} injective on $E_{k,d+1}$, so its adjoint $J_{\text{tot}} : E_{k,d} \rightarrow E_{k,d+1}$ is surjective: $(\text{im } J_{\text{tot}})^\perp = \ker(F_{\text{tot}}|_{E_{k,d+1}}) = 0$. Every integer d satisfies one or both of these inequalities, including the central degree. It follows that

$$\text{rank}(J_{\text{tot}} : E_{k,d} \rightarrow E_{k,d+1}) = \min(h_d, h_{d+1}). \quad (\text{TP.38})$$

No unimodality assumption is used; the necessary dimension inequalities follow from the proved injectivity or surjectivity.

The invariant degrees have $d+k \equiv 0 \pmod{N}$. Their outgoing images have distinct degrees, so the direct-sum decomposition and (TP.38) give the full departure rank and its exact kernel:

$$\begin{aligned} \text{rank}(R_k Q_0) &= \sum_{\substack{0 \leq d \leq K \\ N|d+k}} \min(h_d, h_{d+1}), \\ \dim(\ker R_k \cap \text{im } Q_0) &= \sum_{\substack{0 \leq d \leq K \\ N|d+k}} \max(h_d - h_{d+1}, 0). \end{aligned} \quad (\text{TP.39})$$

Conjugation by P_k , then by the original U_k in (TP.26), preserves these ranks and kernels on the actual marked and original-source invariant planes.

For example, when $k=2, m \geq 2$, the sole invariant degree is $m-1$, and the exact kernel is the line

$$\mathbb{C}\omega_m, \quad \omega_m = \sum_{i=0}^{m-1} (-1)^i y_1^i y_2^{m-1-i}, \quad (y_1 + y_2)\omega_m = 0 \quad \text{in } E_2. \quad (\text{TP.40})$$

The interior coefficients cancel pairwise, and the two end terms have exponent m , so vanish. For a general input $\sum_i c_i y_1^i y_2^{m-1-i}$, the output coefficients are $c_{j-1} + c_j$, $1 \leq j < m$. Its kernel therefore has exactly the stated alternating coefficients and dimension one, and its rank is $m-1$. This is a nonzero arithmetic eigenclass of eigenvalue 2ρ already lying in the boundary-invariant plane. It coexists with the nonzero departure on the remaining $m-1$ directions and with the scalar compressed dilation of (TP.21).

Every constant coefficient map above acts at each original nonbottom split-support label by $(A, x) \mapsto (A, fx)$, and fixes the external absence symbol τ . Its killed coefficient remains the represented zero at that label. The marked specialization $u=0$ is the original represented-zero specialization; none of these projector computations replaces it by external absence. Equations (TP.10), (TP.17), (TP.21) and (TP.26) specify the exact maps relating boundary invariants, the full arithmetic action, its dilation and the original source. The dimension of that invariant space alone supplies no conclusion about the location of the selected zero.

23 Ordered tensors of the actual single-primary finite-field sheaf

23.1 The original coefficient map and the common parameter

Retain the specified coefficient map $R \rightarrow k_0 = \mathbf{F}_{Q_0}$ of characteristic p , the full Taylor class $v_h = j_h(g/h)$ and its inverse in the retained m -dimensional remainder basis, and every other

original coefficient. Write

$$N = m+1 \geq 2, \quad p > N, \quad h(s) = (s-\rho)^m, \quad \Phi(s) = \frac{(s-\rho)^N - (-\rho)^N}{N}, \quad c = -\frac{(-\rho)^N}{N}. \quad (\text{TPF.1})$$

The same coefficient map supplies ρ, c, N^{-1} . In particular, no coefficient of an infinite analytic function is being adjoined in place of its specified full Taylor class. Set

$$d = \text{ord}_{(\mathbf{Z}/N\mathbf{Z})^\times}(Q_0), \quad L = \mathbf{F}_Q, \quad Q = Q_0^d, \quad N \mid Q - 1. \quad (\text{TPF.2})$$

This is defined since $p > N$ implies $\gcd(Q_0, N) = 1$. Fix $\ell \neq p$, coefficients $\overline{\mathbf{Q}}_\ell$, and a nontrivial additive character $\psi_0 : k_0 \rightarrow \overline{\mathbf{Q}}_\ell^\times$. Put $\psi = \psi_0 \circ \text{Tr}_{L/k_0}$. Every Frobenius below is geometric Frobenius. Kummer sheaves $\mathcal{K}_\chi$ have trace χ on $L^\times$; over finite extensions their characters extend by norm. Artin–Schreier characters extend by trace.

Let $\pi : \mathbf{A}_s^1 \times \mathbf{G}_{m,u} \rightarrow \mathbf{G}_{m,u}$ be projection and retain

$$\mathcal{F}_0 = R^1\pi_!\mathcal{L}_{\psi_0}(\Phi(s)/u), \quad \mathcal{T}_{0,k} = \mathcal{F}_0^{\otimes k}, \quad k \geq 1. \quad (\text{TPF.3})$$

The positive integer k has no restriction relative to p or N . The tensor is ordered and taken on this one common u -line. No symmetric, alternating, or sum-coordinate quotient is taken. Write $\mathcal{F} = (\mathcal{F}_0)_L$ and $\mathcal{T}_k = (\mathcal{T}_{0,k})_L$. The complete single-primary calculation SPF gives

$$\mathcal{F} \simeq \mathcal{L}_\psi(c/u) \otimes \bigoplus_{\substack{\chi \in X_N \\ \chi \neq 1}} \mathcal{A}_{a_\chi} \otimes \mathcal{K}_\chi(u), \quad X_N = \{\chi : L^\times \rightarrow \overline{\mathbf{Q}}_\ell^\times : \chi^N = 1\}, \quad (\text{TPF.4})$$

where all constants are exactly

$$G_L(\chi, \psi) = \sum_{z \in L^\times} \chi(z)\psi(z), \quad a_\chi = -\chi(N)G_L(\chi, \psi). \quad (\text{TPF.5})$$

Here $\mathcal{A}_a$ is the geometrically constant line with geometric Fr_Q -eigenvalue a . SPF proves this isomorphism using the finite power-map idempotents, the support-preserving translation $s = y + \rho$, and $w = z/(Nu)$. Its minus sign comes from compact cohomology in degree one. In particular, all $R^i\pi_!$ vanish for $i \neq 1$. We use that proved single-primary result without changing its source or phase.

The relative compact-support Kunneth map consequently gives the precise source realization

$$\mathcal{T}_{0,k} \simeq R^k\pi_!\mathcal{L}_{\psi_0}\left(\frac{\sum_{i=1}^k \Phi(s_i)}{u}\right), \quad \Pi : \mathbf{A}_{s_1, \dots, s_k}^k \times \mathbf{G}_{m,u} \longrightarrow \mathbf{G}_{m,u}. \quad (\text{TPF.6})$$

Indeed the derived relative Kunneth isomorphism tensors the k complexes $R\pi_!\mathcal{L}_{\psi_0}(\Phi/u)$, each concentrated in degree one; their tensor is concentrated in degree k . The external product of the additive sheaves is the sheaf of the sum of their functions, by addition of Artin–Schreier torsor coordinates. No factors are reordered, so no permutation sign is inserted. The source theorem is SGA 4, Expose XVII, 5.4.3 and its finite-family version 5.4.4.1: apply it at the finite coefficient levels of these lisse ℓ -adic sheaves and then pass to the ℓ -adic system and invert ℓ . Here every projection is compactifiable by $\mathbf{P}^1 \times \mathbf{G}_m \rightarrow \mathbf{G}_m$; the product compactification is $(\mathbf{P}^1)^k \times \mathbf{G}_m$. The coordinate isomorphism on this same source is

$$(y_1, \dots, y_k, u) \longmapsto (y_1 + \rho, \dots, y_k + \rho, u), \quad \frac{\sum_i \Phi(s_i)}{u} = \frac{\sum_i y_i^N}{Nu} + \frac{kc}{u}. \quad (\text{TPF.7})$$

Its inverse subtracts ρ in every coordinate, and it preserves compact supports. Each original coefficient function is pulled back by this literal substitution, and its inverse is pulled back by the same map.

23.2 Every ordered character tuple and its exact constant

Define

$$\begin{aligned} \Omega_k &= (X_N \setminus \{1\})^k, & r_k &= (N-1)^k, \\ \eta_{\chi} &= \prod_{i=1}^k \chi_i, & A_{\chi} &= \prod_{i=1}^k (-\chi_i(N) G_L(\chi_i, \psi)), & w_k &= kc \in k_0. \end{aligned} \quad (\text{TPF.8})$$

Thus k is reduced in characteristic p only when it multiplies the field element c . Its integer value still counts all k tensor factors and remains in every cohomological sign. Distributing the ordered tensor in (TPF.4), and using the additive and multiplicative torsor tensor laws, proves

$$\boxed{\mathcal{T}_k \simeq \mathcal{L}_{\psi}(w_k/u) \otimes \bigoplus_{\chi \in \Omega_k} \mathcal{A}_{A_{\chi}} \otimes \mathcal{K}_{\eta_{\chi}}(u).} \quad (\text{TPF.9})$$

Different ordered tuples remain different summands, even when their product characters or constants agree. This sheaf is lisse of rank r_k .

For $L_a = \mathbf{F}_{Q^a}$, put $\psi_a = \psi \circ \text{Tr}_{L_a/L}$ and $\chi_{i,a} = \chi_i \circ N_{L_a/L}$. At $u \in L_a^{\times}$, the eigenvalue on the tuple summand is exactly

$$\psi_a(w_k/u) A_{\chi}^a \prod_i \chi_{i,a}(u). \quad (\text{TPF.10})$$

SPF proves $-G_{L_a}(\chi_{i,a}, \psi_a) = (-G_L(\chi_i, \psi))^a$ by its one-dimensional compact cohomology. Also $\chi_{i,a}(N) = \chi_i(N)^a$. These two equalities prove the stated extension-field constant, without replacing a negative Gauss sum by the negative of its a th power. The trace identity on the actual ordered source is consequently

$$\text{Tr}(\text{Fr}_{Q^a, u} | \mathcal{T}_k) = (-1)^k \sum_{(s_1, \dots, s_k) \in L_a^k} \psi_a \left(\frac{\sum_i \Phi(s_i)}{u} \right). \quad (\text{TPF.11})$$

To check its sign directly, the original sum factors into k one-variable sums, each the negative of the trace on $\mathcal{F}$. Multiplying gives precisely $(-1)^k$.

23.3 Original inertia, its tame projector, and multiplicities

Let I_0 be geometric inertia over $\bar{k}_0((u))$, and put $V_k = (\mathcal{T}_{0,k})_{\bar{\eta}}$. Choose $\vartheta : \mathbf{F}_p \rightarrow \bar{\mathbf{Q}}_{\ell}^{\times}$ nontrivial and the unique $b \in k_0^{\times}$ satisfying

$$\psi_0(x) = \vartheta(\text{Tr}_{k_0/\mathbf{F}_p}(bx)). \quad (\text{TPF.12})$$

Nondegeneracy of the finite-field trace pairing supplies this unique b ; after extension to L the same b represents ψ .

First let $c \neq 0$. Keep the original cover for $\mathcal{F}$, before taking a tensor:

$$u = v^N, \quad z^p - z = \frac{bc}{v^N}. \quad (\text{TPF.13})$$

The Kummer degree is N . Over $\bar{k}_0((v))$ the Artin–Schreier polynomial has no root, since its right side has pole order N , prime to p , while a pole of $f^p - f$ has order divisible by p . Its degree is therefore p . The total extension has degree Np , and the Np displayed automorphisms

$$(\zeta, a) : (v, z) \longmapsto (\zeta v, z + a), \quad \zeta \in \mu_N, \ a \in \mathbf{F}_p \quad (\text{TPF.14})$$

commute and exhaust its Galois group $\Gamma = \mu_N \times \mathbf{F}_p$. This is a quotient of the original geometric inertia: the Kummer part is totally ramified, and if e is the ramification index of the Artin–Schreier extension over $\bar{k}_0((v))$, its defining equation gives $p v_E(z) = -Ne$. Since $\gcd(p, N) = 1$ and $1 \leq e \leq p$, it follows that $e = p$.

For $\chi \in X_N$, let $\lambda_\chi : \mu_N(L) \rightarrow \overline{\mathbf{Q}}_\ell^\times$ be uniquely specified by $\lambda_\chi(x^{(Q-1)/N}) = \chi(x)$. Under the geometric Frobenius convention fixed above, the exact action on the tuple line in (TPF.9) is

$$\rho_k(\zeta, a)|_\chi = \lambda_{\eta_\chi}(\zeta)^{-1} \vartheta(-ka). \quad (\text{TPF.15})$$

The Kummer inverse is the power-map projector convention of SPF: with $D_\zeta e_r = e_{\zeta r}$, its χ -projector is $N^{-1} \sum_\zeta \lambda_\chi(\zeta) D_\zeta$, whose image has eigencharacter λ_χ^{-1} . For the Artin–Schreier torsor $z^p - z = f$, the projector $p^{-1} \sum_a \vartheta(a) D_a$ has eigencharacter ϑ^{-1} . At a finite-field point geometric Frobenius translates the torsor coordinate by $-\text{Tr}(f)$, giving trace $\vartheta(\text{Tr}(f))$. Thus its action is $\vartheta(-a)$, and k ordered tensor factors give $\vartheta(-ka)$, proving every sign in (TPF.15).

When $c = 0$, the original additive torsor is trivial. The geometric inertia action instead factors through the Kummer quotient μ_N , with the exact action $\lambda_{\eta_\chi}^{-1}$; there is no nontrivial wild factor.

Let T_ζ denote the action of $(\zeta, 0)$ in (TPF.15), or its Kummer action when $c = 0$. Its exact tame projector is

$$P_{0,k}^{\text{tame}} = \frac{1}{N} \sum_{\zeta \in \mu_N} T_\zeta, \quad P_{0,k}^{\text{tame}}|_\chi = \begin{cases} 1, & \eta_\chi = 1, \\ 0, & \eta_\chi \neq 1. \end{cases} \quad (\text{TPF.16})$$

Character orthogonality proves the second formula and idempotence. In the case $c \neq 0$, the true original inertia projector is

$$\begin{aligned} P_{I_0,k} &= \frac{1}{Np} \sum_{\zeta \in \mu_N} \sum_{a \in \mathbf{F}_p} \rho_k(\zeta, a) \\ &= P_{0,k}^{\text{tame}} \left(\frac{1}{p} \sum_{a \in \mathbf{F}_p} \vartheta(-ka) \right) = \begin{cases} P_{0,k}^{\text{tame}}, & p \mid k, \\ 0, & p \nmid k. \end{cases} \end{aligned} \quad (\text{TPF.17})$$

Division by Np occurs in the characteristic-zero coefficient field, regardless of the prime ℓ . For $c = 0$, the true projector is $P_{I_0,k} = P_{0,k}^{\text{tame}}$. This calculates it inside the original representation; the tame projector alone is not substituted for full inertia when its wild character is nontrivial.

Here are all tame multiplicities. Identify the cyclic group X_N with the additive group $\mathbf{Z}/N\mathbf{Z}$ using a chosen generator. A tuple is an ordered sequence $(j_1, \dots, j_k)$, each $1 \leq j_i < N$. For $t \in \mathbf{Z}/N\mathbf{Z}$, let $D_{k,t}$ count those with $\sum_i j_i = t$. Choose any primitive complex N th root ω . The elementary root-of-unity filter yields

$$\begin{aligned} D_{k,t} &= \frac{1}{N} \sum_{a=0}^{N-1} \omega^{-at} \left(\sum_{j=1}^{N-1} \omega^{aj} \right)^k \\ &= \begin{cases} \frac{(N-1)^k + (N-1)(-1)^k}{N}, & t = 0, \\ \frac{(N-1)^k - (-1)^k}{N}, & t \neq 0. \end{cases} \end{aligned} \quad (\text{TPF.18})$$

The filter follows by summing $N^{-1} \sum_a \omega^{a(\sum j_i - t)}$, which is one when the congruence holds and zero otherwise. At $a = 0$ the inner sum is $N - 1$; for every other a it is -1 . Finally $\sum_{a=1}^{N-1} \omega^{-at}$ is $N - 1$ for $t = 0$ and -1 otherwise. This proves the count, including all repeated tuple occurrences.

Put $d_{0,k} = D_{k,0}$. Since a field has no nonzero nilpotents and $N \neq 0$, $c = 0$ holds exactly when $\rho = 0$ after the given specialization. Consequently

$$w_k = 0 \iff (c = 0 \text{ or } p \mid k), \quad \boxed{\dim V_k^{I_0} = \begin{cases} d_{0,k} = \frac{(N-1)^k + (N-1)(-1)^k}{N}, & w_k = 0, \\ 0, & w_k \neq 0. \end{cases}} \quad (\text{TPF.19})$$

This allows $d_{0,k} = 0$: for $k = 1$ it is zero, and for $N = 2$ it is zero at odd k , one at even k . For $N \geq 3$ and $k \geq 2$, both formulas in (TPF.18) are strictly positive: at even k , $(N - 1)^k > 1$; at odd $k \geq 3$, $(N - 1)^k > N - 1$.

The action in (TPF.15) specifies the entire original inertia representation, including its kernel. For clarity its image order can also be read off. Define

$$h_k = \begin{cases} 1, & N = 2 \text{ and } k \text{ even,} \\ N, & \text{otherwise.} \end{cases} \quad (\text{TPF.20})$$

The tame image has order h_k . Indeed for $N = 2$ there is only one character, whose k th power has that order. For $N \geq 3, k = 1$ the list includes a primitive character; for $N \geq 3, k \geq 2$ every product character occurs by (TPF.18). In either case the tame kernel is trivial. When $w_k \neq 0$, the wild image has order p , and no nontrivial tame scalar can cancel a nontrivial wild scalar: their orders divide the coprime integers N and p . Thus the full image has order $h_k p$ in that case, and order h_k when $w_k = 0$.

23.4 Swan conductor and the finite-cover logarithm

If $w_k \neq 0$, the nontrivial character $\mathcal{L}_\psi(w_k/u)$ has Swan conductor one. One may verify it directly over $\bar{k}_0((u))$: in $z^p - z = bw_k/u$, the ramification index is p by the valuation argument above, $v_E(z) = -1$, and

$$v_E((z + a)^{-1} - z^{-1}) = 2 \quad (a \in \mathbf{F}_p^\times). \quad (\text{TPF.21})$$

Thus its lower ramification groups are $G_0 = G_1 = \mathbf{F}_p, G_2 = 1$, giving one positive break of value one. Tensoring by a tame character does not change this wild restriction. Every tuple line in (TPF.9) has this same wild character. If $w_k = 0$, every tuple line is tame. Additivity of Swan under direct sum therefore proves

$$\boxed{\text{Swan}_0(\mathcal{T}_k) = \begin{cases} 0, & w_k = 0, \\ (N - 1)^k, & w_k \neq 0. \end{cases}} \quad (\text{TPF.22})$$

The finite cover

$$u = v^N, \quad z_k^p - z_k = \frac{bkc}{v^N} \quad (\text{TPF.23})$$

trivializes all geometric inertia of $\mathcal{T}_k$. The first map kills every product Kummer character. The second trivializes the displayed additive factor by its own torsor. If $w_k = 0$, it is a trivial disjoint torsor and the first map alone suffices. If $w_k \neq 0$, its degree is p , as its pole order N is prime to p . When $c \neq 0$ and $p \nmid k$, it is explicitly the original cover (TPF.13) with $z_k = kz$; this equality follows from $k^p = k$ in $\mathbf{F}_p$. When $p \mid k$, the original cover still kills the representation, and its wild deck subgroup already acts trivially by (TPF.15).

On the subgroup defined by this finite cover every monodromy matrix is the identity. Hence its unipotent logarithm is exactly

$$N_{\text{mon},k} = 0. \quad (\text{TPF.24})$$

There is no conflict with any of the invariant dimensions in (TPF.19). Those concern the original inertia group; after the displayed cover, the restricted representation has the full r_k -dimensional invariant space.

23.5 Frobenius on the invariant space and exact descent

Let $j : \mathbf{G}_m \hookrightarrow \mathbf{A}_u^1$. The ordinary invariant stalk $(j_* \mathcal{T}_{0,k})_{\bar{0}} = V_k^{I_0}$: sections over the punctured strict henselian neighborhood are precisely the invariant vectors of its geometric generic stalk. If $w_k \neq 0$ this stalk is zero. For $w_k = 0$, set

$$\Omega_k^0 = \{\chi \in \Omega_k : \eta_\chi = 1\}. \quad (\text{TPF.25})$$

Over L , the exact Frobenius module is

$$V_k^{I_0} \simeq \bigoplus_{\chi \in \Omega_k^0} \overline{\mathbf{Q}}_\ell(A_\chi), \quad \det(1 - T \operatorname{Fr}_Q | V_k^{I_0}) = \prod_{\chi \in \Omega_k^0} (1 - A_\chi T). \quad (\text{TPF.26})$$

Each surviving summand is the constant sheaf $\mathcal{A}_{A_\chi}$: its product Kummer character and its additive factor are both trivial. This proves the formula for the boundary stalk, as well as for its constant extension to the u -line.

Over k_0 , Frobenius need not preserve the individual tuple lines. Let F_0 denote its geometric Frobenius action. Its exact permutation is

$$F_0 : \chi \mapsto \chi^{Q_0} = (\chi_1^{Q_0}, \dots, \chi_k^{Q_0}). \quad (\text{TPF.27})$$

To prove its direction, over the splitting field the deck operator satisfies $F_0 D_\zeta F_0^{-1} = D_{\zeta^{Q_0^{-1}}}$; the exponent denotes the inverse modulo N . Conjugating the exact projector gives

$$F_0 \left(\frac{1}{N} \sum_{\zeta} \lambda_\chi(\zeta) D_\zeta \right) F_0^{-1} = \frac{1}{N} \sum_{\eta} \lambda_\chi(\eta^{Q_0}) D_\eta = e_{\chi^{Q_0}}. \quad (\text{TPF.28})$$

Tensoring this identity in the retained order proves (TPF.27). The coefficients of the projectors are in the fixed ℓ -adic coefficient field; the conjugation changes their deck operators, not their coefficient values. The subset Ω_k^0 is stable because taking a Q_0 th power preserves the trivial product character.

For each orbit $\mathcal{O}$ in Ω_k^0 , choose a tuple χ in it and let its exact orbit length be e . Then $e \mid d$. Put $L_e = \mathbf{F}_{Q_0^e} \subset L$. Each χ_i uniquely descends by norm to a nontrivial character $\kappa_{i,e} : L_e^\times \rightarrow \overline{\mathbf{Q}}_\ell^\times$:

$$\chi_i = \kappa_{i,e} \circ N_{L/L_e}, \quad \psi_e = \psi_0 \circ \operatorname{Tr}_{L_e/k_0}. \quad (\text{TPF.29})$$

Indeed $\chi_i^{Q_0^e} = \chi_i$, so its order divides $Q_0^e - 1$. If g generates $L^\times$, the kernel of this norm is generated by $g^{Q_0^e - 1}$, on which χ_i is trivial. The norm is surjective, proving existence, uniqueness, and nontriviality of $\kappa_{i,e}$. Also $\prod_i \kappa_{i,e} = 1$, again by surjectivity of the norm.

It is not necessary that all N th roots of unity belong to L_e . Here is the exact source map giving each individual descent. Put $r_i = \operatorname{ord}(\chi_i)$. Then $r_i \mid N$ and $r_i \mid Q_0^e - 1$, so $\mu_{r_i} \subset L_e$. Factor the original power map by

$$\mathbf{A}_y^1 \longrightarrow \mathbf{A}_x^1 \longrightarrow \mathbf{A}_z^1, \quad x = y^{N/r_i}, \quad z = x^{r_i} = y^N.$$

Pullback of functions on geometric fibre roots identifies the direct image for $x^{r_i} = z$ with the μ_{N/r_i} -invariants of the original $y^N = z$ direct image. For $z \neq 0$, the quotient of its root set by μ_{N/r_i} is exactly the root set of $x^{r_i} = z$, and invariant functions are exactly the functions constant on these orbits. At zero both geometric fibres have one point and the map is the identity. The sheaf map is therefore an isomorphism onto those invariants at every geometric stalk.

The character λ_{χ_i} factors through $\mu_N \rightarrow \mu_{r_i}$, $\zeta \mapsto \zeta^{N/r_i}$, as $\lambda_{\kappa_{i,e}}$. Indeed for $a \in L^\times$,

$$\lambda_{\kappa_{i,e}}(a^{(Q-1)/r_i}) = \kappa_{i,e}(a^{(Q-1)/(Q_0^e-1)}) = \chi_i(a) = \lambda_{\chi_i}(a^{(Q-1)/N}),$$

and $a \mapsto a^{(Q-1)/N}$ surjects onto μ_N . Thus the descended individual source summand is precisely the trace- $\kappa_{i,e}$ Kummer line, including its zero stalk. The phase on that source remains $z/(Nu) + c/u$, with the original denominator N , not r_i . Applying the same source isomorphism and trace formula as in SPF over L_e therefore computes the cycle scalar exactly:

$$\Lambda_{\mathcal{O}} = \prod_{i=1}^k (-\kappa_{i,e}(N) G_{L_e}(\kappa_{i,e}, \psi_e)). \quad (\text{TPF.30})$$

Every individual factor, including N and the minus sign, is retained. More explicitly F_0^e acts by this product on the selected tuple line: at $u = 1$ all its Kummer values are one, while the product of its additive values is $\psi_e(kc) = 1$, since $w_k = 0$. The tuple product is a constant sheaf, so this is its action at the invariant boundary stalk too. This proves the scalar over its actual orbit field; it has not been selected as an arbitrary root of an eigenvalue over L .

Choose $v_0 \neq 0$ in the tuple line and set $v_j = F_0^j v_0$ for $0 \leq j < e$. These vectors are in distinct direct-summand lines, so form a basis for the orbit block. In this basis $F_0 v_j = v_{j+1}$ for $j < e - 1$, and $F_0 v_{e-1} = \Lambda_{\mathcal{O}} v_0$. Consequently

$$\boxed{\begin{aligned} \det(T - F_0 \mid V_k^{I_0}) &= \prod_{\mathcal{O} \subset \Omega_k^0} (T^{|\mathcal{O}|} - \Lambda_{\mathcal{O}}), \\ \det(1 - TF_0 \mid V_k^{I_0}) &= \prod_{\mathcal{O} \subset \Omega_k^0} (1 - \Lambda_{\mathcal{O}} T^{|\mathcal{O}|}). \end{aligned}} \quad (\text{TPF.31})$$

These expressions concern $w_k = 0$; when $w_k \neq 0$ both determinants on the zero space are one. Choosing another representative does not change $\Lambda_{\mathcal{O}}$, because F_0^e commutes with F_0 and hence has the same scalar on every line of that orbit. Formula (TPF.31) thus proves descent without asserting diagonal Frobenius in the original tuple coordinates over k_0 .

23.6 Exact weight and the two cancellation cases

The elementary unscaled Gauss calculation in SPF gives, for every complex embedding,

$$|G_L(\chi, \psi)| = Q^{1/2}, \quad |A_{\chi}| = Q^{k/2}. \quad (\text{TPF.32})$$

In (TPF.10) all other factors are roots of unity. At a point of degree a over L , every eigenvalue therefore has absolute value $Q^{ak/2}$. This proves pointwise purity of weight k for the actual ordered tensor. It descends to k_0 : a positive power of any original Frobenius is covered by the calculation over L , and $|\beta^a| = Q_0^{fak/2}$ forces $|\beta| = Q_0^{fk/2}$ at an original point of degree f .

When $w_k = 0$, the same calculation over L_e in (TPF.30) gives

$$|\Lambda_{\mathcal{O}}| = Q_0^{ek/2}, \quad |\beta| = Q_0^{k/2} \quad \text{for every root } \beta^e = \Lambda_{\mathcal{O}}. \quad (\text{TPF.33})$$

Thus every nonzero invariant block has weight k on $\text{Spec } k_0$, including the blocks that occur because $p \mid k$. If $c = 0$, the original single-factor wild character was already trivial. If $c \neq 0$ and $p \mid k$, it becomes trivial by the exact tensor character $\vartheta(-ka)$. Both branches have the same tame multiplicity formula, but the displayed maps account for their different origins. No complex arithmetic zeta zero or comparison of such a zero with these finite-field Frobenius eigenvalues is constructed by this calculation.

23.7 Exact counts of the arithmetic orbit blocks

For $f \mid d$, retain the full greatest common divisor and define

$$g_f = \gcd(N, Q_0^f - 1), \quad D(f) = \frac{(g_f - 1)^k + (g_f - 1)(-1)^k}{g_f}. \quad (\text{TPF.34})$$

The characters fixed by exponent Q_0^f form a cyclic subgroup of X_N of order g_f . A fixed ordered tuple has all entries in this subgroup. Applying the root-of-unity count (TPF.18) inside that subgroup shows that exactly $D(f)$ tuples in Ω_k^0 are fixed. When $g_f = 1$ there are no allowed nontrivial entries and the formula is zero, since $k \geq 1$. If $C(e)$ denotes the number of orbits in Ω_k^0 of exact length $e \mid d$, then

$$D(f) = \sum_{e \mid f} e C(e), \quad C(e) = \frac{1}{e} \sum_{f \mid e} \mu(e/f) D(f). \quad (\text{TPF.35})$$

The first identity counts the elements of every orbit fixed by F_0^f . For the second, multiply that identity by $\mu(e/f)$ and sum over $f \mid e$. The coefficient of $rC(r)$ is $\sum_{r \mid f \mid e} \mu(e/f)$, which is one for $r = e$ and otherwise zero: the latter sum factors as $\prod_{q \mid e/r} (1 - 1)$ over its distinct prime divisors. This proves the orbit count by integer arithmetic. These orbit numbers count the nonzero boundary blocks exactly when $w_k = 0$; when $w_k \neq 0$ the boundary stalk is zero by (TPF.19).

23.8 The terminal character, its orbit, and the original scalar

Fix a generator of the abstract cyclic deck group of order N , and identify its chosen complex N th root of unity and its finite-field root in $\mu_N(L)$ with a chosen N th root in the ℓ -adic coefficient field. These are explicit identifications of finite cyclic groups. Let $\chi_* \in X_N$ be the primitive character whose λ_{χ_*} is the resulting tautological character.

The direction of the deck action matters. For $\delta_\zeta(y) = \zeta y$, direct pullback gives

$$\delta_\zeta^*(y^b dy) = \zeta^{b+1} y^b dy, \quad \delta_\zeta^* e_r = e_{\zeta^{-1}r}, \quad D_\zeta = (\delta_\zeta^{-1})^*. \quad (\text{TPF.36})$$

The middle equality follows by evaluating the indicator function of the point r at ζy . Thus the finite SPF summand labelled χ has D_ζ -eigenvalue $\lambda_\chi(\zeta)^{-1}$ but ordinary deck-pullback eigenvalue $\lambda_\chi(\zeta)$. Comparing the pullback characters with the pullback characters, rather than with the inverse action D_ζ , gives the exact character-label map

$$b \mapsto \chi_b = \chi_*^{b+1}, \quad \chi_{\text{ter}} = \chi_*^m = \chi_*^{-1}, \quad \text{ord}(\chi_{\text{ter}}) = N. \quad (\text{TPF.37})$$

Here the terminal coefficient is y^{m-1} , and its form is $y^{m-1} dy$, so $b = m - 1 = N - 2$. The ordered terminal form has deck-pullback character $\zeta^{km} = \zeta^{-k}$, precisely as the product χ_{ter}^k records. This proves a map of the finite deck-character index sets. It does not assert a de Rham-étale comparison isomorphism or identify a complex arithmetic operator with finite-field Frobenius.

The repeated terminal tuple is $\chi_{\text{ter}} = (\chi_{\text{ter}}, \dots, \chi_{\text{ter}})$. Its actual finite-field summand on the same u -line is exactly

$$\mathcal{L}_\psi(kc/u) \otimes \mathcal{A}_{[-\chi_{\text{ter}}(N)G_L(\chi_{\text{ter}}, \psi)]^k} \otimes \mathcal{K}_{\chi_{\text{ter}}^k}(u). \quad (\text{TPF.38})$$

Its original-inertia invariants have dimension

$$\begin{cases} 1, & N \mid k \text{ and } kc = 0, & c = 0 : \text{ the nonzero case is } N \mid k, \\ 0, & \text{otherwise.} & c \neq 0 : \text{ the nonzero case is } pN \mid k. \end{cases} \quad (\text{TPF.39})$$

Indeed its tame product is trivial exactly when $N \mid k$, since χ_{ter} is primitive. Its wild character is trivial exactly when $kc = 0$. In the second branch $p \mid k$ and $N \mid k$ are equivalent to $pN \mid k$, since $\gcd(p, N) = 1$.

This single terminal line need not descend to k_0 . Its exact arithmetic orbit consists of the d distinct ordered tuples

$$\left(\chi_{\text{ter}}^{Q_0^j}, \dots, \chi_{\text{ter}}^{Q_0^j} \right), \quad 0 \leq j < d.$$

Its length is exactly d : equality of two tuple entries requires equality of their primitive characters, hence $Q_0^e = 1 \bmod N$, whose least positive solution is d . If the criterion in (TPF.39) holds, all these lines are invariant and the descended terminal-orbit block has dimension d , with

$$\begin{aligned} \Lambda_{\text{ter}} &= [-\chi_{\text{ter}}(N)G_L(\chi_{\text{ter}}, \psi)]^k, \\ \det(T - F_0 \mid \text{terminal orbit}) &= T^d - \Lambda_{\text{ter}}, \\ \det(1 - TF_0 \mid \text{terminal orbit}) &= 1 - \Lambda_{\text{ter}} T^d. \end{aligned} \quad (\text{TPF.40})$$

This is (TPF.30)–(TPF.31) with its actual orbit field L . Every root has absolute value $Q_0^{k/2}$. If the criterion fails, the invariant part of this orbit is zero. No Frobenius eigenline over k_0 is inferred from a single summand over L .

Finally retain the original scalar multiplying the terminal coefficient in the full marked packet h_0 , evaluated on a retained stage where its original source coefficients belong to the domain of the actual map α :

$$K = k(m-1), \quad u_0 = (g/h_0)(\rho), \quad C_{h_0,k} = \frac{K!}{((m-1)!)^k} u_0^k. \quad (\text{TPF.41})$$

Here K is an integer, and u_0 is the specified marked coefficient in that actual coefficient model. This calculation does not extend α to a larger cyclic CRT stage by assigning values to new denominators. If such a stage inverts p , no extension of α to it exists: applying a purported extension to $1 = p p^{-1}$ would give $1 = 0$. The absence of a stage specialization is not a vanishing coefficient specialization. Writing a bar for that same specialization gives the exact image

$$\overline{C}_{h_0,k} = \overline{K!} (\overline{(m-1)!})^{-k} \overline{u}_0^k, \quad \overline{C}_{h_0,k} \neq 0 \iff K < p \text{ and } \overline{u}_0 \neq 0. \quad (\text{TPF.42})$$

The denominator is nonzero because $m-1 = N-2 < p$. The numerator $K!$ is nonzero modulo p exactly when $K < p$, and a positive power of a field element is nonzero exactly when that element is nonzero. These facts prove the equivalence. When the original unit class and its inverse are retained at this marked point, their product is one after evaluation at ρ and after specialization; hence $\overline{u}_0 \neq 0$ in that specified model. Even there the factorial can vanish.

In particular, when $m \geq 2$ and $p \mid k$, one has $K = k(m-1) \geq p$, so $\overline{C}_{h_0,k} = 0$. Thus the $c \neq 0$ terminal-invariant degrees $pN \mid k$, on a marked stage admitting the specified α , have zero specialized original scalar for $m \geq 2$, although their sheaf summand and invariant orbit block computed above are nonzero. For $m = 1$, $K = 0$ and the scalar is exactly u_0^k , so only its original coefficient specialization can make it zero. For $c = 0$, the tame condition $N \mid k$ and the separate scalar condition $k(m-1) < p$, $\overline{u}_0 \neq 0$ are both retained.

The last scalar calculation has a precise coefficient-module target. The original translation and coefficient specialization give

$$\begin{aligned} R[y_1, \dots, y_k]/(y_1^m, \dots, y_k^m) &\longrightarrow \overline{E}_k = k_0[y_1, \dots, y_k]/(y_1^m, \dots, y_k^m), \\ \sum_{\mathbf{a}} q_{\mathbf{a}} \prod_i y_i^{a_i} &\longmapsto \sum_{\mathbf{a}} \alpha(q_{\mathbf{a}}) \prod_i y_i^{a_i}, \\ \overline{\tau}_{\text{ter}} = \prod_i y_i^{m-1} &\neq 0, \quad \overline{C}_{h_0,k} \overline{\tau}_{\text{ter}} = 0 \iff \overline{C}_{h_0,k} = 0. \end{aligned} \quad (\text{TPF.43})$$

Successive division by the monic polynomials y_i^m gives the unique remainder basis $\prod_i y_i^{a_i}$, $0 \leq a_i < m$, so $\overline{\tau}_{\text{ter}}$ is one of its nonzero basis vectors and the last equivalence follows. All nilpotent directions of this coefficient algebra are retained. These scalar statements are in characteristic p . There is no unital field map from k_0 to $\mathbf{Q}_\ell$, since such a map would send $p \cdot 1 = 0$ to the nonzero element $p \cdot 1$. Thus no multiplication of an ℓ -adic Gauss line by $\overline{C}_{h_0,k}$ is being asserted. The deck-character map (TPF.37) does not by itself transport coefficient vectors to etale cohomology; the finite-field sheaf is the independently specified compact-support sheaf and does not depend on this scalar.

23.9 The shared integral deck-character model and its two base changes

There is an exact common integral representation behind the character correspondence, which does not require a map between fields of different characteristics. Let

$$\begin{aligned} B &= \mathbf{Z}[\zeta_N, 1/N], \quad C_N = \langle \gamma : \gamma^N = 1 \rangle, \\ e_j &= \frac{1}{N} \sum_{a=0}^{N-1} \zeta_N^{-ja} \gamma^a \in B[C_N] \quad (0 \leq j < N). \end{aligned} \quad (\text{TPF.44})$$

Here ζ_N is a fixed primitive root in the cyclotomic ring. Finite geometric sums give

$$e_i e_j = \delta_{ij} e_j, \quad \sum_{j=0}^{N-1} e_j = 1, \quad \gamma e_j = \zeta_N^j e_j. \quad (\text{TPF.45})$$

For example the coefficient convolution in $e_i e_j$ contains $\sum_a \zeta_N^{a(j-i)}$, which is N for $i = j$ and zero otherwise. The sum of the e_j has coefficient one at γ^0 and zero elsewhere, and reindexing $a + 1$ proves the last identity.

The corresponding free module of polynomial forms with the original degree retained is

$$M_B = B[y] dy / y^m B[y] dy, \quad \gamma(y^b dy) = \zeta_N^{b+1} y^b dy \quad (0 \leq b < m). \quad (\text{TPF.46})$$

It is the quotient of polynomial one-forms by y^m times that free module, not the module of Kahler differentials of the truncated algebra. Its basis is $dy, y dy, \dots, y^{m-1} dy$. Thus the exact $B[C_N]$ -linear isomorphism

$$\bigoplus_{j=1}^{N-1} B e_j \longrightarrow M_B, \quad e_j \longmapsto y^{j-1} dy \quad (\text{TPF.47})$$

follows from (TPF.45) and the displayed bases. In an ordered k -fold tensor, $e_{j_1} \otimes \dots \otimes e_{j_k}$ therefore projects onto the corresponding ordered monomial-form basis line. The simultaneous e_0 -projector selects $\sum_i j_i = 0 \bmod N$, exactly the count in (TPF.18). The terminal line is $(B e_m)^{\otimes k}$.

Choose a primitive root $\zeta_L \in L$ and the primitive ℓ -adic root ζ_ℓ already used in (TPF.37). The two actual ring maps are

$$\iota_p : B \rightarrow L, \quad \zeta_N \mapsto \zeta_L, \quad \iota_\ell : B \rightarrow \overline{\mathbf{Q}}_\ell, \quad \zeta_N \mapsto \zeta_\ell. \quad (\text{TPF.48})$$

They are well defined: $p \nmid N$, $N \mid Q - 1$, and both selected roots are primitive N th roots. The first base change of (TPF.47) is the concrete isomorphism with $L[y] dy / y^m L[y] dy$, under which e_j is the projector onto $L y^{j-1} dy$. The coefficient-map connection is also exact: $R \otimes_{\mathbf{Z}} B \rightarrow L$ sends $r \otimes b'$ to $\alpha(r) \iota_p(b')$. Tensoring its polynomial-form module gives this same characteristic- p module. This does not introduce new inverses from a larger marked stage.

For the second base change, let W_ℓ be the fibre at $u = 1$ of the actual untwisted sheaf $\mathcal{L}_\psi(c/u)^\vee \otimes \mathcal{F}$. Give it the source deck-pullback action $\gamma = D_{\zeta_L}^{-1}$. The image of the same integral projector is

$$\iota_\ell(e_j) = \frac{1}{N} \sum_{a=0}^{N-1} \zeta_\ell^{-ja} D_{\zeta_L}^{-a} = \frac{1}{N} \sum_{a=0}^{N-1} \zeta_\ell^{ja} D_{\zeta_L}^a = e_{\chi_*^j}^{\text{SPF}}. \quad (\text{TPF.49})$$

The second equality reindexes $a \mapsto -a$. The third is precisely the original SPF power-map projector, with $\lambda_{\chi_*^j}(\zeta_L) = \zeta_\ell^j$. Thus it projects onto the actual one-dimensional Gauss line $H_j = e_{\chi_*^j}^{\text{SPF}} W_\ell$, whose geometric Frobenius over L is

$$-\chi_*^j(N) G_L(\chi_*^j, \psi).$$

Keep that entire line, including its Frobenius, rather than choosing a numerical basis for it. If H_j^{triv} denotes the same vector space and Frobenius with only its C_N -action replaced by the trivial action, there is the canonical isotypic reconstruction below. Give each first character-module factor the identity splitting-field Frobenius:

$$\bigoplus_{j=1}^{N-1} (B e_j \otimes_{B, \iota_\ell} \overline{\mathbf{Q}}_\ell) \otimes_{\overline{\mathbf{Q}}_\ell} H_j^{\text{triv}} \longrightarrow W_\ell, \quad (\text{TPF.50})$$

$$(a e_j) \otimes h \longmapsto a h.$$

It is an isomorphism: on the j th summand it is the scalar multiplication isomorphism onto H_j , and the SPF projectors are complete and orthogonal. It preserves the deck action by (TPF.45) and preserves the splitting-field Frobenius by the retained action on H_j^{triv} . No Frobenius action on a characteristic- p coefficient vector has been assumed.

Tensoring these actual maps in the fixed order gives the shared terminal-character projector on both sides: $(e_m)^{\otimes k}$ maps to the characteristic- p line $L(\prod_i y_i^{m-1} dy_i)$ and to the ordered Gauss line $H_m^{\otimes k}$. The full finite-field sheaf reinstates its already specified additive factor $\mathcal{L}_\psi(kc/u)$; the common integral model supplies the finite μ_N representation, not an unconstructed universal additive sheaf. Formulas (TPF.44)–(TPF.50) are the exact common projector and base-change correspondence. They retain the coefficient scalar specialization in (TPF.43) and the independently nonzero ℓ -adic sheaf lines as their stated, different targets.

23.10 The multiplicity-one case and its exact scalar comparison

When $m = 1$, one has $N = 2$, $p > 2$, $d = 1$, and $L = k_0 = \mathbf{F}_Q$, because every odd Q_0 satisfies $Q_0 = 1 \pmod{2}$. There is just the nontrivial quadratic character χ , and it is the terminal character. Retain the original constants

$$c = -\rho^2/2, \quad \gamma_\chi = -\chi(2)G_L(\chi, \psi). \quad (\text{TPF.51})$$

Their exact Gauss-square identity is

$$\gamma_\chi^2 = \chi(-1)Q. \quad (\text{TPF.52})$$

To prove it, take a complex embedding of the cyclotomic field of character values. Quadratic-character values are real, so changing z to $-z$ in the conjugate Gauss sum gives $\overline{G_L(\chi, \psi)} = \chi(-1)G_L(\chi, \psi)$. The proved absolute-square identity $G_L \overline{G_L} = Q$ implies $G_L^2 = \chi(-1)Q$, since $\chi(-1)^2 = 1$. Also $\chi(2)^2 = 1$, proving (TPF.52) with the original minus sign and the factor 2 retained. Both sides lie in the same cyclotomic number field; injectivity of the chosen embedding proves that algebraic identity there and hence in its chosen ℓ -adic realization.

For $c \neq 0$, every positive terminal-invariant degree is exactly $k = 2pr$, $r \geq 1$. Its one-dimensional original-base invariant Frobenius is

$$\gamma_\chi^k = (\chi(-1)Q)^{pr}, \quad |\gamma_\chi^k| = Q^{k/2}. \quad (\text{TPF.53})$$

Its additive factor is exactly $\mathcal{L}_\psi(2prc/u) = \mathcal{L}_\psi(0)$; its Kummer factor is $\mathcal{K}_{\chi^{2pr}} = \mathcal{K}_1$. Thus the line is an actual nonzero invariant line of weight k . Here $K = k(m-1) = 0$ and the original marked coefficient is exactly u_0^k . On an actual coefficient stage retaining the inverse of u_0 , its reduction is nonzero. There is no factorial loss. For $c = 0$, every even k is tame invariant and the same formula reads $\gamma_\chi^k = (\chi(-1)Q)^{k/2}$; no divisibility by p is needed.

There is also an exact comparison calculation for any chosen map between the corresponding one-dimensional complex spaces, without asserting a canonical comparison. The actual arithmetic primary algebra is $\mathbf{C}[s]/(s - \rho)$. In its ordered k -fold tensor, the original sum operator $S = \sum_i s_i$ is multiplication by $k\rho$. For $a > 0$, its dilation operator is therefore

$$A_a = \exp((\log a)S) = a^{k\rho} \text{id}, \quad a^{k\rho} = \exp(k\rho \log a). \quad (\text{TPF.54})$$

Choose a complex embedding of the cyclotomic number field containing γ_χ , and let L_G be the resulting one-dimensional complex Frobenius model, with $F_G = \gamma_\chi^k \text{id}$. This takes a number-field model of the computed eigenvalue to $\mathbf{C}$, not a field map from k_0 to $\mathbf{C}$. For every nonzero complex-linear map $T : L_G \rightarrow L_{\text{arith}}$, scalar multiplication proves

$$\begin{aligned} A_a T &= \frac{a^{k\rho}}{\gamma_\chi^k} T F_G, \\ A_a T - T F_G &= (a^{k\rho} - \gamma_\chi^k) T, \quad \left| \frac{a^{k\rho}}{\gamma_\chi^k} \right| = \frac{a^{k \operatorname{Re} \rho}}{Q^{k/2}}. \end{aligned} \quad (\text{TPF.55})$$

For example such maps exist by sending a nonzero basis vector to a nonzero basis vector; changing either basis multiplies the matrix of T by a nonzero scalar and leaves the displayed ratio unchanged. At $a = Q$ the ratio has modulus $Q^{k(\operatorname{Re} \rho - 1/2)}$. This is the exact intertwining defect for every such map. Existence of a nonzero map does not turn this computed ratio into one, and no identification of the two operators has been assumed.

Sources and dependencies. The sole specialized input is the proved SPF.1–29 single-primary sheaf calculation, with its exact power-map projectors, cohomology, Gauss signs, and local inertia. The foundational curve and trace formulas are in Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, Sections 2.0–2.3 and 4.0–4.3. The torsor constructions are the Stacks Project, tags 03PK and 0A3J. The relative compact-support Kunneth map in (TPF.6) is SGA 4, Expose XVII, Theorem 5.4.3, applied to the common base $\mathbf{G}_{m,u}$. All tensor multiplicities, original inertia projectors, cancellation cases, descent permutations, and orbit scalars are proved here.

24 The retained primary cyclic lattice at its specified coefficient prime

This calculation uses the ordered primary tensor and full Taylor unit of TP.1 and TP.23–25. It computes its original cyclic map over the specified coefficient prime, including the map before reduction, its primitive containing lattice, its cokernel, and the exact reduction sequence. The support carrier used below is the original G_{C_n} of MFC.29 and MFC.35–44.

24.1 The actual coefficient localization and the unchanged coordinates

Use the single-primary coefficient specialization already specified in SPF.1. Denote its retained coefficient subring of $\mathbb{C}$ by A , its map by $\alpha : A \rightarrow k_0$, and its characteristic by the prime p . Retain every coefficient of the full Taylor class and its inverse, and put

$$\mathfrak{q} = \ker \alpha, \quad S = A_{\mathfrak{q}}, \quad \tilde{\alpha} : S \longrightarrow k_0, \quad \tilde{\alpha}(a/b) = \alpha(a)/\alpha(b). \quad (\text{TCL.1})$$

Thus $b \notin \mathfrak{q}$ in this formula. The image of α is a finite subring of a field. For each nonzero element of that image, multiplication on the finite image is injective and therefore surjective; its inverse lies in the image. The image is consequently a field, proving that $\mathfrak{q}$ is maximal. The fraction formula is independent of representatives: equality of two fractions gives equality after cross multiplication in the domain A , and applying α and dividing by the nonzero denominator images gives the asserted equality. It preserves the ring operations by the same calculation. The map extends the specified map, rather than selecting a new one.

Since $A \subset \mathbb{C}$, both A and its localization S are domains of characteristic zero. The element p is a nonzero nonunit: its image is zero whereas the image of the unit is one. Every integer prime to p is outside $\mathfrak{q}$ and is a unit in S . In particular $\mathbb{Z}_{(p)} \rightarrow S$ is a well-defined injective scalar map. No assertion that S is a discrete valuation ring, a principal ideal domain, or a ring with all ideals in a chain is made or used. The algebra below uses $p > m$; the retained SPF specialization has the stronger $p > m + 1$, as required there by its phase denominators.

Keep the selected primary coordinate ρ , its full order $m \geq 1$, and the ordered number of factors $k \geq 1$. Put

$$\begin{aligned} K &= k(m-1), & y_i &= s_i - \rho, & E_S &= S[s_1, \dots, s_k] / ((s_i - \rho)^m)_{i=1}^k = S[y_1, \dots, y_k] / (y_i^m)_{i=1}^k, \\ J &= \sum_{i=1}^k y_i, & M &= \sum_{i=1}^k s_i = k\rho + J, & \mathcal{C}_S &= S[T] / (T^{K+1}), & \mathcal{C}_S^{\text{orig}} &= S[Z] / ((Z - k\rho)^{K+1}). \end{aligned} \quad (\text{TCL.2})$$

The two displayed presentations of E_S are identified by the mutually inverse substitutions $s_i = y_i + \rho$ and $y_i = s_i - \rho$. The cyclic-source coordinate map $\mathcal{C}_S^{\text{orig}} \rightarrow \mathcal{C}_S$ is $Z \mapsto k\rho + T$, with

inverse $T \mapsto Z - k\rho$. These substitutions respect the defining ideals and their composites are the identity on each generator. They preserve the original objects and will be recorded explicitly in the original monomial matrices.

Let $\mathcal{B} = \{0, \dots, m-1\}^k$, ordered lexicographically. Repeated division by the monic relations y_i^m proves that $\{y^{\mathbf{b}} : \mathbf{b} \in \mathcal{B}\}$ is an S -basis: deleting monomials with any exponent at least m gives a remainder, and the relation ideal is spanned by precisely the deleted monomials, so no relation between retained monomials is possible. Every surviving degree is at most K ; hence $J^{K+1} = 0$.

Write the retained one-factor unit and its retained inverse as

$$\begin{aligned} v(y) &= \sum_{j=0}^{m-1} a_j y^j, & a_j &= \frac{(g/h)^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!}, & h &= (s - \rho)^m, \\ w(y) &= \sum_{j=0}^{m-1} b_j y^j, & b_0 &= a_0^{-1}, & b_r &= -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}, \\ U &= \prod_{i=1}^k v(y_i), & U^{-1} &= \prod_{i=1}^k w(y_i) \quad \text{in } E_S. \end{aligned} \tag{TCL.3}$$

These are the actual retained coefficients, all lying in A by its specified construction. The derivative identity follows by expanding $g = (s - \rho)^m (g/h)$ at ρ . The recurrence makes the constant coefficient of vw equal to one and its coefficients of degrees $1, \dots, m-1$ equal to zero. Thus $vw = 1$ modulo y^m , proving the inverse in every factor and then in the tensor. In particular multiplication by U is an actual S -linear automorphism of E_S commuting with multiplication by J and M . No coefficient of this unit has been discarded.

The unchanged unweighted cyclic ring map and the original unit-weighted cyclic coefficient map are respectively

$$\begin{aligned} f : \mathcal{C}_S &\longrightarrow E_S, & f(T^r) &= J^r, \\ f_U : \mathcal{C}_S &\longrightarrow E_S, & f_U(T^r) &= U J^r, \\ f_U^{\text{orig}} : \mathcal{C}_S^{\text{orig}} &\longrightarrow E_S, & f_U^{\text{orig}}(Z^j) &= U M^j. \end{aligned} \tag{TCL.4}$$

The relation $J^{K+1} = 0$ proves that all maps are well-defined. The last two are linear maps, and are maps of modules over the cyclic source ring acting on the target by $T \mapsto J$, respectively $Z \mapsto M$. They are not asserted to be unital ring maps: their value at one is the retained U . Their exact intertwining identities are

$$f_U(Tc) = J f_U(c), \quad f_U^{\text{orig}}(Zc) = M f_U^{\text{orig}}(c). \tag{TCL.5}$$

Both follow by multiplication in the commutative algebra E_S . This is the coefficient realization of the original source map $\mathcal{J}_k \mathcal{T}_k$ in TP.25, restricted to polynomials in the sum of its unchanged s_i coordinates.

24.2 The primitive lattice and the exact factorial inclusion

For $0 \leq r \leq K$, define

$$\mathcal{B}_r = \{\mathbf{b} \in \mathcal{B} : |\mathbf{b}| = r\}, \quad \mathbf{b}! = \prod_{i=1}^k b_i!, \quad V_r = \sum_{\mathbf{b} \in \mathcal{B}_r} \frac{y^{\mathbf{b}}}{\mathbf{b}!}, \quad V_{K+1} = 0. \tag{TCL.6}$$

Every denominator is a unit because $b_i < m < p$. Each set $\mathcal{B}_r$ is nonempty. For $m > 1$ a specified member $\mathbf{b}^*(r)$ is obtained recursively by $b_i^*(r) = \min(m-1, r - \sum_{j < i} b_j^*(r))$. The remaining degree is nonnegative at each step and is zero after k steps because $r \leq k(m-1)$. For $m = 1$ there is only $r = 0$, with $\mathbf{b}^*(0) = (0, \dots, 0)$.

The multinomial expansion and direct multiplication give

$$J^r = r!V_r \quad (0 \leq r \leq K), \quad JV_r = (r+1)V_{r+1} \quad (0 \leq r \leq K). \quad (\text{TCL.7})$$

To prove the first formula, expand the product of r copies of $\sum_i y_i$. For a prescribed exponent vector $\mathbf{b}$ there are $r!/\mathbf{b}!$ assignments of its r ordered factors to the variables. Exponents at least m give zero in the quotient. This leaves exactly the displayed coefficients. For the second formula, the coefficient of a retained $y^{\mathbf{b}}$ of degree $r+1$ is $\sum_{i:b_i>0} 1/(b_i-1)! \prod_{j \neq i} 1/b_j! = (\sum_i b_i)/\mathbf{b}! = (r+1)/\mathbf{b}!$. For $r = K$ every resulting monomial is zero, which agrees with the convention for V_{K+1} .

Define coefficient functionals, Λ containing lattice and its projection by

$$\begin{aligned} \lambda_r(x) &= \mathbf{b}^*(r)! [y^{\mathbf{b}^*(r)}]x, \\ \Lambda &= \bigoplus_{r=0}^K SV_r \subset E_S, \quad \pi(x) = \sum_{r=0}^K \lambda_r(x) V_r, \\ F &= \text{span}_S \{y^{\mathbf{b}} : \mathbf{b} \notin \{\mathbf{b}^*(0), \dots, \mathbf{b}^*(K)\}\}. \end{aligned} \quad (\text{TCL.8})$$

Distinct V_r have distinct degrees, and the pivot coefficient of V_r is the unit $1/\mathbf{b}^*(r)!$. Therefore $\lambda_r(V_t) = \delta_{rt}$, proving both the direct sum and $\pi^2 = \pi$. Subtracting $\pi(x)$ deletes exactly all pivot coordinates, so $x - \pi(x) \in F$. Conversely all pivot coordinates vanish on F ; hence $\pi(F) = 0$. We have proved the explicit splitting

$$E_S = \Lambda \oplus F, \quad \ker \pi = F. \quad (\text{TCL.9})$$

In particular each V_r is primitive in the sense that it is part of the displayed basis of a direct summand, with no division by $r!$ in the lattice construction.

Write $L = f(\mathcal{C}_S)$. Equations (TCL.7)–(TCL.9) imply

$$\begin{aligned} L &= \bigoplus_{r=0}^K r!SV_r \subset \Lambda, \quad \Lambda/L \simeq \bigoplus_{r=0}^K S/(r!), \\ W_r &= UV_r, \quad \Lambda_U = U\Lambda, \quad L_U = f_U(\mathcal{C}_S) = UL, \quad F_U = UF, \\ E_S &= \Lambda_U \oplus F_U, \quad \pi_U = U\pi U^{-1}, \quad L_U = \bigoplus_{r=0}^K r!SW_r. \end{aligned} \quad (\text{TCL.10})$$

Since S is a characteristic-zero domain, each $r!$ is nonzero and multiplication by it is injective. Thus f and f_U are injective. The quotient isomorphism sends $\sum_r c_r V_r$ to $(c_r \bmod r!)_r$; its kernel and surjectivity follow coordinate by coordinate. Applying the automorphism U proves every transported identity in the display. There is no claim that $U\Lambda = \Lambda$; it is the pair of lattices $L \subset \Lambda$ that is transported.

Let $Q = \text{Frac}(S)$. The exact saturation statement is

$$(L_U \otimes_S Q) \cap E_S = \Lambda_U \quad \text{inside } E_S \otimes_S Q. \quad (\text{TCL.11})$$

Indeed the nonzero $r!$ are invertible in Q , so $L_U \otimes Q = \Lambda_U \otimes Q$. In the splitting $E_S = \Lambda_U \oplus F_U$, an integral element belonging to this rational subspace has zero F_U -component, because the free module F_U injects into $F_U \otimes Q$. This proves the stated intersection, including both inclusions.

Both lattices are stable under J . On their indicated bases, the full action and its covariant inclusion are

$$\begin{aligned} JW_r &= (r+1)W_{r+1}, \quad J(r!W_r) = (r+1)!W_{r+1}, \\ \mathcal{C}_S &\xrightarrow{f_U} \Lambda_U, \quad T^r \mapsto r!W_r, \quad f_U \circ M_T = M_J \circ f_U. \end{aligned} \quad (\text{TCL.12})$$

At $r = K$ both actions are zero. Commutation of U and J and (TCL.7) prove all entries. This displays the actual source multiplication and the containing-lattice multiplication together with their exact intertwining map.

For clarity, the same map in all original monomial coordinates is

$$\begin{aligned} P_{ab} &= \begin{cases} \binom{b}{a} \rho^{b-a} & a \leq b, \\ 0 & a > b, \end{cases} \quad (0 \leq a, b < m), \quad P_k = P^{\otimes k}, \\ B_{rj} &= \begin{cases} \binom{j}{r} (k\rho)^{j-r} & r \leq j, \\ 0 & r > j, \end{cases} \quad (0 \leq r, j \leq K), \\ \mathbf{V}_{\mathbf{br}} &= \begin{cases} 1/\mathbf{b}! & |\mathbf{b}| = r, \\ 0 & \text{otherwise,} \end{cases} \quad \Delta = \text{diag}(0!, 1!, \dots, K!), \\ [f_U^{\text{orig}}]_{(s^{\mathbf{b}}), (Z^j)} &= P_k^{-1} [U]_{(y^{\mathbf{b}})} \mathbf{V} \Delta B. \end{aligned} \tag{TCL.13}$$

The binomial theorem gives the two translation matrices P_k and B . Their inverses replace ρ and $k\rho$ by their negatives, respectively, as verified by the inverse substitutions above; their diagonal entries are one. The j -th column on the right is consequently the unchanged polynomial $U \sum_{r=0}^j \binom{j}{r} (k\rho)^{j-r} r! V_r = U(k\rho + J)^j$ written in the original $s^{\mathbf{b}}$ -basis. This proves the matrix formula without replacing the original source map or its unit.

24.3 Exact integer depths, quotient coordinates, and reduction

For each integer $r \geq 0$ define

$$d_r = v_p(r!) = \sum_{a \geq 1} \left\lfloor \frac{r}{p^a} \right\rfloor, \quad r! = p^{d_r} \varepsilon_r, \quad \varepsilon_r \in \mathbb{Z}_{(p)}^\times \subset S^\times. \tag{TCL.14}$$

Here $0! = 1$, and the sum is finite because its terms vanish when $p^a > r$. Each integer j between one and r contributes one factor of p for each positive a with $p^a \mid j$. Interchanging these two finite counts yields $\sum_a \#\{1 \leq j \leq r : p^a \mid j\} = \sum_a \lfloor r/p^a \rfloor$, proving the valuation formula. The remaining integer factor is prime to p , hence is a unit in S . Therefore $(r!) = p^{d_r} S$ exactly. The depth cannot increase in S : an equality $p^d = p^{d+1}s$ would permit cancellation in the domain and give $1 = ps$, contradicting $\tilde{\alpha}(1) = 1$ and $\tilde{\alpha}(p) = 0$. This argument proves strictness of every consecutive pair of principal p -power ideals, without assigning a valuation to arbitrary elements of S .

Set $C = \Lambda_U/L_U$. The following are actual quotient maps:

$$\begin{aligned} q_r : E_S &\longrightarrow S/(p^{d_r}), \quad q_r(x) = \lambda_r(U^{-1}x) \bmod p^{d_r}, \\ C &\xrightarrow{\sim} \bigoplus_{r=0}^K S/(p^{d_r}), \quad \left[\sum_r c_r W_r \right] \longmapsto (c_r \bmod p^{d_r})_r, \\ E_S/L_U &\xrightarrow{\sim} F_U \oplus \bigoplus_{r=0}^K S/(p^{d_r}). \end{aligned} \tag{TCL.15}$$

The convention is $S/(p^0) = 0$. Each q_r is linear, vanishes on F_U and on L_U , and takes W_t to δ_{rt} in its indicated target. Equations (TCL.9)–(TCL.10) prove the two isomorphisms and all these claims. For the coordinate class $\nu_r = [W_r] \in C$, its exact annihilator in S is $p^{d_r} S$, including the unit ideal when $\nu_r = 0$ and $d_r = 0$.

Let B denote either S/pS with its quotient map, or the specified target k_0 with the map $\tilde{\alpha}$. In both cases $B \neq 0$ and has characteristic p . Append a bar to each base-changed module, element and map. Equations (TCL.8)–(TCL.10) base change directly, because their projections

and inverse changes of basis have their coefficients in S . Thus $\overline{W}_0, \dots, \overline{W}_K$ remain a basis of the direct summand $\overline{\Lambda}_U \subset \overline{E}$. The original map becomes

$$\begin{aligned} \overline{f}_U(\overline{T}^r) &= \begin{cases} \overline{r!} \overline{W}_r & 0 \leq r < p, \\ 0 & p \leq r \leq K, \end{cases} \\ \ker \overline{f}_U &= \bigoplus_{r=p}^K B \overline{T}^r = (\overline{T}^p), \\ \operatorname{im} \overline{f}_U &= \bigoplus_{r=0}^{\min(K, p-1)} B \overline{W}_r. \end{aligned} \tag{TCL.16}$$

An empty direct sum is zero, and the ideal $(\overline{T}^p)$ is zero when $p > K$. For $r < p$, every factor of $r!$ is a nonzero element of the prime field and therefore a unit in B . For $r \geq p$ the factorial has a factor p and maps to zero. Applying these facts to the independent $\overline{W}_r$ proves exactly the displayed kernel and image.

The nilpotency indices of the indicated actions are

module and action	nilpotency index
$\overline{\mathcal{C}}_S = B[T]/(T^{K+1}), M_T$	$K + 1$
$\operatorname{im} \overline{f}_U, M_{\overline{J}}$	$\min(p, K + 1)$
$\overline{E}, M_{\overline{J}}$	$\min(p, K + 1)$
$\overline{\Lambda}_U, M_{\overline{J}}$	$\min(p, K + 1).$

(TCL.17)

The nilpotency index is the least positive integer whose power is zero; thus it is one for the zero action on a nonzero module. In the first row, $T^K \neq 0$ in its free monomial basis and $T^{K+1} = 0$. On $\overline{E}$, the binomial coefficients $\binom{p}{i}$ for $0 < i < p$ are divisible by p : their factorial expression has one factor of p in its numerator and none in its denominator. The commuting-variable binomial formula therefore gives $\overline{J}^p = \sum_i \overline{y}_i^p = 0$, since $p > m$. Also $\overline{J}^{K+1} = 0$. If $d = \min(p - 1, K)$, then $\overline{J}^d = \overline{d!} \overline{V}_d \neq 0$, by its unit pivot coordinate. This proves the third row. Its application to $\overline{W}_0 = \overline{U}$ gives $\overline{J}^d \overline{W}_0 = \overline{d!} \overline{W}_d \neq 0$, proving the second and fourth rows together with their inherited upper bounds. In particular the abstract reduced cyclic source has not acquired the smaller index: its exact map to the target has lost the directions in (TCL.16).

The reduced containing lattice has the more precise chain decomposition

$$\overline{\Lambda}_U = \bigoplus_{\substack{j \geq 0 \\ jp \leq K}}^{\min(p-1, K-jp)} \bigoplus_{t=0} B e_{j,t}, \quad e_{j,t} = \left(\prod_{a=1}^t \overline{jp + a} \right) \overline{W}_{jp+t}. \tag{TCL.18}$$

The empty product is one. All displayed products are units because $1 \leq a \leq p - 1$. These are therefore changes of basis on disjoint intervals of the basis $\overline{W}_r$. Formula (TCL.12) gives $\overline{J} e_{j,t} = e_{j,t+1}$ except at the last position of a displayed interval. There it is zero, either because the next coefficient is $\overline{(j+1)p} = 0$, or because the next index is $K + 1$. This proves each full chain, including every chain that is absent from the image in (TCL.16).

The lost directions are also the exact connecting image in the base-change sequence

$$\begin{aligned} 0 &\longrightarrow \mathcal{C}_S \xrightarrow{f_U} \Lambda_U \longrightarrow C \longrightarrow 0, \\ 0 &\longrightarrow \operatorname{Tor}_1^S(C, B) \xrightarrow{\partial} \overline{\mathcal{C}}_S \xrightarrow{\overline{f}_U} \overline{\Lambda}_U \longrightarrow C \otimes_S B \longrightarrow 0, \\ \operatorname{Tor}_1^S(C, B) &= \bigoplus_{r=p}^K B \epsilon_r, \quad \partial(\epsilon_r) = \overline{T}^r, \quad C \otimes_S B \simeq \bigoplus_{r=p}^K B[\overline{W}_r]. \end{aligned} \tag{TCL.19}$$

Here Tor_1 can be computed directly, without a flatness assertion about B . The first line is a free resolution of C of length one, whose differential in the displayed bases is Δ . Tensoring this two-term complex with B gives differential $\bar{\Delta}$. By definition its degree-one homology is $\text{Tor}_1^S(C, B)$. The factorial calculation in (TCL.16) gives its kernel with the displayed basis and gives its cokernel with the displayed basis. The degree-one differential was chosen as $+f_U$, so the map of the kernel into the degree-one free module is precisely $\epsilon_r \mapsto \bar{T}^r$, with no additional sign. This proves the entire sequence and its maps explicitly.

For the top original vector the formula, including its full scalar, is

$$f_U(T^K) = \frac{K!}{((m-1)!)^k} a_0^k \prod_{i=1}^k y_i^{m-1}. \quad (\text{TCL.20})$$

Only $\mathbf{b} = (m-1, \dots, m-1)$ has total degree K , so (TCL.7) gives its multinomial coefficient. Every nonconstant term of U annihilates that monomial; its constant coefficient is exactly a_0^k , proving the formula. Its scalar is nonzero in the actual characteristic-zero domain, and its exact factorial depth is d_K , because $(m-1)!$ and a_0 are units. The displayed vector maps to zero under the specified characteristic p coefficient map exactly when $K \geq p$. This statement concerns the already specified local single-primary source. It does not extend that specialization to a larger coefficient ring containing additional inverse CRT denominators. A ring in which p is a unit cannot map unitally to characteristic p , since applying such a map to $pp^{-1} = 1$ would give $0 = 1$.

24.4 The exact original chain-support map for these depths

Fix an integer $L \geq \max(1, d_K)$ and put

$$\begin{aligned} R_{p,L} &= \mathbb{Z}/p^L\mathbb{Z}, & S_L &= S/p^L S, \\ \iota_L : R_{p,L} &\longrightarrow S_L, & [z] &\longmapsto [z], \\ I_a &= p^a R_{p,L}, & H_a &= p^a S_L \quad (0 \leq a \leq L). \end{aligned} \quad (\text{TCL.21})$$

The scalar map is well-defined and unital. It is injective. Indeed, if an integer z not divisible by p^L has image zero, write $z = p^a u$ with $0 \leq a < L$ and u prime to p . Then $p^a u = p^L s$ in S . Cancellation and inversion of u imply $1 = p^{L-a} s u^{-1}$, contradicting the specified residue map. Thus precisely $p^L \mathbb{Z}$ maps to zero, proving injectivity. The same argument proves $p^a \notin p^{a+1} S_L$ for $a < L$: a purported equality lifts to $p^a = p^{a+1} s + p^L t$, and cancellation gives $1 = ps + p^{L-a} t$, again impossible in the residue field. Consequently all H_a are distinct.

For completeness, every ideal of $R_{p,L}$ is some I_a . For a nonzero ideal, choose a nonzero element with least exponent of p among its integer representatives. It has the form $p^a u$ with u a unit modulo p^L , by the integer Euclidean algorithm. Multiplying by u^{-1} puts p^a in the ideal, and all its other elements have exponent at least a by choice, proving that the ideal is I_a . The zero ideal is I_L . This argument is not applied to S_L : there we retain exactly the principal subchain $\{H_a\}$.

Extension and contraction on this actual subchain are

$$\iota_L(I_a)S_L = H_a, \quad \iota_L^{-1}(H_a) = I_a, \quad (0 \leq a \leq L). \quad (\text{TCL.22})$$

Extension follows because the image of the principal generator is p^a . For contraction, an integer of exponent $b < a$ belonging to $p^a S_L$ would give $p^b u = p^a s + p^L t$. Cancelling p^b contradicts the nonzero residue of the unit u . Thus only the integers divisible by p^a contract, and they all do, proving the second formula.

The original ideal laws are retained exactly:

$$\begin{aligned} I_a + I_b &= I_{\min(a,b)}, & I_a \cap I_b &= I_{\max(a,b)}, & I_a I_b &= I_{\min(L, a+b)}, \\ H_a + H_b &= H_{\min(a,b)}, & H_a \cap H_b &= H_{\max(a,b)}, & H_a H_b &= H_{\min(L, a+b)}. \end{aligned} \quad (\text{TCL.23})$$

For sum and intersection the proof is inclusion: when $a \leq b$, the ideal indexed by b is contained in the one indexed by a . Their sum is the larger and their intersection the smaller. For product,

every sum of products of multiples of p^a and p^b is a multiple of p^{a+b} , and the generator p^{a+b} itself is such a product. It becomes zero for $a + b \geq L$. These arguments apply in both displayed rings and prove all six laws, including the boundary indices.

Let $C_L = \{0 < 1 < \dots < L\}$. The original lattice map and its actual coefficient extension are

$$\theta_p : C_L \longrightarrow \text{Id}(R_{p,L}), \quad j \longmapsto I_{L-j}, \quad \theta_S : C_L \longrightarrow \{H_a : 0 \leq a \leq L\}, \quad j \longmapsto H_{L-j}. \quad (\text{TCL.24})$$

They are bijections by the distinctness and classification just proved. Each takes bottom to the zero ideal, top to the unit ideal, maximum to ideal sum and minimum to ideal intersection, by (TCL.23). Thus they are bounded-lattice isomorphisms onto their stated codomains. Extension $I_a \mapsto H_a$ commutes with these two maps and preserves all three ideal operations of (TCL.23). In the j -coordinates, ordinary ideal multiplication is $j \odot t = \max(0, j + t - L)$, since its exponent is $\min(L, (L - j) + (L - t))$. It is not identified with lattice meet: for $L \geq 2$, $j = t = L - 1$ gives respective values $L - 2$ and $L - 1$.

For either coefficient ring $B = R_{p,L}$ or $B = S_L$, use precisely the original split-support carrier

$$\begin{aligned} G_{C_L}(B) &= \{(0_B, j) : 0 \leq j \leq L\} \cup \{(b, L) : b \in B\}, \\ (b, j) + (c, t) &= (b + c, \max(j, t)), \quad (b, j)(c, t) = (bc, \min(j, t)), \\ \tau_a &= (0_B, L - a) \quad (0 \leq a \leq L), \quad e = \tau_0 = (0_B, L), \quad \tau_L = (0_B, 0). \end{aligned} \quad (\text{TCL.25})$$

The union is inside $B \times C_L$, so the two occurrences of $(0_B, L)$ give one element. A nonzero amplitude occurs only at top support. A nonzero sum has a nonzero summand, so has top support; a nonzero product has both amplitudes nonzero, so has top support. This proves closure. Associativity and commutativity hold in each coordinate. Distributivity follows from ring distributivity and $\min(j, \max(t, v)) = \max(\min(j, t), \min(j, v))$, verified by ordering t, v . The zero is τ_L , and the unit is $(1_B, L)$. These facts prove the stated semiring structure directly.

The exact support-preserving coefficient morphism is

$$\begin{aligned} G_{C_L}(\iota_L) : G_{C_L}(R_{p,L}) &\longrightarrow G_{C_L}(S_L), \quad (b, j) \longmapsto (\iota_L(b), j), \\ \chi_B : G_{C_L}(B) &\longrightarrow C_L, \quad (b, j) \longmapsto j, \\ \theta_S \chi_{S_L} G_{C_L}(\iota_L) &= \text{ext}_{\iota_L} \theta_p \chi_{R_{p,L}}. \end{aligned} \quad (\text{TCL.26})$$

The first map is well-defined because nonzero target amplitude has nonzero source amplitude and hence top support. Applying the ring homomorphism to amplitudes and leaving the support coordinate fixed proves preservation of both operations, zero and unit. Its injectivity follows from injectivity of ι_L and the unchanged support coordinate. The support projection preserves the two operations by their defining formulas. The last equality follows on every entry (b, j) from $I_{L-j}S_L = H_{L-j}$; it is an equality into the principal ideal subchain with ideal sum and ideal intersection. In particular each τ_a maps to the distinct τ_a , then to I_a or H_a . The supported zero e remains distinct from the absent zero τ_L .

Finally the connection of this support diagram to the actual cyclic quotient is not an identification of a torsion module with a semiring. The precise coordinate and annihilator maps are

$$\begin{aligned} C &\simeq \bigoplus_{r=0}^K S/(p^{d_r}), \quad p^L C = 0, \\ \text{Ann}_{S_L}(\nu_r) &= H_{d_r} = \theta_S(L - d_r) = \theta_S \chi_{S_L}(\tau_{d_r}), \\ q_r(x) &= \lambda_r(U^{-1}x) \bmod p^{d_r}. \end{aligned} \quad (\text{TCL.27})$$

The first and last lines are the proved maps of (TCL.15), with p^L annihilating every summand because $d_r \leq L$. An element of S_L kills the unit class of the r -th summand exactly when a representative in S lies in $p^{d_r}S$; this proves the annihilator formula and all its equalities using

(TCL.24)–(TCL.26). Thus the depths of the original factorial inclusion have explicit labels in the original chain-support carrier, with their arithmetic ideal product as well as their lattice sum and intersection still recorded. The entry τ_{d_r} labels the annihilator; it does not replace the vector ν_r . In particular, when $d_r = L$ the vector ν_r is the nonzero unit class in S_L , whereas its annihilator is the zero ideal labelled by τ_L .

24.5 The retained boundary-character labels and the full-stage obstruction

The action on the lost source directions is also exact. For $K \geq p$, the kernel in (TCL.16), equivalently the connecting image of the Tor group in (TCL.19), has the ordered basis $\bar{T}^p, \dots, \bar{T}^K$, with

$$M_T \bar{T}^r = \bar{T}^{r+1} \quad (p \leq r < K), \quad M_T \bar{T}^K = 0, \quad \text{ind}(M_T|_{\ker \bar{J}_U}) = K - p + 1. \quad (\text{TCL.28})$$

These equalities follow in the free monomial source basis. In particular the $(K - p)$ -th power takes $\bar{T}^p$ to the nonzero $\bar{T}^K$, and the next power is zero on every basis element. This index can exceed p . It is an action on the exact kernel in the source, not a subspace of the ambient target. Thus it is compatible with $\bar{J}^p = 0$ on $\bar{E}$: the original map is zero on every one of these kernel vectors.

Put $N = m + 1$. For a residue class $c \in \mathbb{Z}/N\mathbb{Z}$ define the original coefficient character masks in y -coordinates and their full-unit transports by

$$\begin{aligned} Q_c(y^{\mathbf{b}}) &= \mathbf{1}_{\{|\mathbf{b}|+k \equiv c \pmod{N}\}} y^{\mathbf{b}}, & \widehat{Q}_c &= U Q_c U^{-1} : E_S \longrightarrow E_S, \\ D_c(T^r) &= \mathbf{1}_{\{r+k \equiv c \pmod{N}\}} T^r & \text{on } \mathcal{C}_S. \end{aligned} \quad (\text{TCL.29})$$

Each diagonal entry defining Q_c, D_c is zero or one in S , so these maps exist over this exact coefficient ring without adjoining roots of unity. On every indicated basis vector exactly one of the masks is one. Hence each family consists of pairwise orthogonal algebraic idempotents summing to the identity; conjugation by the actual U proves the same assertion for $\widehat{Q}_c$. The word orthogonal here means that distinct idempotents compose to zero; no inner product or orthogonal projection on a metric space is introduced.

The full original boundary-character comparison is

$$\begin{aligned} Q_c V_r &= \mathbf{1}_{\{r+k \equiv c\}} V_r, & \widehat{Q}_c W_r &= \mathbf{1}_{\{r+k \equiv c\}} W_r, \\ \widehat{Q}_c f_U &= f_U D_c, & \widehat{Q}_c J &= J \widehat{Q}_{c-1}, & D_c M_T &= M_T D_{c-1}. \end{aligned} \quad (\text{TCL.30})$$

Every summand of V_r has degree r , proving the first equality, and conjugation proves the second. Apply these formulas to $f_U(T^r) = r! W_r$ to prove the third equality on a source basis. Multiplication by each y_i raises degree by one for every surviving monomial and gives zero at the truncation boundary. It follows on the complete monomial basis that $Q_c J = J Q_{c-1}$. Conjugation and $UJ = JU$ prove the fourth identity; the identical degree argument on T^r proves the last. These are covariant character shifts, not a claim that a fixed character mask commutes with J .

Over $\mathbb{C}$, this is exactly the coefficient mask of the boundary monodromy in TP.9–10: that period monodromy has eigenvalue $\exp(2\pi i(|\mathbf{b}| + k)/N)$ on its period character indexed by $\mathbf{b}$. In the unchanged original $s^{\mathbf{b}}$ -coordinates the coefficient mask is $P_k^{-1} Q_c P_k$, and the source mask is its conjugate by $P_k^{-1} [U]_{(y^{\mathbf{b}})} P_k$. The complex period matrix is $\Pi_k = e^{k c_0 / u} W_k D_k(u) P_k$, with the original phase constant here denoted $c_0 = -(-\rho)^N / N$, and $D_k(u)$ diagonal in these same $\mathbf{b}$ indices. Consequently

$$\mathcal{P}_c \Pi_k = \Pi_k (P_k^{-1} Q_c P_k), \quad \mathcal{P}_c \Pi_k [U]_{(s^{\mathbf{b}})}^{-1} = \Pi_k [U]_{(s^{\mathbf{b}})}^{-1} [\widehat{Q}_c]_{(s^{\mathbf{b}})}, \quad \mathcal{P}_c = W_k Q_c W_k^{-1}. \quad (\text{TCL.31})$$

Indeed Q_c commutes with the diagonal $D_k(u)$, which proves the first equality by multiplication, and substitution of the conjugation formula for $\widehat{Q}_c$ proves the second. The common scalar $e^{k c_0 / u}$, the original P_k and the entire U are present in these identities.

The masks preserve both L_U and Λ_U , as follows from their diagonal action on $r!W_r$ and W_r . They therefore act on the quotient C , and the exact free resolution in (TCL.19) commutes with D_c on its source and $\widehat{Q}_c$ on its target. Its base change and its kernel inherit precisely those same masks. Thus the simultaneously retained depth and character of every quotient coordinate and lost direction are

$$\begin{aligned} \nu_r = [W_r] : \quad \text{Ann}_{S_L}(\nu_r) = H_{d_r}, \quad \widehat{Q}_c \nu_r = \mathbf{1}_{\{r+k \equiv c\}} \nu_r, \\ \epsilon_r \in \text{Tor}_1^S(C, B) \quad (r \geq p) : \quad \overline{D}_c \partial \epsilon_r = \mathbf{1}_{\{r+k \equiv c\}} \overline{T}^r. \end{aligned} \quad (\text{TCL.32})$$

The first line combines the proved annihilator in (TCL.27) with the induced quotient map. The second follows from the explicit connecting map $\partial \epsilon_r = \overline{T}^r$. This proves both labels on the actual maps, including the zero coordinates when $d_r = 0$. It does not assert an inertia-invariant description of a different sheaf after reduction.

There is an exact obstruction to carrying this local specialization through the full amplified quartet CRT stage. In the counterfactual quartet of PAM.1–4 retain $\rho = \frac{1}{2} + \delta + i\gamma$, with $0 < \delta < \frac{1}{2}$, $\gamma > 0$, and take $k \geq p$. Two actual ordered tensor centres in that construction are

$$\lambda_0 = k\rho, \quad \lambda_1 = (k-p)\rho + p(1-\bar{\rho}), \quad \lambda_0 - \lambda_1 = p(\rho + \bar{\rho} - 1) = 2p\delta \neq 0. \quad (\text{TCL.33})$$

The first chooses ρ in every factor; the second chooses $1-\bar{\rho}$ in exactly p factors and ρ in the other $k-p$ factors. Both choices are permitted in the retained quartet, and the difference follows by subtraction. In the grid coordinates of PAM.2 they have indices respectively $(a, b) = (k, k)$ and $(k-p, k)$.

Let $A_{\text{CRT}} \subset \mathbb{C}$ be the actual coefficient ring carrying those quartet coordinates and every coefficient of the retained terminal CRT idempotent polynomial representative $e(Z) = \sum_{r=0}^d e_r Z^r$ at λ_0 . By its specified CRT residues, $e(\lambda_0) = 1$ and $e(\lambda_1) = 0$. Define the following element of this same ring:

$$\mathcal{D} = \sum_{r=1}^d e_r \sum_{j=0}^{r-1} \lambda_0^{r-1-j} \lambda_1^j \in A_{\text{CRT}}, \quad 1 = p(\rho + \bar{\rho} - 1)\mathcal{D}, \quad p^{-1} = (\rho + \bar{\rho} - 1)\mathcal{D} \in A_{\text{CRT}}. \quad (\text{TCL.34})$$

For every $r \geq 1$, multiplying the inner sum by $\lambda_0 - \lambda_1$ cancels all intermediate monomials and leaves $\lambda_0^r - \lambda_1^r$. After multiplication by e_r and summation the result is $e(\lambda_0) - e(\lambda_1) = 1$. Substitution of (TCL.33) proves the second equality and then the displayed inverse of p . This proof uses the original CRT polynomial itself and does not depend on a selected formula for its coefficients or on an assumed list of inverted pairwise differences. Since p is a unit, no prime ideal of A_{CRT} lies over (p) , and there is no unital characteristic- p specialization of that complete stage. A prime containing p would contain 1, and any such specialization would send the proved identity $pp^{-1} = 1$ to $0 = 1$.

Thus for $k \geq p$ the actual local prime used in (TCL.1) does not extend to this full quartet stage with its retained terminal CRT idempotent. This is a proved obstruction for that specified stage, not a statement that the local and global characteristic-zero objects are unrelated. Their local cyclic coefficient map is still exactly (TCL.4), their unchanged monomial translation is (TCL.13), and the characteristic-zero terminal coefficient is (TCL.20). The local characteristic- p quotient and its lost directions are the maps (TCL.16)–(TCL.19); a missing extension of their coefficient map is not the image of a global vector under a nonexistent specialization.

More generally let $A_{\text{CRT},k} \subset \mathbb{C}$ denote the retained coefficient ring of the complete split corner at degree k . The argument (TCL.33)–(TCL.34), applied to every rational prime $q \leq k$ in place of p , proves

$$q^{-1} \in A_{\text{CRT},k} \quad (q \text{ prime, } q \leq k). \quad (\text{TCL.35})$$

Indeed the second centre uses q reflected factors, which are available for every such q , and the same retained idempotent separates it from the terminal centre. Its divided difference gives the

inverse without changing the idempotent. For a directed coefficient ring containing all these actual split stages one may take the explicit subring

$$A_{\text{CRT},\infty} = \bigcup_{n \geq 2} \mathbb{Z}[A_{\text{CRT},2}, \dots, A_{\text{CRT},n}] \subset \mathbb{C}, \quad \mathbb{Q} \subset A_{\text{CRT},\infty}. \quad (\text{TCL.36})$$

The finite generated-stage subrings in this union form an increasing sequence, so the union is a subring containing every stage with its actual coefficient maps. For any prime q , the stage $k = q$ supplies q^{-1} by (TCL.35). Factoring a nonzero integer into primes then puts its reciprocal in the union, proving the asserted copy of $\mathbb{Q}$. Every residue field of this union therefore has characteristic zero: a characteristic- q ring homomorphism would send the unit identity $qq^{-1} = 1$ to $0 = 1$. Equivalently the scalar morphism $\text{Spec } A_{\text{CRT},\infty} \rightarrow \text{Spec } \mathbb{Z}$ has no point over any (q) . This follows directly by contraction of primes and the proved invertibility of every q . These conclusions concern the full split corner diagram with its retained CRT idempotents. They do not remove the finite single-primary models or the local maps (TCL.1)–(TCL.32).

There are also exact surviving character-zero vectors in the reduced local cyclic image. Take $m \geq 2$ and $k = p\ell$ with $\ell \geq 1$, so $K = k(m-1) \geq p$. Its final nonzero cyclic image basis vector and its unchanged characteristic-zero precursor satisfy

$$\begin{aligned} \bar{J}\bar{W}_{p-1} &= 0, & \bar{W}_{p-1} &\neq 0, & JW_{p-1} &= pW_p \neq 0, \\ \bar{Q}_0 \bar{W}_{p-1} &= \bar{W}_{p-1} & \iff & & p(\ell+1) &\equiv 1 \pmod{N}, & N &= m+1. \end{aligned} \quad (\text{TCL.37})$$

The first line follows from (TCL.12), the unit pivot after reduction, and the nonzero scalar p on the free characteristic-zero containing lattice. The vector belongs to the original reduced cyclic image because $\bar{f}_U(\bar{T}^{p-1}) = \overline{(p-1)!} \bar{W}_{p-1}$, whose scalar is a unit. Formula (TCL.30) gives the second line as $p-1+k \equiv 0 \pmod{N}$, and substitution of $k = p\ell$ gives the stated congruence. In the retained coefficient datum $p > N$, so p is invertible modulo N . Choosing the single residue class $\ell \equiv p^{-1} - 1 \pmod{N}$ and then its infinitely many positive representatives proves that the displayed character-zero case occurs for infinitely many tensor orders $k = p\ell$. These vectors are killed by J only after base change, as their exact characteristic-zero image pW_p shows. This is a calculation in the local coefficient module with its retained character masks, not an identification with a finite-field sheaf vector or a recovery of the original terminal jet.

Every tensor coefficient, the original monomial translations, the full unit and inverse, and the specified coefficient map occur in the displayed morphisms. The characteristic- p kernel in (TCL.16) is a statement about these coefficient modules. It neither sets an integer scalar to zero in $\bar{\mathbb{Q}}_\ell$ nor identifies this reduction with the inertia or Frobenius action on the finite-field sheaf. No verdict about the location of an actual zero of ξ follows from this lattice calculation alone.